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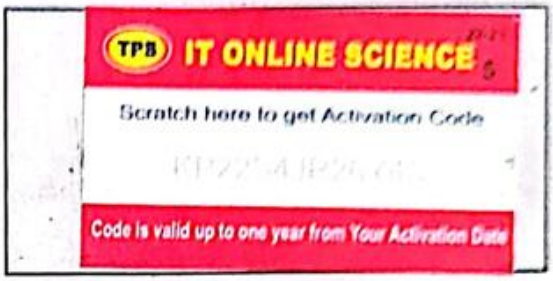

TPS INFORMATION TECHNOLOGY

ONLINE

STD. XII SCIENCE

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BALBHARATI REGISTRATION NO. 2019MH0032

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PLEASE FOLLOW THIS INSTRUCTIONS BEFORE USING THIS SOFTWARE

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Instructions for the students to be followed while appearing for online exam

1. Do not use short forms in answers you write.	**Example :** Write Hyper Text Markup Language not HTML
2. HTML tags should be compulsory written in <>.	**Example :** Write <form> not form
3. Write digits answers and not in word	**Example :** Write 6 not six
4. Write Symbols in answer not words.	**Example :** Write # not hash
5. Write: (Dot) for file extension.	**Example :** Write .jpg not jpg
6. Write answers in uppercase or lower case	
7. Write **alt+p** not alt p	

Information Technology

Maharashtra State Board of Secondary and Higher Secondary Education, Pune 411 004

Annual Examination Question Paper Format for XII (Science) IT

Marks : 80 **Duration : 2 Hrs : 30 Mins**

Q. No.	Sub Q. No	Description	Number of Questions	Marks
1.		Fill in the Blanks	10	10
2.		State True or False	10	10
3.		Multiple choice SINGLE answer	10	10
4.		Multiple choice TWO correct answer	10	20
5.		Multiple choice THREE correct answer	02	06
6.		Match the following	04	04
7.		Answer briefly (any 5)	08	10
8.	A A	Write Codes in HTML OR Write codes in HTML	02	05
8	B B	Write a program in JavaScript OR Write a program in JavaScript	02	05
		Total		**80**

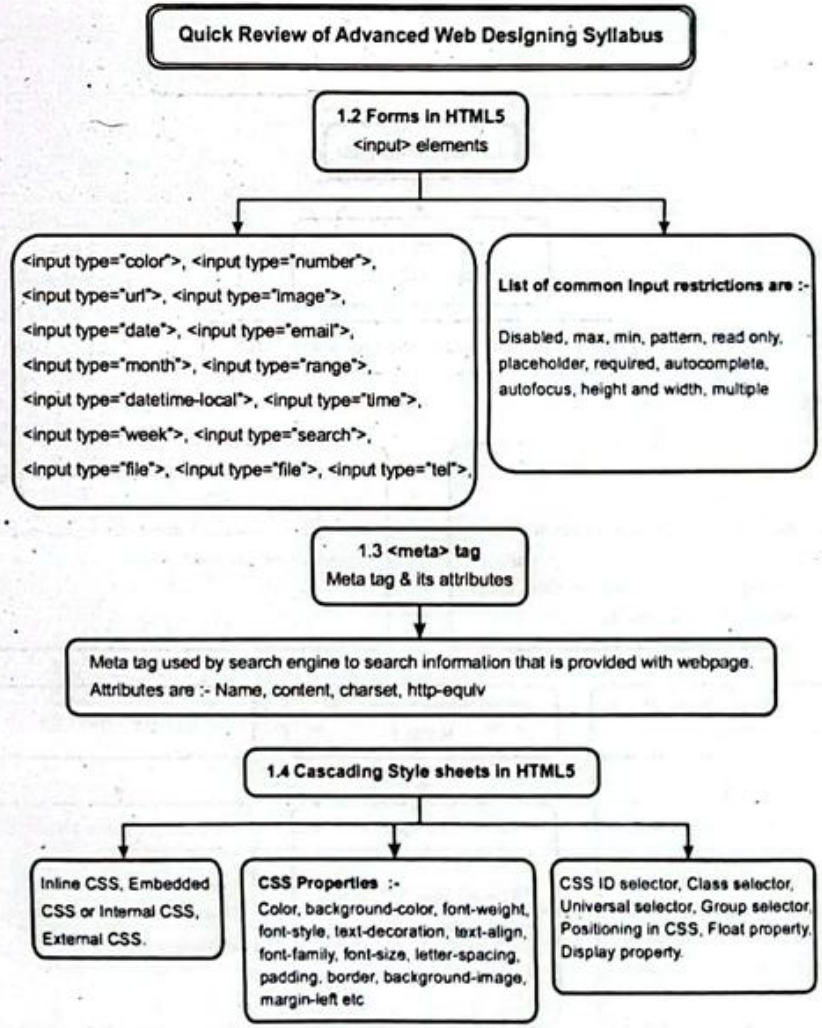
Topic wise Mark Distribution / Weightage for Annual Examination XII SCIENCE - IT (2022-23)

Chapter No.	Chapter Name	Marks	Marks with Options
1.	Advanced Web Designing	20	25
2.	Introduction to SEO (Search Engine Optimization)	10	12
3.	Advanced JavaScript	15	20
4.	Emerging Technologies (IOT, cloud Computing, AI, 5G)	10	12
5.	Server-Side Scripting (PHP)	15	15
6.	E-Commerce and E-governance	10	12
	Total Marks	80	96

MARKS DISTRIBUTION (THEORY) XII SCIENCE - IT									
Chapter Name	Q1 (10)	Q2 (10)	Q3(10)	Q4 (20)	Q5 (6)	Q6 (4)	Q7 (10)	Q8 (10)	Total
Advanced Web Designing	2	1	2	6	3	1	–	5 OR 5	20 + 5 Option
Introduction to SEO (Search Engine Optimization)	2	1	2	2	–	1	4	–	10 + 2 Option
Advanced JavaScript	2	2	1	4	–	1	–	5 OR 5	20 + 5 Option
Emerging Technologies (IOT, cloud Computing, AI, 5G)	2	1	1	2	–	–	6	–	10 + 2 Option
Server-Side Scripting (PHP)	1	3	3	4	3	1	–	–	15
E-Commerce and E-governance	1	2	1	2	–	–	6	–	10 + 2 Option
Total Marks	10	10	10	20	6	4	10 + 6 Option	10 + 10 Option	80 + 16 Option

1 Advanced Web Designing

Chapter at a Glance



1.5 Ordered list or Numbered list

 tag defines ordered list.
Attributes of <ol> tag are:- Type, reversed, start.

1.6 Unordered list or bulleted list

 tag defines unordered list.
Attributes of <ul> tag are:- Type.

1.7 Defination list

<dl> tag
<dd>, <dt> tag

1.8 Inserting audio and video in HTML5

<audio> element enables to embed audio files on web pages.
Attributes are: - src, autoplay, controls, loop, muted.
<source> tag is used to specify multiple media resources for media elements.

<video> tag is used to embed video into your web page.
Attributes are: - src, autoplay, controls, loop, muted, poster, preload, width.

1.9 Image map in HTML5

<img> tag is used to insert an image on web page.
Usemap attribute acts as a pointer. <map> tag , <area> tag specified clickable regions.
Attributes are:- href, shape, cords, alt.

1.10 Inline Frames in HTML5

<iframe> tag creates inline frames.
Attributes are:- src, height, width, name, srcdoc.

1.11 Web Hosting

Web hosting is the service of providing storage space.
Types of Web hosting are :- Shared hosting, Free hosting, Dedicated hosting.

Textbook Exercise

Fill in the Blanks

1. The.....element is a staring element in an HTML, it indicates that document type definition being used by the document. **March 2022**

Ans. : <!DOCTYPE html>

2. The..... is a tag in html that describe some aspects of contents of a webpage.

Ans. : <meta>

3. The <ol> tag defines an.....

Ans. : Ordered List

4. An unordered list created using the ..... tag. **March 2022**

**Ans. : **

5. The ..... element creates an inline frame.

Ans. : <iframe>

6. .....tag is used to specify video on an HTML document.

Ans. : <video>

7. If a web developer wants to add the description to an image he must use ..... attribute of <img> tag.

Ans. : alt

8. The.....property is used to set position for an element.

Ans. : Position

9. The float property defines the..of content. **March 2022**

Ans. : Flow

10. .....is used with elements that overlap with each other.

Ans. : Positioning

State whether the following statement is True or False

1. HTML is an Object Oriented Programming Language.

Ans. : False

2. Charset is used for character encoding declaration. **March 2022**

Ans. : True

3. An unordered list can be numerical or alphabetical.

Ans. : False

4. Multilevel list can be created in HTML 5.

Ans. : True

5. Srccode specifies the HTML content of the page to show in the <iframe>

Ans. : True

6. The 'controls' attribute is not used to add play, pause, and volume.

Ans. : False

7. .cs is the extension of CSS file. **March 2022**

Ans. : False

Choose Single correct answer from the given options.

1. .....element used to create a linking image
   (a) <img> (b) <td> (c) <map> (d) <usemap>

Ans. : (a)

2. The .....tag is used to embed audio files on Webpages.
   (a) <sound> (b) <audio> (c) <video> (d) <embeded>

Ans. : (b)

3. A programmer wants to define range for age between 18 to 50, he will use a form with following appropriate control.
   (a) number (b) compare (c) range (d) Textboxes

Ans. : (c)

4. .....character is used to create id in CSS.
   (a) % (b) $ (c) @ (d) #

Ans. : (d)

Choose Two correct answers from the given options.

1. List within another list either.....list or.....list is called nested list.
   (a) multilevel (b) ordered
   (c) unordered (d) general (e) cascading

Ans. : (b,c)

2. Image maps are of two types .....and .....
   (a) Network side (b) Client Side
   (c) Computer side (d) Server Side (e) n-computing

Ans. : (b,d)

3. A CSS rule set contains..... and.....
   (a) Set (b) selector
   (c) post (d) declaration (e) block

Ans. : (b,d)

4. Client-side image map can be created using two elements .....and.....
   (a) <area> (b) <image>
   (c) <usemap> (d) <map> (e) <server>

Ans. : (a,d)

Choose Three correct answers from the given options.

1. Attributes of <area> tag is.....
   (a) href (b) src (c) coords
   (d) data (e) alt (f) usemap

Ans. : (a,c,e)

2. Attributes used with iframe are.....
   (a) srcdoc (b) name (c) alt
   (d) src (e) href (f) loop

Ans. : (a,b,d)

3. Following are the Form controls.....
   (a) email (b) search (c) label
   (d) video (e) tel (f) audio

Ans. : (a,b,e)

4. Attributes used with <audio> tag.....
   (a) autoplay (b) href (c) controls
   (d) cntrl (e) loop (f) bgsound

Ans. : (a,c,e)

5. CSS types are ..... and.....
   (a) internal (b) external (c) control
   (d) inline (e) loop (f) style

Ans. : (a,b,d)

6. Positioning types in CSS are.....
   (a) Static (b) fixed (c) absolute
   (d) position (e) dynamic (f) nested

Ans. : (a,b,c)

7. Types of floating properties are.....
   (a) left (b) zero (c) right
   (d) all (e) none (f) dock

Ans. : (a,c,e)

Match the pair

A		B	
1)		a)	Client side image map
2)	usemap	b)	CSS property
3)	color	c)	bulleted list
4)	<img>	d)	Image as a submit button
5)	<input type=image>	e)	inserts an image

Ans. : 1-c , 2-a , 3-b , 4-e , 5-d

Programs

1. Write a program using html with following CSS specification-
   (1) The background colour of the company name should be in green.
   (2) The text colour of the company name should be red.
   (3) The heading should be large with font "comic sans ms"
   (4) The description of the company should be displayed in blue color in a paragraph.

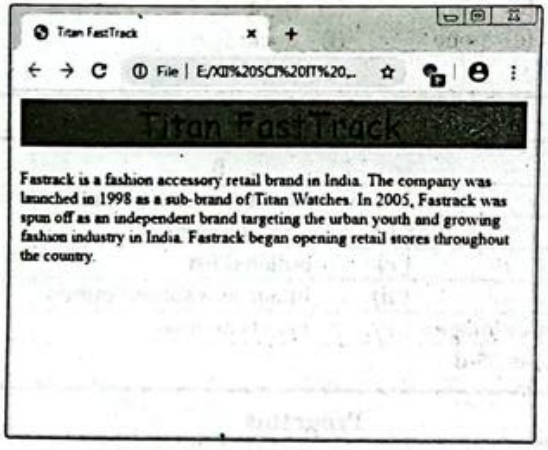
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
```

```
Titan FastTrack
</title>
<style>
h1{background-color:green;color:red;font-family:Comic Sans MS}
p{color:blue}
</style>
</head>
<body>
<h1 align="center">Titan FastTrack</h1>
<p>
Fastrack is a fashion accessory retail brand in India. The company was launched in 1998 as a sub-brand of Titan Watches. In 2005, Fastrack was spun off as an independent brand targeting the urban youth and growing fashion industry in India. Fastrack began opening retail stores throughout the country.
</p>
</body>
</html>
```

Output :



2. Write Html5 code with CSS as follows :

1. To create form to accept name,age, email address, from the user.
2. Create a submit button to send the data.
3. The heading of the form should have a background colour and a different font style.

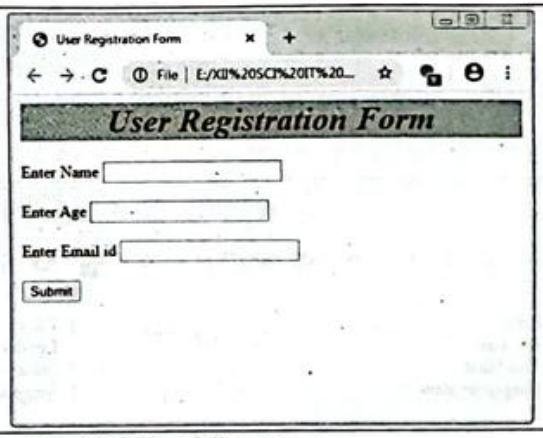
Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
User Registration Form
```



```
</title>
<style>
h1{background-color:orange;font-style:italic}
</style>
</head>
<body>
<h1 align="center">User Registration Form</h1>
<form>
Enter Name
<input type="text" name="t1"><br><br>
Enter Age
<input type="number" name="age"><br><br>
Enter Email id
<input type="email" name="email"><br><br>
<input type="submit" name="b1" value="Submit">
</form>
</body>
</html>
```

Output :



3. Write Html5 code with CSS as follows :

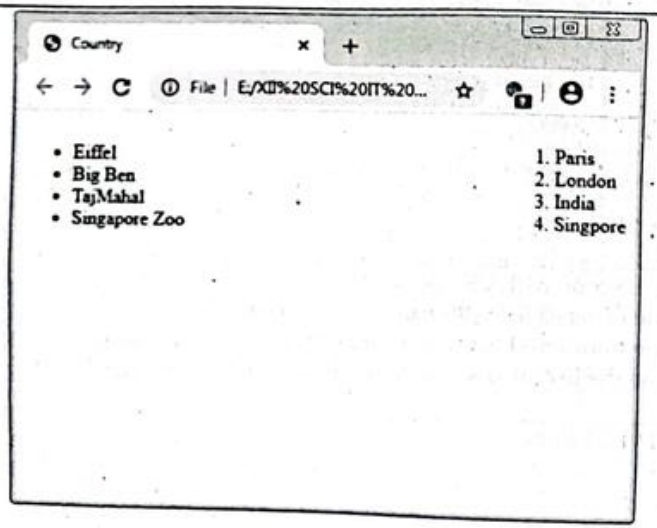
1. Create ordered list with names of tourist Cities.
2. Create unordered list with tourist PLACES of those cities.
3. Divide the list into two sections left and right by using CSS.

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Country
</title>
```

```
<style>
ul{float:left}
ol{float:right}
</style>
</head>
<body>
<ol>
<li>Paris</li>
<li>London</li>
<li>India</li>
<li>Singpore</li>
</ol>
<ul>
<li>Eiffel</li>
<li>Big Ben</li>
<li>TajMahal</li>
<li>Singapore Zoo</li>
</ul>
</body>
</html>
```

Output :



Fill in the Blanks

1.1 Advanced Web Designing

1. HTML stands for .....
Ans. : Hyper Text Markup Language
2. The ..... is an evolving language with different version supporting different features.
Ans. : HTML (Hyper Text Markup Language)
3. ..... is currently used because it supports mobile technology.
Ans. : HTML5

1.2 Forms in HTML5

4. The ..... tag is used to create a form. **March 2022**
Ans. : <form> </form>
5. <input type="....."> defines a color picker.
Ans. : color
6. <input type="....."> defines a field for entering a number. **March 2022**
Ans. : Number
7. <input type= "....."> defines a field for entering a URL.
Ans. : URL (Uniform Resource Locator)
8. <input type="....."> defines an image as a submit button. **March 2022**
Ans. : Image
9. <input type= "....."> defines a date picker with the year, month and day.
Ans. : Date
10. <input type="....."> defines a field for an email address. **March 2022**
Ans. : email
11. <input type= "....."> defines a month and year control in format as 'YYYY-MM'.
Ans. : Month
12. <input type= "....."> defines the range control. **March 2022**
Ans. : Range
13. <input type= "....."> defines a date picker that includes the year, month day and time. **March 2022**
Ans. : datetime –local
14. <input type="....."> defines a control for entering a time.
Ans. : time
15. <input type="....."> defines a week and year control.
Ans. : week
16. <input type= "....."> defines a text field for entering a search string.
Ans. : search
17. <input type= "....."> defines a file select field and a 'Browse' button for file uploads. **March 2022**
Ans. : file
18. Default range of range control is 0 to _____.
Ans. : 100

19. <input type= "....."> used to define input fields that should contain a telephone number. March 2022

Ans. : tel

20. The .....attribute of <input> tag specifies that an input field should be disabled.

Ans. : disabled

21. The .....attribute of <input> tag specifies the maximum value for an input field. March 2022

Ans. : max

22. The ..... attribute of <input> tag specifies the minimum value for an input field.

Ans. : min

23. The..... attribute of <input> tag specifies a regular expression to check the input values. March 2022

Ans. : Pattern

24. The..... attribute of <input> tag specifies that an input field is read only. March 2022

Ans. : read only

25. The ..... attribute of <input> tag acts as a temporary label showing the purpose of a text field without requiring a label tag.

Ans. : Placeholder

26. The ..... attribute of <input> tag specifies that an input field is required.

Ans. : required

27. The ..... attribute of <input> tag specifies whether a form or input field should have auto completed on or off.

Ans. : autocomplete

28. The ..... attribute of <input> tag specifies that the input field should automatically get focus when the page loaded. March 2022

Ans. : autofocus

29. The ..... attribute of <input> tag specifies the height of an <input type= "image">

Ans. : Height

30. The ..... attribute of <input> tag specifies the width of an <input type= "image"> March 2022

Ans. : Width

31. The ..... attribute of <input> tag specifies that the user is allowed to enter more than one value in the <input> element.

Ans. : multiple

32. The ..... attribute of <input> tag is used to identify the html element uniquely through the document object model.

Ans. : ID

33. The ..... attribute of <input> tag is used to apply CSS style to the individual input element. March 2022

Ans. : Class

1.3 <meta> tag

34. The .....tag describes aspects of contents of a web page.

Ans. : <meta>

35. The .....tag is used by search engines to search information that is provided with the web page. March 2022

Ans. : <meta>

36. .....will not be displayed on the web page.

Ans. : Metadata

37. .....tag is a singular tag which carries information within its attributes.

Ans. : <meta>

38. The ..... attribute of <meta> tag specifies the name of the meta data. March 2022

Ans. : Name

39. The .... attribute of <meta> tag can have any textual matter related to the name.

Ans. : Content

40. The ..... attribute of <meta> tag specifies the character encoding used by the document.

Ans. : Charset

41. The ..... attribute of <meta> tag is used for http response message headers.

Ans. : http-equiv

42. <meta http-equiv= ".....">, the page will get refresh after every five seconds.

Ans. : refresh

43. <meta http-equiv= ".....">, browser sends the cookies back to the server. March 2022

Ans. : set-cookies

44. <meta http-equiv= ".....">, specifies the character encoding for the document.

Ans. : content-type

45. <meta http-equiv= ".....">, page session will get expire at specified date and time.

Ans.: expires

1.4 Cascading Style Sheets HTML5

46. CSS stands for.....

Ans. : Cascading Style Sheets.

47. ..... can control the layout of multiple web pages all at once.

Ans. : CSS (Cascading Style Sheets)

48. ..... allows you to control the look and feel of several pages by changing a single source.

Ans. : CSS (Cascading style Sheets)

49. A ..... rule set contains a selector and a declaration blocks.

Ans. : CSS (Cascading Style Sheets)

50. ..... indicates the HTML element you want to style.

Ans. : Selector

51. The ..... block can contain one or more declarations separated by a semicolon. March 2022

Ans. : Declaration

52. Each declaration contains a property name and value, separated by .....
Ans. : Colon
53. A ..... is a type of attribute of HTML element.
Ans. : Property
54. ..... are assigned to CSS properties.
Ans. : Values
55. There are ..... methods of implementing styling information to an HTML document.
Ans. : Three
56. ..... style sheet uses the style attribute in the HTML start tag.
Ans. : Inline
57. ..... CSS is used to apply CSS on a single line of element. **March 2022**
Ans. : Inline
58. ..... style sheet is used to apply CSS on a single document or page.
Ans. : Embedded/Internal
59. Internal CSS is written inside ..... tag within head section of html.
Ans. : <style>
60. The ..... style sheet is used when you want to make changes on multiple pages. **March 2022**
Ans. : External
61. ..... style sheet facilitates to change the look of the entire web site by changing just one file.
Ans. : External
62. External Style Sheet uses ..... tag on every page. **March 2022**
Ans. : <link>
63. In External CSS ..... tag is placed inside head section.
Ans. : <link>
64. External style sheet can be written in text editor and must be saved with ..... extension. **March 2022**
Ans. : • CSS
65. The ..... CSS file should not contain any HTML tags.
Ans. : External
66. In CSS, ... property changes the color of the text.
Ans. : color
67. The ..... property is used to set background color in your webpage. **March 2022**
Ans. : Background-color
68. The ..... property is used make the text bold.
Ans. : Font-weight
69. The ..... property is used to make the text italic.
Ans. : Font-style
70. The ..... property is used to add strike through marks, underline, overline etc. **March 2022**
Ans. : Text-decoration

71. The ..... property is used to control the horizontal alignment of any block level text. **March 2022**
Ans. : Text-Align
72. The ..... property is used to control the fonts.
Ans. : font-family
73. The ..... property allows you to control the size of the font. **March 2022**
Ans. : font-size
74. The ..... property helps in controlling the horizontal spacing between characters of text.
Ans. : Letter-spacing
75. The ..... property is used when you want to add padding around the contents of an element.
Ans. : Padding
76. The ..... property adds a border to a web page element.
Ans. : Border
77. The ..... property is used to set an image as the background of webpage. **March 2022**
Ans. : Background-image
78. The ..... property sets margin area on the left side of the element.
Ans. : Margin-left
79. The ..... selector selects the id attribute of an HTML element to select a specific element. **March 2022**
Ans. : ID
80. An ..... is always unique within the page so it is unique element.
Ans. : ID
81. ID is written with the ..... character followed by the ID name.
Ans. : Hash
82. The ..... selector selects HTML elements with a specific class attribute.
Ans. : Class
83. The Class selector is used with ..... character followed by the class name. **March 2022**
Ans. : Period
84. The ..... selector is used when you want to change a group of elements within HTML page.
Ans. : Class
85. The _____ name should not start with number.
Ans. : Class
86. The ..... selector is used as a wildcard character.
Ans. : Universal
87. The ..... selector is used to select all the elements with the same style definition.
Ans. : Grouping
88. In grouping selector ..... are used to separate each selector.
Ans. : Commas
89. In CSS, ... property is used to set position for an element.
Ans. : Position

90. There are ..... types of positioning in CSS. March 2022
Ans. : Four
91. The ..... positioning is not affected by the top, bottom, left and right properties.
Ans. : Static
92. The ..... positioning is the default position for HTML element. March 2022
Ans. : Static
93. The ..... positioning helps to put the text fixed on the browser.
Ans. : Fixed
94. The ..... property forces an element into a fixed positioning relative to the browser window.
Ans. : Fixed
95. The ..... positioning property is used to set the element related to its normal position.
Ans. : Relative
96. The ..... positioning property sets an element in a specific location and it is not affected by the flow of the page.
Ans. : Absolute
97. _____ properly positions the element at the coordinates relative to your screen top-left corner.
Ans. : Absolute
98. ..... is a CSS property written in CSS file or directly in the style of an element.
Ans. : Float
99. The _____ property defines the flow of content.
Ans. : float
100. ..... keeps the element float on left side of the container. March 2022
Ans. : float : left
101. ..... keeps the element float on right side of container. March 2022
Ans. : float : right
102. ..... is a default property which shows the element as it is.
Ans. : float : none
103. The ..... property in CSS defines how the components are going to be placed on the web page.
Ans. : Display
104. ..... property is used to display an element as an inline element.
Ans. : Display : Inline
105. ..... property is used to display an element as an block element.
Ans. : Display : Block
106. ..... property is used to set the width and height.
Ans. : Block-inline
107. Using ..... property the element is completely removed.
Ans. : Display : None

1.5 Ordered list or numbered list

108. The ..... tag defines an Ordered list.
Ans. : <OL>
109. The ..... list can be numerical or alphabetical.
Ans. : Ordered
110. The ..... attribute of <ol> tag specifies the numbering type for the used items. March 2022
Ans. : type
111. The ..... attribute of <ol> tag specifies that the items of the list are specified in the reverse order.
Ans. : Reversed
112. The ..... attribute of <ol> tag specifies the starting number of the first item.
Ans. : start
113. ..... list is also called as numbered list.
Ans. : Ordered

1.6 Unordered list or bulleted list

114. An unordered list is created using tag.
Ans. : <UL>
115. ..... list is also called as bulleted list. March 2022
Ans. : Unordered
116. Each list item starts with ..... tag.
Ans. :
117. The ..... attribute of <ul> tag sets the list item marked to disc, circle or square.
Ans. : type
118. ..... list can be numerical or alphabetical. March 2022
Ans. : Ordered

1.7 Definition list

119. To define a definition list ..... tag is used. March 2022
Ans. : <dl>
120. The ..... tag is used to define the term.
Ans. : <dt>
121. The ..... is used to define term's definition. March 2022
Ans. : <dd>
122. The list within another list is called ..... list.
Ans. : Nested

1.8 Inserting audio and video in HTML5

123. MPEG stands for .....
Ans. : Moving Pictures Experts Group / Motion Pictures Experts Group
124. AAC stands for .....
Ans. : Advanced Audio Coding

125. The ..... element enables to embed or add audio files an webpages.
Ans. : <audio>
126. Using ... attribute of <audio> tag the audio will start playing as soon as it is ready.
Ans. : Autoplay
127. The ..... attribute of <audio> tag defines the audio controls should be displayed.
Ans. : controls
128. The ..... attribute of <audio> tag sets the audio will start over again, every time it is finished. March 2022
Ans. : loop
129. The ..... attribute of <audio> tag specifies that the audio output should be muted.
Ans. : Muted
130. The ..... attribute of <audio> tag specifies the URL of the audio file.
Ans. : Src
131. The ..... attribute is used to add audio controls such as play, pause and volume.
Ans. : Controls
132. The ..... tag is used to specify multiple media resources for media elements.
Ans. : <source>
133. The ..... tag is used to embed video into web page.
Ans. : <video>
134. The ..... attribute defines the link to video file.
Ans. : Src
135. The ..... attribute specifies that the video will start playing as soon as it is ready.
Ans. : Autoplay
136. The ..... attribute specifies that video controls should be displayed.
Ans. : Controls
137. The ..... attributes sets the height of the video player.
Ans. : height
138. The ..... attribute specifies that the video will start over again every time it is finished.
Ans. : loop
139. The ..... attribute specifies that the audio output of the video should be muted.
Ans. : muted
140. The ..... attribute specifies an image to be shown while the video is downloading.
Ans. : poster
141. The ..... attribute specifies if and how the author thinks the video should be loaded when the web page loads.
Ans. : preload
142. The ..... attribute sets the width of the video player.
Ans. : width

1.9 Image map in HTML5

143. _____ value of shape attribute of <area> tag specifies rectangular area.
Ans. : Rect
144. Rect specifies rectangular area with _____ co-ordinates.
Ans. : four
145. _____ value of shape attribute of <area> tag defines a circular region.
Ans. : Circle
146. Circle defines a circular region which requires _____ co-ordinates.
Ans. : three
147. _____ value of shape attribute of <area> tag defines polygon region.
Ans. : Poly
148. An image with multiple hyperlinks is called an ..... March 2022
Ans. : Image map
149. ..... is used to connect links to different regions on the web page.
Ans. : Image Map
150. The clickable regions of image map are called as .....
Ans. : hotspots
151. ..... tag is used to insert an image on a web page.
Ans. : <img>
152. To create a client Side Image Map ..... attribute of <img> tag is used. March 2022
Ans. : Usemap
153. Usemap attribute of <img> tag is used with value which is preceded with a ..... symbol.
Ans. : #
154. The ..... attribute acts as a pointer which indicates that the image is a Client Side Image Map. March 2022
Ans. : Usemap
155. The ..... tag is used to map the image.
Ans. : <map>
156. The ..... attribute of <map> tag specifies name of the image used for client side image map. March 2022
Ans. : Name
157. ..... tag defines the specific clickable regions.
Ans. : <area>
158. The ..... attribute of <area> tag defines the URL to which the clickable region within the image map navigates. March 2022
Ans. : Href

159. The ..... attribute defines the shape as rect, circle or poly.
Ans. : Shape
160. The ..... attribute specifies co-ordinates of the clickable regions on the image map. March 2022
Ans. : Coords
161. The ..... attribute specifies extra information about clickable area.
Ans. : alt

1.10 Inline Frame in HTML5

162. The ..... element creates an inline frame.
Ans. : <iframe>
163. ..... frames are often used in online advertising.
Ans. : Inline
164. The ..... attribute specifies the address of the document to embed in the <iframe>.
Ans. : Src
165. The ..... attribute specifies the height of an <iframe>.
Ans. : Height
166. The ..... attribute specific the width of an <iframe>.
Ans. : width
167. The ..... attribute specifies name of an <iframe>.
Ans. : Name
168. The ..... attribute specifies the HTML content of the page to show in the <iframe>.
Ans. : Srcdoc

1.11 Website Hosting

169. Website is made available on the Internet with the help of ..... March 2022
Ans. : Web Hosting
170. ..... is the service of providing storage space.
Ans. : Web Hosting
171. The Companies that provides web hosting services are called .....
Ans. : Web hosts
172. ..... own and manage web servers.
Ans. : Web hosts
173. ..... hosting gives domain name to your website.
Ans. : Shared
174. ..... are paid hosting servers for large web site.
Ans. : Dedicated hosting
175. ..... provides you free hosting of the website for limited period of time.
Ans. : Free hosting
176. Cost effective method of web hosting is ..... March 2022
Ans. : Shared Hosting

True or False

1.1 Advanced Web Designing

1. HTML stands for Hyper Text Markup language.
Ans.: True
2. HTML is an evolving language, with different versions supporting different features.
Ans. : True
3. Google chrome does not support the features of HTML5.
Ans. : False
4. Microsoft Edge, Safari browsers support the features of HTML5.
Ans. : True

1.2 Forms in HTML5

5. Text, radio, checkbox etc. are the controls used to collect user inputs.
Ans. : True.
6. <input type = "color"> defines a color picker. March 2022
Ans. : True
7. <input type = "number"> defines a field for entering URL.
Ans. : False
8. <input type= "URL"> defines a field for entering URL.
Ans. : True
9. <input type = "image"> defines an image as a submit button.
Ans. : True
10. <input type = "date"> defines a date picker with year, month and day.
Ans. : True
11. <input type = "email"> defines a field for entering number.
Ans. : False
12. <input type = "month"> defines a month and year control. March 2022
Ans. : True
13. <input type = "range"> defines a range control.
Ans. : True
14. Default range of range control is 100-200.
Ans. : False
15. <input type = "datetime–local"> defines a date picker that includes the year, month, day and time. March 2022
Ans. :True
16. <input type = "time"> defines the control for entering a week.
Ans. : False
17. <input type = "week"> defines a week and year control. March 2022
Ans. : True

18. <input type = "search"> defines a text field for entering a search string.
Ans. : True
19. <input type = "file"> defines a file-select field and a browse button for file uploads.
Ans. : True
20. <input type = "tel"> used to define input fields should be disabled. **March 2022**
Ans. : False
21. Enabled attribute specifies that an input field should be disabled.
Ans. : False
22. Max attribute specifies the maximum value for an input field.
Ans. : True
23. Min attribute specifies the minimum value for an input field.
Ans. : True
24. Pattern attribute specifies a regular expression to check the input values.
Ans. : True
25. Read only attribute specifies that an input field is Read only. **March 2022**
Ans. : True
26. Placeholder attribute acts as a temporary label showing the purpose of a text field without requiring a label tag.
Ans. : True
27. Required attribute specifies that an input field must be filled out.
Ans. : True
28. Auto-complete attribute specifies that the input field should automatically get focus when the page loads.
Ans. : False
29. Autofocus attribute specifies whether input field should have autocomplete ON or OFF.
Ans. : False
30. Height and width attribute specifies the height and width of an <input type = "image">. **March 2022**
Ans. : True
31. Multiple attribute works with input type text and image.
Ans. : False
32. Multiple attribute specifies that the user is allowed to enter more than one value in the <input> element.
Ans. : True
33. ID attribute is used to identify the html element uniquely through the Document Object Model.
Ans. : True
34. Class attribute is used to apply CSS style to the individual input element.
Ans. : True

1.3 <meta> tag

35. <meta> tag is used by search engines to search information that is provided with the webpage. **March 2022**
Ans. : True
36. <meta> tag is a paired tag.
Ans. : False
37. <meta> tag is placed between <title> and </title> tag.
Ans. : False
38. Metadata will not be displayed on the web page.
Ans. : True
39. Name attribute of <meta> tag specifies the name of metadata.
Ans. : True
40. Content attribute of <meta> tag cannot have any textual matter related to the name.
Ans. : False
41. Charset attribute of <meta> tag specifies the character encoding use by the document.
Ans. : True
42. <meta charset = "UTF-8"> is for Indian characters.
Ans.: True
43. <meta charset = "Big5"> is for Chinese characters.
Ans. : True
44. http-equiv attribute of <meta> tag is used for http response message headers.
Ans. : True
45. <meta http-equiv = "refresh" content = "5">, here the page will get refresh after every 5 seconds.
Ans. : True
46. <meta http-equiv = "set-cookies">, the browser sends the cookies back to the server. **March 2022**
Ans. : True
47. <meta http-equiv = "Content-type" content = "text/html" charset = "Big5"> specifies the character decoding of the document.
Ans. : False

1.4 Cascading Style Sheets HTML5

48. CSS stands for Cascading Style Sheets. **March 2022**
Ans. : True
49. CSS describes how HTML elements are to be displayed on screen.
Ans. : True
50. CSS cannot control the layout of multiple web pages all at once.
Ans. : False

51. CSS allows you to control the look and feel of several pages by changing a single source. **March 2022**

Ans. : True

52. A CSS rule set contains a selector and a declaration block.

Ans. : True

53. Selector indicates the HTML element you want to style.

Ans. : True

54. The declaration block can contain one or more declarations separated by a semicolon.

Ans. : True

55. Each declaration contains a property name and value separated by a semicolon. **March 2022**

Ans. : False

56. A property is a type of attribute of HTML element.

Ans. : True

57. Values are assigned to CSS properties.

Ans. : True

58. There are four methods of implementing styling information to an HTML document.

Ans. : False

59. Inline style-sheet uses the style attribute in the HTML start tag.

Ans. : True

60. Inline CSS is used to apply CSS in multiple lines or elements. **March 2022**

Ans. : False

61. CSS is the example of inline CSS.

Ans. : True

62. Internal style sheet is also known as embedded style sheet.

Ans. : True

63. Internal CSS is used to apply CSS on a single document or page.

Ans. : True

64. Internal CSS cannot affect all the elements of the page.

Ans. : False

65. Internal CSS is written inside <style> tag within head section.

Ans. : True

66. The color property changes the background color of the web page.

Ans. : False

67. Background-color property is used to set the background color in your web page.

Ans. : True

68. Font-weight property is used to make the text bold.

Ans. : True

69. Font-style property is used to italicize text. **March 2022**

Ans. : True

70. Text-decoration property is used to add strike-through marks, underline, overstrike etc.

Ans. : True

71. Text-align property is used to control the vertical alignment of any block-level text.

Ans. : False.

72. Font-family property is used to control the fonts.

Ans. : True

73. Font-size property allows you to control the size of the font.

Ans. : True

74. Letter-spacing property helps in controlling the horizontal spacing between characters of text.

Ans. : True

75. Padding property is used when you want to remove blank spaces around the content of an element.

Ans. : False

76. Border property adds a border to a web page element.

Ans. : True

77. Background-image property is used to set an image as the background of your webpage.

Ans. : True

78. Margin-left property sets the margin area on the left side of the element.

Ans. : True

79. External style sheet is used when you want to make changes on multiple pages.

Ans. : True

80. Internal style sheet facilitates to change the look of entire web site by changing just one file.

Ans. : False.

81. External CSS uses <link> tag.

Ans. : True

82. <link> tag is placed inside body section.

Ans. : False

83. External CSS file must be saved with .css extension. **March 2022**

Ans. : True

84. External CSS file should contain HTML tags.

Ans. : False

85. The ID selector selects the id attribute of an HTML element to select a specific element.

Ans. : True

86. An ID is unique element. **March 2022**

Ans. : True

87. ID is written with hash (#) character followed by id name.

Ans. : True

88. #abc {color : red} is example of ID selector.

Ans. : True

89. Class selector selects HTML elements with a specific class attribute.
Ans. : True
90. Class selector is used with a period character.
Ans. : True
91. Class name should start with a number. **March 2022**
Ans. : False
92. The class selector is used when you want to change a group of elements within your HTML page.
Ans. : True
93. •abc {color : blue} is example of class selector. **March 2022**
Ans.: True
94. The universal selector is used as wildcard character.
Ans. : True
95. Universal selector selects single element on the web page.
Ans. : False
96. Grouping selector is used to select element with the different style definitions.
Ans. : False
97. Commas are used to separate each selector in grouping. **March 2022**
Ans. : True
98. h1, h2, b { color : pink} is example of class selector.
Ans. : False
99. The position property is used to set position for an element.
Ans. : True
100. The element can be positioned using the top, bottom, left and right properties.
Ans. : True
101. Values in positions are fixed, absolute and relative.
Ans. : True
102. There are four types of positioning in CSS.
Ans. : True
103. Static positioning is the default position for HTML elements. **March 2022**
Ans. : True
104. Static positioning is affected by the top, bottom, left and right properties.
Ans. : False
105. Fixed Positioning property helps to put the text fixed on the browser.
Ans. : True
106. The fixed element will move even when the page is scrolled.
Ans. : False.
107. The static property forces an element into a fixed position relative to the browser window.
Ans. : False
108. Relative Positioning property is used to set the element relative to its normal position. **March 2022**
Ans. : True

109. Absolute positioning property sets an element in a specific location and it is not affected by the flow of the page.
Ans. : True
110. Absolute positioning property positions the element at the specified coordinates relative to screen top right corner.
Ans. : False
111. Float property is written in CSS file or directs in the style of an element.
Ans. : True
112. The float property defines the flow of content. **March 2022**
Ans. : True
113. Float : left keeps the element float on right side of the container.
Ans. : False
114. Float : right keeps the element float on right side of the container.
Ans. : True
115. Float : none is the default property. **March 2022**
Ans. : True
116. Float : name does not shows the element as it is.
Ans. : False
117. The Display property in CSS defines how the components are going to be placed on the web page.
Ans. : True
118. Display property specifies how the element is float.
Ans. : False.
119. Display property is used to define the display of different parts of a web page.
Ans. : True
120. Inline value is used to display an element as an block element. **March 2022**
Ans. : False
121. Block value is used to display an element as an inline element.
Ans. : False
122. Block value starts on a new line, and takes up the whole width of the browser window.
Ans. : True
123. The difference between Inline and block inline value is that you are able to set the width and height.
Ans. : True
124. The element is completely removed using none value.
Ans. : True

1.5 Ordered list or numbered list

125. tag defines ordered list. **March 2022**

Ans. : True

126. Ordered list cannot be numerical or alphabetical.

Ans. : False

127. Type attribute is used to define number type for the used items.

Ans. : True

128. The default value of type attribute in ordered list is 1. **March 2022**

Ans. : True

129. Reversed attribute specifies that the items of the list are specified in the reverse order.

Ans. : True

130. Start attribute specifies the ending number of the first item in an ordered list.

Ans. : False

1.6 Unordered list or bulleted list

131. An unordered list is created using <ul> tag.

Ans. : True

132. Each list item starts with <li> tag.

Ans. : True

133. The list items in unordered lists are marked with numbers. **March 2022**

Ans. : False

134. By default, the list items in unordered list are marked with bullets.

Ans. : True

135. <ul type = "disc"> sets the list item marker to a bullet.

Ans. : True

136. <ul type = "circle"> sets the list item marker to a square.

Ans. : False

137. <ul type = "square"> sets the list item marker to a square. **March 2022**

Ans. : True

138. Using <ul type = "none"> the list items will not be marked.

Ans. : True

139. HTML5 does not support bullets, circle and square value of type attribute.

Ans. : True

140. HTML5 does not support type attribute of <ul> tag so you can use CSS style.

Ans. : True

1.7 Definition list

141. To define a definition list <dl> tag is used.

Ans. : True

142. Items can be created in definition list with <dt> and <dd> tags.

Ans. : True

143. The <dd> tag is used to define the term whereas the <dt> tag is used to define the term's definition.

Ans. : False

144. List within list is called nested list.

Ans.: True

145. List can be Single level nested list or multi level list.

Ans. : True

1.8 Inserting audio and video in HTML5

146. HTML features includes native audio and video support without the need for flash.

Ans. : True

147. The <audio> element enables you to embed, audio files on webpages.

Ans. : True

148. The audio will stop playing as soon as it is ready using auto-play attribute.

Ans. : False

149. Controls attribute is used to display audio controls. **March 2022**

Ans. : True

150. Controls attribute display play, pause buttons etc.

Ans. : True

151. Using loop attribute the audio will stop over again, every time it is finished.

Ans. : False

152. Muted attributes specifies that the audio output should be muted.

Ans. : True

153. Src attribute specifies the URL of the audio file. **March 2022**

Ans. : True

154. Autoplay, controls, muted are the attributes without any values.

Ans. : True

155. The source element is used to specify the audio files which the browser may use.

Ans. : True

156. The <source> tag is used to specify multiple media resources for media elements.

Ans. : True

157. Multiple sources of audios are specified so that if the browser is unable to play the first source then if will jump to the second source. **March 2022**

Ans. : True

158. The <video> tag is used to embed video in web page.
Ans. : True
159. Src attribute defines link to video file.
Ans. : True
160. Autoplay attribute specifies that the video will start playing as soon as it is ready.
Ans. : True
161. Controls attribute specifies that video controls should not be displayed.
Ans. : False
162. Height attribute sets the height of the video-player. March 2022
Ans. : True
163. Loop attribute specifies that the video will start over again every time it is finished.
Ans. : True
164. Muted attribute specifies that the video output of the video should be muted.
Ans. : False
165. Poster attribute specifies on image to be shown while the video is downloading.
Ans. : True
166. Preload attribute specifies if and how the author thinks the video should be loaded when the webpage loads.
Ans. : True
167. Width attribute sets the width of the video player.
Ans. : True
168. Width attribute sets the value in pixels. March 2022
Ans. : True
169. The <source> tag cannot be used to specify multiple media resources for video as well as audio media elements.
Ans. : False

1.9 Image map in HTML5

170. An image with multiple hyperlinks is called as image map.
Ans. : True
171. Image map is used to connect links to different regions on the webpage.
Ans. : True
172. Clickable regions in Image map are called as hotspots.
Ans. : True
173. Image Maps are of four types.
Ans. : False
174. <img> tag is used to insert an image on a web page. March 2022
Ans. : True
175. Image map is created by marking certain regions on an image clickable.
Ans. : True

176. To create a server side image map use map attribute of <img> is used.
Ans. : False
177. The usemap attribute acts as a pointer which indicates that the image is a client side image map.
Ans. : True
178. Name attribute specifies name of the image used for client side image map.
Ans. : True
179. The attribute shape = poly is used when we wish to give hotspot of pentagon shape. March 2022
Ans. : False
180. <map> tag is used to create client side image map.
Ans. : True
181. Usemap is the attribute used with <map> tag.
Ans. : False
182. <area> tag defines specifies clickable regions.
Ans. : True
183. <area> is paired tag.
Ans. : False
184. <map> tag is paired tag. March 2022
Ans. : True
185. A given <map> element can contain multiple <area> element within it.
Ans. : True
186. Src attribute defines URL to which the clickable region within the image map navigates.
Ans. : False
187. Shape attribute can value rect, circle or poly.
Ans. : True
188. Coords attribute specifies co-ordinates of the clickable region on the image map.
Ans. : True
189. Rect value specifies rectangular area within four co-ordinates.
Ans. : True
190. Circle value specifies a circular region. March 2022
Ans. : True
191. Circle requires two co-ordinates.
Ans. : False
192. Poly value defines a polygon region with co-ordinates specifying each point on the polygon.
Ans. : True
193. No coordinates are required in default value.
Ans. : True

194. Alt attribute specifies extra information about clickable area. March 2022
Ans. : True
195. Href attribute gives alternative text to the clickable region.
Ans. : False

1.10 Inline Frame in HTML5

196. The <iframe> element creates an inline frame.
Ans. : True
197. Inline frames are used in online advertising.
Ans. : True
198. Src attribute specifies the address of the document to embed in the <iframe>.
Ans. : True
199. Height attribute specifies the width of an <iframe>.
Ans. : False
200. Width attribute specifies the width of an <iframe>. March 2022
Ans. : True
201. Height and width is defined in pixels.
Ans. : True
202. Name attribute specifies name of an <iframe>.
Ans. : True
203. Srcdoc specifies the HTML content of the page to show in the <iframe>.
Ans. : True

1.11 Website Hosting

204. Web server is the service of providing storage space.
Ans. : False
205. The website is made available on the internet with the help of web hosting.
Ans. : True
206. The companies that provides web hosting services are called web host. March 2022
Ans. : True
207. Web host own and manage web servers.
Ans. : True
208. Web servers offer interrupted Internal Connectivity.
Ans. : False.
209. Shared hosting is cost effective.
Ans.: True
210. Free hosting gives domain name to your website.
Ans. : False
211. Websites which provide free hosting for limited period of time is called free hosting
Ans. : True

212. Dedicated hosting are paid hosting servers for large website.
Ans. : True
213. You can buy your own web server space, but it is most expensive way to publish your website. March 2022
Ans. : True

MCQ (One Correct Answers)

1.1 Advanced Web Designing

1. HTML stands for .....
(a) Hyper Transfer Markup Language (b) Hyper Text Markup Language
(c) Hyper Transform Markup (d) Hyper Text Made Language
Ans. : (b)
2. The ..... is an evolving language with different versions supporting different features.
(a) HTML (Hyper Text Markup Language)
(b) ASP (Active Server Pages)
(c) DOM (Document Object Model)
(d) HTTP (Hyper Text Transfer Protocol)
Ans. : (a)
3. ..... is currently used because it supports mobile technology.
(a) DOM (b) HTTP (c) HTML5 (d) FTP
Ans. : (c)

1.2 Forms in HTML5

4. The ..... tag is used to create a form. March 2022
(a) <input> (b) <text area> (c) <select> (d) <form>
Ans. : (d)
5. <input type = "....."> defines a color picker.
(a) Number (b) URL (c) Color (d) Date
Ans. :(c)
6. <input type= "....."> defines a field for entering a number. March 2022
(a) color (b) Number (c) Date (d) email
Ans. : (b)
7. <input type="....."> defines a field for entering a URL.
(a) color (b) image
(c) email (d) URL (Uniform Resource Locator)
Ans. : (d)

8. <input type="....."> defines an image as a submit button.
(a) Image (b) Button (c) date (d) radio
Ans. : (a)

9. <input type="....."> defines a date picker with the year, month and day.
(a) Month (b) Range (c) Date (d) Number
Ans. : (c)

10. <input type="....."> defines a field for an email address. **March 2022**
(a) Image (b) email (c) URL (d) Color
Ans. : (b)

11. <input type="....."> defines a month and year control in format as 'YYYY-MM'.
(a) Date (b) datetime –local
(c) Week (d) month
Ans. : (d)

12. <input type= "....."> defines the range control. **March 2022**
(a) time (b) search (c) range (d) file
Ans. : (c)

13. <input type= "....."> defines a date picker that includes the year, month day and time.
(a) datetime-local (b) date
(c) week (d) month
Ans. : (a)

14. <input type= "....."> defines a control for entering a time.
(a) week (b) time (c) date (d) date time-local
Ans. : (b)

15. <input type="....."> defines a week and year control.
(a) week (b) date
(c) datetime-control (d) Month
Ans. : (a)

16. <input type= "....."> defines a text field for entering a search string.
(a) File (b) Range (c) Search (d) tel
Ans. : (c)

17. <input type="....."> defines a field select field and a 'Browse' button for file uploads. **March 2022**
(a) tel (b) Range (c) Search (d) File
Ans. : (d)

18. <input type= "....."> used to define input fields that should contain a telephone number.
(a) tel (b) search (c) number (d) color
Ans. : (a)

19. The default range of range control is .....
(a) 0 to 100 (b) 100 to 150 (c) 90 to 100 (d) 200 to 300
Ans. : (a)

20. The ..... attribute of <input> tag specifies that an input field should be disabled. **March 2022**
(a) max (b) required (c) disabled (d) auto-complete
Ans. : (c)

21. The ..... attribute of <input> tag specifies the maximum value for an input field.
(a) disabled (b) max (c) read only (d) min
Ans. : (b)

22. The ..... attribute of <input> tag specifies the minimum value for an input field. **March 2022**
(a) min (b) max (c) pattern (d) autofocus
Ans. : (a)

23. The ..... attribute of <input> tag specifies a regular expression to check the input values. **March 2022**
(a) Required (b) Multiple (c) Pattern (d) Max
Ans. : (c)

24. The ..... attribute of <input> tag specifies that an input field is read only.
(a) read only (b) placeholder
(c) disabled (d) required
Ans. : (a)

25. The ..... attribute of <input> tag acts as a temporary label showing the purpose of a text field without requiring a label tag. **March 2022**
(a) Pattern (b) Placeholder
(c) Multiple (d) Read only
Ans. : (b)

26. The ..... attribute of <input> tag specifies that an input field is required.
(a) required (b) disabled (c) multiple (d) auto focus
Ans. : (a)

27. The ..... attribute of <input> tag specifies whether a form or input field should have autocomplete on or off. **March 2022**
(a) Required (b) max (c) autofocus (d) autocomplete
Ans. : (d)

28. The ..... attribute of <input> tag specifies that the input field should automatically got focus when the page loaded.
(a) autocomplete (b) multiple
(c) autofocus (d) required
Ans. : (c)

29. The ..... attribute of <input> tag specifies the height and width of an <input type ="image">
(a) Multiple (b) Height and width
(c) disabled (d) min

Ans. : (b)

30. The ..... attribute of <input> tag specifies that the user is allowed to enter more than one value in the <input> element.
(a) Read only (b) required (c) multiple (d) pattern

Ans. : (c)

31. ..... attribute works with input types like email and file. March 2022
(a) Required (b) Multiple (c) Pattern (d) Read only

Ans. : (b)

32. The ..... attribute of <input> tag is used to identify the html element uniquely through the document object model.
(a) ID (b) Class (c) Relative (d) Grouping

Ans. : (a)

33. The ..... attribute of <input> tag is used to apply CSS style to the individual input element.
(a) Relative (b) Class (c) ID (d) Absolute

Ans. : (b)

34. A placement agency wants to collect resume of the students of renowned collage, which of the following is the correct option they should use in their form ?
(a) <input type = "url"> (b) <input type = "image">
(c) <input type = "file"> (d) <input type = "search">

Ans. : (c)

35. Sangeeta wants to collect mobile numbers of her friends in a specific format. Which of the following correct attribute of 'tell' she should use while designing a form ?
(a) <input type = "tel" pattern = "[0-9] {2} – [0-9] {10}">
(b) <input type = "tel" format = "[0-9] {2} – [0-9] {10}">
(c) <input type = "tel" max = "[0-9] {2} – [0-9] {10}">
(d) <input type = "tel" multiple = "[0-9] {2} – [0-9] {10}">

Ans. : (a)

1.3 <meta> tag

36. The .....tag is used by search engines to search information that is provided with the web page. March 2022
(a) <input> (b) <body> (c) <form> (d) <meta>

Ans. : (d)

37. The .....tag is placed between <head> tag.
(a) <body> (b) <meta> (c) <form> (d) <audio>

Ans. : (b)

38. .....will not be displayed on the web page. March 2022
(a) Head data (b) Form data (c) Metadata (d) Table data

Ans. : (c)

39. The ..... attribute specifies the name of the metadata.
(a) Name (b) http-equiv (c) Content (d) Charset

Ans. : (a)

40. The .....attribute of <meta> tag can have any textual matter related to the name. March 2022
(a) http-equiv (b) Name
(c) Content (d) Charset

Ans. : (c)

41. The .....attribute of <meta> tag specifies the character encoding used by the document.
(a) Metadata (b) Name (c) Content (d) Charset

Ans. : (d)

42. <meta charset = "UTF – 8"> is for .....characters.
(a) Indian (b) Japan (c) America (d) Chinese

Ans. : (a)

43. <meta charset = "Big5" is for .....language.
(a) Indian (b) USA (c) Chinese (d) Italy

Ans. : (c)

44. The ..... attribute of <meta> tag is used for http response message headers.
(a) Name (b) http-equiv (c) content (d) Charset

Ans. : (b)

45. The page will get refresh using .....value of http-equiv attribute. March 2022
(a) expires (b) set-cookie
(c) content-type (d) Refresh

Ans. : (d)

46. The browser sends the cookies back to the server using .....value of http-equiv attribute.
(a) set – cookies (b) expires
(c) content-type (d) refresh

Ans. : (a)

47. The .....value specifies the character encoding for the document. March 2022
(a) expires (b) content-type
(c) charset (d) refresh

Ans. : (c)

48. The page session will get expire at specified date and time using .....value of http-equiv attribute.
(a) Expires (b) Refresh
(c) set-cookie (d) content-type
Ans. : (a)

1.4 Cascading Style Sheets HTML5

49. CSS stands for ..... **March 2022**
(a) Content Style Sheets (b) Commanding Style Sheets
(c) Cascading Style Sheets (d) Context Style Sheets
Ans. : (c)

50. ..... allows you to control the look and feel of several pages by changing a single source.
(a) ASP (b) CSS (c) PHP (d) DOM
Ans. : (b)

51. ..... indicates the HTML element you want to style. **March 2022**
(a) Selector (b) Declaration
(c) Property (d) Value
Ans. : (a)

52. The ..... block can contain one or more declarations separated by a semi colon.
(a) Declaration (b) Selector (c) Property (d) Value
Ans. : (a)

53. Each declaration contains a property name and value, separated by .....
(a) Colon (b) Comma (c) Semicolon (d) Period
Ans. : (a)

54. A ..... is a type of attribute of HTML element. **March 2022**
(a) Property (b) Value (c) Declaration (d) Selector
Ans. : (a)

55. ..... are assigned to CSS properties.
(a) Methods (b) Values (c) Selector (d) Property
Ans. : (b)

56. There are ..... methods of implementing styling information to an HTML document.
(a) Four (b) Five (c) Three (d) Six
Ans. : (c)

57. ..... style sheet uses the style attribute in the HTML start tag.
(a) Internal (b) Inline (c) External (d) Selector
Ans. : (b)

58. ..... CSS is used to apply CSS as an single line of element.
(a) Internal (b) External (c) Embedded (d) Inline
Ans. : (d)

59. Example of Inline CSS is .....
(a) <body background – color : red >
(b) <body style = "background – color : red">
(c) <body class = "background – color : red">
(d) <body id = "background" – color : red">
Ans. : (b)

60. Internal CSS is also called an ..... CSS. **March 2022**
(a) Embedded (b) Inline (c) Inbuilt (d) External
Ans. : (a)

61. ..... style sheet is used to apply CSS on a single document or page.
(a) Internal (b) Inline (c) External (d) Inbuilt
Ans. : (a)

62. Internal CSS is written inside ..... tag within head section of html.
(a) <body> (b) <style> (c) <link> (d) <title>
Ans. : (b)

63. The ..... style sheet is used when you want to make changes on multiple pages. **March 2022**
(a) Internal (b) External (c) Inline (d) Embedded
Ans. : (b)

64. ..... style sheet facilitates to change the look of the entire web site by changing just one file.
(a) External (b) Inline (c) Internal (d) Embedded
Ans. : (a)

65. In external CSS ..... tag should be put inside head section. **March 2022**
(a) <style> (b) <link> (c) <title> (d) <body>
Ans. : (b)

66. An external style sheet must be saved with ..... extension.
(a) .txt (b) .html (c) .JS (d) .CSS
Ans. : (d)

67. The ..... CSS file should not contain any HTML tags.
(a) Internal (b) Inline (c) External (d) Embedded
Ans. : (c)

68. The ..... property changes the color of text. **March 2022**
(a) color (b) background-color
(c) font – color (d) text-color
Ans. : (a)

69. The ..... property sets background color in your webpage.
(a) background – color (b) background-bg color
(c) back color (d) color
Ans. : (a)

70. The ..... property is used to make the text bold. March 2022
(a) font-color (b) font-size (c) font-weight (d) font-style
Ans. : (c)

71. The ..... property is used to make the text italics.
(a) font-weight (b) font-size
(c) font-height (d) font-style
Ans. : (d)

72. The ..... property is used to decorate the text.
(a) Text-align (b) Text-color
(c) Text-decoration (d) Decoration
Ans. : (c)

73. The ..... value of text-decoration property is used to add strike through marks. March 2022
(a) line-through (b) underline
(c) strike-through (d) overline
Ans. : (a)

74. The ..... value of text-decoration property is used to underline the text.
(a) Overline (b) line-through
(c) underline (d) strike
Ans. : (c)

75. The ..... value of text-decoration property is used to overstrike the text.
(a) Overline (b) underline (c) overstrike (d) line-through
Ans. : (a)

76. The .....value of text-decoration property is used to remove underlines from lines. March 2022
(a) Overline (b) Underline (c) none (d) line through
Ans. : (c)

77. The ..... property is used to control the horizontal alignment of any block level text.
(a) Text-align (b) text- decoration
(c) font size (d) text size
Ans. : (a)

78. The ..... property is used to control the fonts.
(a) font-size (b) font-family
(c) font-style (d) font-weight
Ans. : (b)

79. The ..... property allows you to control the size of the font.
(a) font-style (b) font-weight
(c) font-size (d) font family
Ans. : (c)

80. The ..... property helps in controlling the horizontal spacing between characters of text. March 2022
(a) letter-spacing (b) spacing
(c) padding (d) border
Ans. : (a)

81. The ..... property is used when you want to add blank spaces around the content of an element.
(a) Border (b) letter-spacing
(c) Padding (d) Margin-left
Ans. : (c)

82. The ..... property adds a border to a web page element. March 2022
(a) padding (b) border (c) font-size (d) marginal
Ans. : (b)

83. The ..... property is used to set an image as the background of webpage.
(a) Background-color (b) Background-image
(c) Background-img (d) Image
Ans. : (b)

84. The ..... property sets margin area on the left side of the element. March 2022
(a) Margin-left (b) Padding (c) Border (d) font-style
Ans. : (a)

85. The ..... selector selects the id attribute of an HTML element to select a specific element. March 2022
(a) ID (b) Class (c) Grouping (d) Universal
Ans. : (a)

86. The ID selector is written with the .....character followed by the id name. March 2022
(a) Period (•) (b) asterisks (*) (c) hash (#) (d) Copyright ©
Ans. : (c)

87. The ..... selector selects HTML elements with a specific class attribute. March 2022
(a) ID (b) Class (c) Grouping (d) Universal
Ans. : (b)

88. A Class selector is used with ..... character followed by the class name.
(a) Period (•) (b) asterisks (*) (c) hash (#) (d) Copyright ©
Ans. : (a)

89. The class name should not start with a ..... March 2022
(a) Text (b) Character (c) Number (d) Special symbol
Ans. : (c)

90. The ..... Selector is used as a wildcard character.
(a) Grouping (b) Class (c) ID (d) Universal
Ans. : (d)

91. The ..... selector is used to select all the elements with the same style definitions. **March 2022**
(a) Class (b) Grouping (c) ID (d) Universal
Ans. : (b)

92. ..... are used to separate each selector in grouping.
(a) Period (b) color (c) commas (d) semicolon
Ans. : (c)

93. The ..... property is used to select position for an element.
(a) Absolute (b) Position (c) Relative (d) Fixed
Ans. : (b)

94. There are ..... types of positioning in CSS. **March 2022**
(a) One (b) Four (c) Six (d) Three
Ans. : (b)

95. The ..... is the default position for HTML element.
(a) Static (b) Fixed (c) Relative (d) Absolute
Ans. : (a)

96. The ..... property helps to put the text fixed on the browser.
(a) Relative (b) Static (c) Fixed (d) Absolute
Ans. : (c)

97. The ..... property forces an element into a fixed position relative to the browser. **March 2022**
(a) Fixed (b) Static (c) Absolute (d) Relative
Ans. : (a)

98. The ..... element will not move even when the page is scrolled. **March 2022**
(a) Absolute (b) Relative (c) Static (d) Fixed
Ans. : (d)

99. The ..... positioning property is used to set the element relative to its normal position.
(a) Absolute (b) Relative (c) Fixed (d) Static
Ans. : (b)

100. The ..... property gets an element in a specific location and it is not affected by the flow of the page. **March 2022**
(a) Absolute (b) Fixed (c) Relative (d) Static
Ans. : (a)

101. Absolute positioning property positions the element at the specified coordinates relative to screen ..... corner.
(a) Top-left (b) Top-right (c) Top-bottom (d) Top-bottom.
Ans. : (a)

102. ..... is a CSS property written in CSS file or directly in the style of an element.
(a) Absolute (b) Fixed (c) Float (d) Static **March 2022**
Ans. : (c)

103. ..... keeps the element float on left side of container.
(a) Float : left (b) Float : right
(c) Float : none (d) Float : top
Ans. : (a)

104. ..... keeps the element float on right side of container.
(a) Float : left (b) Float : right
(c) Float : none (d) Float : top
Ans. : (b)

105. ..... is default property which shows the element as it is. **March 2022**
(a) Float : left (b) Float : right
(c) Float : none (d) Float : top
Ans. : (c)

106. The ..... Property is CSS defines how the components are going to be placed on the web page.
(a) Static (b) Fixed (c) Display (d) Relative
Ans. : (c)

107. The ..... Property is used to define the display of different parts of a web-page.
(a) display (b) static (c) Fixed (d) absolute **March 2022**
Ans. : (a)

108. ..... Value is used to display an element as an inline element.
(a) Inline (b) Block (c) None (d) Block-inline
Ans. : (a)

109. ..... is used to display an element as an block element.
(a) Inline (b) Block (c) None (d) Block-inline
Ans. : (b)

110. ..... Value is used to set width and height.
(a) Inline (b) Block (c) None (d) Block-inline
Ans. : (d)

111. Using ..... Value element is completely removed. **March 2022**
(a) Inline (b) Block (c) None (d) Block-inline
Ans. : (c)

112. Rucha has designed a website for her college but her principal told her to change the font color and background color of all pages by doing changes in one file only. Which of the following CSS she should use ?
(a) Internal CSS (b) Inline CSS
(c) Embedded CSS (d) External CSS
Ans. : (d)

113. Nikhil has designed a website for a firm, but after checking the output, his friend suggested him to remove the underline of all the links, he has used in his website. Which of the following change he should do in his coding ?
(a) a {text-decoration : underline} (b) a {text-decoration : remove}
(c) a {text-decoration : none} (d) a {text-decoration : delete}

Ans. : (c)

1.5 Ordered list or numbered list

114. The ..... tag defines an Ordered list.
(a) (b) (c) <dt> (d)

Ans. : (b)

115. A ..... list can be numerical or alphabetical. March 2022
(a) Ordered (b) Unordered
(c) Definition (d) None

Ans. : (a)

116. The ..... attribute is used to specify the numbering type for the used items.
(a) Reversed (b) start (c) Type (d) Name

Ans. : (c)

117. The default value of type attribute of ordered list is ..... March 2022
(a) I (b) a (c) 1 (d) 10

Ans. : (c)

118. The ..... Attribute specifies that the items of the lists are specified in the reverse order.
(a) type (b) reversed (c) Start (d) Name

Ans. : (b)

119. The ..... specifies the starting number of the first item in an ordered list.
(a) Type (b) Reversed (c) start (d) Name

Ans. : (c)

120. Suraj has designed a form to print the numbers 1 to 10 by using ordered list. Now he wants to print the numbers in descending order (i.e. from 10 to 1), which of the following attribute he should use with <∂L>.
(a) <OL decrease> (b) <OL descending>
(c) <OL ascending> (d) <OL reversed>

Ans. : (d)

121. A Geography teacher told the IT students to design a webpage that should contain the map of India with all states. She told the students to display the information of all the states after clicking on the area of that particular state. Which of the following students should use for designing such a webpage ?
(a) Image mapping (b) Website mapping
(c) Area mapping (d) Media mapping

Ans. : (a)

1.6 Unordered list or bulleted list

122. An Unordered list is created using .....tag. March 2022
(a) (b) (c) <dd> (d) <dt>

Ans. : (b)

123. ..... list is also called as bulleted list.
(a) Unordered (b) Ordered (c) Dataset (d) lists

Ans. : (a)

124. Each list item starts with ..... tag.
(a) <dd> (b) <dt> (c) (d)

Ans. : (d)

125. The ..... attribute of <ul> tag sets the list item marked to disc, circle or square.
(a) type (b) start (c) name (d) reversal

Ans. : (a)

126. The .....is the default value of type attribute of <ul> tag.
(a) Disc (b) square (c) circle (d) bullet

Ans. : (a)

1.7 Definition list

127. To define a definition list ..... tag is used. March 2022
(a) <dd> (b) <dl> (c) <dt> (d)

Ans. : (b)

128. The ..... tag is used to define the term.
(a) (b) <dd> (c) <dt> (d) <dl>

Ans. : (c)

129. The ..... is used to define term's definition.
(a) <dd> (b) <dt> (c) <dl> (d)

Ans. : (a)

130. The list within another list is called ..... list.
(a) listed (b) tested (c) none (d) nested.

Ans.: (d)

1.8 Inserting audio and video in HTML5

131. MPEG stands for ..... March 2022
(a) Making Pictures Expert Groups (b) Moving pictures Experts Group
(c) Managing Pictures Expert Groups (d) Merging Pictures Expert Groups

Ans. : (b)

132. AAC stands for .....
(a) Auto Audio coding (b) Arrange Audio Coding
(c) Advanced Audio Coding (d) Advanced Arrange Coding
Ans. : (c)

133. The ..... element enables to embed or add audio files an webpages.
(a) <audio> (b) <video> (c) <style> (d) <img>
Ans. : (a)

134. Using ..... attribute of <audio> tag the audio will start playing as soon as it is ready. March 2022
(a) Controls (b) Src (c) loop (d) Autoplay
Ans. : (d)

135. The ..... attribute of <audio> tag defines the audio controls should be displayed.
(a) loop (b) muted (c) controls (d) Src
Ans. : (c)

136. The ..... attribute of <audio> tag sets the audio will start over again, every time it is finished. March 2022
(a) loop (b) controls (c) Src (d) muted
Ans. : (a)

137. The ..... attribute of <audio> tag specifies that the audio output should be muted.
(a) Src (b) Controls (c) Autoplay (d) muted
Ans. : (d)

138. The ..... attribute of <audio> tag specifies the URL of the audio file.
(a) muted (b) loop (c) Src (d) controls
Ans. : (c)

139. The ..... attribute is used to add audio controls such as play, pause and volume. March 2022
(a) loop (b) controls (c) autoplay (d) Src
Ans. : (b)

140. The ..... tag is used to specify multiple media resources for media elements.
(a) (b) <img> (c) <style> (d) <source>
Ans. : (d)

141. The ..... tag is used to embed video into web page. March 2022
(a) <video> (b) <audio> (c) <style> (d) <img>
Ans. : (a)

142. The ..... attribute defines the link to video file.
(a) Autoplay (b) Height (c) Src (d) loop
Ans. : (c)

143. The ..... attribute specifies that the video will start playing as soon as it is ready.
(a) Controls (b) Autoplay (c) loop (d) Muted
Ans. : (b)

144. The ..... attribute specifies that video controls should be displayed.
(a) controls (b) loop (c) autoplay (d) muted
Ans. : (a)

145. The ..... attribute sets the height of the video player.
(a) Width (b) loop (c) Muted (d) height
Ans. : (d)

146. The ..... attribute specifies that the video will start over again every time it is finished.
(a) Muted (b) loop (c) Controls (d) Autoplay
Ans. : (b)

147. The ..... attribute specifies that the audio output of the video should be muted. March 2022
(a) controls (b) loop (c) Muted (d) Height
Ans. : (c)

148. The ..... attribute an image to be shown while the video is downloading.
(a) Poster (b) URL (c) Autoplay (d) Muted
Ans. : (a)

149. The ..... attribute specifies if and how the author thinks the video should be loaded when the web page loads.
(a) Poster (b) width (c) controls (d) preload
Ans. : (d)

150. The ..... attribute sets the width of the video player. March 2022
(a) Muted (b) width (c) loop (d) controls
Ans. : (b)

1.9 Image map in HTML5

151. An image with multiple hyperlinks is called an .....
(a) links (b) Navigation
(c) Hyperlinks (d) Image map
Ans. : (d)

152. ..... is used to connect links to different regions on the web page.
(a) links (b) hotspots (c) Image map (d) Hyper links
Ans. : (c)

153. The clickable regions of image map are called as .....
(a) hotspots (b) Image map (c) Hyperlinks (d) links
Ans. : (a)

154. ..... tag is used to insert an image on a web page. March 2022
(a) <image> (b) <img> (c) <body> (d) <head>
Ans. : (b)

155. To create a Client Side Image Map ..... attribute of <img> tag is used.
(a) Ismap (b) Nomap (c) usemap (d) Mymap
Ans. : (c)

156. Usemap attribute of <img> tag is used with value which is preceded with a ..... symbol.
(a) # (b) @ (c) $ (d) •
Ans. : (a)

157. The ..... attribute acts as a pointer which indicates that the image is a Client Side Image Map.
(a) Ismap (b) Nomap (c) Mymap (d) usemap
Ans. : (d)

158. The ..... tag is used to map the image. March 2022
(a) <img> (b) <map> (c) <area> (d) <body>
Ans. : (b)

159. The ..... attribute of <map> tag specifies name of the image used for client side image map.
(a) src (b) href (c) Name (d) alt
Ans. : (c)

160. ..... tag defines the specific clickable regions.
(a) <map> (b) <img> (c) <body> (d) <area>
Ans. : (d)

161. The ..... attribute of <area> tag defines the URL to which the clickable region within the image map navigates. March 2022
(a) Href (b) Src (c) URL (d) Link
Ans. : (a)

162. The ..... attribute defines the shape as rect, circle or poly.
(a) Coords (b) shape (c) alt (d) Href
Ans. : (b)

163. The ..... attribute specifies co-ordinates of the clickable regions on the image map. March 2022
(a) shape (b) alt (c) coords (d) Href
Ans. : (c)

164. The ..... attribute specifies extra information about clickable area.
(a) Coords (b) shape (c) href (d) alt
Ans. : (d)

165. ..... value of shape attribute of <area> tag specifies rectangular area.
(a) Rect (b) Circ (c) Poly (d) triangle
Ans. : (a)

166. Rect specifies rectangular area with ..... co-ordinates.
(a) three (b) six (c) eight (d) four
Ans. : (d)

167. ..... value of shape attribute of <area> tag defines a circular region.
(a) Rect (b) Circ (c) Poly (d) Star
Ans. : (b)

168. Circle defines the circular region which requires ..... co-ordinates.
(a) three (b) two (c) six (d) seven March 2022
Ans. : (a)

169. ..... value of shape attribute of <area> tag defines polygon region.
(a) Circ (b) Rect (c) Triangle (d) Poly
Ans. : (d)

1.10 Inline Frame in HTML5

170. The ..... element creates an inline frame.
(a) <body> (b) <frame> (c) <link> (d) <iframe>
Ans. : (d)

171. ..... frames are often used in online advertising. March 2022
(a) Inline (b) Internal (c) External (d) Embedded
Ans. : (a)

172. The ..... attribute specifies the address of the document to embed in the <iframe>.
(a) Height (b) Width (c) Src (d) Name
Ans. : (c)

173. The ..... attribute specifies the height of an <iframe> tag.
(a) Name (b) Height (c) Width (d) Srcdoc
Ans. : (b)

174. The ..... attributes specific the width of an <iframe> tag.
(a) Width (b) Height (c) Name (d) Src
Ans. : (a)

175. The ..... attribute specifies name of an <iframe> tag. March 2022
(a) Srcdoc (b) Src (c) Width (d) Name
Ans. : (d)

176. The ..... attribute specifies the HTML content of the page to show in the <iframe> tag.
(a) Src (b) Srcdoc (c) Name (d) Height
Ans. : (b)

1.11 Website Hosting

177. Website is made available on the Internet with the help of ..... March 2022
(a) Website (b) Web space
(c) Web world (d) Web hosting
Ans. : (d)

178. ..... is the service of providing storage space.
(a) Website (b) Web hosting
(c) Web world (d) web page
Ans. : (b)

179. The Companies that provides web hosting services are called .....
(a) Web hosts (b) Web page (c) Web site (d) Browser
Ans. : (a)

180. ..... own and manage web servers.
(a) Website (b) Web page (c) Web Hosts (d) Web link
Ans. : (c)

181. ..... hosting gives domain name to your website.
(a) free (b) fixed (c) Dedicated (d) shared
Ans. : (d)

182. ..... are paid hosting servers for large web site. March 2022
(a) free (b) dedicated hosting
(c) fixed (d) shared
Ans. : (b)

183. ..... provides you free hosting of the website for limited period of time.
(a) free hosting (b) fixed
(c) shared (d) Dedicated hosting
Ans. : (a)

MCQ (Two Correct Answers)

1.1 Advanced Web Designing

1. The major browsers which support features of HTML5 are
(a) Windows Explorer (b) Microsoft Edge
(c) Google Chrome (d) My files
Ans. : (b), (c)

1.2 Forms in HTML5

2. Default range of range control is ..... to ..... March 2022
(a) 0 (b) 100 (c) 120 (d) 200
Ans. : (a), (b)

3. Following are the values of type attribute of <input> tag
(a) Minimum (b) Maximum (c) date (d) email
Ans. : (c), (d)

4. Following attributes of <input> tag are used to specify minimum and maximum value for input field ..... March 2022
(a) min (b) disabled (c) multiple (d) max
Ans. : (a), (d)

5. Multiple value of type attribute of <input> tag works with ..... and .....
(a) email (b) file (c) week (d) month
Ans. : (a), (b)

6. <input type = "month"> defines a _____ and _____ control in "yyyy-mm" format.
(a) Time (b) Month (c) Year (d) Week
Ans. : (b), (c)

7. Following are the attributes of <input type = "image">. March 2022
(a) width (b) min (c) required (d) height
Ans. : (a), (d)

8. Auto complete attribute of <input> tag has _____ and _____ values.
(a) on (b) off (c) start (d) stop
Ans. : (a), (b)

1.3 <meta> tag

9. Following are the attributes of <meta> tag .....
(a) Charset (b) email (c) http-equiv (d) file
Ans. : (a), (c)

10. Following are the values of http-equiv attribute of <meta> tag .....
(a) month (b) refresh (c) image (d) expires March 2022
Ans. : (b), (d)

11. Charset attribute of <meta> tag use ..... for Indian characters and ..... for Chinese characters.
(a) UTF-8 (b) Big 5 (c) UTF-2 (d) Big B
Ans. : (a), (b)

1.4 Cascading Style Sheets HTML5

12. A CSS rule set contains ..... and ..... block. March 2022
(a) value (b) selector (c) Declaration (d) Property
Ans. : (b), (c)

13. Examples of declaration are .....
(a) H1 {color : green} (b) b {font – size : 11px}
(c) h1 (color : green) (d) b (font-style = 11px)
Ans. : (a), (b)

14. Text decoration property is used to add .....
(a) line-through (b) strike-through
(c) underline (d) justify
Ans. : (a), (c)

15. Text-align property has following values ..... March 2022
(a) Middle (b) Center (c) top (d) Justify
Ans. : (b), (d)

16. In CSS following properties are used to position the element.
(a) Top (b) Bottom (c) Middle (d) Mid
Ans. : (a), (b)

17. Following are the values of border property.
(a) Solid (b) Liquid (c) Groove (d) Gentle
Ans. : (a), (c)

18. ID selector is written with ..... character followed by id name and class selector is written with ..... character followed by class name.
(a) asterisk (*) (b) Dollar ($) (c) hash (#) (d) period (•)
Ans. : (c), (d)

19. Types of positioning in CSS are ..... March 2022
(a) Form (b) Constant (c) Fixed (d) static
Ans. : (c), (d)

20. Following are the types of floating properties .....
(a) float : left (b) float : right (c) float : top (d) float : bottom
Ans. : (a), (b)

21. Following are the values of Display properties. March 2022
(a) Middle (b) Inline (c) Block (d) top
Ans.: (b), (c)

22. Font style property has following values _____.
(a) solid (b) italic (c) groove (d) oblique
Ans.: (b), (d)

23. Types of positioning are _____. March 2022
(a) static (b) logical (c) absolute (d) disabled
Ans.: (a), (c)

24. Value of font size property can be in _____.
(a) pt (b) mp (c) px (d) mx
Ans.: (a), (c)

1.5 Ordered list or numbered list

25. An ordered list can be ..... or ..... March 2022
(a) Static (b) Numerical (c) Constant (d) Alphabetical
Ans. : (b), (d)

26. Following are the values of type attribute of <ol> tag .....
(a) I (b) 1 (c) 5 (d) z
Ans. : (a), (b)

27. Following are the attributes of <ol> tag ..... March 2022
(a) Alphabet (b) start (c) Number (d) Reversed
Ans. : (b), (d)

1.6 Unordered list or bulleted list

28. List item starts with <li> tag in ..... and ..... tags
(a) <dd> (b) <dt> (c) (d)
Ans. : (c), (d)

29. Following are the values of type attribute of <ul> tag ..... March 2022
(a) circle (b) triangle (c) square (d) bold
Ans. : (a), (c)

1.7 Definition list

30. To create items in definition list ..... and ..... tag are used.
(a) <dt> (b) <dd> (c) <dm> (d)
Ans. : (a), (b)

31. Examples of nested list are ..... and ..... March 2022
(a) Double list (b) Single level list
(c) Multi level list (d) Grouping list
Ans. : (b), (c)

1.8 Inserting audio and video in HTML5

32. Common audio formats are .....
(a) .aac (b) .mp3 (c) .mov (d) .webm
Ans. : (a), (b)

33. Following are attributes of <audio> tag ..... March 2022
(a) width (b) autoplay (c) muted (d) height
Ans. : (b), (c)

34. Following attributes of <audio> tag does not have any values. March 2022
(a) controls (b) arc (c) loop (d) muted
Ans. : (a), (d)

35. Following are common video formats .....
(a) .ogg (b) .aac (c) .mp4 (d) .webM
Ans. : (c), (d)

36. Following are the attributes of <video> tag ..... March 2022
(a) href (b) Src (c) loop (d) link
Ans. : (b), (c)

37. Preload attribute of <video> tag have following values .....
(a) Src (b) width (c) auto (d) Metadata
Ans. : (c), (d)

38. To set the width and height of video following attributes of <video> tag are used .....
(a) top (b) bottom (c) width (d) height
Ans. : (c), (d)

1.9 Image map in HTML5

39. Image maps are of two types ..... March 2022
(a) My side (b) client side (c) Host side (d) server side
Ans. : (b), (d)

40. Tags used to define Client Side Image Map are .....
(a) <img> (b) <link> (c) <map> (d) <alt>
Ans. : (a), (c)

41. A given ..... element can contain multiple ..... element with it.
(a) <map> (b) <area>
(c) <link> (d) <body>
Ans. : (a), (b)

42. Following are the attributes of <area> tag. March 2022
(a) Href (b) Src (c) link (d) coords
Ans. : (a), (d)

43. Shape attribute of <area> tag can have following values.
(a) Square (b) Rect (c) Circle (d) Star
Ans. : (b), (c)

1.10 Inline Frame in HTML5

44. Attributes of <iframe> tag are .....
(a) Src (b) href (c) loop (d) Srcdoc
Ans. : (a), (d)

1.11 Website Hosting

45. Types of Web Hosting are ..... March 2022
(a) Fixed hosting (b) Shared Hosting
(c) Dedicated hosting (d) My Hosting
Ans. : (b), (c)

MCQ (Three Correct Answers)

1.1 Advanced Web Designing

1. The major browsers which support features of HTMLs are .....
(a) Google Chrome (b) Window explorer
(c) My Documents (d) Safari
(e) Opera (f) My files
Ans. : (a), (d), (e)

2. Following form controls are used to collect different kinds of user inputs. March 2022
(a) text (b) textbox (c) radio button
(d) checkbox (e) radio (f) option button
Ans. : (a), (d), (e)

1.2 Forms in HTML5

3. <input type = "date"> defines a date picker with the ....., ... and .....
(a) week (b) Date (c) year
(d) month (e) day (f) Mthname
Ans. : (c), (d), (e)

4. Following are the valid values of type attribute of <input> tag .....
(a) link (b) date (c) list
(d) email (e) file (f) start March 2022
Ans. : (b), (d), (e)

5. Following are the valid attributes of <input> tag .....
(a) autofocus (b) link (c) pattern
(d) list (e) multiple (f) start
Ans. : (a), (c), (e)

1.3 <meta> tag

6. Following are the attributes of <meta> tag ..... March 2022
(a) Content (b) Charset (c) Name
(d) Min (e) Max (f) Minimum
Ans. : (a), (b), (c)

7. Name attribute of <meta>tag specifies the name of meta data like ....., ... or .....
(a) Content (b) author (c) Keywords
(d) Src (e) Description (f) Desc
Ans. : (b), (c), (e)

8. Following are the valid values of http-equiv attribute of <meta> tag .....
(a) Refresh (b) expires (c) get-cookie (d) make-cookie
(e) set-cookie (f) get-value
Ans. : (a), (b), (e)

1.4 Cascading Style Sheets HTML5

9. Following are the types of CSS .....
(a) Inline (b) Internal (c) Object (d) External
(e) Multiple (f) Multilevel
Ans. : (a), (b), (d)

10. Text-decoration property is used to add ..... **March 2022**
(a) Strike-through (b) line-through
(c) underline (d) overline
(e) middle line (f) Line between
Ans. : (b), (c), (d)

11. Text-align property has following value .....
(a) left (b) top (c) right (d) Center
(e) middle (f) Joint
Ans. : (a), (c), (d)

12. Following are valid CSS properties.
(a) Mg-color (b) Bg-color (c) Color (d) Background-image
(e) Background – color (f) Fg-color
Ans. : (c), (d), (e)

13. Types of selectors used in CSS are ..... **March 2022**
(a) group (b) grouping (c) universal
(d) block (e) class (f) static
Ans. : (b), (c), (e)

14. Following are the valid CSS properties used to control fonts .....
(a) font-figure (b) font-family (c) font-style
(d) font-weight (e) font-color (f) font-face
Ans. : (b), (c), (d)

15. The elements can be positioned using ..... properties. **March 2022**
(a) top (b) bottom (c) middle (d) right
(e) up (f) down
Ans. : (a), (b), (d)

16. Types of Positioning in CSS are .....
(a) Static (b) Fixed (c) Absolute (d) Mixed
(e) Firm (f) Constant
Ans. : (a), (b), (c)

17. Following are the type of floating properties **March 2022**
(a) Float : mid (b) Float : top (c) Float : left
(d) Float : none (e) Float : right (f) Float : min
Ans. : (c), (d), (e)

18. Following are the values of Display Property .....
(a) Internal (b) Inline (c) Block (d) Block-inline
(e) Outline (f) Between
Ans. : (b), (c), (d)

1.5 Ordered list or numbered list

19. Following are the valid values of type attribute of <ol> tag .....
(a) I (b) 1 (c) a (d) 2
(e) 0 (f) C
Ans. : (a), (b), (c)

20. Following are the attributes of <ol> tag .....
(a) loop (b) type (c) Name (d) Reversed
(e) start (f) src
Ans. : (b), (d), (e)

1.6 Unordered list or bulleted list

21. Following are the valid values of type attribute of <ul> tag .....
(a) rectangle (b) square (c) triangle (d) circle
(e) Disc (f) diamond **March 2022**
Ans. : (b), (d), (e)

1.8 Inserting audio and video in HTML5

22. Following are the common Audio format .....
(a) .mp3 (b) .aac (c) .mov (d) .ogg
(e) .Webm (f) .www
Ans. : (a), (b), (d)

23. Following are the attribute of <audio> tag ..... **March 2022**
(a) Width (b) Controls (c) loop (d) muted
(e) Height (f) Depth
Ans. : (b), (c), (d)

24. Following attributes of <audio> and <video> tags does not have any values.
(a) Autoplay (b) Controls (c) Src (d) Muted
(e) loop (f) Href
Ans. : (a), (b), (d)

25. Following are the common video formats ..... March 2022
(a) .AV (b) .mp4 (c) .ogg (d) .webM
(e) .aac (f) .MIDI

Ans. : (b), (c), (d)

26. Following are the attributes of <video> tag .....
(a) Height (b) links (c) href (d) loop
(e) muted (f) video

Ans. : (a), (d), (e)

27. Following are the valid values of preload attribute of <video> tag .....
(a) Auto (b) Metadata (c) None (d) Autoplay
(e) Autopause (f) SetData

Ans. : (a), (b), (c)

1.9 Image map in HTML5

28. Tags used to define Client Side Image Map are ..... March 2022
(a) <link> (b) <img> (c) <href> (d) <map>
(e) <area> (f) <image>

Ans. : (b), (d), (e)

29. Following are the values of shape attribute of <area> tag .....
(a) rect (b) circle (c) square (d) star
(e) poly (f) triangle

Ans. : (a), (b), (e)

30. Following are the attributes of <area> tag.
(a) href (b) coords (c) width (d) alt
(e) height (f) src doc

Ans. : (a), (b), (d)

1.10 Inline Frame in HTML5

31. Following are the attributes of <iframe> tag
(a) Src (b) Name (c) Srcdoc (d) loop
(e) href (f) width

Ans. : (a), (b), (c)

1.11 Website Hosting

32. Types of Web hosting are ..... March 2022
(a) Fixed hosting (b) Dedicated hosting
(c) formal hosting (d) shared hosting
(e) free hosting (f) No hosting

Ans. : (b), (d), (e)

Match the pair

1.2 Forms in HTML5

(I)

	A		B
(1)	<input type = "color"> March 2022	(a)	Defines image as submit button
(2)	<input type = "number">	(b)	Define the color picker
(3)	<input type = "url">	(c)	Defines date picker
(4)	<input type = "image">	(d)	Defines a field for entering number
		(e)	Defines a field for entering URL

Ans. : (1) – (b), (2) – (d), (3) – (e), (4) – (a)

(II)

	A		B
(1)	<input type = "date">	(a)	Defines month and year control
(2)	<input type = "email"> March 2022	(b)	Defines a range control
(3)	<input type = "month">	(c)	Defines week control
(4)	<input type = "range">	(d)	Defines a date picker
		(e)	Defines field for email address

Ans. : (1) – (d), (2) – (e), (3) – (a), (4) – (b)

(III)

	A		B
(1)	<input type = "datetime – local">	(a)	Defines a control for entering a time
(2)	<input type = "time">	(b)	Defines a date picker
(3)	<input type = "week">	(c)	Defines a file select field
(4)	<input type = "search">	(d)	Defines a text field for entering a search string
(5)	<input type = "file">	(e)	Defines a week and year control
(6)	<input type = "tel">	(f)	Defines color picker
		(g)	Defines input field that contains telephone number.

Ans. : (1) – (b), (2) – (a), (3) – (e), (4) – (d), (5) – (c), (6) – (g)

(IV)

	A		B
(1)	Disabled	(a)	Specifies regular expression to check input values
(2)	Max	(b)	Specifies that an input field should be disabled
(3)	Min	(c)	Specifies maximum value for an input field
(4)	Pattern	(d)	Specifies height and width
		(e)	Specifies minimum value for an input field

Ans. : (1) – (b), (2) – (c), (3) – (e), (4) – (a)

(V)

	A		B
(1)	Read only	(a)	Specifies that an input field is required
(2)	Placeholder	(b)	Specifies that input field is read only
(3)	Required	(c)	This acts as a temporary label
(4)	Autocomplete	(d)	Specifies whether a form should have auto-complete ON or OFF
		(e)	Defines a control for entering time

Ans. : (1) – (b), (2) – (c), (3) – (a), (4) – (d)

1.3 <meta> tag

(I)

	A		B
(1)	Name	(a)	Used for http response message headers
(2)	Content	(b)	Specifies the name of meta data
(3)	Charset	(c)	It can have any textual matter related to the name
(4)	http-equiv **March 2022**	(d)	Specifies the character encoding used by the document
		(e)	Changes color of the text

Ans. : (1) – (b), (2) – (c), (3) – (d), (4) – (a)

1.4 Cascading Style Sheets HTML5

(I)

	A		B
(1)	Selector	(a)	It contains declarations separated by semicolon
(2)	Declaration Block	(b)	It is a type of attribute of HTML document
(3)	Property **March 2022**	(c)	These are assigned to CSS properties
(4)	Value	(d)	It indicates the HTML element you want to style
		(e)	Inline CSS

Ans. : (1) – (d), (2) – (a), (3) – (b), (4) – (c)

(II)

	A		B
(1)	Font-weight	(a)	Used to control fonts
(2)	Font- style	(b)	Used to bold text
(3)	Font –family	(c)	Used to italicize text
(4)	Font-size	(d)	Used to control size of the font
		(e)	Changes the color of the text

Ans. : (1) – (b), (2) – (c), (3) – (a), (4) – (d)

(III)

	A		B
(1)	Background-color	(a)	Changes the color of the text
(2)	Color	(b)	Sets the image as background of the web page
(3)	Background image	(c)	Sets the background color of the web page
(4)	Border	(d)	Use to bold text
		(e)	Adds the border to the webpage

Ans. : (1) – (c), (2) – (a), (3) – (b), (4) – (e)

(IV)

	A		B
(1)	Text-decoration	(a)	This property is used to control horizontal alignment of text
(2)	Text- align **March 2022**	(b)	This property is used to add blank spaces around the content of element.
(3)	Letter-spacing	(c)	This property is used to control horizontal spacing between characters.
(4)	Padding	(d)	This property is used to add underline, overline effects.
		(e)	Used to italicize text

Ans. : (1) – (d), (2) – (a), (3) – (c), (4) – (b)

(V)

	A		B
(1)	Class selector **March 2022**	(a)	Used as wild character
(2)	ID Selector	(b)	Written with hash (#) character
(3)	Universal Selector	(c)	Written with period (•) character
(4)	Grouping Selector	(d)	Use to select all the elements with same style definitions
		(e)	Sets the position of the text

Ans. : (1) – (c), (2) – (b), (3) – (a), (4) – (d)

(VI)

	A		B
(1)	Static Positioning **March 2022**	(a)	This property forces an element into fixed position
(2)	Fixed Positioning	(b)	This property sets an element in a specific location
(3)	Relative Positioning	(c)	Selects Id attribute of HTML element
(4)	Absolute positioning	(d)	This property is used to set the element relative to its normal position
		(e)	This property is default position for HTML elements

Ans. : (1) – (e), (2) – (a), (3) – (d), (4) – (b)

1.5 Ordered list or numbered list

(I)

	A		B
(1)		(a)	Defines values for the list
(2)	Type	(b)	Displays list items in reverse order
(3)	Reversed	(c)	Specifies starting number of list
(4)	Start	(d)	Specifies list items
		(e)	Defines ordered list

Ans. : (1) – (e), (2) – (a), (3) – (b), (4) – (c)

1.6 Unordered list or bulleted list

(I)

	A		B
(1)	**March 2022**	(a)	Create unordered list
(2)		(b)	Sets the list item marker to a circle
(3)	Type = "circle">	(c)	Sets the list item marker to square
(4)	Type = "square">	(d)	The list items will not be marked
		(e)	Specifies list items

Ans. : (1) – (a), (2) – (e), (3) – (b), (4) – (c)

1.7 Definition list

(I)

	A		B
(1)	<dl>	(a)	Tag used to define term
(2)	<dd>	(b)	Tag used to define term's definition
(3)	<dt> **March 2022**	(c)	Used to define unordered list
(4)	Nested list	(d)	Defines definitions list
		(e)	List within another list

Ans. : (1) – (d), (2) – (b), (3) – (a), (4) – (e)

1.8 Inserting audio and video in HTML5

(I)

	A		B
(1)	<audio>	(a)	Advanced Audio Coding
(2)	Mp3	(b)	Automated Arrange Control
(3)	Aac	(c)	Open container and free audio format
(4)	Ogg	(d)	This element enables you to add audio files an webpages
		(e)	Motion picture expert group

Ans. : (1) – (d), (2) – (e), (3) – (a), (4) – (c)

(II)

	A		B
(1)	Autoplay	(a)	Specifies URL of audio/video file
(2)	Controls	(b)	The audio-video controls should be displayed
(3)	Loop	(c)	This audio will start playing as soon as it is ready
(4)	Src **March 2022**	(d)	Sets the height of the audio/video player
		(e)	Video / audio will start over again every time it is finished.

Ans. : (1) – (c), (2) – (b), (3) – (e), (4) – (a)

(III)

	A		B
(1)	Muted	(a)	Sets the height of the video player
(2)	Height	(b)	Specifies if and how the author should be loaded when webpage loads.
(3)	Poster	(c)	Sets the width of the video player
(4)	Preload	(d)	Specifies that the audio output of the video should be muted
		(e)	Specifies on image to be shown while video is downloading

Ans. : (1) – (d), (2) – (a), (3) – (e), (4) – (b)

1.9 Image map in HTML5

(I)

	A		B
(1)	<img>	(a)	It is used to connect links to different regions on the web page
(2)	<map>	(b)	Defines the clickable regions
(3)	<area>	(c)	It specifies name of image used for client side image map
(4)	Image map	(d)	Clickable regions are called as hotspots
		(e)	It is used to insert image on the web page

Ans. : (1) – (e), (2) – (c), (3) – (b), (4) – (a)

(II)

	A		B
(1)	Href	(a)	It can have value rect, circle and poly
(2)	Shape	(b)	Image with multiple hyperlinks
(3)	Coords **March 2022**	(c)	Defines URL to which the clickable region within the image map navigators
(4)	Alt	(d)	Specifies coordinates of the clickable regions
		(e)	Specifies extra information about clickable area

Ans. : (1) – (c), (2) – (a), (3) – (d), (4) – (e)

1.10 Inline Frame in HTML5

(I)

	A		B
(1)	Name	(a)	Specifies the HTML content of the page to show the <iframe>
(2)	Src	(b)	Specifies name of the <iframe>
(3)	Height	(c)	Specifies the address of the document to embed in the <iframe>
(4)	Srcdoc	(d)	Specifies the width of an <iframe>
		(e)	Specifies the height of an <iframe>

Ans. : (1) – (b), (2) – (c), (3) – (e), (4) – (a)

1.11 Website Hosting

(I)

	A		B
(1)	Web host **March 2022**	(a)	It gives domain name to your website
(2)	Shared Hosting	(b)	Websites providing free hosting for limited period
(3)	Dedicated Hosting	(c)	Computer with internet domain system
(4)	Free hosting	(d)	Owns and manages web servers
		(e)	These are paid hosting servers for large websites

Ans. : (1) – (d), (2) – (a), (3) – (e), (4) – (b)

Answer in Brief

1.2 Forms in HTML5

1. Explain advanced <input> elements in HTML5 ?

Ans. :

Input type	Description
<input type="color">	Defines a color picker
<input type="number">	Defines a field for entering a number
<input type="url">	Defines a field for entering a URL.
<input type="image">	Defines an image as a submit button.
<input type="date">	Defines a date picker with the year, month and day
<input type="email">	Defines a field for an e-mail address
<input type="month">	Defines a month and year control in format is "YYYY-MM"
<input type="range">	Define a range control. Default range is 0 to 100.
<input type="datetime-local">	Defines a date picker that includes the year, month, day and time.
<input type="time">	Defines a control for entering a time.
<input type="week">	Defines a week and year control.
<input type="search">	Defines a text field for entering a search string like a site search or Google search.
<input type="file">	Defines a file-select field and a "Browse" button for file uploads.
<input type="tel">	Used to define input fields that should contain a telephone number.

2. Explain the list of common input restrictions which can be used for validation purpose?

Ans. : Following is the list of some common input restrictions is given below, few of which can be used for validation purpose.

Attribute	Description
Disabled	Specifies that an input field should be disabled.
Max	Specifies the maximum value for an input field.
Min	Specifies the minimum value for an input field.

Attribute	Description
Pattern	Specifies a regular expression to check the input values.
read only	Specifies that an input field is read only (cannot be changed).
placeholder	This acts as a temporary label showing the purpose of a text field without requiring a label tag.
Required	Specifies that an input field is required (must be filled out).
autocomplete	Specifies whether a form or input field should have autocomplete On or Off.
autofocus	Specifies that the input fjeld should automatically get focus when the page loads.
height and width	Specifies the height and width of an <input type="image">
Multiple	Specifies that the user is allowed to enter more than one value in the <input> element. This works with input types like email and file.

1.3 <meta> tag

3. Explain <meta> tag?

Ans. :

1) HTML <meta> tag is used to represent the metadata about the HTML document. It specifies page description, keywords, copyright, language, author of the documents, etc.
2) The metadata does not display on the webpage, but it is used by search engines, browsers and other web services which scan the site or webpage to know about the webpage.
3) With the help of meta tag, you can experiment and preview that how your webpage will render on the browser. The <meta> tag is placed within the <head> tag, and it can be used more than one times in a document.

4. Explain attributes of <meta> tag?

Ans. : Attributes of <meta> tag

Attribute	Values	Description
Name	The value of the name attribute can be related to any of the following : i) Author ii) Description iii) Keywords iv) copyright e.g. <meta name = "author" >	Specifies the Name of the meta- data like the author, keywords or description.

Attribute	Values	Description
Content	It can have any textual matter related to the name as in eg. (i) <meta name = "author" content = "Balbharti"> (ii) <meta name = "description" content = "Advance web de- signing"> (iii) <meta name = "keywords" content = "html5, learn html5, list in html 5">	Here content of author is balbharati. Here the value for content attribute specifies name of the topic advance web designing Here the values for content attribute are given as keywords like html5 , learn html5 etc.
Charset	UTF-8, Big5 e.g <meta charset="UTF-8"> <meta charset="Big5">	Specifies the character encoding used by the document, This is called a character encoding declaration. UTF-8 For Indian characters Big5 – for Chinese characters
http-equiv	refresh , set-cookie, content-type, expires, e.g. <meta http- equiv="refresh" content="5"> <meta http-equiv="set-cookies"> <meta http-equiv="content-type"content="text/ html"charset="Big5"> <meta http-equiv="expires" content="userid=pqr; expires=Wednesday, 8-feb-2018 23:59:59 GMT;">	Used for http response message headers. Here the page will get refresh after every 5 seconds. The browser sends the cookies back to the server. Specifies the character encoding for the document Here page session will get expire at specified date and time.

1.4 Cascading Style Sheets HTML5

5. Explain Cascading Style Sheets.

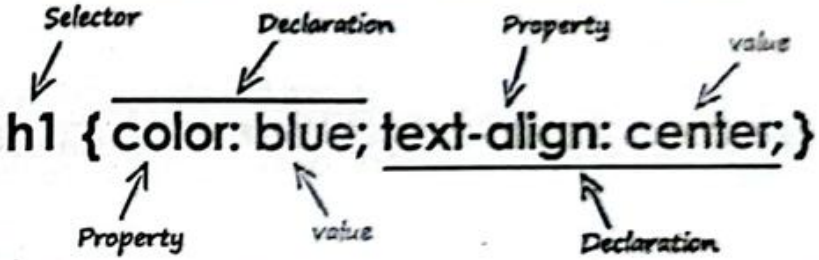
Ans. : CSS stands for Cascading Style Sheets. CSS is a standard style sheet language used for describing the presentation (i.e. the layout and formatting) of the web pages. CSS can either be attached as a separate document or embedded in the HTML document itself. There are three methods of including CSS in an HTML document. Internal, External and Inline Style Sheets.

6. Explain Cascading Style Sheets (CSS) syntax.

Ans. : CSS Syntax :-

1) A CSS style sheet consists of a set of rules that are interpreted by the web browser and then applied to the corresponding elements such as paragraphs, headings, etc. in the document.

2) A CSS rule have two main parts, a selector and one or more declarations :



3) The selector specifies which element or elements in the HTML page the CSS rule applies to. Whereas, the declarations within the block determines how the elements are formatted on a webpage.

4) Each declaration consists of a property and a value separated by a colon (:) and ending with a semicolon (;), and the declaration groups are surrounded by curly braces {}

7. Explain Inline CSS.

Ans. :

1) Inline style sheet uses the style attribute in the HTML tag.

2) Inline CSS is used to apply CSS on a single line or element.

3) For example XII Information Technology

8. Explain Internal/Embedded CSS.

Ans. :

1) Internal CSS is used to apply CSS on a single document or page.

2) It is written inside <head> section of html.

9. Explain External CSS.

Ans. :

1) External CSS is used when we want to make changes on multiple pages.

2) It uses <link> tag which is placed inside head section.

10. Explain various CSS properties.

Ans. :

Property	Use	Value	Example
Color	Changes The Color Of The Text	Color Name	H1{Color: Maroon}

Property	Use	Value	Example
Background-Color	To Set The Background Color In Your Webpage	Color Name	Body{Background-Color:Yellow}
Font-Weight	Used To Bold Text	Bold Or 100, 200..900	P{Font-Weight:300}
Font-Style	Used To Italicize Text	Italic, Oblique Or Normal	P{Font-Style:Italic}
text-decoration	This Property Is Used To Add 1. Strike-Through Marks 2. Underline 3. Overstrike 4. To Remove Underlines From Links	1. Line-Through 2. Underline 3. Overline 4. None	P{Text-Decoration: Underline} A{Text-Decoration: None}
Text-Align	This Property Is Use To Control The Horizontal Alignment Of Any Block-Level Text That Are Paragraphs, Tables And Other Elements	Left, Right, Center Or Justify	H1{Text-Align:Center}
Font-Family	This Is Used To Control The Fonts	Font Name	P{Font-Family:Arial}
Font-Size	This Property Allows You To Control The Size Of The Font	Px, In, Mm, Cm, Pt	P{Font-Size:10px}
FONT-STYLE	This property changes the style of the font	Italic	B{font-style:italic}
Letter-Spacing	This Helps In Controlling The Horizontal Spacing Between Characters Of Text	Px, In, Mm, Cm, Pt	H1{Letter-Spacing 5pt}
Padding	This Property Is Used When You Want To Add Padding (Blank Spaces) Around The Content Of An Element.	Pixel	H1{Padding:30px}
Border-STYLE	This Property Adds A Border To A Webpage Element	Solid, Double, Groove, Ridge, Inset, Outset, Dotted Or Dashed	H1{Border-style:groove}

Property	Use	Value	Example
Border-COLOR	This property changes the color of the border	Red	H1{border-color:red}
BORDER-WIDTH	This property increases the width of the border	PX, PT	H1{border-width:10pt}
Background-Image	To Set An Image As The Background Of Your Webpage	Url("X") Where X Is The Path Of Image File	Body{Background-Image: Url("Background.Jpg")}
Margin-Left	Sets Margin Area On The Left Side Of The Element.	Px,Pt,Cm Etc.	H1{Margin-Left;10px}

11. Explain CSS Id Selector.

Ans. : The id selector is used to define style rules for a *single* or *unique* element. The id selector is defined with a hash sign (#) immediately followed by the id value.

12. Explain CSS Class Selector.

Ans. : The class selectors can be used to select any HTML element that has a class attribute. All the elements having that class will be formatted according to the defined rule. The class selector is defined with a period sign (.) immediately followed by the class value. Class name should not start with a number.

13. Explain CSS Universal Selector.

Ans. : The universal selector is used as wildcard character, denoted by an asterisk (*), matches every single element on the page.

14. Explain CSS Group Selector.

Ans. : Often several selectors in a style sheet share the same style rules declarations. You can group them into a comma-separated list to minimize the code in your style sheet. It also prevents you from repeating the same style rules over and over again.

15. Explain Positioning in CSS.

Ans. : Positioning elements appropriately on the web pages is a necessity for a good layout design.

Syntax is :-

Selector(position:value; top:value;bottom:value;left:value;right:value}

Following are four(4) methods in CSS that you can use for positioning elements.

1) Static Positioning :

A static position is the default position for html element, A static positioned element is always positioned according to the normal flow of the page. HTML elements are positioned static by default. Static positioned elements are not affected by the top, bottom, left, right, and z-index properties.

2) Fixed Positioning

This property helps to put the text fixed on the browser.

3) Relative Positioning :

The relative positioning property is used to set the element relative to its normal position.

4) Absolute Positioning

This property sets an element in a specific location and it is not affected by the flow of the page.

16. Explain Float Property

Ans. : Float is a CSS property written in CSS file . It defines the flow of the content.

Following are the types of floating properties :

1. **float : left :** this keeps the element float on left side of the container
2. **float : right :** this keeps the element float on right side of container
3. **float : none :** this is default property i.e. this shows the element as it is.

17. Explain Display Property.

Ans. : The display property defines how the components are going to be placed on the web page.

Syntax :

display : value;

where values are :

inline : it is used to display an element as an inline element.

block : it is used to display an element as an block element. it starts on a new line, and takes up the whole width of the browser window.

block-inline : this value is very similar to inline element but the difference is that you are able to set the width and height.

none : the element is completely removed.

1.5 Ordered List or Numbered List

18. Explain Ordered list or Numbered List.

Ans. : The <ol> tag defines the ordered list. It can be numbered or alphabetical Attributes used are :-

Attribute	Values	Description
Type	"1" /"a"/"I"/"i"	1 is default value and other values specify the numbering type for the used items.
Reversed	Reversed	This attribute specifies that the items of the list are specified in the reverse order.
Start	Number	Specifies the starting number of the first item in an ordered list.

1.6 Unordered List or Bulleted List

19. Explain Unordered list or Bulleted List.

Ans. : Unordered list is created by <ul> tag. It is also called as bulleted list.

Attribute	Values	Description
Type	Disc, Square, circle	Defines the type of the list as circ le, square , disc. Default is disc.

1.7 Definition List

20. Explain Definition List & Nested list.

Ans. : Definition List :-

The definition list is created using <dl> element. The <dl> element is used in conjunction with the <dt> element which specify a term, and the <dd> element which specify the term's definition.

Nested List

List within another list either ordered or unordered list is called nested list.

1.8 Inserting Audio and Video in HTML5

21. Explain <audio> tag with attributes.

Ans. : The newly introduced HTML5 <audio> element provides a standard way to embed audio in web pages. File formats supported in audio are mp3, aac, ogg etc. The <source> tag is sued to add multiple files for media elements.

Syntax is :-

<audio src="music.mp3" type="audio/mpeg" controls> </audio>

Attributes of <audio> tag :-

Attribute	Values	Description
Autoplay	-	The audio will start playing as soon as it is ready
Controls	-	The audio controls should be displayed (i.e. play/pause button etc.)
loop	Value	The audio will start over again, every time it is finished
muted	-	This Specifies that the audio output should be muted
src	URL	Specifies the URL of the audio file

22. Explain <video> tag with attributes.

Ans. : <video> tag

The newly introduced HTML5 <video> element provides a standard way to embed video in web pages. File formats supported in video are mp4, ogg, WebM etc. The <source> tag is sued to add multiple files for media elements.

Syntax is :-

<video src="video.mp4" controls> </video>

Attributes of <video> tag are :-

Attribute	Values	Description
Src	URL	Defines link to video file
autoplay	-	Specifies that the video will start playing as soon as it is ready
controls	-	Specifies that video controls should be displayed (such as a play/pause button etc).
height	Pixels	Sets the height of the video player
loop	Value	Specifies that the video will start over again, every time it is finished
muted	-	Specifies that the audio output of the video should be muted
poster	URL	Specifies an image to be shown while the video is downloading, or until the user hits the play button
preload	1. Auto 2. metadata 3. none	Specifies if and how the author thinks the video should be loaded when the webpage loads
width	Pixels	Sets the width of the video player

1.9 Image map in HTML5

23. Explain Image Mapping.

Ans. :

1. An image map allows you to define hotspots on an image that acts just like a hyperlink. Image maps are of two type Client side Image Maps and server Side Image Maps.
2. Tags used to define Client Side Image Maps are :
   1. **<img>** → It is used to insert image on the web page.
   2. **<map>.. </map>** → It specifies name of the image used for client side image map.
   3. **<area>** → It defines the clickable region.

Attributes of <area> tag are :

Attribute	Description
Href	Defines the URL to which the clickable region within the image-map navigates.

Attribute	Description
Shape	It can value rect, circle or poly.
coords	Specifies co-ordinates of the clickable regions on the image-map.
Alt	Specifies extra information about clickable area. It is the alternative text to the clickable region.

1.10 Inline Frame in HTML5

24. Explain <iframe> tag.

Ans. : An iframe or inline frame is used to display external objects including other web pages within a web page, its is sued for online advertisements etc. The <iframe> tag is used to create inline frames.

Attributes of <iframe> tag are :

Attribute	Values	Description
Src	URL	Specifies the address of the document to embed in the<iframe>
Height	Pixel	Specifies the height of an<iframe>
Width	Pixels	Specifies the width of an<iframe>
Name	Text	Specifies name of an <iframe>
srcdoc	HTML_code	Specifies the HTML content of the page to show in the <iframe>

1.11 Web Hosting

25. What is web hosting?

Ans. : Web hosting is the service of providing storage space. The website is made available on the internet with the help of web hosting.

26. What is web host?

Ans. : The companies that provides web hosting services are called web hosts. Web hosts own and manage web servers. These web servers offer uninterrupted internet connectivity.

Types of web hosting :

Types of web hosting are :

1. **Shared hosting :** It is cost effective. It gives domain name to your website.
2. **Free hosting :** There are some hosting websites which provide you free hosting of the website for limited period of time.
3. **Dedicated hosting :** These are paid hosting servers for large websites.

Programs for Practice

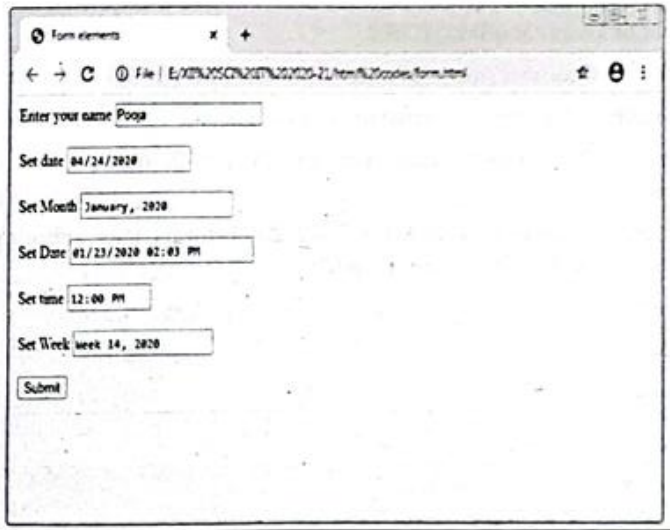
Examples of advanced input elements

1. **Design a web page that should accept Personal Details of the user i.e. name of the user along with date and time values. The Page must contain submit button.**

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Form Date & Time elements
</title>
</head>
<body>
<form name="f1">
Enter your name
<input type="text" name="t1" autocomplete="on"><br><br>
Set date
<input type="date"><br><br>
Set Month
<input type="month"><br><br>
Set Date
<input type="datetime-local"><br><br>
Set time
<input type="time"><br><br>
Set Week
<input type="week"><br><br>
<input type="submit" name="b1" value="Submit">
</form>
</body>
</html>
```


Output :



2. **Design a web page that should accept name of the user, Email ID, Number of years completed in office, Office phone number(compulsory), image with submit button.**

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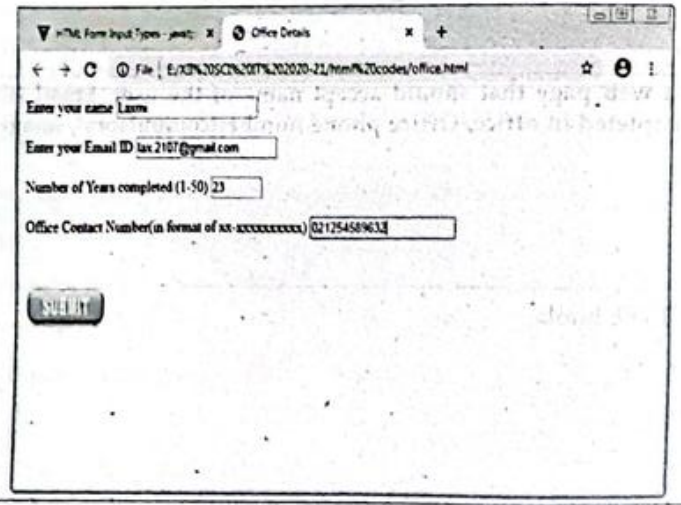
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Office Details
</title>
</head>
<body>
<form name="f1">
Enter your name
<input type="text" name="t1" autocomplete="off"><br><br>
Enter your Email ID
```

```
<input type="email" name="emailid"><br><br>
Number of Years completed (1-50)
<input type="number" min="1" max="50"><br><br>
Office Contact Number(in format of xx-xxxxxxxxxx)
<input type="tel" name="contact" pattern="[0-9] {2}-[0-9] {10}"
required><br><br>
<input type="image" src="E:\XII SCI IT 2020-21\html codes\submit.jpg"
alt="Submit" width="100" height="100">
</form>
</body>
</html>
```

Output :



3. Design a web page that should accept name of the user, select file for upload, color picker tool, website URL, search and submit button.

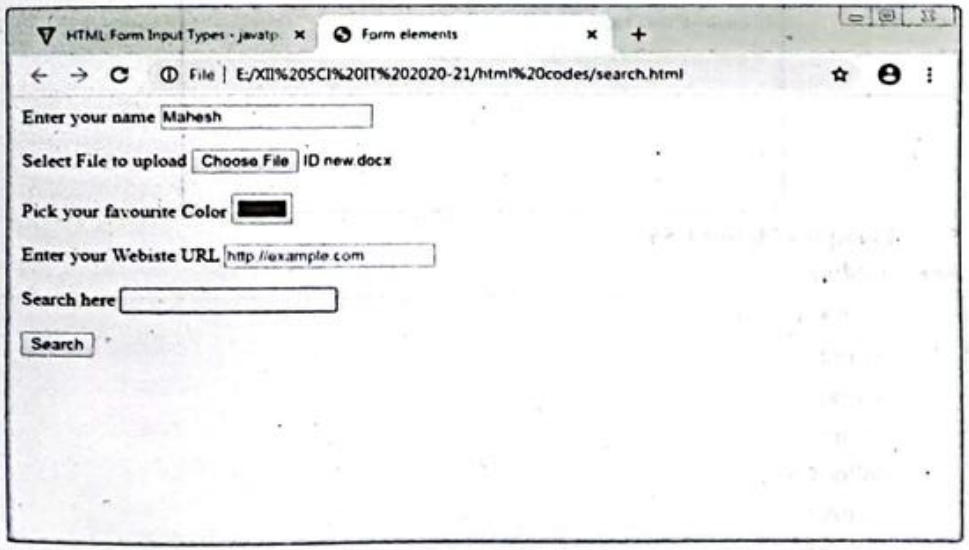
Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Form elements
</title>
```



```
</head>
<body>
<form name="f1">
Enter your name
<input type="text" name="t1" autocomplete="on"><br><br>
Select File to upload
<input type="file" name="newfile"><br><br>
Pick your favourite Color
<input type="color" name="color"><br><br>
Enter your Webiste URL
<input type="url" name="website" placeholder="http://example.com"><br><br>
Search here
<input type="search" name="s"><br><br>
<input type="submit" name="b1" value="Search">
</form>
</body>
</html>
```

Output :

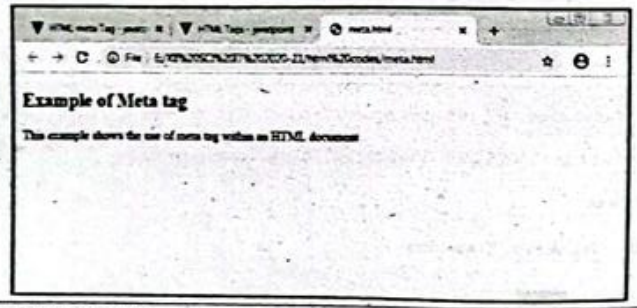


4. Example of Meta tag

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<meta name="keywords" content="Form elements">
<meta name="description" content="input elements">
<meta name="author" content="thisauthor">
<meta http-equiv="refresh" content="5 url=file:///E:/XII%20SCI
%20IT%202020-21/html%20codes/form.html">
</head>
<body>
<h2>Example of Meta tag</h2>
<p>This example shows the use of meta tag within an HTML document</p>
</body> </html>
```

Output :



5. Example of Inline CSS

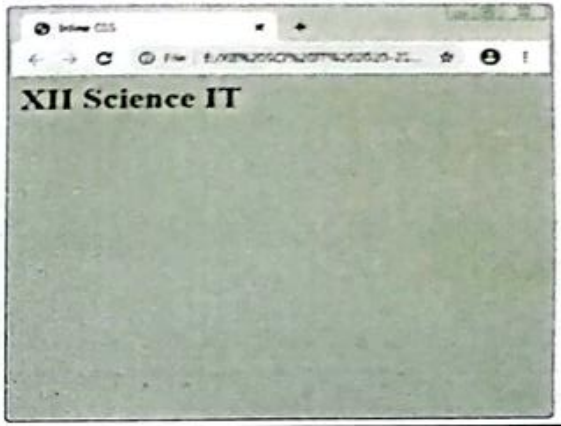
Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Inline CSS
</title>
</head>
<body style="background-color:pink">
```



```
<b style="color:red,font-size:25pt">XII Science IT</b>
</body>
</html>
```

Output :



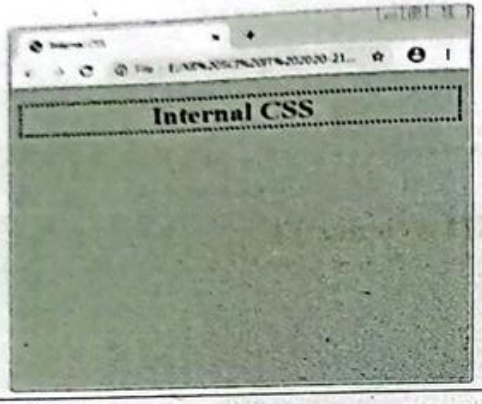
6. Example of Internal CSS.

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Internal CSS
</title>
<style>
h1{border-style:dotted}
body{background-color:skyblue}
</style>
</head>
<body>
<h1 align="center">Internal CSS</h1>
</body>
</html>
```

Output :



7. Example of External CSS

Ans. :

Coding :

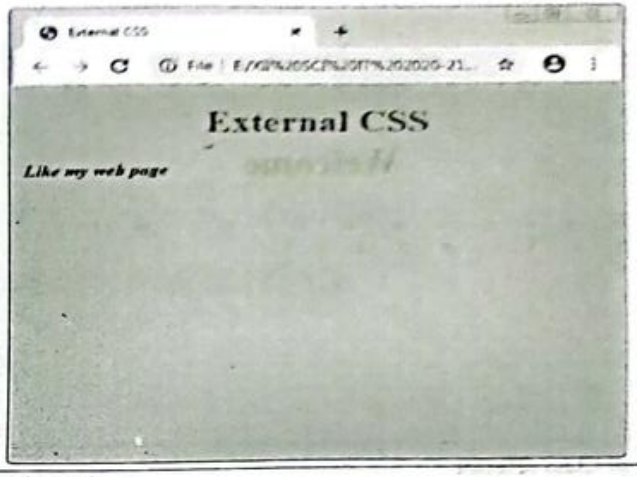
Externalcss.html

```
<!DOCTYPE html>
<html>
<head>
<title>
External CSS
</title>
<link rel="stylesheet" type="text/css" href="style.css">
</head>
<body>
<h1 align="center">External CSS</h1>
<b>Like my web page</b>
</body>
</html>
```

Style.css

```
h1{color:green}
b{font-style:italic}
body{background-color:pink}
```


Output :

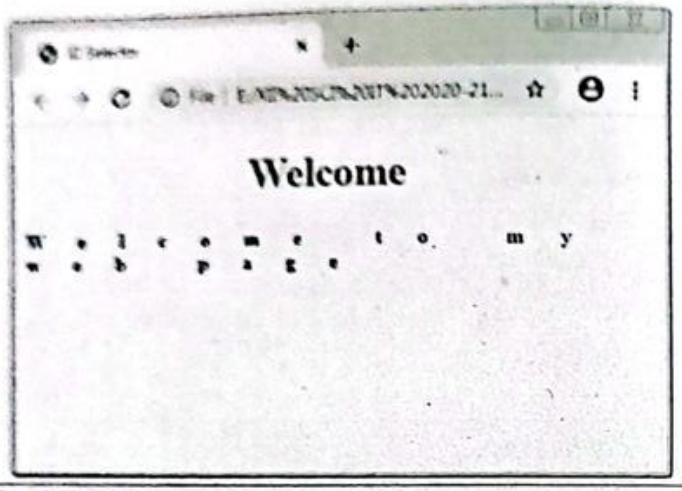


8. Example of Id selector

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
ID Selector
</title>
<style>
#abc{letter-spacing:20pt;color:red}
</style>
</head>
<body>
<h1 align="center">Welcome</h1>
<b id="abc">Welcome to my web page</b>
</body>
</html>
```

Output :

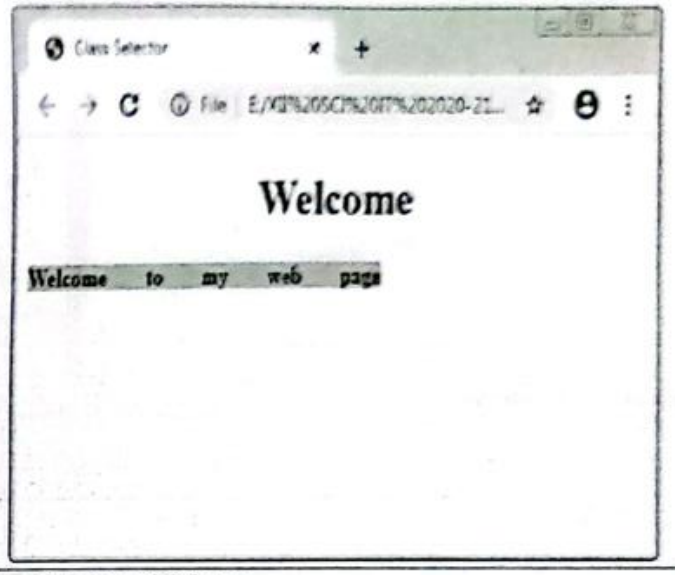


9. Example of Class selector

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
Class Selector
</title>
<style>
.xyz{word-spacing:20pt;background-color:aqua}
</style>
</head>
<body>
<h1 align="center">Welcome</h1>
<b class="xyz">Welcome to my web page</b>
</body>
</html>
```


Output :

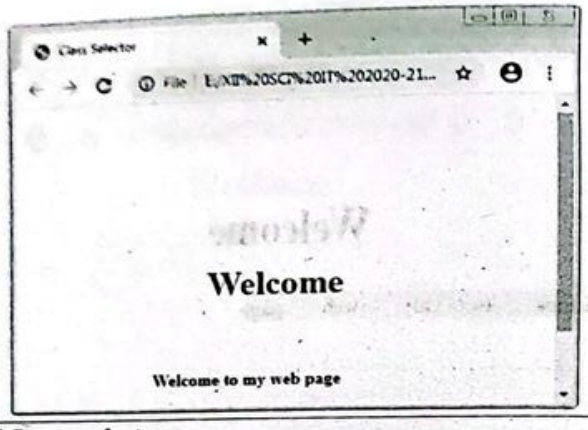


10. Example of Universal Selector

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
Universal Selector
</title>
<style>
*{padding:30pt;color:green}
</style>
</head>
<body>
<h1 align="center">Welcome</h1>
<b>Welcome to my web page</b>
</body> </html>
```

Output :



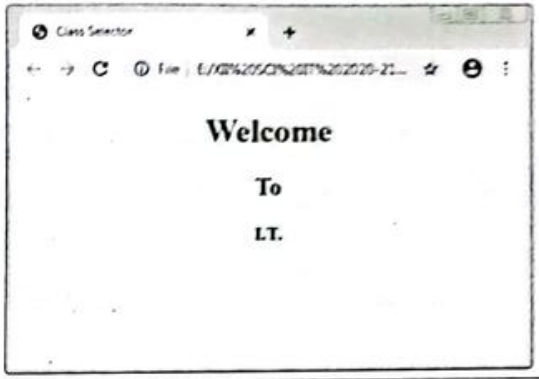
11. Example of Group selector

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Grouping Selector
</title>
<style>
h1,h2,h3{color:green}
</style>
</head>
<body>
<h1 align="center">Welcome</h1>
<h2 align="center">To</h2>
<h3 align="center">I.T.</h3>
</body>
</html>
```


Output :

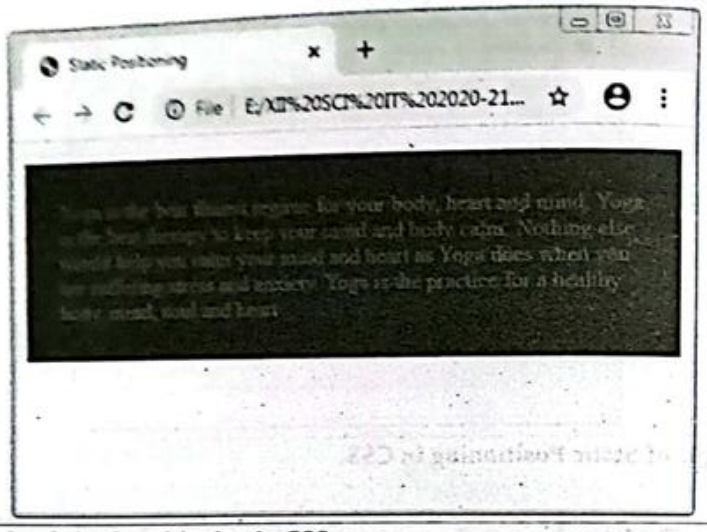


12. Example of Static Positioning in CSS.

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
Static Positioning
</title>
<style>
.static{position:static;color:green;background-color:red;padding:20pt}
</style>
</head>
<body>
<p class="static">
Yoga is the best fitness regime for your body, heart and mind, Yoga is the best therapy to keep your mind and body calm. Nothing else would help you calm your mind and heart as Yoga does when you are suffering stress and anxiety. Yoga is the practice for a healthy body, mind, soul and heart
</p>
</body>
</html>
```

Output :



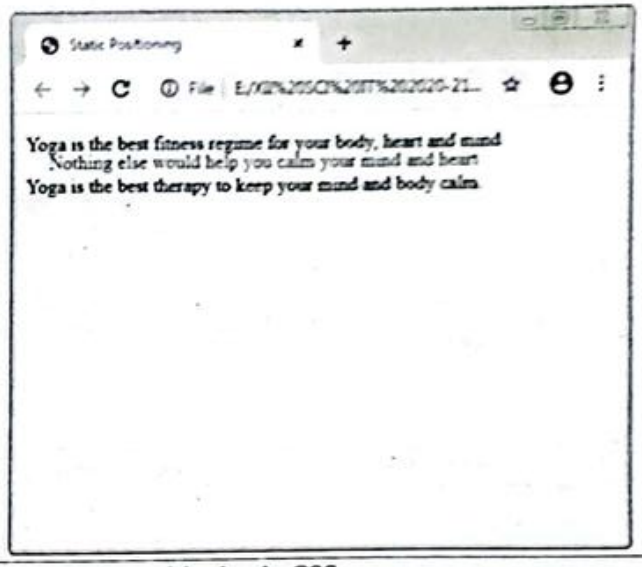
13. Example of Fixed Positioning in CSS.

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Fixed Positioning
</title>
<style>
.fixed{position:fixed;color:green;top:15px;right:120px}
</style>
</head>
<body>
<p>Yoga is the best fitness regime for your body, heart and mind</p>
<p>Yoga is the best therapy to keep your mind and body calm. </p>
<p class ="fixed">Nothing else would help you calm your mind and heart </p>
</body>
</html>
```


Output :



14. Example of Relative Positioning in CSS.

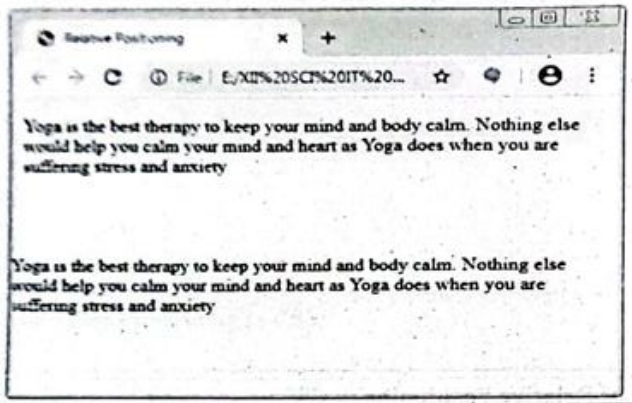
Ans. :

Coding :-

```
<!DOCTYPE html>
<html>
<head>
<title>
Relative Positioning
</title>
<style>
.rel{position:relative;top:55px;right:10px}
</style>
</head>
<body>
<p>
Yoga is the best therapy to keep your mind and body calm. Nothing else would
help you calm your mind and heart as Yoga does when you are suffering stress
and anxiety </p>
<p class="rel">
```

```
Yoga is the best therapy to keep your mind and body calm. Nothing else would
help you calm your mind and heart as Yoga does when you are suffering stress
and anxiety </p>
</body>
</html>
```

Output :



15. Example of Absolute Positioning in CSS.

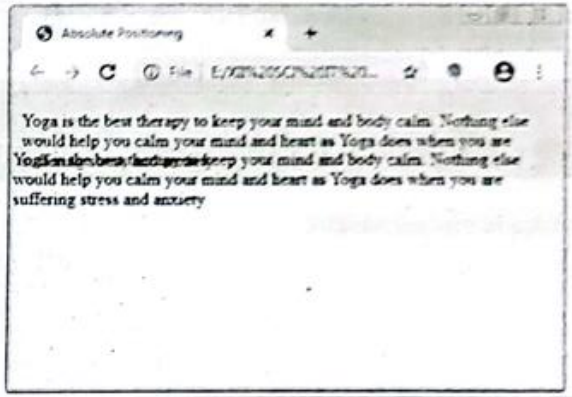
Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
Absolute Positioning
</title>
<style>
.abs{position:absolute;top:35px;right:10px}
</style>
</head>
<body>
<p>
Yoga is the best therapy to keep your mind and body calm. Nothing else would
help you calm your mind and heart as Yoga does when you are suffering stress
and anxiety </p>
<p class="abs">
```



```
Yoga is the best therapy to keep your mind and body calm. Nothing else would
help you calm your mind and heart as Yoga does when you are suffering stress
and anxiety </p>
</body>
</html>
```

Output :



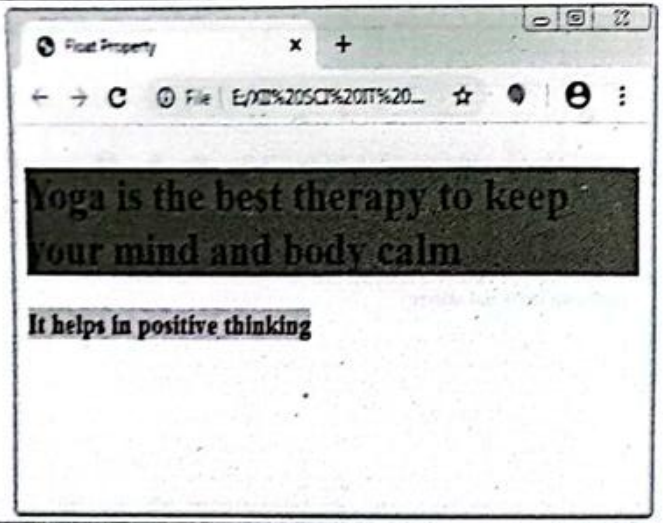
16. Example of Float property.

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
Float Property
</title>
<style>
.left{float:left;background-color:green}
.right{float-right;font-size:15pt;background-

color:gold}
</style>
</head>
<body>
<h1 class="left">Yoga is the best therapy to keep your
```

```
mind and body calm</h1>
<b class="right">It helps in positive thinking</b>
</body>
</html>
```



17. Example of Display Property

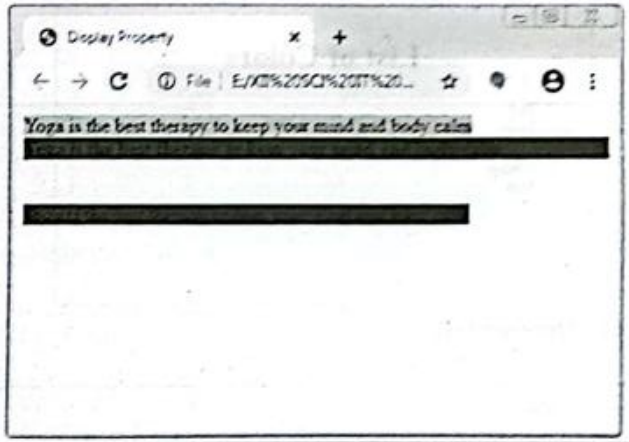
Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
Display Property
</title>
<style>
p{display:inline;background-color:aqua}
b{display:block;background-color:red}
i{display:block-inline;background-color:red}
</style>
</head>
<body>
<p>
Yoga is the best therapy to keep your mind and body calm </p>
```



```
<b>Yoga is the best therapy to keep your mind and body calm</b><br><br>
<i>Yoga is the best therapy to keep your mind and body calm</i>
</body>
</html>
```

Output :



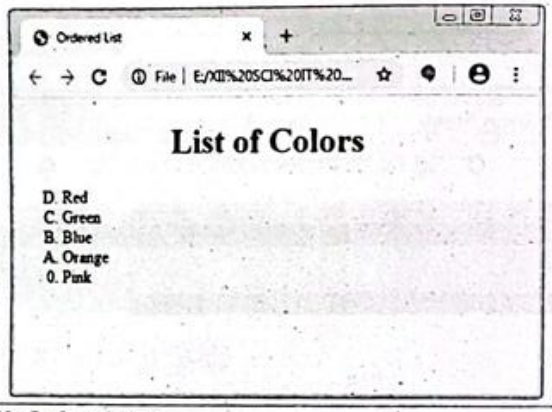
18. Example of Ordered List

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
Ordered List
</title>
</head>
<body>
<h1 align="center">List of Colors</h1>
<ol start="4" type="A" reversed>
<li>Red</li>
<li>Green</li>
<li>Blue</li>
<li>Orange</li>
<li>Pink</li>
</ol>
```

```
</body>
</html>
```

Output :

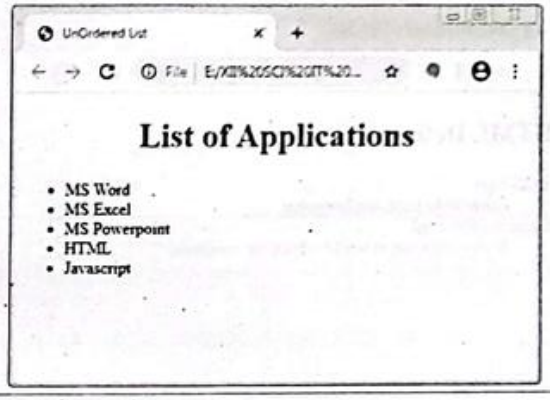


19. Example of UnOrdered List

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>
UnOrdered List
</title>
</head>
<body>
<h1 align="center">List of Applications</h1>
<ul>
<li>MS Word</li>
<li>MS Excel</li>
<li>MS Powerpoint</li>
<li>HTML</li>
<li>Javascript</li>
</ul>
</body>
</html>
```


Output :



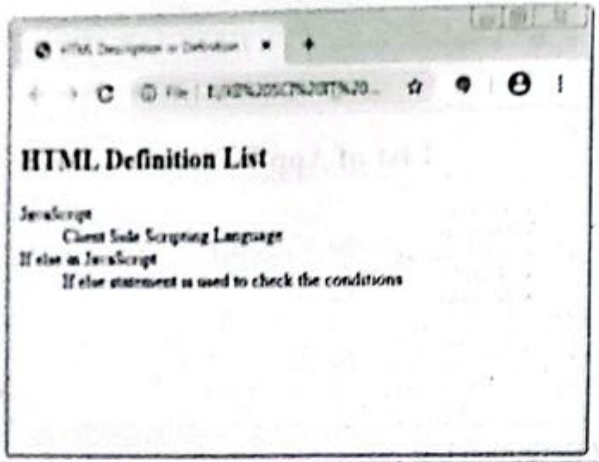
20. Example of Definition List

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Description or Definition List</title>
</head>
<body>
<h2>HTML Definition List</h2>

<dl>
<dt>JavaScript</dt>
<dd>Client Side Scripting Language</dd>
<dt>If else in JavaScript</dt>
<dd>If else statement is used to check the conditions</dd>
</dl>
</body>
</html>
```

Output :



21. Example of Nested List

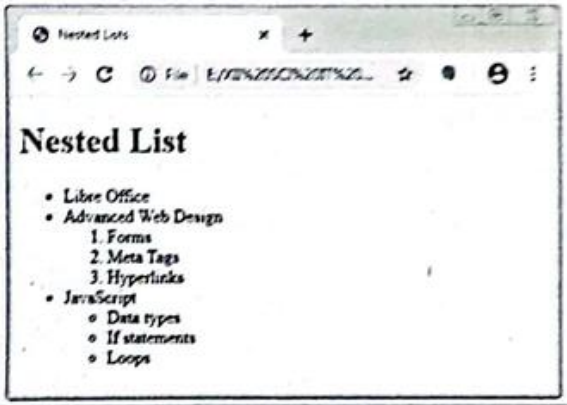
Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>Nested Lists</title>
</head>
<body>
<h1>Nested List</h1>
<ul>
<li>Libre Office</li>
<li>Advanced Web Design</li>
<ol>
<li>Forms</li>
<li>Meta Tags</li>
<li>Hyperlinks</li>
</ol>
<li>JavaScript</li>
<ul style="list-style-type:circle">
<li>Data types</li>
```



```
<li>If statements</li>
<li>Loops</li>
</ul>
</ul>
</body>
</html>
```

Output :



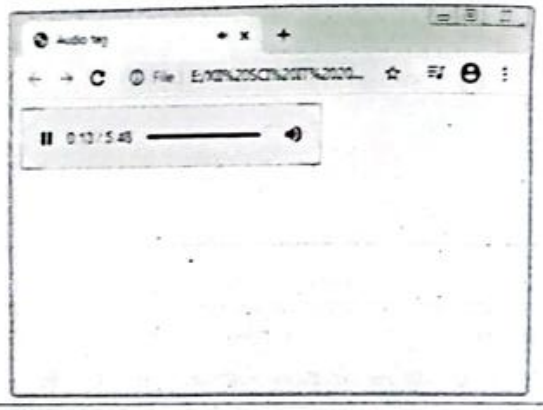
22. Example of <audio> tag

Ans. :

```
<!DOCTYPE html>
<html>
<head>
<title>Audio tag</title>
</head>
<body>
<audio controls muted autoplay loop="5">
<source src="C:\Users\Public\Music\Sample Music

\Kalimba.mp3" type="audio/mpeg">
</audio>
</body>
</html>
```

Output :



23. Example of <video> tag

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>Video tag</title>
</head>
<body>
<video src="C:\Users\Public\Videos\Sample Videos \shuttle.mp4" controls
width="500" height="300" loop="4" muted poster="C:\Users\Public\Pictures\Sample
Pictures\Penguins.jpg" autoplay>  </video>
</body>
</html>
```

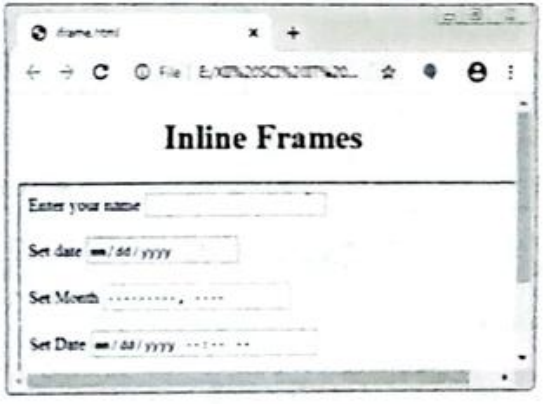
Output :

24. Example of <iframe> tag

Ans. :

```
<!DOCTYPE html>
<html>
<body>
<h1 align="center">Inline Frames</h1>
<iframe src="E:\XII SCI IT 2020-21\html codes \form.html" width="500"
height="250"></iframe>
</body>
</html>
```

Output :



Extra CSS Examples

1. Write a html program to display "Cyber world" having Arial font and background colour cyan. Add any two advantages having red color for the text. **(March 2022)**

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Digital India
</title>
<style>
```

```
h1{font-family:arial;background-color:cyan}
b{color:red}
</style>
</head>
<body>
<h1 align="center">Digital India</h1>
<h2>Advantages</h2>
<b>Infrastructure as a utility to each citizen</b><br>
<b>High-speed internet will be made available in all the gram panchayats</b>
</body>
</html>
```

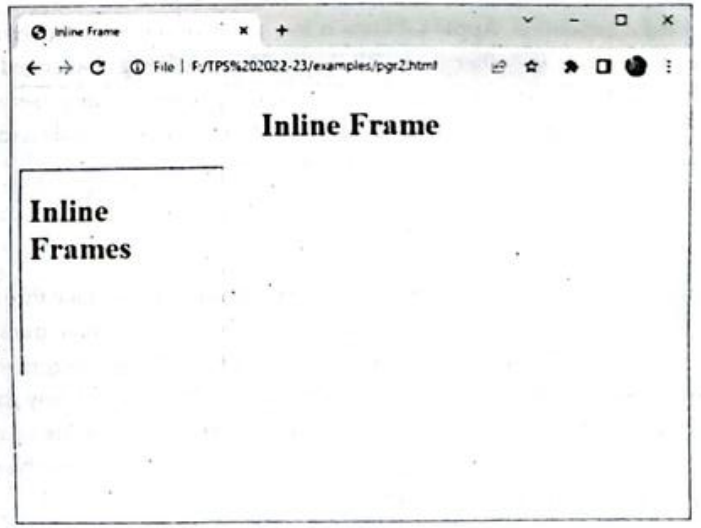
Output :

2. Write a html program to insert inline frame on web page. Use abc.html file as source for inline frame Size of Inline frame should be 200 x 200 pixels. (March 2022)

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Inline Frame
</title>
</head>
<body>
<h1 align="center">Inline Frame</h1>
<iframe src="abc.html" width="200" height="200"></iframe>
</body>
</html>
```

Output

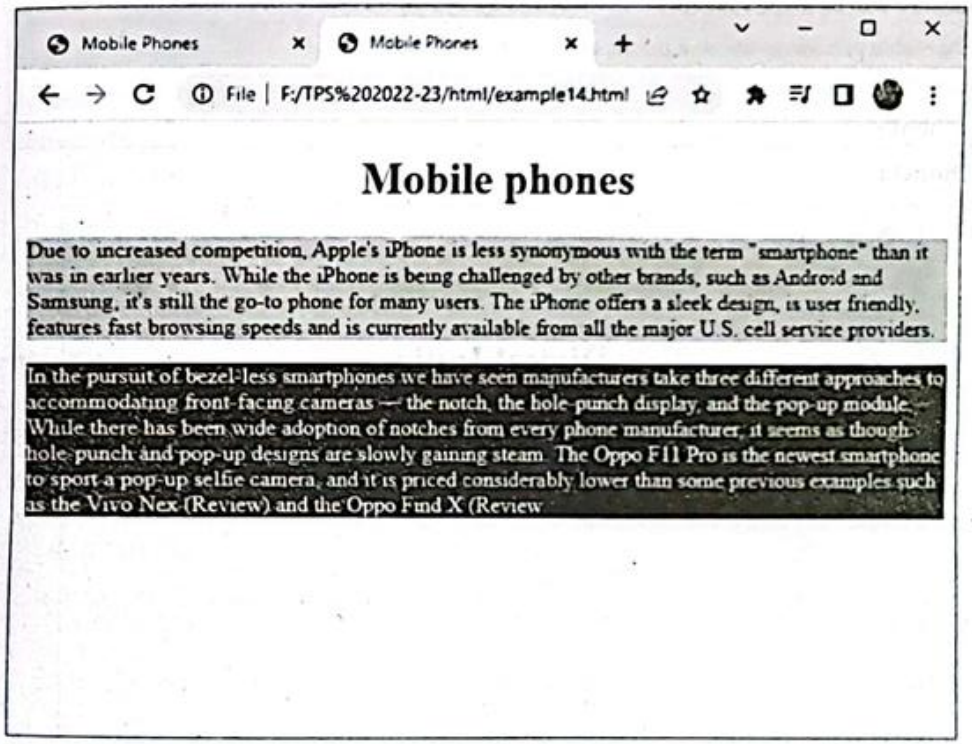


3. Write a html program using internal CSS to display two paragraphs. The first paragraph should have cyan background color and second should have yellow text color with blue background color. (Write topic of your choice in paragraph (March 2022)

Ans. : Coding :

```
<!doctype html>
<html>
<head>
<title>
Mobile Phones
</title>
<style>
.abc{background-color:cyan}
.xyz{color:yellow;background-color:blue}
</style>
</head>
<body>
<h1 align="center">Mobile phones</h1>
<p class="abc">
Due to increased competition, Apple's iPhone is less synonymous with the term "smartphone" than it was in earlier years. While the iPhone is being challenged by other brands, such as Android and Samsung, it's still the go-to phone for many users. The iPhone offers a sleek design, is user friendly, features fast browsing speeds and is currently available from all the major U.S. cell service providers.
</p>
<p class="xyz">
In the pursuit of bezel-less smartphones we have seen manufacturers take three different approaches to accommodating front-facing cameras — the notch, the hole-punch display and the pop-up module. While there has been wide adoption of notches from every phone manufacturer, it seems as though hole-punch and pop-up designs are slowly gaining steam. The Oppo F11 Pro is the newest smartphone to sport a pop-up selfie camera, and it is priced considerably lower than some previous examples such as the Vivo Nex (Review) and the Oppo Find X (Review)
</p>
</body>
</html>
```


Output



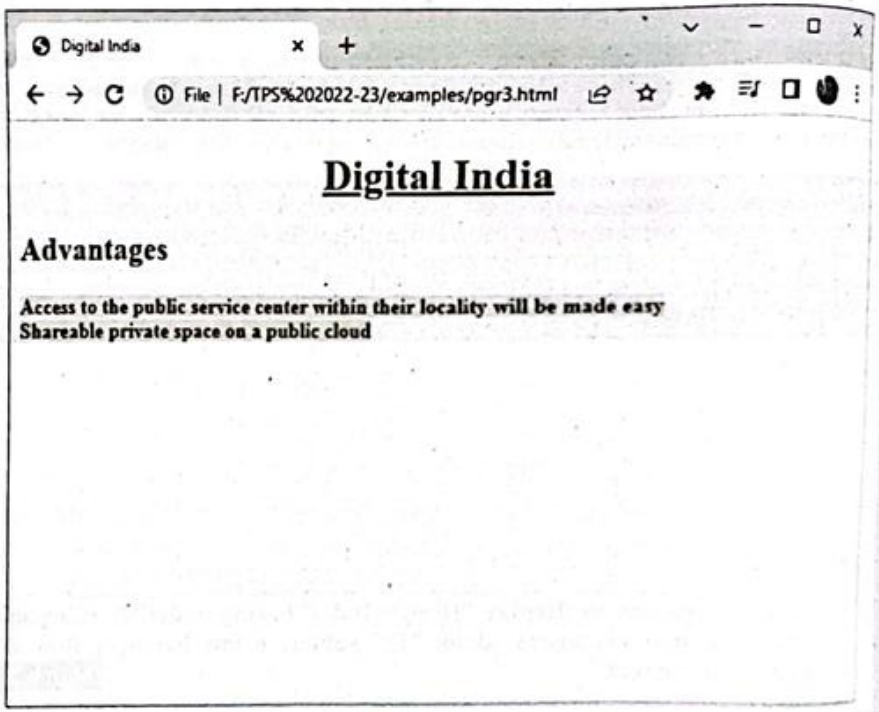
4. Write a html program to display "Digital India" having underline using inline CSS. Add any two sentences about "IT" subject below having yellow color background for the text. (March 2022)

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Digital India
</title>
</head>
<body>
<h1 align="center" style="text-decoration:underline">Digital India</h1>
<h2>Advantages</h2>
```

```
<b style="background-color:yellow">Access to the public service center within their locality will be made easy<br>
Shareable private space on a public cloud</b><br>
</body>
</html>
```

Output :



5. Write a html program to display "Ecommerce" having underline using inter... CSS. Add any two sentences about "Ecommerce" subject below having cyan col... background for the text.

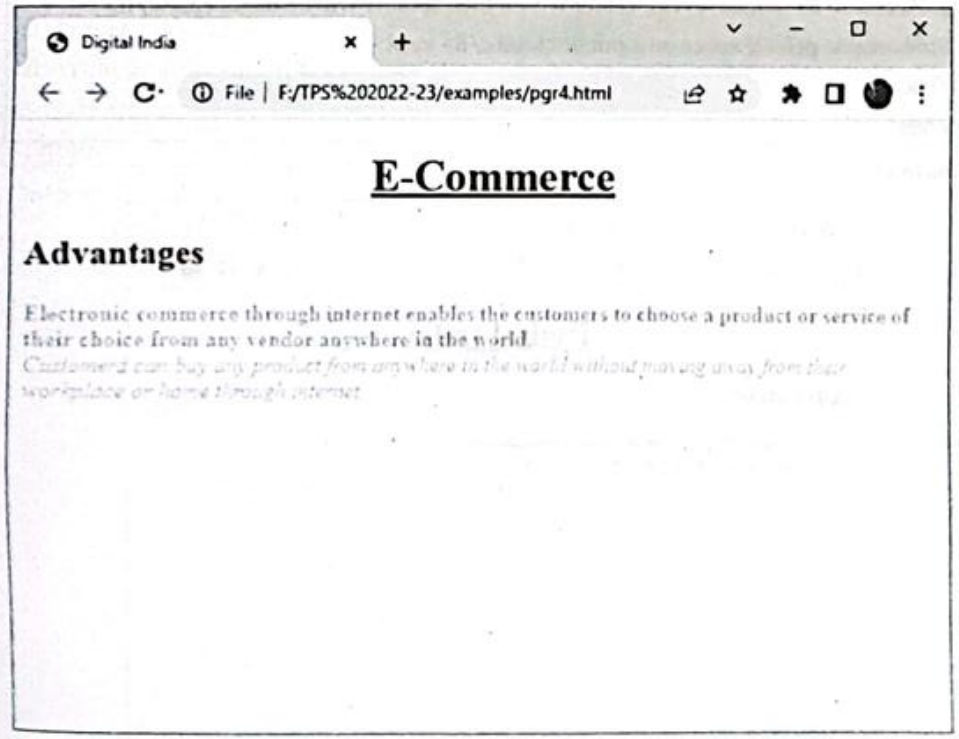
Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
E Commerce
</title>
<style>
```



```
h1{text-decoration:underline}
i,b{color:cyan}
</style>
</head>
<body>
<h1 align="center">E-Commerce</h1>
<h2>Advantages</h2>
<b>Electronic commerce through internet enables the customers to choose a product or service of their choice from any vendor anywhere in the world.</b><br>
<i>Customers can buy any product from anywhere in the world without moving away from their workplace or home through internet.</i>
</body>
</html>
```

Output :



6. Write a html program to display "Digital world" having underline using inline CSS. Add an ordered list having any two advantages of it. (March 2022)

Ans.: Coding:

```
<!DOCTYPE html>
<html>
<head>
<title>
Digital India
</title>
</head>
<body>
<h1 align="center" style="text-decoration:underline">Digital India</h1>
<h2>Advantages</h2>
<ol>
<li>Access to the public service center within their locality will be made easy</li>
<li>Shareable private space on a public cloud</li> </ol>
</body>
</html>
```

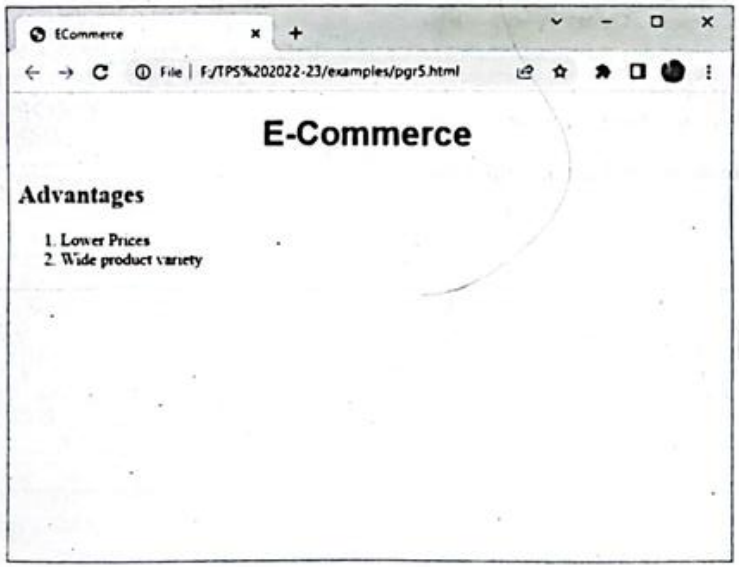
Output:

7. Write a html program to display "Ecommerce" having "Arial" font using inline CSS. Add an ordered list having any two advantages of it. (March 2022)

Ans.: Coding:

```
<!DOCTYPE html>
<html>
<head>
<title>
ECommerce
</title>
</head>
<body>
<h1 align="center" style="font-family:arial">E-Commerce</h1>
<h2>Advantages</h2>
<ol>
<li>Lower Prices</li>
<li>Wide product variety</li> </ol>
</body>
</html>
```

Output:

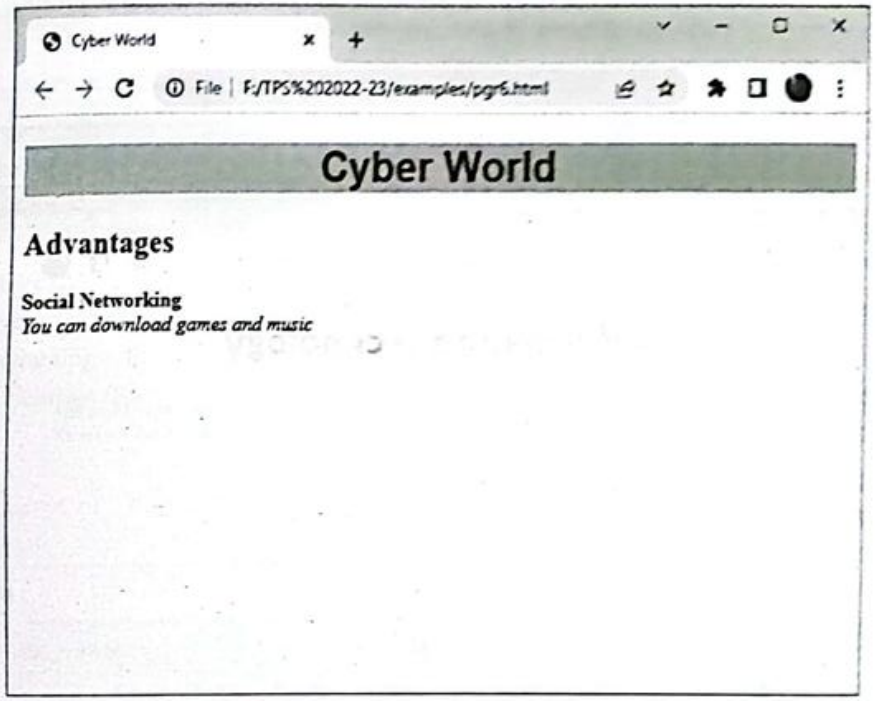


8. Write a html program to display "Cyber world" having Arial font and background colourcyan. Add any two advantages having blue color for the text. (March 202

Ans.: Coding:

```
<!DOCTYPE html>
<html>
<head>
<title>
Cyber World
</title>
<style>
h1{font-family:arial;background-color:cyan}
i,b{color:blue}
</style>
</head>
<body>
<h1 align="center">Cyber World</h1>
<h2>Advantages</h2>
<b>Social Networking</b><br>
<i>You can download games and music</i>
</body>
</html>
```


Output:



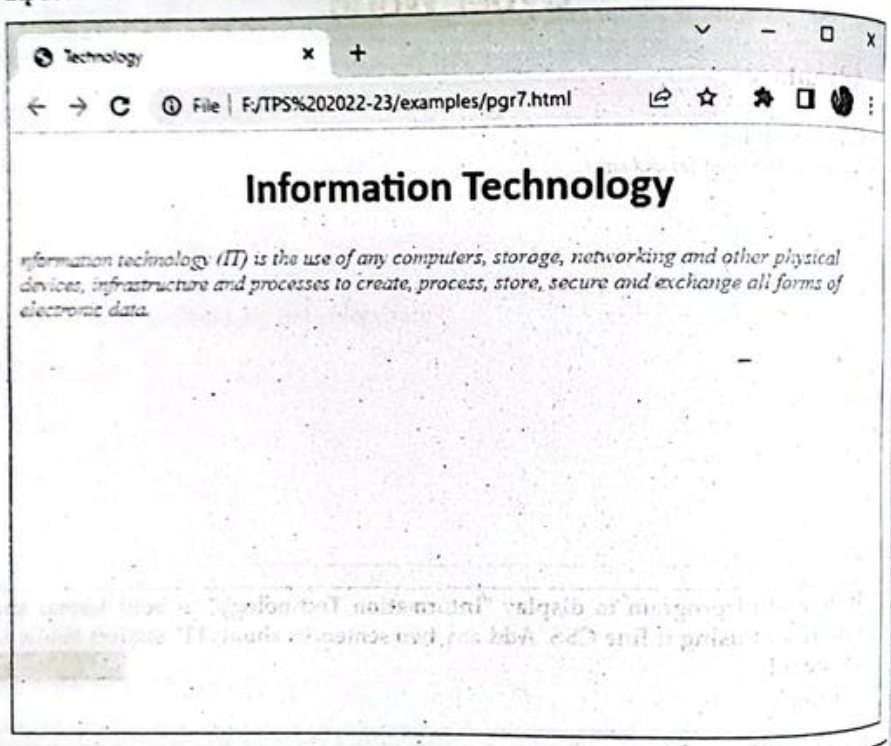
9. Write a html program to display "Information Technology" in bold format and Calibri font using inline CSS. Add any two sentences about "IT" subject below in orange color. (March 2022)

Ans.: Coding:

```
<!DOCTYPE html>
<html>
<head>
<title>
Technology
</title>
</head>
<body>
<h1 align="center" style="font-family:Calibri;font-style:bold">Information
Technology</h1>
```

```
<i style="color:orange">nformation technology (IT) is the use of any computers, storage, networking and other physical devices, infrastructure and processes to create, process, store, secure and exchange all forms of electronic data.</i>
</body>
</html>
```

Output :



10. Write a html program to create an ordered list having names of two friends. Add unordered list of their hobbies under each name as shown below. 1. Anil (March 20

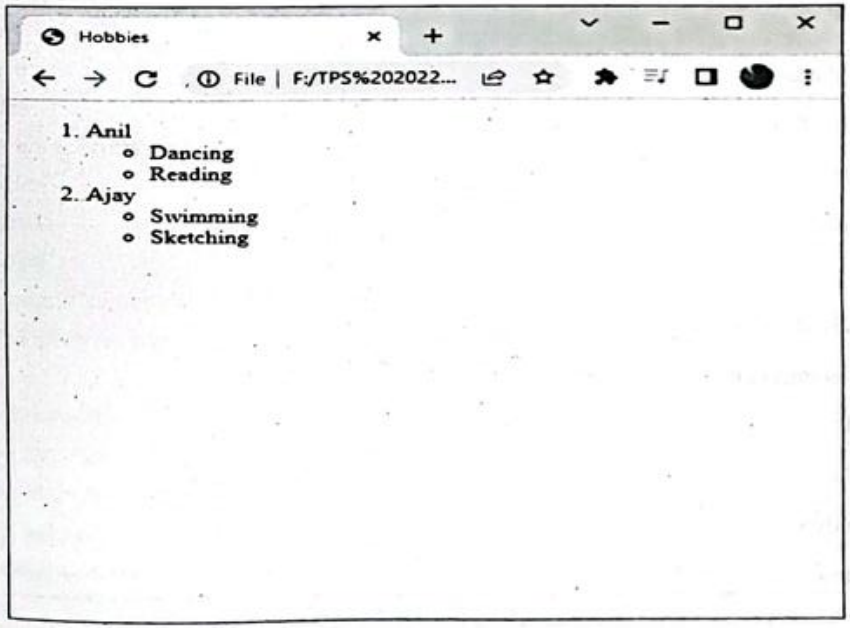
Ans. : Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Hobbies
</title>
```



```
</head>
<body>
<ol>
<li>Anil</li>
<ul type = "circle">
<li>Dancing</li>
<li>Reading</li>
</ul>
<li>Ajay</li>
<ul type = "circle">
<li>Swimming</li>
<li>Sketching</li>
</ul>
</ol>
</body>
</html>
```

Output :



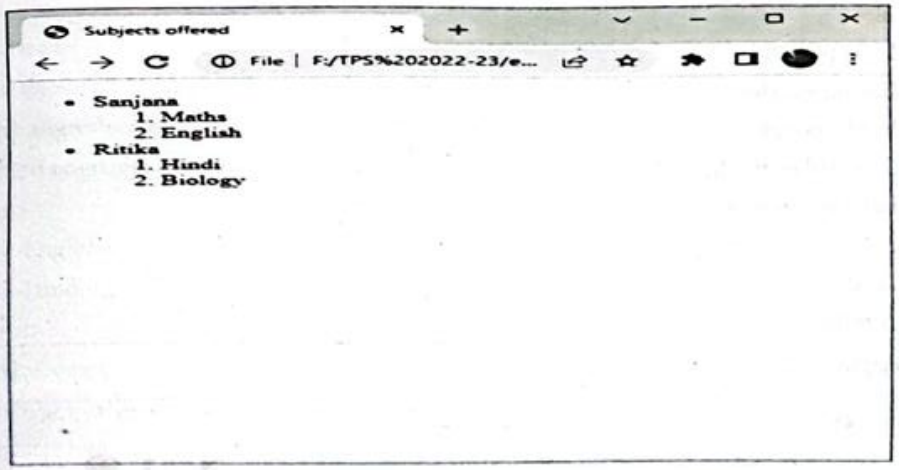
11. Write a html program to create an unordered list having names of two students. Add ordered list of subjects they selected as shown below. • Rupali

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<head>
<title>
Subjects offered
</title>
</head>
<body>
<ul type="disc">
<li>Sanjana</li>
<ol>
<li>Maths</li>
<li>English</li>
</ol>
<li>Ritika</li>
<ol>
<li>Hindi</li>
<li>Biology</li>
</ol>
</ul>
</body>
</html>
```


Output :



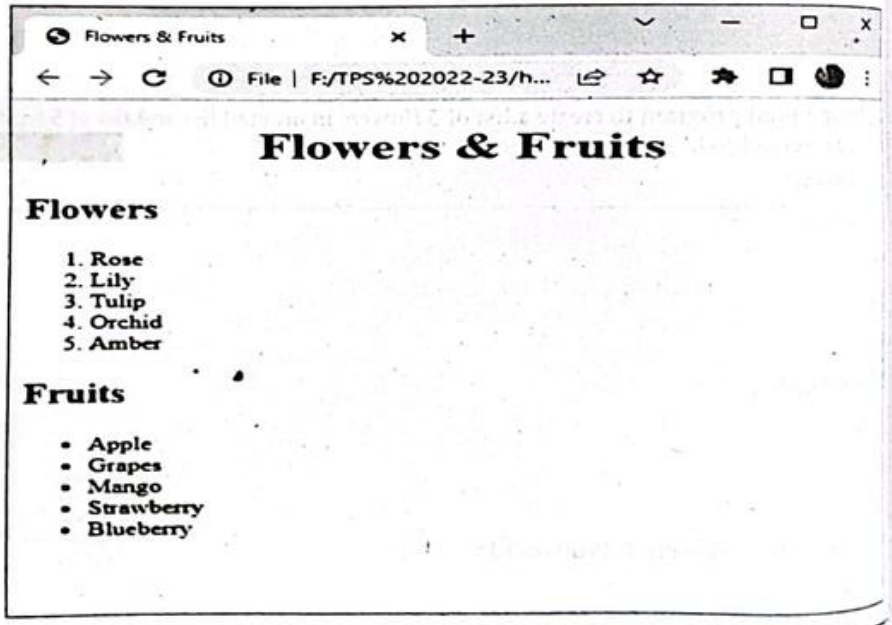
12. Write a html program to create a list of 5 flowers in ordered list and list of 5 fruits in unordered list. (March 2022)

Ans. : Coding :

```
<!doctype html>
<html>
<head>
<title>
Flowers & Fruits
</title>
</head>
<body>
<h1 align="center">Flowers & Fruits</h1>
<h2>Flowers</h2>
<ol>
<li>Rose</li>
<li>Lily</li>
<li>Tulip</li>
<li>Orchid</li>
<li>Amber</li>
</ol>
```

```
<h2>Fruits</h2>
<ul>
<li>Apple</li>
<li>Grapes</li>
<li>Mango</li>
<li>Strawberry</li>
<li>Blueberry</li>
</ul>
</body>
</html>
```

Output :



13. Write a html program to create an ordered list of 2 languages used for speaki and unordered list having 3 computer languages. (March 2

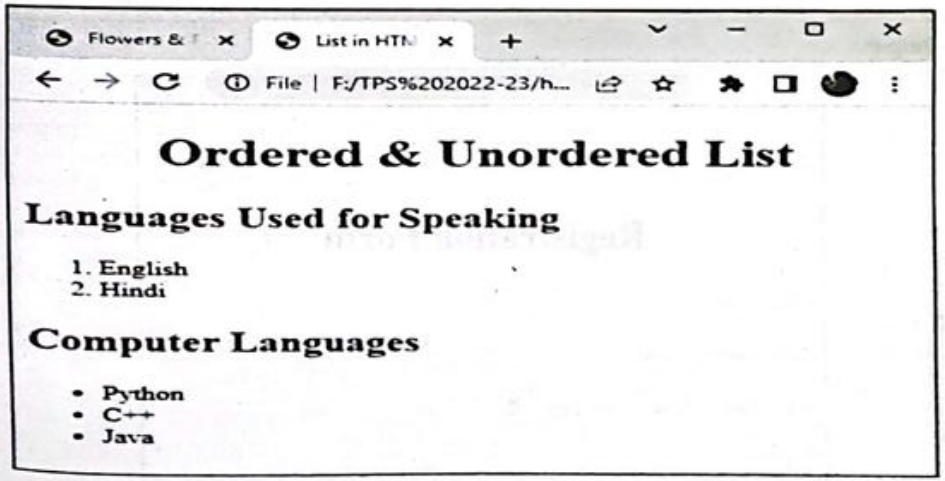
Ans. : Coding :

```
<!doctype html>
<html>
<head>
<title>
```



```
List in HTML
</title>
</head>
<body>
<h1 align="center">Ordered & Unordered List </h1>
<h2>Languages Used for Speaking</h2>
<ol>
<li>English</li>
<li>Hindi</li>
</ol>
<h2>Computer Languages</h2>
<ul>
<li>Python</li>
<li>C++</li>
<li>Java</li>
</ul>
</body>
</html>
```

Output :

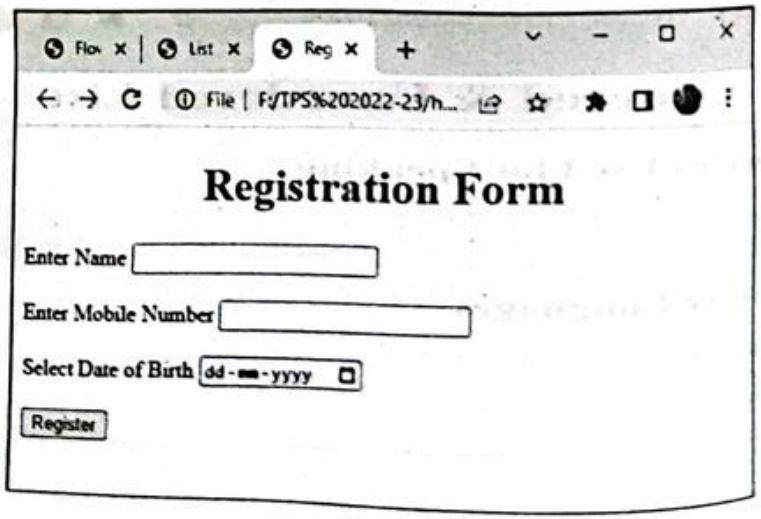


14. Write a html program to create registration form to accept name, mobile no, date of birth .The form should have register caption on the button to submit the (March 2

Ans. : Coding :

```
<!doctype html>
<html>
<head>
<title>
Registration Form
</title>
</head>
<body>
<h1 align="center">Registration Form</h1>
<form name="f1">
Enter Name
<input type="text" name="t1"><br><br>
Enter Mobile Number
<input type="tel" name="t2" pattern="[0-9]{10}"><br><br>
Select Date of Birth
<input type="date" name="t3"><br><br>
<input type="submit" name="b1" value="Register">
</form>
</body>
</html>
```

Output :

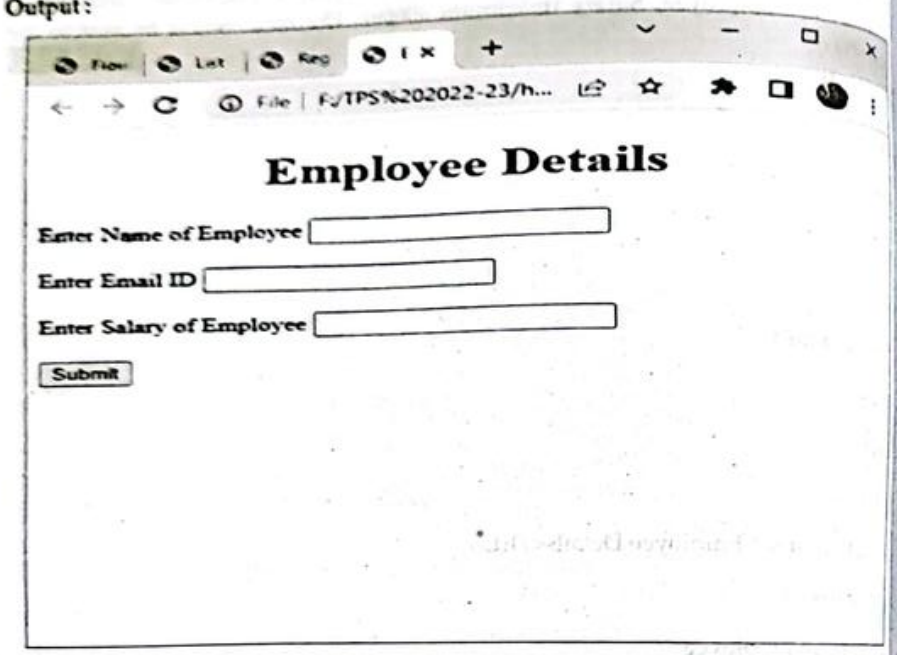


15. Write a html program to accept Name of the Employee (cannot be blank), Email Id of the Employee, Salary (maximum 40000). The data should be sent to the server. (March 2022)

Ans. : Coding :

```
<!doctype html>
<html>
<head>
<title>
Employee Details
</title>
</head>
<body>
<h1 align="center">Employee Details</h1>
<form name="f1">
Enter Name of Employee
<input type="text" name="t1" required><br><br>
Enter Email ID
<input type="email" name="t2"><br><br>
Enter Salary of Employee
<input type="number" name="t3" max="50000"><br><br>
<input type="submit" name="b1">
</form>
</body>
</html>
```

Output :



16. Write a html program to create a form to accept students roll no(In numb format) , Unit test marks(maximum 30 marks), Terminal exam marks(maximu 80 marks). Include the name of the Subject teacher and send the data to the serv (March 20

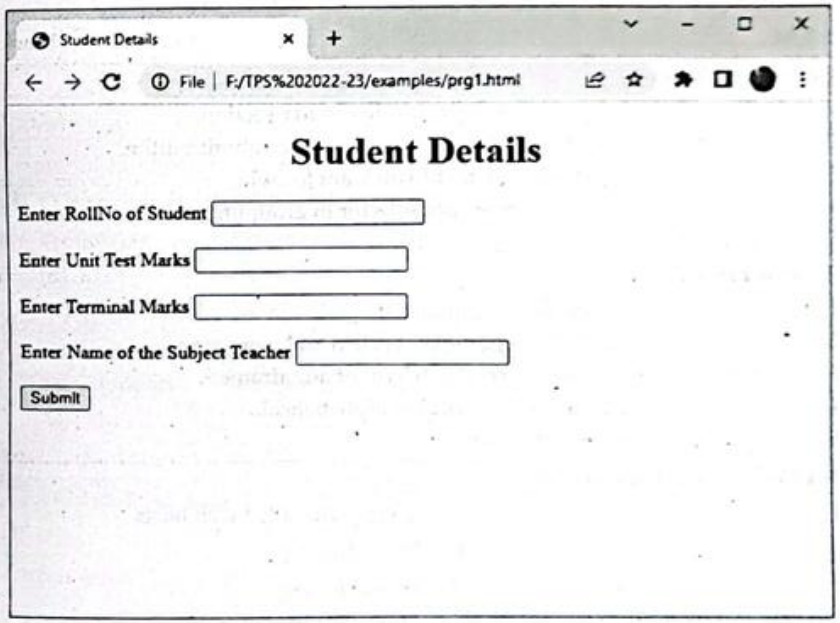
Ans. : Coding :

```
<!doctype html>
<html>
<head>
<title>
Student Details
</title>
</head>
<body>
<h1 align="center">Student Details</h1>
<form name="f1">
Enter RollNo of Student
<input type="number" name="r1"><br><br>
```



```
Enter Unit Test Marks
<input type="number" name="t1" max="30"><br><br>
Enter Terminal Marks
<input type="number" name="t2" max="80"><br><br>
Enter Name of the Subject Teacher
<input type="text" name="t3"><br><br>
<input type="submit" name="b1">
</form>
</body>
</html>
```

Output :



Examples for Practice

1. Write a html program to display "Digital India" in Verdana font using Internal CSS. Add any two sentences about Digital India below in orange color.
2. Write a html program to insert Inline frame on web page. Use xyz.html file as a source file for Inline frame , size of Inline frame should be 300 × 300 pixels.

3. Write a html program to create a form to accept Doctor's Name, Number of pat (maximum 20), date of examining the patients.
4. Write a html program to create a form to accept Student's Name, numbe practical he has completed and provide facility to upload his completion certifica
5. Write a html program to accept name of the college, total number of students in college, total number of halls (range till 100), The data should be sent to the serve
6. Write a html program to display names of the two departments and also disp course link B.sc, M.sc, B.A, M.A under department name in unordered list.
7. Write a html program to display "Maharashtra State Board" in font size 50 pix using Internal CSS> Give background color yellow for the sane text.

Topic Test

Time :- 1 hr Marks

Q. 1 Fill in the blanks (05)

1. In ____ CSS <link> tag is written inside <head> section.
2. <input type="image"> defines an _____ as a submit button.
3. _____ indicates the HTML element you want to style.
4. _____ are used to separate each selector in grouping.
5. The <ol> tag defines _____ list.

Q. 2 True or False (05)

1. <dl> tag is used to define definition list.
2. An image with multiple hyperlinks is called an image map.
3. Length attribute is used to specify height of an <iframe>.
4. An unordered list can be numerical or alphabetical.
5. Shared hosting is cost effective.

Q. 3 MCQ one correct alternative. (05)

1. The companies that provides ____ services are called web hosts.
   a) Web Hosting b) New Hosting
   c) Frame Hosting d) Work Hosting
2. _____ attribute of <area> tag can value rect, circle or poly.
   a) Coords b) Shape
   c) href d) alt
3. The _____ element is used to specify the audio files which the browser may use.
   a) Audio b) Video
   c) Source d) Href

4. The _____ property defines the flow of the content.
   a) Inline b) Block
   c) Heading d) float
5. The _____ selector selects HTML elements with a specific class attribute.
   a) ID b) Class
   c) Float d) Display

Q. 4 MCQ two correct alternative. (06)

1. Common audio file formats are_____
   a) Mp3 b) aac
   c) WebM d) mov e) mp6
2. Items in definition list can be created with _____ and _____tags
   a) b) <dl>
   c) <dt> d) <dd> e)
3. ____ attribute specifies the height and width of the image.
   a) Autofocus b) Autocomplete
   c) placeholder d) height e) width

Q. 5 MCQ three correct alternative. (06)

1. Following are the attributes of <input> tag.
   a) Max b) min c) length
   d) pattern e) range f) Email
2. Following are the attributes of <video> tag.
   a) Autoplay b) length c) Loop
   d) Muted e) source f) Href

Q. 6 Match the Following (03)

Column A	Column B
1) Disabled	a) A selector and a declaration block
2) <input type="tel">	b) used as a wildcard character
3) CSS rule set contains	c) Defines how the components are placed on the web page
4) Universal Selector	d) Enables to embed audio files on the web page
5) Display property	e) specifies that an input field should be disabled
6) <audio>	f) Used to define input filed that should contain a telephone number

Q. 7 Write a HTML program

a) Write html-5 code to create a drop down list of five Airways name. Multiple selections can be done by the user. Provisions should be made to submit and clear the contents.

b) Write a html program to display "Digital India" having underline using inline CSS. Add any two sentences about I.T. subject below having yellow color background for the text.

Answer of Questions

Q.1 Fill in the blanks

1) External 2) image 3) Selector
4) Commas 5) Ordered

Q.2 True or False

1) True 2) True 3) False
4) False 5) True

Q.3 MCQ one correct alternative

1) A 2) B 3) C 4) D 5) B

Q.4 MCQ two correct alternative

1) (a, b) 2) (c, d) 3) (d, e)

Q.5 MCQ three correct alternative

1) (a, b, d) 2) (a, c, d)

Q.6 Match the Following

1 - e 2 – f 3 - a 4 - b 5 - c 6 - d

Q.7 Write a HTML program

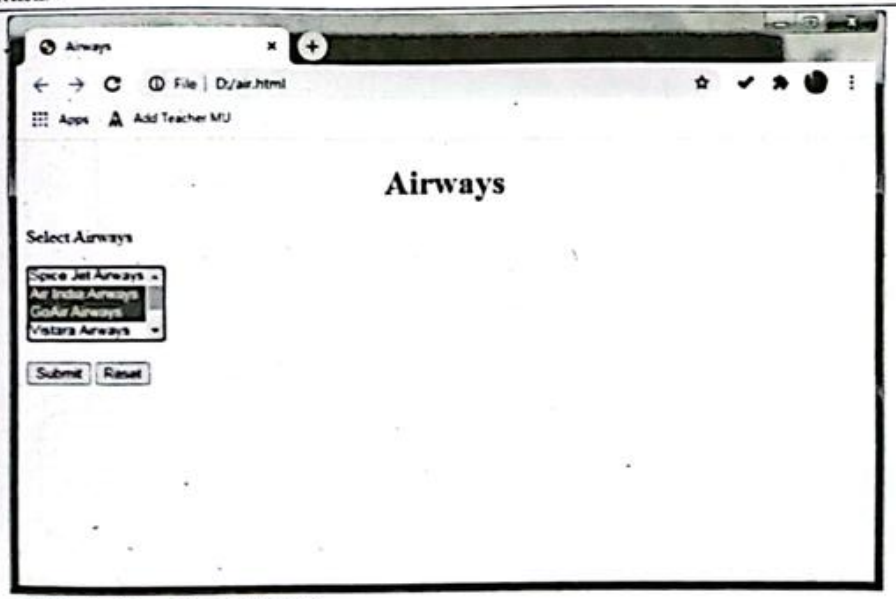
1)

Coding:-

```
<!DOCTYPE html>
<html>
<head>
<title>
Airways
</title>
</head>
<body>
```



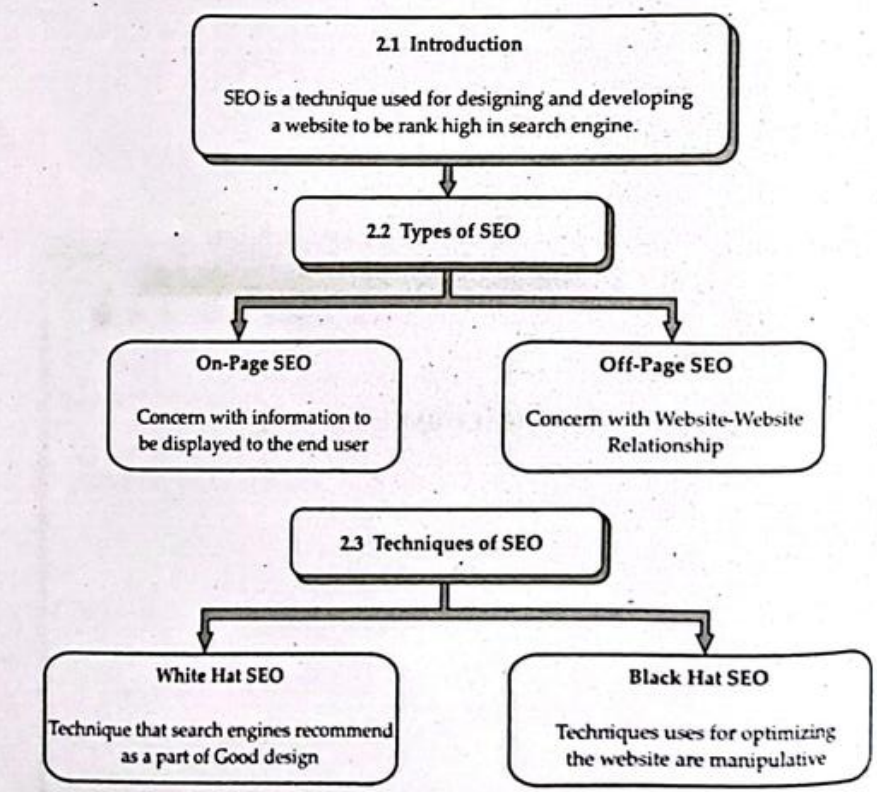
```
<h1 align="center">Airways</h1>
Select Airways <br><br>
<select name="s1" multiple>
<option>Spice Jet Airways</option>
<option>Air India Airways</option>
<option>GoAir Airways</option>
<option>Vistara Airways</option>
<option>IndiGo Airways</option>
<option>Air Asia</option>
</select>
<br><br>
<input type="submit" name="b1">
<input type="reset" name="b2">
</body>
</html>
```

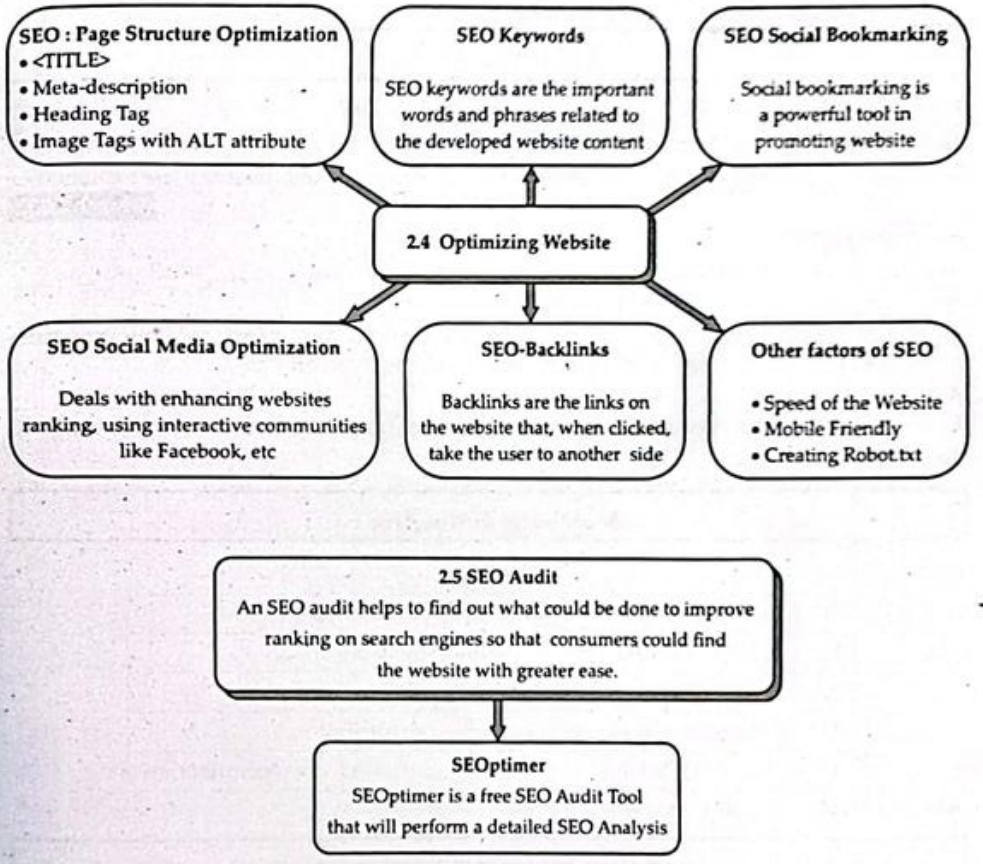


2) Refer Page No. 1-103 Q. 4

2 Introduction to SEO (Search Engine Optimization)

Chapter at a Glance





Exercise

Fill in the Blanks

1. The full form of SEO is _____.

Ans.: Search Engine Optimization

2. _____ are the links on the website that, when clicked, take the user to anot side. (March 20

Ans. : Backlinks

3. One of the optimizing factor to improve images is by using _____ attribute.

Ans. : alt

4. An _____ helps to find out what you can be done to improve ranking on sear engines, so that consumers can find the website with greater ease.

Ans. : SEO Audit

5. _____ file tells search engine which page you want to exclude from indexing

Ans. : Robots.txt

Match the Following

(1)

'A'		'B'	
1.	Image Optimization	(a)	meta tag
2.	Links to Twitter, Facebook	(b)	SEO technique
3.	Description about author	(c)	alt attribute
4.	White Hat Strategy	(d)	Social Media Optimization

Ans. : 1. - (c), 2. - (d), 3. – (a), 4. – (b)

Choose correct answers from the following (2 correct)

1. The page content optimization can be done by using keywords in (March 20

(a) <font> (b) <title>
(c) <hr> (d) <meta> (e) <input>

Ans. : (b), (d)

2. Other factor to rank the website higher are

(a) Desktop friendly (b) Speed of the website
(c) Creating robots.txt (d) No backlinks
(e) No external links

Ans. : (b), (c)

Answer Briefly

1. Define SEO. State types of SEO.

Ans. :

(i) SEO stands for Search Engine Optimization, which is the practice of increasing the quantity and quality of traffic to your website through organic search engine results.

(ii) Thus SEO is useful in increasing the number of visitors to a website.

(iii) There are many strategies and techniques adopted to optimize the webpage.

(iv) **Types of SEO :** There are two types of optimization:-

(v) **On-Page SEO :** This includes Provision of good content, good keywords selection, putting keywords on correct places, giving appropriate title to every page etc.

(vi) On-Page SEO is concerned with information that is displayed to the end user, such as text, images and website navigation.

(vii) **Off-Page SEO :** It includes link building, increasing link popularity by submitting open directories, search engines, link exchange etc.

(viii) Off-Page SEO is concerned with Website-Website relationship.

2. Which are the different SEO techniques.

Ans. :

(i) SEO techniques are classified into two broad categories, which are White Hat SEO and Black Hat SEO.

(ii) **White Hat SEO :** Techniques that search engines suggest as a part of good design are called White Hat SEO.

(iii) It contains proper and well labeled images according to the content .

(iv) It is in the form of Standard-compliant HTML.

(v) **Black Hat SEO :** Techniques that search engines do not approve and those techniques which are used for optimizing the website are called Black Hat SEO.

(vi) It mainly contains duplicate contents. Such websites are mainly used to redirect users to other websites and cause traffic.

3. Explain in short White Hat technique. (March 2022)

Ans. :

(i) Techniques that search engines suggest as a part of good design are called White Hat SEO.

(ii) It contains proper and well labeled images according to the content.

(iii) It is in the form of Standard-compliant HTML.

(iv) The page titles are unique but relevant at the same time.

(v) It has Complete sentences with good spelling and grammar

4. Explain in short Black Hat technique.

Ans. :

(i) Techniques that search engines do not approve and those techniques which are used for optimizing the website are called Black Hat SEO.

(ii) It mainly contains duplicate contents which mostly are stolen from other original creators mainly without any credits and permissions.

(iii) Such websites are mainly used to redirect users to other websites and cause traffic.

(iv) Links from such sites are irrelevant and don't show the actual search content. Websites like these contain invisible text and stuffed keywords

5. Explain in short SEO Keyword search. (March 2022)

Ans. :

(i) SEO keywords are the important words and phrases related to the developed website content.

(ii) A list of keywords, need to be carefully chosen to optimize the search. Keyword Research and Analysis is an important part of the Search Engine Optimization (SEO) as well as "Search Engine Marketing".

(iii) For "Search Engine Marketing", one can pay for certain keywords.

(iv) There are various online Keyword Planner tools (such as Google Add Words Keyword Planner) to shortlist keywords.

(v) This is more useful to compare the cost of various keywords, if the website designer opts for Paid Search Engine Marketing.

6. Explain any two Page content optimization points. (March 2022)

Ans. :

(i) TITLE : Title is the first thing that user notices in search result list while using Google for searching. Contents should be up to 60 characters (spaces included)

(ii) Meta-description Tag : While creating a web page, Meta description should be used to boost your On-Page SEO. Meta Description should be between 70 and 160 characters (spaces included) that others are interested to know about the contents form your website.

7. Explain how could you optimize images.

Ans. :

(i) Image Tags with ALT attribute : The alt attribute gives information about the image so even if the image does not download, it can make the viewers understand the image related information.

(ii) Image Tag with Title attribute : The title tag of images show the keyword of the image when mouse is hovered above the image.

(iii) Image compression : The images which are compressed load faster which give user good friendly experience.

(iv) Images can have Caption keywords : Certain images can have keywords on the image to make it relevant for user to know about it.

Fill in the Blanks

2.1 Introduction

1. SEO stands for _____.
Ans. : Search Engine Optimization
2. SEO is subset of _____.
Ans. : Search Engine Marketing
3. _____ is a subset of Search Engine Marketing.
Ans. : SEO
4. _____ is useful in increasing number of visitors to a website.
Ans. : SEO
5. _____ is a technique is used for designing and developing a website be to rank high in search engine result.
Ans. : Search Engine Optimization
6. _____ is an online program or a software that helps users to search for information on world wide web.
Ans. : Search Engine

2.2 Types of SEO

7. _____ SEO include Provision of good content good keyword.
Ans. : On-Page
8. _____ SEO include link building, increasing link popularity by submitting open directories. (March 2022)
Ans. : Off-Page
9. _____ SEO is concerned with information that is displayed to the end user.
Ans. : On-Page
10. _____ SEO is concerned with website-website relationship.
Ans. : Off-Page
11. _____ is concerned with how good is your network.
Ans. : Off-Page SEO

2.3 Techniques of SEO

12. _____ SEO techniques that search engines do not approved.
Ans. : Black Hat.
13. _____ SEO Techniques that search engines recommends as apart of good design.
Ans. : White Hat
14. White hat strategies include _____ content.
Ans. : relevant

15. Black hat strategies include _____ content.
Ans. : duplicate
16. _____ strategies include complete sentences with good spelling and grammar.
Ans. : White hat
17. Well-labeled images are the part of_____ strategies.
Ans. : White hat
18. Invisible text and stuffed keywords are part of _____ strategies.
Ans. : Black hat

2.4 Optimizing Website

19. Html programs should start with_____. (March 2022)
Ans. : <!DOCTYPE html>
20. Singular tag should be _____.
Ans. : Self enclosed
21. One of the optimizing factor to improve images is by using _____attribute.
Ans. : alt
22. <img> tag with _____ attribute also contribute to optimization.
Ans. : alt
23. _____ tag with alt attribute also contribute to optimization. (March 2022)
Ans. : <img>
24. _____ is the first thing that user notices in search result list while using Google for search.
Ans. : Title
25. Title contents should be up to ___ characters.
Ans. : 60
26. Meta description should be used to boost your _____ SEO.
Ans. : On-Page
27. Meta Description should be between 70 and ____ characters. (March 2022)
Ans. : 160
28. Meta Description should be between ___ and 160 characters.
Ans. : 70
29. _____ Description should be between 70 and 160 characters.
Ans. : Meta
30. _____ are the important words and phrases related to the developed website content. (March 2022)
Ans. : SEO Keywords
31. Keyword Research and _____ is an important part of the Search Engine Optimization.
Ans. : Analysis
32. SEM stands for_____ .
Ans. : Search Engine Marketing

33. Keyword _____ and Analysis is an important part of the Search Engine Optimization.
Ans. : Research
34. _____ is a powerful tool in promoting a website.
Ans. : Social bookmarking
35. SEO social bookmarking is very useful for _____ SEO optimization. (March 2022)
Ans. : Off-Page
36. _____ are website links that are stored for future references.
Ans. : Bookmarks
37. _____ Service allows you to save , organize and share bookmarks with other users.
Ans. : Bookmarking
38. _____ are the important SEO factors to rank a website higher in the search engine.
Ans. : External links
39. Using _____ you can quickly create high-quality backlinks.
Ans. : Social bookmarking
40. SMO Stands for _____.
Ans. : Social Media Optimization
41. _____ deals with enhancing the website's ranking , using interactive communities.
Ans. : Social Media Optimization
42. _____ are the links on the website that, when clicked, take the user to another side. (March 2022)
Ans. : Backlinks
43. Website's _____ has become a ranking factor for Google.
Ans. : mobile friendliness
44. _____ file tells search engine which page you want to exclude from indexing.
Ans. : Robots.txt
45. _____ is a powerful tool in promoting a website.
Ans. : Social bookmarking
46. Social bookmarking is very useful for _____ SEO optimization. (March 2022)
Ans. : Off-Page
47. _____ are website links that are stored for future references.
Ans. : Bookmarks

2.5 SEO Audit

48. An _____ helps to find out what you can be done to improve ranking on search engines, so that consumers can find the website with greater ease.
Ans. : SEO Audit
49. _____ is used to analyze the websites health.
Ans. : SEO Audit

50. _____ is a free SEO Audit Tool that will perform a detailed SEO Analysis.
Ans. : SEOptimer
51. _____ is a process by which search engines scan a website. (March 2022)
Ans. : Crawling

True or False

2.1 Introduction

1. SEO stands for Search Engine Optimization. (March 2022)
Ans. : True
2. SEO stands for Search Engine Optimizing.
Ans. : False
3. SEO is a technique used for: designing and developing a website to be rank high in search engine results. (March 2022)
Ans. : True
4. SEO is a subset of search engine marketing.
Ans. : True
5. SEO is useful in decreasing the number of visitors to a website.
Ans. : False
6. Search engine is an offline program that helps users to search for information on www. (March 2022)
Ans. : False
7. Examples of search engine include Google, Yahoo and Bing.
Ans. : True

2.2 Types of SEO

8. Off-Page SEO include provision of good content, good keyword.
Ans. : False
9. On-Page SEO include provision of good content, good keyword. (March 2022)
Ans. : True
10. On-Page SEO include link building, increasing link popularity.
Ans. : False
11. Off-Page SEO include link building, increasing link popularity.
Ans. : True
12. On-Page SEO is concerned with information that is displayed to the end user. (March 2022)
Ans. : True
13. Off-Page SEO is concerned with website-website relationship.
Ans. : True

2.3 Techniques of SEO

14. White Hat strategies include duplicate content.
Ans. : False
15. White Hat strategies include relevant content. (March 2022)
Ans. : True
16. Invisible text and stuffed keyword are part of Black Hat Strategies.
Ans. : True
17. Well-labeled images are part of White Hat Strategies.
Ans. : True
18. White Hat SEO techniques that search engine recommends as apart of good design.
Ans. : True
19. SEO techniques are classified into three categories. (March 2022)
Ans. : False
20. Black Hat SEO techniques are used for optimizing the website. (March 2022)
Ans. : True

2.4 Optimizing Websites

21. Singular tag like <img> is should be self enclosed like this-</img>.
Ans. : False
22. Singular tag like <img> is should be self enclosed like this-<img/>. (March 2022)
Ans. : True
23. Most of the times recently opened tag will be closed first.
Ans. : True
24. Meta-description is the first thing that user notices in search result list while using Google for search.
Ans. : False
25. Title content should be up to 60 characters. (March 2022)
Ans. : True
26. Meta description should be used to boost your Of-Page SEO.
Ans. : False
27. Meta Description should be between 70 and 160 characters.
Ans. : True
28. The src attribute gives information about image.
Ans. : False.
29. <img> tag with alt attribute also contributes to optimization. (March 2022)
Ans. : True
30. Keyword Research and Analysis is an important part of SEO.
Ans. : True
31. SEO Social Bookmarking is very useful for Off-Page SEO Optimization.
Ans. : True

32. Bookmarks are website links that are stored for future references.

Ans. : True

33. Backlinks are links on one website that, when clicked, take the user to another site

Ans. : True

34. Social media links and backlinks are not used for optimizing the webpage.

Ans. : False

35. Website's mobile friendliness is not important factor in ranking. (March 202

Ans. : False

36. Mobile friendliness is only used for ranking the websites.

Ans. : False

37. Robots.txt file tells search engine which pages you want to include from indexing.

Ans. : False

38. Social Media Optimization deals with enhancing the website's ranking, using interactive communities.

Ans. : True

39. SEO keywords are the important words and phrases related to the developed website content.

Ans. : True

40. Social bookmarking is a powerful tool in promoting a website. (March 202

Ans. : True

41. With social bookmarking you can quickly create high quality backlinks.

Ans. : True

42. Spread also rank the websites higher.

Ans. : True

43. SEOptimer provides clear and actionable recommendations that can be taken to improve your online presence. (March 2022)

Ans. : True

2.5 SEO Audit

44. SEO Audit does not help to analyze the websites health. (March 202

Ans. : False

45. An SEO Audit helps to find out what could be done to improve ranking on search engines, so that consumers could find the website with greater ease.

Ans. : True

46. SEOptimer is paid SEO Audit Tool. (March 202

Ans. : False

MCQ (One Correct Answers)

2.1 Introduction

1. SEO Stands for _____. (March 2022)
   (a) Search Engine Optimization (b) Search Engine Option
   (c) Search Engine Optimizing (d) Search Engine Organizing

Ans. : (a) Search Engine Optimization

2. SEO is a subset of _____.
   (a) Search Engine Optimization (b) Search Engine Marketing
   (c) Social Media Optimization (d) Search Engine

Ans. : (b) Search Engine Marketing

3. SEO is useful in _____ the number of visitors to a website. (March 2022)
   (a) decreasing (b) increasing
   (c) removing (d) analyzing

Ans. : (b) increasing

4. _____ is an online program or software that helps users to search for information on World Wide Web.
   (a) Search Engine (b) Website
   (c) Application (d) API

Ans. : (a) Search Engine

2.2 Types of SEO

5. _____ is concern with Website-Website relationship. (March 2022)
   (a) On-Page SEO (b) Off-Page SEO
   (c) White Hat SEO (d) Black Hat SEO

Ans. : (b) Off-Page SEO

6. _____ is concern with information that is displayed to the end user.
   (a) On-Page SEO (b) Off-Page SEO
   (c) White Hat SEO (d) Black Hat SEO

Ans. : (b) On-Page SEO

7. _____ is includes provision of good content, good keywords selection etc.
   (a) Off-Page SEO (b) On-Page SEO
   (c) Over-Page SEO (d) Above-Page SEO (March 2022)

Ans. : (b) On-Page SEO

8. _____ includes link building, search link exchange etc.
   (a) Off-Page SEO (b) On-Page SEO
   (c) Over-Page SEO (d) Above-Page SEO

Ans. : (a) On-Page SEO

2.3 Techniques of SEO

9. _____ include relevant content. (March 20
(a) On-Page SEO (b) Off-Page SEO
(c) White Hat SEO (d) Black Hat SEO
Ans. : (c) White Hat SEO

10. _____ include duplicate content.
(a) On-Page SEO (b) Off-Page SEO
(c) White Hat SEO (d) Black Hat SEO
Ans. : (d) Black Hat SEO

11. _____ is concern with White Hat SEO.
(a) Well-labeled images (b) Duplicate content
(c) Invisible text (d) Stuffed Keyword
Ans. : (a) Well-labeled images

2.4 Optimizing Websites

12. If it is singular tag for eg<img> it should be self enclosed like _____.
(a) <img> (b) </img> (c) <img/> (d) </img/> (March 202
Ans. : (c) <img/>

13. Title contents should be up to _____. (March 202
(a) 50 (b) 60 (c) 70 (d) 80
Ans. : (b) 60

14. Meta Description should be between _____.
(a) 70 and 100 (b) 100 and 200 (c) 70 and 160 (d) 50 and 100
Ans. : (c) 70 and 160

15. Image tag with _____ attribute gives information about the image
(a) alt (b) src (c) height (d) width (March 202
Ans. : (a) alt

16. _____ keywords are the important words and phrases related to the developed website content.
(a) SEO (b) Structure (c) Search (d) Web Design
Ans. : (a) SEO

17. _____ is a powerful tool in promoting a website. (March 202
(a) Black links (b) Social bookmarking (c) Speed (d) Media
Ans. : (b) Social bookmarking

18. _____ are website links that are stored for future references.
(a) Webpage (b) Bookmarks (c) Website speed (d) Media optimization
Ans. : (b) Bookmarks

19. SMO stands for _____.
(a) Social Media Optimization (b) Service Media Optimization
(c) Search Media Optimization (d) Sound Media Optimization
Ans. : (a) Social Media Optimization

20. _____ deals with enhancing the websites ranking, using interactive communities like facebook, twitter etc. (March 2022)
(a) Social Media Optimization (b) Search Engine Optimization
(c) Backlinks (d) Speed of website
Ans. : (a) Social Media Optimization

21. _____ are links on one website that, when clicked take the user to another site. (March 2022)
(a) Backlinks (b) Frontlinks (c) Blacklinks (d) Whitelinks
Ans. : (a) Backlinks

22. _____ represent a "vote of confidence" from one site to another.
(a) Frontlinks (b) Blacklinks (c) Backlinks (d) Whitelinks
Ans. : (c) Backlinks

23. _____ file tells search engine which pages you want to exclude from indexing.
(a) myfile.txt (b) file.txt
(c) search.txt (d) Robots.txt (March 2022)
Ans. : (d) Robots.txt

2.5 SEO Audit

24. SEO Audit analyze websites_____.
(a) users (b) health (c) visitors (d) rank
Ans. : (b) health

25. _____ helps to find out what could be done to improve ranking on search engines. (March 2022)
(a) SEO Search (b) SEO Audit
(c) SEO Speed (d) SEO optimization
Ans. : (b) SEO Audit

26. _____ is a free SEO Audit Tool that will perform detailed SEO analysis.
(a) SEO Audit (b) SEO Search (c) SEOptimer (d) SEO Tool
Ans. : (c) SEOptimer

MCQ (Two Correct Answers)

2.1 Introduction

1. The SEO is a technique used for _____ and _____ a website to be rank high in a search engine. (March 2022)
(a) Designing (b) Developing (c) Monitoring (d) Analyzing
Ans. : (a), (b)

2.2 Types of SEO

2. Types of SEO are _____. (March 2022)
(a) On-Page SEO (b) White Hat SEO (c) Off-Page SEO (d) Black Hat SEO
Ans. : (a), (c)

2.3 Techniques of SEO

3. Techniques of SEO are classified into _____ and _____. (March 20
   (a) On-Page SEO (b) White Hat SEO
   (c) Off-Page SEO (d) Black Hat SEO

Ans. : (b), (d)

2.4 Optimizing Websites

4. The page content optimization can be done by using keywords in _____ and _____
   (a) <font> (b) <title>
   (c) <hr> (d) <meta>

Ans. : (b), (d)

5. Other factor to rank the website higher are _____ and _____. (March 20
   (a) Desktop friendly (b) Speed of the website
   (c) Creating robots.txt (d) No backlinks

Ans. : (b), (c)

6. Social Media Optimization deals with enhancing the website's ranking using _____ and _____.
   (a) Blogs (b) E Commerce website's
   (c) Facebook (d) Banking

Ans. : (a), (c)

7. Meta Tag description should be between _____ and _____ characters.
   (a) 70 (b) 80 (c) 160 (d) 120

Ans. : (a), (c)

8. The page should include proper heading tag from _____ to _____ wherever required. (March 20
   (a) <head> (b) <h1> (c) </h1> (d) <title>

Ans. : (b), (c)

9. _____ and _____ is an important part of the Search Engine Optimization.
   (a) Keyword Research (b) Analysis
   (c) SEO Structure (d) Website Speed

Ans. : (a), (b)

10. Various online keyword Planner Tools are _____. (March 20
    (a) Browser (b) Keyword Planner
    (c) URL (d) Google AdWords

Ans. : (b), (d)

11. Factors for SEO are _____.
    (a) Speed of the website (b) Web Browser
    (c) Mobile friendly (d) URL

Ans. : (a), (c)

2.5 SEO Audit

12. SEO Audit Tool are _____ and _____. (March 2022)
    (a) SEOptimer (b) SEOptimizer
    (c) Seomator (d) SEO Auditor

Ans. : (a), (c)

MCQ (Three Correct Answers)

2.2 Types of SEO

1. On-Page SEO includes provision of _____.
   (a) good content (b) good keywords selection
   (c) putting keywords on correct places (d) link building
   (e) directories (f) search

Ans. : (a), (b), (c)

2. Off-Page SEO includes _____. (March 2022)
   (a) link building (b) link exchange
   (c) search engines (d) good content
   (e) keywords (f) title

Ans. : (a), (b), (c)

3. On-Page SEO is concerned with information that is displayed to the end user such as _____, _____ and _____.
   (a) Network (b) text (c) Speed
   (d) URL (e) image (f) website navigation

Ans. : (b), (e), (f)

4. Example of search engine are _____. (March 2022)
   (a) Google (b) Yahoo (c) Bing
   (d) Java (e) Internet Explorer

Ans. : (a), (b), (c)

2.3 Techniques of SEO

5. Black Hat Strategies includes _____. (March 2022)
   (a) duplicate content (b) relevant content (c) invisible text
   (d) stuffed keywords (e) page titles (f) relevant keywords

Ans. : (a), (c), (d)

6. White Hat Strategies include _____. (March 2022)
   (a) duplicate content (b) relevant content (c) well-labeled images
   (d) relevant links (e) duplicate keywords (f) stuffed keywords

Ans. : (b), (c), (d)

2.4 Optimizing Website

7. Important parts of Search Engine Optimization. (March 20
   (a) Keywords (b) Research (c) Analysis
   (d) Information (e) Conversion (f) Parameters

Ans. : (a), (c), (d)

8. Other factors to rank the website higher are
   (a) Desktop friendly (b) Speed of website (c) Mobile friendly
   (d) Creating robots.txt (e) Broken links (f) On page links

Ans. : (b), (c), (d)

9. Following are the ways to optimize the webpage for SEO are _____.
   (a) Page structure optimization (b) labeled images
   (c) SEO keywords (d) Duplicate content
   (e) SEO Social Bookmarking (f) Unique page titles (March 202

Ans. : (a), (c), (e)

Match the Following

A	B
1. Image optimization	(a) Duplicate content
2. Links to Twitter, Facebook	(b) file that tell which pages you want to exclud from indexing
3. Description about author	(c) links when clicked , take the user to anothe site
4. White hat Strategy	(d) meta tag
5. Black hat Strategy	(e) alt attribute
6. Robots.txt	(f) Social Media Optimization
7. SEOptimer	(g) Relevant content
8. Backlinks	(h) determine the health of the website
9. SEO Audit	(i) SEO Audit Tool.

Ans. :

1. = (e) 2. = (f) 3. = (d) 4. = (g) 5. = (a)
6. = (b) 7. = (i) 8. = (c) 9. = (h)

Answer in Brief

2.1 Introduction

1. Define SEO. (March 202

Ans. :

(i) **SEO** stands for Search Engine Optimization. SEO is a technique used for designi and developing a website results. It is a subset of search engine marketing.

(ii) SEO is useful in increasing the number of visitors to a website.

(iii) There are two types of optimization :-

(iv) **On-Page SEO** : This includes Provision of good content, good keywords selection, putting keywords on correct places, giving appropriate title to every page etc.

(v) **Off-Page SEO** : It includes link building, increasing link popularity by submitting open directories, search engines, link exchange etc.

2.2 Types of SEO

2. Define SEO. State types of SEO. (March 2022)

Ans. :

(i) SEO stands for Search Engine Optimization, which is the practice of increasing the quantity and quality of traffic to your website through organic search engine results.

(ii) Thus SEO is useful in increasing the number of visitors to a website.

(iii) There are many strategies and techniques adopted to optimize the webpage.

(iv) **Types of SEO-** There are two types of optimization:-

(v) **On-Page SEO** : This includes Provision of good content, good keywords selection, putting keywords on correct places, giving appropriate title to every page etc.

(vi) On-Page SEO is concerned with information that is displayed to the end user, such as text, images and website navigation.

(vii) **Off-Page SEO** : It includes link building, increasing link popularity by submitting open directories, search engines, link exchange etc.

(viii) Off-Page SEO is concerned with Website-Website relationship.

2.3 Techniques of SEO

3. Which are the different SEO techniques. (March 2022)

Ans. :

(i) SEO techniques are classified into two broad categories, which are White Hat SEO and Black Hat SEO.

(ii) **White Hat SEO :** Techniques that search engines suggest as a part of good design are called White Hat SEO.

(iii) It contains proper and well labelled images according to the content .

(iv) It is in the form of Standard-compliant HTML.

(v) **Black Hat SEO :** Techniques that search engines do not approve and those techniques which are used for optimizing the website are called Black Hat SEO.

(vi) It mainly contains duplicate contents. Such websites are mainly used to redirect users to other websites and cause traffic.

4. Explain in short White Hat technique.

Ans. :

(i) Techniques that search engines suggest as a part of good design are called White Hat SEO.

(ii) It contains proper and well labelled images according to the content.

(iii) It is in the form of Standard-compliant HTML.

(iv) The page titles are unique but relevant at the same time.

(v) It has Complete sentences with good spelling and grammar

5. Explain in short Black Hat technique.

Ans. :

(i) Techniques that search engines do not approve and those techniques which are used for optimizing the website are called Black Hat SEO.

(ii) It mainly contains duplicate contents which mostly are stolen from other original creators mainly without any credits and permissions.

(iii) Such websites are mainly used to redirect users to other websites and cause traffic.

(iv) links from such sites are irrelevant and don't show the actual search content. Websites like these contain invisible text and stuffed keywords.

2.4 Optimizing Website

6. Explain SEO Page content optimization points?

Ans. :

(i) **TITLE :** Title is the first thing that user notices in search result list while using Google for searching. Contents should be up to 60 characters (spaces included)

(ii) **Meta-description Tag :** While creating a web page, Meta description should be used to boost your On-Page SEO. Meta Description should be between 70 and characters (spaces included) that others are interested to know about the content form your website.

(iii) **Heading Tag :** Page should include proper heading tag whenever required.

(iv) **Image tag with ALT attributes :** ALT attribute gives information about the image.

7. Explain Page Structure Optimization?

Ans. :

(i) Html programs should start with<!doctype html>.

(ii) If it is singular tag eg <img> it should be self enclosed like this-<img />

(iii) Order of tags should be proper.

(iv) Most of the time recently opened tag will be closed first.

(v) Title is the first thing that user notices in search result list while using Google for searching

(vi) While creating a web page, Meta description should be used to boost your On-Page SEO.

(vii) The pages should include proper heading tags from <h1> to <h6>wherever required.

(viii) The alt attribute gives information about the image, so even if the image does not download, it can make the viewers understand the image related information.

(ix) **Example :**

```
<!doctype html>
<html>
<head><title> Page structure with
appropriate order of opening and
closing tags</title>
</head>
<body>
<h1>Let us understand SEO</h1>
<p>Webpage optimization helps for higher ranking in search engines.
<b>There are number of ways to optimize the pages
.<i>Even the Social media plays an important role!!!</i>
</b>
</p>
</body>
</html>
```

8. Explain in short SEO Keyword search? (March 2022)

Ans. :

(i) SEO keywords are the important words and phrases related to the developed website content.

(ii) A list of keywords, need to be carefully chosen to optimize the search. Keyword Research and Analysis is an important part of the Search Engine Optimization (SEO) as well as "Search Engine Marketing".

(iii) For "Search Engine Marketing", one can pay for certain keywords.

(iv) There are various online Keyword Planner tools (such as Google Add Words Keyword Planner) to shortlist keywords.

(v) This is more useful to compare the cost of various keywords, if the website designer opts for Paid Search Engine Marketing.

9. Explain SEO-Social Media Optimization and SEO-Backlinks ? (March 2022)

Ans. :

(i) It deals with enhancing the website's ranking, using interactive communities like facebook, twitter, blogs, forums and so on.

(ii) When these communities have links to the created website it builds familiarity and trust about the website.

(iii) Backlinks are links on one website that, when clicked, take the user to another site.

(iv) Backlinks are especially valuable for SEO because they contribute to the overall strength and value of the content.

(v) They also represent a "vote of confidence" from one site to another
(vi) Backlinks to your website are a signal to search engines that others are interested know about the contents form your website.

10. Explain how could you optimize images.

Ans. :

(i) Image Tags with ALT attribute : The alt attribute gives information about the image, so even if the image does not download, it can make the viewers understand the image related information.

(ii) Image Tag with Title attribute : The title tag of images show the keyword of the image when mouse is hovered above the image.

(iii) Image compression : The images which are compressed load faster which give user good friendly experience.

(iv) Images can have Caption keywords : Certain images can have keywords on the image to make it relevant for user to know about it.

11. Explain SEO Social Bookmarking. (March 2018)

Ans. :

(i) Social Bookmarking is a powerful tool in promoting a website.
(ii) It is very useful for off-page SEO optimization.
(iii) Bookmarks are website links that are stored for future references.
(iv) Bookmarking service allows you to save, organize and share bookmarks with other users.
(v) With social bookmarking, you can quickly create high quality backlinks.

12. Explain following factors for SEO.
(1) Speed of the website
(2) Mobile friendly
(3) Creating Robot$.txt.

Ans. :

(1) Speed of the website

(i) While creating a website, attention is given on the design, content and as many visuals as possible.
(ii) However, all this can slow down the website and obstruct usability.

(2) Mobile friendly

(i) Website's mobile friendliness has become a ranking factor for google.
(ii) It is very important to ensure that the created website is mobile friendly for maximum exposure and to improve the websites ranking.

(3) Creating Robots.txt.

(i) This file tells search engine which pages you want to exclude from indexing.
(ii) For example, if you do not want a certain page or image being indexed, then you can restrict that page from being indexed.

2.5 SEO Audit

13. Explain SEO Audit in detail?

Ans. :

(i) An SEO Audit helps to find out what could be done to improve ranking on search engines, so that consumers could find the website with greater ease.
(ii) It analyses the websites health. SEO audit helps to identify tags in the SEO process.
(iii) Audit is a checklist created for SEO to be followed.
(iv) Once the list is complete we may find our website moving up the rank.
(v) Audit Results will have following details :

• Overall Ranking of the site	• Recommendations for improvements
• Search Engine Optimization	• Usability of the website
• Performance of website	• Social links
• Connected social accounts	• Security parameters
• SSL Enabled	• HTTPS Redirect
• Email Privacy	• Header content recommendations
• Title and Meta description	• Body Content
• Header Tags	• Body content recommendations
• Keyword Consistency	• Amount of content in the pages
• Links	• Number of backlinks
• Broken links	• Friendly links
• On page links	• Technology list which is used in the website

Self Assessment

Time : 1 Hour **Marks: - 40**

1. Fill in the blanks **(05)**

1. _____ is the process of boosting content and technical setup of the website.
2. Companies communicate with their audience through _____ marketing,
3. _____ SEO involves looking for ways to improve user experience ethically and genuinely.
4. _____ strategies include relevant content.
5. One of the optimizing factor to improve images is by using _____attribute.

2. True or False **(05)**

1. Device Rendering is an interactive media technology that makes sites interesting.
2. SEO stands for Search Engine Optimization.

3. Page Size in the SEO refers to the downloaded file size of a given web page.
4. Black Links are the links that send a message to its visitors that the web p no longer exists.
5. Social media marketing is the use of social media platforms and website promote a product or service.

3. MCQ one correct alternative.

1. The ____ includes provision of good content.
   a) SEO b) Google Crawler
   c) On Page SEO d) Off Page SEO
2. _____ is an software that helps the users to search for information world wide web.
   (a) Mobile Marketing (b) Paid Search
   (c) Search Engine (d) email Marketing
3. _____ is a subset of search engine marketing.
   a) MIS b) GPS c) SEO d) CRM
4. _____ is a powerful tool in promoting a website.
   a) Social Distancing b) Social impact
   c) Social Gathering d) Social Bookmarking
5. _____ file tells search engine which page you want to exclude from indexing
   (a) Robots.docx (b) Robots.pptx
   (c) Robots.txt (d) Tobots.txt

4. MCQ two correct alternative.

1. Types of SEO are _____
   (a) No Page SEO (b) On Page SEO
   (c) Off Page SEO (d) All Page SEO (e) One Page SEO
2. Keywords _____ and _____ is an important part of SEO.
   (a) Module (b) Resource (c) Planning
   (d) Research (e) Analysis
3. _____ and _____ are the major factors to optimize the web page.
   (a) Crawling (b) Hyperlinks (c) Social media links
   (d) No Links (e) Back links

5. MCQ three correct alternative.

1. Page structure optimization includes _____.
   (a) <title> (b) bold (c) Conversion
   (d) Heading Tags (e) Image Tags (f) No tags

2. Examples of search engines are_____.
   (a) Website (b) Google (c) C++
   (d) Yahoo (e) Java (f) Bing

6. Match the Following. (03)

A	B
1. SEO	a) concerned with Website-Website relationship
2. Off-Page SEO	b) tool used in promoting website
3. Social Bookmarking	c) Free SEO Audit Tool
4. SEO optimizer	d) Search Engine Optimization
5. Backlinks	e) Helps to find out what could be done to improve ranking on search engines
6. SEO Audit	f) Links when clicked , take the user to another site

7. Answer the following (10)

a) Explain White Hat technique. **(Refer Page No 2.19, Q.4).**
b) Explain SEO keyword search. **(Refer Page No 2.20, Q.8)**
c) Define SEO. **(Refer Page No 2.17, Q.1)**
d) Explain SEO technique. **(Refer Page No 2.18, Q.3)**
e) Explain On Page SEO. **(Refer Page No 2.18, Q.2)**

Answer of Questions

1. Fill in the blanks
1) Search Engine Optimization (SEO) 2) Email
3) White Hat 4) White Hat 5) Alt

2. True or False
1) False 2) True 3) True 4) False 5) True

3. MCQ one correct alternative
1) C 2) C 3) C 4) D 5) C

4. MCQ two correct alternative
1) (b, c) 2) (d, e) 3) (c, e)

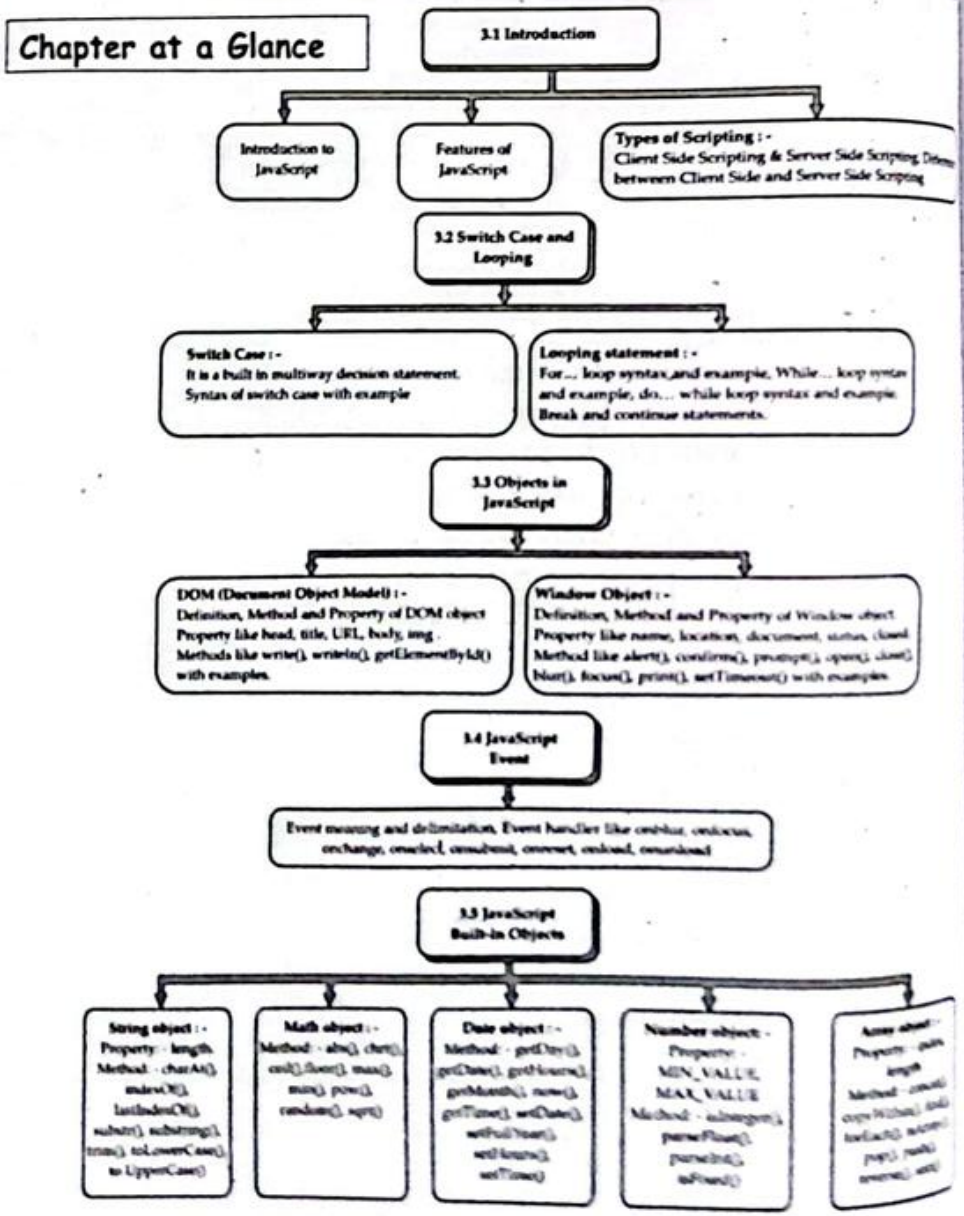
5. MCQ three correct alternative
1) (a, d, e) 2) (b, d, f)

6. Match the Following
1 - d 2 - a 3 - b 4 - c 5 - f 6 - e

□□□

3 Advanced JavaScript

Chapter at a Glance



Exercise

Fill in the blanks

1. _____ script resides onserver computer.

Ans. : Server Side

2. _____ statement is used to jump out of loop.

Ans. : Break

3. _____ defines logical structure of document.

Ans. : DOM (Document Object Model)

4. _____ property of window object returns Boolean value in- dictating whether window is closed or not.

Ans. : closed

5. _____ event occurs when an element looses its focus.

Ans. : onblur

State whether given statement is true or false

1. JavaScript is case sensitive language. **(March 2022)**

Ans. : True

2. Math.ceil() function is used to return the nearest integer less than or equal to given number.

Ans. : False

3. MAX_VALUE property of number object returns smallest possible value.

Ans. : False

4. getDay() method of Date object returns month in number.

Ans. : False

5. onKeydown event occurs when user moves mouse pointer.

Ans. : False

Multiple choice questions. Select one correct answer

1. JavaScript is _____ language.

(a) Compiled (b) Interpreted

(c) Both a and b (d) None of the above

Ans. : (b)

2. Select correct method name of String object _____.

(a) charAt() (b) characterAt()

(c) valueAt() (d) lengthAt()

Ans. : (a)

3. _____ method displays message box with Ok and Cancel button.
   (a) Confirm() (b) Alert()
   (c) both a and b (d) None of these

Ans. : (a)

4. We can declare all types of variables using keyword _____.
   (a) var (b) dim (c) variable (d) declare

Ans. : (a)

5. Trace output of following JavaScript code.

```
var str="Information Technology";
document.write(str. lastIndexOf("o");
```

   (a) 18 (b) 19 (c) 20 (d) 21

Ans. : (b)

Multiple choice questions. Select two correct answer

1. Valid two methods of Date object are _____ and _____.
   (a) setTime() (b) getValidTime()
   (c) getTime() (d) setValidTime()

Ans. : (a, c)

2. Properties of document object are _____ and _____.
   (a) URL (b) title (c) name (d) status

Ans. : (a, b)

3. _____ and _____ are event / event handler used with text object in JavaScript.
   (a) onBlur (b) onMove (c) onFocus (d) onAction

Ans. : (a, c)

Multiple choice questions. Select three correct answers

1. Select three correct methods of window object _____.
   (a) write() (b) alert() (c) writeln() (d) close()
   (e) open() (f) charAt()

Ans. : (b, d, e)

2. JavaScript features are _____, _____ and _____.
   (a) supports event based facilities (b) is platform dependent language
   (c) case insensitive scripting language (d) provide inbuilt objects
   (e) can handle date and time effectively (f) requires special software to run

Ans. : (a, d, e)

3. Inbuilt objects in JavaScript are _____, _____ and _____.
   (a) Time (b) Date (c) Inheritance
   (d) Array (e) Number (f) function

Ans. : (b, d, e)

Explain the following

1. What are similarities and differences between client side scripting and server side scripting.

Ans. : Refer answer the following Q. 2, 3.1.

2. Briefly explain features of JavaScript.

Ans. : Refer answer the following Q. 1, 3.1.

3. Explain switch case conditional statement in JavaScript with example.

Ans. : Refer answer the following Q. 3, 3.2.

Write event driven JavaScript program for the following

1. **Display Addition, substraction, multiplication, division of two numbers, which were accepted from user.**

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Ist Number
<input type="number" name="n1"><br><br>
Enter IInd Number
<input type="number" name="n2"><br><br>
<input type="button" name="b1" value="Calculate" onClick="chk()">
</form>
</body>
<script type="text/javascript">
function chk()
{
var a,b,mul,sub,add,div;
a=parseInt(f1.n1.value); or a = parseInt (document.fl.nl.value);
b=parseInt(f1.n2.value); or b = parseInt (document.fl.nl.value);
add=a+b;
```

```
mul=a*b;
sub=a-b;
div=a/b;
alert("Addition is "+add);
alert("Substraction is "+sub);
alert("Multiplication is "+mul);
alert("Division is "+div);
}
</script>
</html>
```

2. Display number sequence from 100 to 150 in following format.
(100 101 102.....150)

Coding :

```
<!DOCTYPE html>
<html>
<script type="text/javascript">
var i;
document.write("<br>Numbers from 100-150 are <br>");
for(i=100;i<=150;i++)
{
document.write("\t"+i);
}
</script>
</html>
```

3. Find and display factorial of given number.

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
<input type="number" name="t1"><br>
<input type="button" name="b1" value="Factorial" onMouseOver="fact()">
</form>
</body>
```



```
<script type="text/javascript">
function fact()
{
var a,f=1,i;
a=parseInt(f1.t1.value);
for(i=a;i>=1;i--)
{
f=f*i;
}
alert("Factorial of a Number is "+f);
}
</script>
</html>
```

4. Accept any string from user and count and display number of vowels occurs in it.

Coding :

```
<!DOCTYPE html>
<html>
<script type="text/javascript">
var n,i,ch,cnt=0;
n=prompt("Enter a String");
for(i=0;i<n.length;i++)
{
ch=n.charAt(i);
if(ch=='a' || ch=='A' ||ch=='e' || ch=='E' ||ch=='i' || ch=='i' ||
ch=='o' || ch=='O' || ch=='u' || ch=='U')
{
cnt=cnt+1;
}
}
document.write("Number of vowels in string are "+cnt);
</script>
</html>
```

Match the following

A		B
1.	ceil()	(a) Writes HTML expression or JavaScript code to a document.
2.	floor()	(b) Sets focus to current window.
3.	write()	(c) Removes white spaces from both sides of string.
4.	focus()	(d) Returns next integer greater than or equal to given number.
5.	trim()	(e) Returns the next integer less than or equal to given number.

Ans. : 1.-d, 2.-e, 3.-a, 4.-b, 5.-c

Fill in the Blanks

3.1 Introduction

1. _____ is an interpreted scripting language.

Ans. : JavaScript

2. _____ is a set of instructions used to produce various kinds of outputs.

Ans. : Program

3. The programs in JavaScript language are called .....

Ans. : Scripts

4. _____ is an object oriented scripting language and supports event ba... programming facility. (March 20...

Ans. : JavaScript

5. _____ is platform independent scripting language.

Ans. : JavaScript

6. _____ is knows as Universal client side scripting language.

Ans. : JavaScript

7. In _____ scripting the script resides on the client computer.

Ans. : Client side

8. In _____ scripting the script resides on web server.

Ans. : Server side

9. _____ scripting does not need any server interaction.

Ans. : Client side

10. _____ scripting communicates to the server.

Ans. : Server side

11. _____ scripts are used for validation purpose.

Ans. : Client side

12. _____ scripts requires web browser as an interface.

Ans. : Client side

13. _____ scripts requires web server software to execute.

Ans. : Server side

14. _____ is a JavaScript based open source frontend web framework developed for single page application.

Ans. : Angular JS

15. _____ is JavaScript based framework for building interactive user interface.

Ans. : Vue Js

16. In JavaScript function is declared using _____ keyword.

Ans. : function

17. _____ consists of JavaScript libraries for building UI for single page application and mobile application..

Ans. : React

3.2 Switch case and Looping Structures

18. JavaScript has a built-in multiway decision statement known as _____ (March 2022)

Ans. : Switch

19. _____ refers to the execution of statement or a group of statements of code for a fixed number of times.

Ans. : Iteration

20. _____ loop combines initialization, condition and loop iteration in single statement. (March 2022)

Ans. : For

21. _____ means increment or decrement value of a running variable.

Ans. : Iteration

22. _____ statement is used to jump out of loop. (March 2022)

Ans. : Break

23. When it is necessary to skip statement block and take the control at the beginning for next iteration _____ statement is used. (March 2022)

Ans. : Continue

3.3 Objects in JavaScript

24. The _____ keyword is used to create new object in JavaScript. (March 2022)

Ans. : new

25. An _____ can group data together with functions needed to manipulate it.

Ans. : Object

26. All tangible things are known as _____ (March 2022)

Ans. : Objects

27. DOM stands for .....
Ans. : Document Object Model
28. The way in which HTML document content is accessed and modified is called .....
Ans. : Document Object Model
29. ..... property of DOM object returns the <head> element of a document. (March 2022)
Ans. : Head
30. ..... is an object based scripting language.
Ans. : JavaScript
31. ..... property of DOM object sets or returns title of the document.
Ans. : Title
32. ..... property of DOM object returns URL of the HTML document.
Ans. : URL
33. ..... method of DOM object writes HTML expressions or JavaScript code to a document. (March 2022)
Ans. : write()
34. ..... method of DOM object adds a newline character after each statement.
Ans. : writeln()
35. Using ..... method id property is used to find an element.
Ans. : getElementById()
36. The ..... property is useful for getting html element and changing its content. (March 2022)
Ans. : innerHTML
37. ..... object is parent of all other objects.
Ans. : window
38. ..... object represents an open window in a browser.
Ans. : window
39. ..... property of window object sets or returns the name of the window.
Ans. : Name
40. ..... property of window object returns the location object for the window.
Ans. : location
41. ..... property of window object returns the document object for the window.
Ans. : document
42. ..... property of window object sets or returns the fact in the status bar of a window.
Ans. : status

43. ..... property of window object returns a Boolean value indicating whether a window has been closed or not.
Ans. : closed
44. ..... method of window object display the alert box containing message with OK button.
Ans. : alert ()
45. ..... method of window object displays the confirm dialog box containing message with ok and cancel button.
Ans. : confirm()
46. ..... method of window object displays a dialog box to get input from the user. (March 2022)
Ans. : prompt()
47. ..... method of window object opens the new window.
Ans. : open()
48. ..... method of window object closes the current window.
Ans. : close()
49. ..... method of window object removes focus from the current window.
Ans. : blur()
50. ..... method of window object sets focus to the current window.
Ans. : focus()
51. ..... method of window object prints the content of current window.
Ans. : print()
52. ..... method of window object calls a function or evaluates an expression after a specified number of milliseconds.
Ans. : setTimeout()

3.4 JavaScript Events

53. ..... are the actions done by users or an application that occurs on the webpage . (March 2022)
Ans. : Events
54. ..... event handler occurs when user leaves field or looses focus of an element .
Ans. : onblur
55. ..... event handler occurs when an element gets focus .
Ans. : onfocus
56. ..... event handler occurs when user changes content of an element or selects drop down list . (March 2022)
Ans. : onchange
57. ..... event handler occurs when user selects some text of an element .
Ans. : onselect

58. ..... event handler occurs when user clicks submit button .
Ans. : onsubmit
59. ..... event handler occurs when user clicks reset button .
Ans. : onreset
60. ..... event handler occurs when page/image has been loaded .
Ans. : onload
61. ..... event handler occurs when document page has been unloaded / closes .
Ans. : onunload

3.5 JavaScript built-in Objects

62. ..... object helps us to work with number.
Ans. : Number
63. ..... property of Number object returns the largest minimum value. (March 2022)
Ans. : MIN_VALUE
64. ..... property of number object returns the largest maximum value. (March 2022)
Ans. : MAX_VALUE
65. ..... property of Number object represents not a number value.
Ans. : NaN
66. ..... method of number object determines whether the given value is a Integer. (March 2022)
Ans. : isInteger()
67. ..... method of number object converts the given string into a floating point number.
Ans. : parseFloat()
68. ..... method of Number object converts the given string into a integer number.
Ans. : parseInt()
69. ..... method of number object returns the string that represents a number with exact digits after a decimal point.
Ans. : isFixed()
70. ..... object is used to store or manipulate text . (March 2022)
Ans. : String
71. ..... property of string object returns number of characters in a string. (March 2022)
Ans. : Length
72. ..... method of string object returns the character at the specified position.
Ans. : charAt()

73. ..... method of string object returns the index of the first occurance of specified character in given string. (March 2022)
Ans. : indexOf()
74. ..... method of string object returns the index of last occurance of specified character in given string.
Ans. : lastIndexOf()
75. ..... method of string object removes whitespace from both sides of a string.
Ans. : trim()
76. ..... method of string object converts a string to lower case. (March 2022)
Ans. : toLowerCase()
77. ..... method of string object convert a string to upper case.
Ans. : toUpperCase()
78. The built in ..... object includes mathematical constants and functions.
Ans. : math
79. ..... method of math object returns the absolute value of a number.
Ans. : abs()
80. ..... method of Math object returns the cube root of a number.
Ans. : cbrt()
81. ..... method of math object returns the next integer grater than or equal to a given number.
Ans. : ceil()
82. ..... method of math object returns the next integer less than or equal to a given number.
Ans. : floor()
83. ..... method of math object returns the highest valued number in a list of numbers.
Ans. : max()
84. ..... method of math object returns the lowest valued number in a list of numbers.
Ans. : min()
85. ..... method of math object returns the base to the exponent power.
Ans. : pow()
86. ..... method of math object returns a random number between 0 and 1.
Ans. : random()
87. ..... method of math object returns the square root of a number.
Ans. : sqrt()
88. ..... object is used to create date and time values.
Ans. : date
89. ..... method of date object returns the day of the month.
Ans. : getDate()
90. ..... method of date object returns the day of the week.
Ans. : getDay()

91. ..... method of date object returns the year.
Ans. : getFullYear()
92. ..... method of date object returns the hour.
Ans. : getHours()
93. ..... method of date object returns the minutes.
Ans. : getMinutes()
94. ..... method of date object returns the month.
Ans. : getMonth()
95. ..... method of date object returns the seconds.
Ans. : getSeconds()
96. ..... method of date object sets the day of the month.
Ans. : setDate()
97. ..... method of date object sets the full year.
Ans. : setFullYear()
98. ..... method sets the hours of a date object.
Ans. : setHours()
99. ..... method sets the minutes of a date object.
Ans. : setMinutes()
100. ..... method sets the month of a date object.
Ans. : setMonth()
101. ..... method sets the seconds of a date object.
Ans. : setSeconds()
102. ..... method sets a date to a specified number of millisecond after/before Jan 1, 1970.
Ans. : setTime()
103. An ..... is an object that can store a collection of items.
Ans. : Array
104. An ..... is a special variable which can hold more than one value a time.
Ans. : Array
105. The index of the first element of an array is .....
Ans. : Zero
106. ..... is referred to access and set the items in an array.
Ans. : Index Number
107. ..... property of Array object represents the zero based index of the match in the string.
Ans. : Index
108. ..... property of Array object reflect number of elements in array.
Ans. : length
109. ..... method of array object joins two or more arrays and returns a copy of the joined arrays.
Ans. : concat()

110. ..... method of Array object copies array elements within the array, to and from specified position.
Ans. : copyWithin()
111. ..... method of array object returns the value of the first element in an array that satisfies a test in testing.
Ans. : find()
112. ..... method of Array object calls a function for each array element.
Ans. : forEach()
113. ..... method of Array object search the array for an element and returns its position.
Ans. : indexOf()
114. ..... method of array object checks whether an object is an array.
Ans. : isArray()
115. ..... method of array object removes the last element of an array and returns that element.
Ans. : pop()
116. ..... method of array object adds new elements to the end of an array.
Ans. : push()
117. ..... method of array object reverses the order of the elements in an array.
Ans. : reverse()
118. ..... method of array object sorts the elements of an array.
Ans. : sort()

True or False

3.1 Introduction

1. JavaScript is an involved scripting language.
Ans. : False
2. An interpreted language is a type of programming language that executes its instructions directly without compiling machine language.
Ans. : True
3. Program is a set of instructions used to produce various kinds of outputs.
Ans. : True
4. JavaScript was created to destroy web pages.
Ans. : False
5. There is no need of special software to run JavaScript program.
Ans. : True

6. Java script is an object oriented scripting language.

Ans. : True

7. JavaScript is not case sensitive language. (March 2022)

Ans. : False

8. JavaScript helps the browser to perform input validation.

Ans. : True

9. JavaScript cannot handle date and time.

Ans. : False

10. JavaScript has the ability to create new functions within scripts.

Ans. : True

11. Functions are declared in JavaScript using var keyword. (March 2022)

Ans. : False

12. Software that can run on any hardware platform is called as platform independent software.

Ans. : True

13. JavaScript is platform independent scripting language.

Ans. : True

14. There are two types of scripting. (March 2022)

Ans. : True

15. In client side scripting, the script resides on client computer and that can run on the client.

Ans. : True

16. Client side scripts are placed inside HTML document.

Ans. : True

17. In Server side scripting the scripts resides on web server.

Ans. : True

18. Server side scripting is used at the front end. (March 2022)

Ans. : False

19. Client side scripting is used at the front end.

Ans. : True

20. When a server side script is processed it communications to the client.

Ans. : False

21. Client side scripting does not need any server interaction.

Ans. : True

22. PHP, ASP.net etc are server side scripting languages. (March 2022)

Ans. : True

23. Client side scripts are generally used for validation purpose.

Ans. : True

3.2 Switch case and Looping Structures

24. JavaScript has a built-in multiway decision statement known as switch.

Ans. : True

25. There should not be duplicity between the cases. (March 2022)

Ans. : True

26. The value for the case must be similar data type as the variable in switch.

Ans. : True

27. Iteration refers to the execution of statements of code for a fixed number of times till the condition is satisfied.

Ans. : True

28. For loop combines initializing, condition and loop iteration in single statement.

Ans. : True

29. Loop will execute statement in statements block will the condition is false. (March 2022)

Ans. : False

30. Iteration means increment or decrement value of a running variable. (March 2022)

Ans. : True

31. While loop executes statements as long as the condition is false.

Ans. : False

32. Break statement is used to jump out of loop. (March 2022)

Ans. : True

33. Continue statement is used to make early exit from a loop.

Ans. : False

34. Continue statement is used to skip statement block and take the control at the beginning for next iteration.

Ans. : True

3.3 Objects in JavaScript

35. JavaScript is an object based scripting language.

Ans. : True

36. A JavaScript object is an entity having properties and objects. (March 2022)

Ans. : False

37. Properties and methods of object are accessed with "." Operator.

Ans. : True

38. JavaScript supports 4 types of objects.

Ans. : False

39. JavaScript gives facility to create user defined objects.

Ans. : True

40. The new keyword is used to create new object in JavaScript.
Ans. : True
41. DOM stands for Document Original Model. (March 2022)
Ans. : False
42. The way in which HTML document content is accessed and modified is called a DOM.
Ans. : True
43. Head property of DOM object returns the <head> element of the document.
Ans. : True
44. Title property is DOM object returns the name of the document.
Ans. : False
45. URL property of DOM object returns full URL of the HTML document. (March 2022)
Ans. : True
46. Body property of DOM object returns <body> elements respectively.
Ans. : True
47. write() method of DOM object write Java script code to a document.
Ans. : True
48. writeln() method is same as write ( ) method of DOM object. (March 2022)
Ans. : False
49. writeln() method of DOM object adds a new line character after each statement.
Ans. : True
50. The outerHTML property is used to change any HTML element.
Ans. : False
51. window object is parent object of all other objects. (March 2022)
Ans. : True
52. An object of window is created automatically by the browser.
Ans. : True
53. name property of window object sets or returns the name of a window.
Ans. : True
54. location property of window object returns a Boolean value indicating whether a window has been closed or not.
Ans. : False
55. document property of window object returns the document object for the window.
Ans. : True
56. status property of window object sets the name of a window. (March 2022)
Ans. : False
57. alert() method of window object displays the alert box containing message with ok button.
Ans. : True

58. prompt() method of window object displays a dialog box to set output for the user.
Ans. : False
59. confirm() method of window object displays the confirm dialog box containing message with ok and cancel button.
Ans. : True
60. open( ) method of window object removes the focus from the current window. (March 2022)
Ans. : False
61. close() method of window object closes the current window.
Ans. : True
62. blur() method of window object removes focus from the current window.
Ans. : True
63. focus() method of window object removes focus from the current window. (March 2022)
Ans. : False
64. print() method of window object prints the content of current window.
Ans. : True
65. setTimeout() method of window object calls a function after a specified number of milliseconds.
Ans. : True

3.4 JavaScript Events

66. Events are the actions done by the user or an application that occurs on the web page. (March 2022)
Ans. : True
67. onKeyPress, onKeyDown are keyboard events.
Ans. : True
68. onClick, onMouseUp are mouse events. (March 2022)
Ans. : True
69. onblur event handler occurs when an element gets focus.
Ans. : False
70. onfocus event handler occurs which an element gets focus.
Ans. : True
71. onchange event handler occurs when user changes content of an element.
Ans. : True
72. onselect event handler occurs when page image has been loaded. (March 2022)
Ans. : False
73. onsubmit event handler occurs when user clicks submit button.
Ans. : True

74. onreset event handler occurs when user clicks reset button.
Ans. : True
75. onload event handler occurs when document/page has been loaded.
Ans. : True
76. onunload event handler occurs when user clicks on submit button. (March 20
Ans. : False

3.5 JavaScript built-in Objects

77. Number object helps us to work with numbers.
Ans. : True
78. MIN_VALUE property of number object returns the largest minimum value. (March 20
Ans. : True
79. MAX_VALUE property of number objects returns the largest minimum value.
Ans. : False
80. NaN property of number of object returns Not a Number value. (March 20
Ans. : True
81. isInteger() method of number of object determine whether the given value is character.
Ans. : False
82. parseFloat() method of number object converts the given string into a floating po number.
Ans. : True
83. parseInt() method of number object converts the given string into a integer numbe
Ans. : True
84. isFixed() method of number object returns the string that represents a number wi exact digits after a decimal point.
Ans. : True
85. String is used to store zero or more character of text within single or double quote
Ans. : True
86. String object is used to store and manipulate numbers.
Ans. : False
87. length property returns the number of characters in a string.
Ans. : True
88. charAt() method of string object returns the character at the specified position.
Ans. : True
89. indexOf() method of string object returns the index of the middle occurrence of a specified character in the given string.
Ans. : False
90. lastIndexOf() method of string object returns the index of the last occurrence specified character in given string.
Ans. : True

91. substr() method of string object returns the characters you specified.
Ans. : True
92. trim() method of string objects adds the white spaces from both sides of a string.
Ans. : False
93. toLowerCase() method of string object converts a string to capitals.
Ans. : False
94. toUpperCase() method of string object converts the string to upper case.
Ans. : True
95. The Math object includes mathematical constants and functions.
Ans. : True
96. There is no need to create a Math object before using it.
Ans. : True
97. abs() method of math object returns square root of a number.
Ans. : False
98. cbrt() method of math object returns cube root of a number.
Ans. : True
99. ceil() method of math object returns the next inter greater than or equal to a given number.
Ans. : True
100. floor() method of math object returns the next integer less than or equal to a given number.
Ans. : True
101. max() method of math object returns the smallest valued number of a list of numbers.
Ans. : False
102. min() method of math object returns the power of the number.
Ans. : False
103. pow() method of math object returns the base to the exponent power.
Ans. : True
104. random() method of math object returns a random number between 0 and one.
Ans. : True
105. sqrt() method of math object returns the square root of a number.
Ans. : True
106. The time object is used to create date and time values.
Ans. : False
107. Date object is created using new keyword.
Ans. : True
108. getDate() method of date object returns the day of month.
Ans. : True
109. getDay() method of date object returns the day of the month.
Ans. : False
110. getFullYear() method of date object returns the year.
Ans. : True

111. getHours() method of Date object returns the month.
Ans. : False
112. getMinutes() method of Date object returns minutes.
Ans. : True
113. getMonth() method of Date object returns month name.
Ans. : False
114. getSeconds() method of Date object returns seconds.
Ans. : True
115. getTime() method of Date object returns date.
Ans. : False
116. now() method of Date object returns the number of a milliseconds.
Ans. : True
117. setDate() method of Date object set day of the month of a date object.
Ans. : True
118. setFullYear() method of Date object sets year.
Ans. : True
119. setHours() method of Date object set the seconds.
Ans. : False
120. setMinutes() method of Date object sets minutes.
Ans. : True
121. setMonth() method of Date object sets month.
Ans. : True
122. setSeconds() method of Date object sets hours.
Ans. : False
123. setTime() method of Date object sets a date to a specified number of milliseconds
Ans. : True
124. An array is an object that can store a collection of items.
Ans. : True
125. Arrays are used to store single value in single variable.
Ans. : False
126. Items of array can be accessed by referring to its index number.
Ans. : True
127. Index of first element of an array is one.
Ans. : False
128. marks[0] is the first element of the array.
Ans. : True
129. Index property of Array object represents two based index of the matching string.
Ans. : False
130. Length property of Array object reflects number of elements in array.
Ans. : True

131. concat() method of Array object joins two or more arrays and returns a copy of joined array.
Ans. : True
132. copyWithin() method of Array object adds array elements within the array to and from a specified positions.
Ans. : False
133. find() method of Array object returns the value of the first element in an array that satisfies a test in testing.
Ans. : True
134. forEach() method of Array object search the array and returns it's position.
Ans. : False
135. indexOf() method of Array object search the array for an element and returns its position.
Ans. : True
136. isArray( ) method of Array object removes element from array.
Ans. : False
137. pop() method of Array objects removes the last element of an array and returns that element.
Ans. : True
138. push() method of Array object add new elements to the end of an array and returns the new length.
Ans. : True
139. reverse() method of Array object returns the elements in an array.
Ans. : False
140. sort() method of Array object sorts the elements of an array.
Ans. : True

MCQ (One Correct Answers)

3.1 Introduction

1. ..... is an interpreted scripting language.
(a) PHP (b) C++ (c) HTML (d) JavaScript
Ans. : (d)
2. ..... is a set of instructions used to produce various kinds of outputs.
(a) Design (b) Software (c) Program (d) Hardware
Ans. : (c)

3. The programs in JavaScript language are called ..... (March
(a) Website (b) Scripts (c) Web server (d) Software
Ans. : (b)

4. ..... is an object oriented scripting language and supports event programming facility.
(a) JavaScript (b) C++ (c) Excel (d) Word
Ans. : (a)

5. ..... is platform independent scripting language. (March
(a) MSWord (b) JavaScript (c) PHP (d) C++
Ans. : (b)

6. ..... is knows as Universal client side scripting language.
(a) Word (b) Powerpoint (c) Tally (d) JavaScript
Ans. : (d)

7. In ..... scripting the script resides on the client computer.
(a) Client Side (b) Server Side (c) My Side (d) Website
Ans. : (a)

8. In ..... scripting the script resides on the web server.
(a) Client side (b) My side (c) Server Side (d) Website
Ans. : (c)

9. ..... scripting does not need any server interaction.
(a) Server Side (b) Website (c) My side (d) Client Side
Ans. : (d)

10. ..... scripting communicates to the server.
(a) Client Side (b) Software Side
(c) Server Side (d) Website
Ans. : (c)

11. ..... scripts are used for validation purpose. (March
(a) Client Side (b) Server Side (c) Website (d) Web Browser
Ans. : (a)

12. ..... scripts requires web browser as an interface.
(a) Server Side (b) Webpage (c) Website (d) Client Side
Ans. : (d)

13. ..... scripts requires web server software to execute.
(a) Client Side (b) Server Side (c) Peer (d) Home
Ans. : (b)

14. ..... is a JavaScript based open source frontend web framework developed for single page application.
(a) Vue JS (b) Response (c) Angular JS (d) React
Ans. : (c)

15. ..... is JavaScript based framework for building interactive user interface.
(a) Vue JS (b) Response (c) React (d) Uue JS
Ans. : (a)

16. ..... consists of JavaScript libraries for building UI for single page application and mobile application..
(a) Angular JS (b) Vue JS (c) Respond (d) React
Ans. : (d)

17. Functions in JavaScript are declared using ..... Keyword.
(a) Function (b) Object (c) Method (d) var
Ans. : (a)

3.2 Switch case and Looping Structures

18. JavaScript has a built-in multiway decision statement known as .....
(a) Condition (b) Switch (c) Break (d) Continue (March 2022)
Ans. : (b)

19. ..... refers to the execution of statement or a group of statements of code for a fixed number of times. (March 2022)
(a) Statement (b) Function (c) Iteration (d) Method
Ans. : (c)

20. ..... loop combines initialization, condition and loop iteration in single statement.
(a) for (b) Break (c) Switch (d) Continue
Ans. : (a)

21. ..... means increment or decrement value of a running variable.
(a) Switch (b) Method (c) Object (d) Iteration (March 2022)
Ans. : (d)

22. ..... statement is used to jump out of loop. (March 2022)
(a) Respond (b) Continue (c) Break (d) React
Ans. : (c)

23. What it is necessary to skip statement block and take the control at the beginning for next iteration ..... statement is used. (March 2022)
(a) Break (b) Continue (c) React (d) Response
Ans. : (b)

3.3 Objects in JavaScript

24. The ..... keyword is used to create new object in JavaScript.
(a) Next (b) Wend (c) Loop (d) New (March 2022)
Ans. : (d)

25. An ..... can group data together with functions needed to manip... it.
(a) Method (b) Function (c) Object (d) Response
Ans. : (c)

26. All tangible things are known as .....
(a) Method (b) Objects (c) Function (d) Variable
Ans. : (b)

27. DOM stands for ..... (March 2...
(a) Document Object Model (b) Document One Model
(c) Design One Model (d) Document Object Manage
Ans. : (a)

28. The way in which HTML document content is accessed and modified is calle... .....
(a) Design Object Model (b) Develop Object Model
(c) Do Object Model (d) Document Object Model
Ans. : (d)

29. ..... property of DOM object returns the <head> element of ... document.
(a) head (b) body (c) title (d) link
Ans. : (a)

30. ..... property of DOM object sets or returns title of the document.
(a) head (b) body (c) source (d) Title
Ans. : (d)

31. ..... property of DOM object returns URL of the HTML docume...
(a) SRC (b) HREF (c) URL (d) LINK
Ans. : (c)

32. ..... method of DOM object writes HTML expressions or JavaScr... code to a document. (March 2...
(a) write() (b) read() (c) close() (d) ReadOnly()
Ans. : (a)

33. ..... method of DOM object adds a new line character after ea... statement.
(a) close() (b) writeln() (c) read() (d) readOnly()
Ans. : (b)

34. Using ..... method id property is used to find an element.
(a) write (b) URL (c) writeln (d) getElementById()
Ans. : (d)

35. The ..... property is useful for getting html element and changing its content.
(a) write (b) URL (c) innerHTML (d) writeln
Ans. : (c)

36. ..... object is parent of all other objects. (March 2022)
(a) window (b) math (c) string (d) number
Ans. : (a)

37. ..... object represents an open window in a browser. (March 2022)
(a) Math (b) Array (c) String (d) Window
Ans. : (d)

38. ..... property of window object sets or returns the name of the window.
(a) location (b) name (c) document (d) status
Ans. : (b)

39. ..... property of window object returns the location object for the window.
(a) Window (b) Array (c) Math (d) Location
Ans. : (d)

40. ..... property of window object returns the location object for the window.
(a) name (b) location (c) status (d) closed
Ans. : (b)

41. ..... property of window object sets or returns the text in the status bar of a window.
(a) name (b) location (c) status (d) closed
Ans. : (c)

42. ..... property of window object returns a Boolean valve indicating whether a window has been closed or not. (March 2022)
(a) closed (b) status (c) document (d) name
Ans. : (a)

43. ..... method of window object display the alert box containing message with OK button.
(a) open() (b) close() (c) alert() (d) prompt()
Ans. : (c)

44. ..... method of window object displays the confirm dialog box containing message with ok and cancel button.
(a) blur() (b) confirm() (c) print() (d) focus()
Ans. : (b)

45. ..... method of window object displays a dialog box to get input from ... user.
(a) alert() (b) open() (c) confirm() (d) prompt() (March ...)
Ans. : (d)

46. ..... method of window object opens the new window. (March ...)
(a) open() (b) close() (c) focus() (d) blur()
Ans. : (a)

47. ..... method of window object closes the current window.
(a) open() (b) close() (c) blur() (d) alert()
Ans. : (b)

48. ..... method of window object removes focus from the current window.
(a) print() (b) open() (c) prompt() (d) blur()
Ans. : (d)

49. ..... method of window object sets focus to the current window.
(a) focus() (b) open() (c) blur() (d) close()
Ans. : (a)

50. ..... method of window object prints the content of current window.
(a) alert() (b) setTimeout()
(c) print() (d) open() (March ...)
Ans. : (c)

51. ..... method of window object calls a function or evaluates an expression after a specified number of milliseconds.
(a) open() (b) setTimeout()
(c) blur() (d) focus()
Ans. : (b)

3.4 JavaScript Events

52. ..... are the actions done by users or an application that occurs on the webpage .
(a) Events (b) Methods (c) Object (d) Functions
Ans. : (a)

53. ..... event occurs when user leaves or looses focus of an element. (March ...)
(a) onchange (b) onblur (c) onfocus (d) onload
Ans. : (b)

54. ..... event occurs when an element gets focus .
(a) onload (b) onblur (c) onfocus (d) onchange
Ans. : (c)

55. ..... event occurs when user changes content of an element or selects drop down value .
(a) onblur (b) onfocus (c) onload (d) onchange
Ans. : (d)

56. ..... event occurs when user selects some text of an element . (March 2022)
(a) onselect (b) onblur (c) onchange (d) onsubmit
Ans. : (a)

57. ..... event occurs when user clicks submit button .
(a) onblur (b) onchange (c) onfocus (d) onsubmit
Ans. : (d)

58. ..... event occurs when user clicks reset button . (March 2022)
(a) onload (b) onreset (c) onchange (d) onsubmit
Ans. : (b)

59. ..... event occurs when page/image has been loaded .
(a) onblur (b) onsubmit (c) onload (d) onunload
Ans. : (c)

60. ..... event occurs when document page has been unloaded or closes .
(a) onsubmit (b) onunload (c) onselect (d) onload (March 2022)
Ans. : (b)

3.5 JavaScript built-in Objects

61. ..... object helps us to work with number.
(a) Math (b) Number (c) Array (d) Date
Ans. : (b)

62. ..... property of Number object returns the largest minimum value.
(a) MAX_VALUE (b) NaN
(c) MIN_VALUE (d) Fixed (March 2022)
Ans. : (c)

63. ..... property of number object represents not a number value.
(a) NaN (b) Value (c) Fixed (d) Index (March 2022)
Ans. : (a)

64. ..... Method of Number object determines whether the given value is a integer.
(a) parseFloat() (b) isFixed() (c) isInteger() (d) parseInt()
Ans. : (c)

65. ..... method of number object determines whether the given value is floating point number. (March 2022)
(a) parseFloat() (b) parseInt() (c) isFixed() (d) NaN
Ans. : (a)

66. ..... method of number object converts the given string into a ... number.
(a) isInteger() (b) isFixed() (c) parseFloat() (d) parseInt()
Ans. : (d)

67. ..... method of Number object returns the string that represe... number with exact digits after a decimal point.
(a) isInteger() (b) isFixed() (c) parseInt() (d) parseFloat()
Ans. : (b)

68. How to declare string variable ? (March 20...
(a) var str – new str ( ); (b) var a = I. T.;
(c) a = str. Str; (d) var str = "Information Technology";
Ans. : (d)

69. What will be the output of following program <script type = text/Javascript>. mystr = "Hello" document · write (mystr·length), </script>
(a) 4 (b) 5 (c) 6 (d) 7
Ans. : (b)

70. ..... object is used to store or manipulate text .
(a) Math (b) Date (c) Number (d) String
Ans. : (d)

71. ..... property of string object returns number of characters in a string
(a) value (b) length (c) object (d) len
Ans. : (b)

72. ..... method of string object returns the character at the specif... position.
(a) indexOf() (b) substr() (c) charAt() (d) trim()
Ans. : (c)

73. ..... method of string object returns the index of the first occurrenc... of specified character in given string. (March 20...
(a) indexOf() (b) substr() (c) substring() (d) trim()
Ans. : (a)

74. ..... method of string object returns the index of last occurrence ... specified character in given string.
(a) substr() (b) charAt() (c) lastIndexOf() (d) trim()
Ans. : (c)

75. ..... method of string object removes white space from both sides of ... string.
(a) trim() (b) substr() (c) indexOf() (d) substring()
Ans. : (a)

76. ..... method of string object converts a string to lower case.
(a) toUpperCase() (b) substr()
(c) trim() (d) toLowerCase() (March 20...
Ans. : (d)

77. ..... method of string object converts a string to upper case.
(a) toLowerCase() (b) toUpperCase()
(c) trim() (d) substring()
Ans. : (b)

78. The built in ..... object includes mathematical constants and functions.
(a) String (b) Date (c) Math (d) Math
Ans. : (c)

79. ..... method of math object returns the absolute value of a number.
(a) abs() (b) cbrt() (c) min() (d) sqrt()
Ans. : (a)

80. ..... method of Math object returns the cube root of a number.
(a) sqrt() (b) max() (c) ceil() (d) cbrt()
Ans. : (d)

81. ..... method of math object returns the next integer greater than or equal to a given number.
(a) abs() (b) cbrt() (c) ceil() (d) floor()
Ans. : (c)

82. ..... method of math object returns the next integer less than or equal to a given number.
(a) abs() (b) floor() (c) min() (d) max()
Ans. : (b)

83. ..... method of math object returns the highest valued number in a list of numbers.
(a) min() (b) pow() (c) sqrt() (d) max()
Ans. : (d)

84. ..... method of math object returns the lowest valued number in a list of numbers.
(a) pow() (b) max() (c) min() (d) floor()
Ans. : (c)

85. ..... method of math object returns the base to the exponent power.
(a) pow() (b) floor() (c) abs() (d) cbrt()
Ans. : (a)

86. ..... method of math object returns a random number between 0 and 1.
(a) abs() (b) random() (c) pow() (d) sqrt()
Ans. : (b)

87. ..... method of math object returns the square root of a number.
(a) cbrt() (b) ceil() (c) sqrt() (d) abs()
Ans. : (c)

88. ..... object is used to create date and time values.
(a) math (b) Date (c) string (d) Number
Ans. : (b)

88. _____ method of date object returns the day of the month.
(a) getDay() (b) getHours() (c) getMonth() (d) getDate()
Ans. : (d)

89. _____ method of date object returns the day of the week.
(a) getDay() (b) getHours() (c) getMonth() (d) getDate()
Ans. : (a)

91. _____ method of date object returns the year.
(a) getDate() (b) setSeconds() (c) getFullYear() (d) setDate()
Ans. : (c)

92. _____ method of date object returns the hour.
(a) getHours() (b) now() (c) getTime() (d) setHours()
Ans. : (a)

93. _____ method of date object returns the minutes.
(a) getHours() (b) getMonth() (c) now() (d) getMinutes()
Ans. : (d)

94. _____ method of date object returns the month.
(a) getSeconds() (b) getMonth()
(c) setHours() (d) getTime()
Ans. : (b)

95. _____ method of date object returns the seconds.
(a) getSeconds() (b) getDate()
(c) getMonth() (d) now()
Ans. : (a)

96. _____ method of date object sets the day of the month.
(a) setHours() (b) setTime() (c) setDate() (d) setMinutes()
Ans. : (c)

97. _____ method of date object sets the full year.
(a) setTime() (b) setFullYear() (c) now() (d) getMonth()
Ans. : (b)

98. _____ method sets the hours of a date object.
(a) now() (b) getDay() (c) getMonth() (d) setHours()
Ans. : (d)

99. _____ method sets the minutes of a date object.
(a) setMinutes() (b) getDay() (c) now() (d) setHours()
Ans. : (a)

100. _____ method sets the month of a date object.
(a) setTime() (b) getDay() (c) setMonth() (d) getDate()
Ans. : (c)

101. _____ method sets the seconds of a date object.
(a) setHours() (b) setSeconds() (c) setMonth() (d) setTime()
Ans. : (b)

102. _____ method sets a date to a specified number of milliseconds after/before Jan 1, 1970.
(a) setTime() (b) setSeconds() (c) now() (d) setHours()
Ans. : (a)

103. An _____ is an object that can store a collection of items.
(a) Array (b) String (c) Number (d) Math
Ans. : (a)

104. _____ are used to hold more than one value at a time.
(a) Number (b) Math (c) Array (d) Date
Ans. : (c)

105. To create an array in JavaScript the correct method is _____.
(a) var d = new d [one, two, three] (b) var d = (one, two three)
(c) var arr = ["One", "Two", "Three"] (d) var d = (One), (Two), (Three)
Ans. : (c)

106. To access and set the items in any array _____ is referred.
(a) indexnumber (b) element
(c) array (d) object
Ans. : (a)

107. The index of the first element of an array is _____
(a) One (b) Two (c) Three (d) Zero
Ans. : (d)

108. _____ property of Array object represents the zero based index of the match in the string.
(a) index (b) length (c) NaN (d) Close
Ans. : (a)

109. _____ property of Array object reflect number of elements in array.
(a) index (b) length (c) NaN (d) Value
Ans. : (b)

110. _____ method of Array object joins two or more arrays, and returns a copy of the joined arrays.
(a) find() (b) concat() (c) indexOf() (d) pop()
Ans. : (b)

111. _____ method of Array object copies array elements within the array, to and from specific positions.
(a) find() (b) search() (c) copyWithin() (d) indexOf()
Ans. : (c)

112. .....method of Array object returns the value of the first element in an array t satisfies a test in testing.
(a) concat() (b) pop() (c) push() (d) find()
Ans. : (d)

113. .....method of Array object calls a function for each array element.
(a) concat() (b) forEach() (c) pop() (d) push()
Ans. : (b)

114. .....method of Array object search the array for an element and returns its posi
(a) concat() (b) forEach() (c) indexOf() (d) pop()
Ans. : (c)

115. .....method of Array object checks whether an object is an array.
(a) find() (b) pop() (c) forEach() (d) isArray()
Ans. : (d)

116. ..... method of Array object removes the last element of an array and returns t element.
(a) pop() (b) push() (c) find() (d) reverse()
Ans. : (a)

117. .....method of Array object add new elements to the end of an array, and retu the new length.
(a) pop() (b) push() (c) sort() (d) find()
Ans. : (b)

118. .....method of Array object reverses the order of the elements in an array.
(a) find() (b) isArray() (c) pop() (d) reverse()
Ans. : (d)

119. .....method of Array object sorts the elements of an array.
(a) concat() (b) sort() (c) pop() (d) find ()
Ans. : (b)

MCQ (Two Correct Answers)

3.1 Introduction

1. Features of JavaScript are .....
(a) It needs special software
(b) JavaScript is light weight scripting language
(c) Cannot create new function
(d) It is object oriented scripting
Ans. : (b), (d)

2. There are two types of Scripting .....
(a) Server side (b) My side (c) Browser side (d) Client side
Ans. : (a), (d)

3. Client Side Scripting .....
(a) These scripts are placed inside HTML document
(b) In this type the script resides on Client Computer.
(c) In this type, the script resides on web server
(d) To execute script it must be activated by client then it is executed on web server.
Ans. : (a), (b)

3.2 Switch case and Looping Structures

4. Types of loops in JavaScript are ..... March 2022
(a) for ....loop (b) while .....loop
(c) for ....next (d) while .....wend
Ans. : (a), (b)

5. Correct method of for loop are .....

(a)
```
for (i = 1; i< = 5; i++)
{
document . write(i);
}
```

(b)
```
for (i = 1, i! = 4);
{
document.write (i);
}
```

(c)
```
for (i = 1; i < = 5; i + +)
{
document.writeln(i);
}
```

(d)
```
[i – 1, ic = 5, i ++)
{
document.write (i),
}
```

Ans. : (a), (c)

3.3 Objects in JavaScript

6. JavaScript supports following types of objects .....
(a) Variables (b) built-in objects
(c) User defined objects (d) loops
Ans. : (b), (c)

7. Following are the built in objects in JavaScript .....
(a) Math (b) Time (c) Array (d) Month
Ans. : (a), (c)

8. Following are the properties of Document Object Model .....
(a) Src (b) head (c) URL (d) title
Ans. : (b), (c), (d)

9. Following are the methods of Document Object Model .....
(a) body (b) write() (c) URL (d) writeln()
Ans. : (b), (d)

10. Following are the properties of Window object ..... (March 20
(a) name (b) location (c) alert() (d) confirm()
Ans. : (a), (b)

11. Following are the methods of Window object .....
(a) open() (b) status (c) closed (d) close()
Ans. : (a), (d)

12. confirm() method of window object displays confirm dialog box containing messa with .....and .....button. (March 20
(a) GO (b) OK (c) Cancel (d) Click
Ans. : (b), (c)

3.4 JavaScript Events

13. Following are the event handlers in JavaScript .....
(a) onsubmit (b) write() (c) onchange (d) writeln()
Ans. : (a), (c)

3.5 JavaScript built-in Objects

14. In JavaScript following methods of String object returns the characters fr specified.
(a) substring() (b) indexOf() (c) substr() (d) lastIndexOf()
Ans. : (a), (c)

15. Following properties of Number Object returns the largest minimum and maximu value .....
(a) MIN_VALUE (b) MAX_VALUE
(c) toLowerCase() (d) toUpperCase()
Ans. : (a), (b)

16. Following are the properties of Number Object in Javascript ..... (March 20
(a) lowercase (b) Uppercase (c) NaN (d) MAX_VALUE
Ans. : (c), (d)

17. Following are the methods of Number of object in JavaScript .....
(a) NaN (b) parseInt() (c) isFixed() (d) toLowercase()
Ans. : (b), (c)

18. Following are the methods of string objects ..... (March 2022)
(a) chartAt() (b) width (c) length (d) trim()
Ans. : (a), (d)

19. To find the highest and lowest valued number in a list of numbers which Math Object methods are used in JavaScript .....
(a) ceil() (b) sqrt() (c) max() (d) min()
Ans. : (c), (d)

20. Following are the methods of Math objects .....
(a) substr() (b) abs() (c) trim() (d) random()
Ans. : (b), (d)

21. Different ways to create new Date object are .....
(a) var d = new Date(); (b) var d = Date()
(c) var d = month() (d) var d = new Date (DateString);
Ans. : (a), (d)

22. Following are the methods of Date object .....
(a) setDateTime() (b) getDateTime()
(c) getTime() (d) getDay()
Ans. : (c), (d)

23. Following methods are used to create Array variable in JavaScript .....
(a) var a = "One, two",
(b) var a = ["One", "Two", "Three"];
(c) array a = ("One", "Two"),
(d) var a = new Array ("One", "Two", "Three");
Ans. : (b), (d)

24. Following are the properties of Array object in JavaScript .....
(a) index (b) isFixed (c) concat (d) length
Ans. : (a), (d)

25. Following are the methods of Array object in JavaScript .....
(a) index (b) length (c) forEach() (d) isArray()
Ans. : (c), (d)

26. Following methods of Date object returns the number of miliseconds
(a) getTime() (b) getDate() (c) getTime() (d) now()
Ans. : (c), (d)

27. The Date object is used to create .....and .....values.
(a) date (b) string (c) Time (d) Number
Ans. : (a), (c)

28. The built-in Math object includes mathematical .....and .....
(a) constants (b) functions (c) objects (d) values
Ans. : (a), (b)

MCQ (Three Correct Answers)

3.1 Introduction

1. Features of JavaScript are .....
   (a) Need of special software (b) It can handle date and time effectively
   (c) It cannot create new functions (d) It can not create new functions
   (e) It supports event based programming
   (f) It is platform dependent scripting language

Ans. : (b), (c), (e)

2. Server Side Scripting .....
   (a) It is used as front end (b) Does not need any server interactions
   (c) Involves languages such as HTML5, JavaScript etc.
   (d) It is used as back end (e) Special software is required to execute
   (f) Script resides on web server

Ans. : (d), (e), (f)

3. Popular Framework /libraries .....
   (a) HTML (b) Angular JS (c) Network Frame
   (d) Vue Js (e) React (f) Switch Case

Ans. : (b), (d), (e)

3.2 Switch case and Looping Structures

4. Java Script looping statements are .....
   (a) for ....loop (b) switch case
   (c) while ....loop (d) if
   (e) if else if (f) Do while ....loop

Ans. : (a), (c), (f)

3.3 Objects in JavaScript

5. JavaScript built in objects are .....
   (a) Math (b) String (c) Number
   (d) Time (e) Month (f) Year

Ans. : (a), (b), (c)

6. Properties of Document Object Model are .....
   (a) write (b) writeln (c) title
   (d) URL (e) writeln () (f) head

Ans. : (c), (d, (f)

7. Methods of Document Object Model are .....
   (a) write() (b) getElementById() (c) URL
   (d) head (e) SRC (f) writeln()

Ans. : (a), (b), (f)

8. Methods of Window Object are .....
   (a) Name (b) blur() (c) location (d) focus()
   (e) closed (f) setTimeOut()

Ans. : (b), (d), (f)

9. Select three correct properties of Window Object .....
   (a) close (b) name (c) print (d) status
   (e) open (f) closed

Ans. : (b), (d), (f)

3.4 JavaScript Events

10. Select three correct event handlers in JavaScript .....
    (a) close (b) open (c) onsubmit (d) onchange
    (e) onunload (f) write

Ans. : (c), (d),(e)

3.5 JavaScript built-in Objects

11. Properties of Number objects are .....
    (a) length (b) onblur
    (c) onfocus (d) MIN_VALUE
    (e) MAX_VALUE (f) NaN

Ans. : (d), (e), (f)

12. Select three correct methods of Number object .....
    (a) NaN (b) parseFloat() (c) isFixed()
    (d) length (e) index (f) isInteger()

Ans. : (b), (c), (f)

13. Methods of string object are .....
    (a) substr() (b) length (c) toLowerCase()
    (d) abs (e) indexOf() (f) ceil

Ans. : (a), (c), (e)

14. Methods of Math object are .....
    (a) sqrt() (b) substr() (c) indexOf() (d) random ( )
    (e) close() (f) floor ( )

Ans. : (a), (d), (f)

15. Methods of Date objects are .....
    (a) trim() (b) getDay() (c) getHours() (d) indexOf()
    (e) getTime() (f) charAt()

Ans. : (b), (c), (e)

16. Different ways to create new date object are .....

(a) var d = new Date();
(b) var d = Date()
(c) var d = date new()
(d) var d = new Date(DateString);
(e) var d = new Date (date String)
(f) var d = new Date(miliseconds);

Ans. : (a), (d), (f)

17. Methods of Array object are .....

(a) sort()
(b) indexof()
(c) isFixed()
(d) forEach()
(e) isInteger()
(f) parseInt()

Ans. : (a), (b), (d)

Match the Following

3.2 Switch Case and Looping Structures

(I) (March 2022)

	A		B
(1)	Switch case	(a)	Combines initialization, condition and loop iteration
(2)	For loop	(b)	Skip statement block and take the control at the beginning for next iteration
(3)	Break	(c)	Server side script
(4)	Continue	(d)	Used to jump out of loop
		(e)	Decision statement

Ans. : (1) – (e), (2) – (a), (3) – (d), (4) – (b)

3.3 Objects in JavaScript

(I)

	A		B
(1)	Head	(a)	Returns title of the document
(2)	Title	(b)	Returns <body> <img> elements
(3)	URL	(c)	Returns URL of document
(4)	Body, img	(d)	Returns <head> element of the document
		(e)	Writes expression

Ans. : (1) – (d), (2) – (a), (3) – (c), (4) – (b)

(II)

	A		B
(1)	DOM	(a)	Defines logical structure of document
(2)	write()	(b)	Client Side Script
(3)	writeln()	(c)	In which id property is used to find element
(4)	getElementById()	(d)	Writes JavaScript Code
		(e)	Writes JavaScript code to a document by adding new line after each statement

Ans. : (1) – (a), (2) – (d), (3) – (e), (4) – (c)

(III)

	A		B
(1)	Location	(a)	Sets name of the window
(2)	Document	(b)	Returns Boolean value indicating whether a window
(3)	Status	(c)	Returns document object of window
(4)	Closed	(d)	Returns the text in the status bar of a window
(5)	Name	(e)	Returns the location object for the window

Ans. : (1) – (e), (2) – (c), (3) – (d), (4) – (b), (5) – (a)

(IV)

	A		B
(1)	Window object (March 2022)	(a)	Displays dialog box containing message with ok and cancel button
(2)	alert()	(b)	Opens the new window
(3)	confirm()	(c)	Displays dialog box to get input from the user
(4)	prompt()	(d)	Parent object of all other objects
(5)	open()	(e)	Displays box with ok button

Ans. : (1) – (d), (2) – (e), (3) – (a), (4) – (c), (5) – (b)

(V)

	A		B
(1)	close()	(a)	Removes focus from the current window
(2)	blur()	(b)	Prints the content of current window
(3)	focus()	(c)	Closes the current window
(4)	print()	(d)	Calls a function after specified number of miliseconds
(5)	setTimeOut()	(e)	Sets focus to the current window

Ans. : (1) – (c), (2) – (a), (3) – (e), (4) – (b), (5) – (d)

3.4 JavaScript Events

(I)

	A		B
(1)	On blur	(a)	Occurs when user changes content of an element
(2)	On focus	(b)	Occurs when user selects some text of an element
(3)	On change	(c)	Occurs when an element get focus
(4)	On select	(d)	Occurs when user leaves field or losses focus of an element
		(e)	Closes window

Ans. : (1) – (d), (2) – (c), (3) – (a), (4) – (b)

(II)

	A		B
(1)	Onsubmit	(a)	Occurs when user clicks on reset button
(2)	Onreset	(b)	Prints the contents of the window
(3)	Onload	(c)	Occurs when document page has been unloaded or closes
(4)	Onunload	(d)	Occurs when user clicks submit button
		(e)	Occurs when page image has been loaded

Ans. : (1) – (d), (2) – (a), (3) – (e), (4) – (c)

3.5 JavaScript built-in Objects

(I)

	A		B
(1)	MIN_VALUE	(a)	Represent not a Number value
(2)	MAX_VALUE	(b)	Determines whether the given value is a Integer
(3)	NaN	(c)	Returns the largest minimum value
(4)	isInteger( )	(d)	Converts string into Floating point number
		(e)	Returns the largest maximum value

Ans. : (1) – (c), (2) – (e), (3) – (a), (4) – (b)

(II)

	A		B
(1)	parseFloat()	(a)	Used to create date and time values
(2)	parseInt()	(b)	Returns the string that represents a number with exact digits after a decimal point
(3)	isFixed()	(c)	Converts a given string into a floating point number
(4)	Date object	(d)	Converts the given string into integer number

Ans. : (1) – (c), (2) – (d), (3) – (b), (4) – (a)

(III)

	A		B
(1)	Length	(a)	Returns the character at the specified position
(2)	chartAt()	(b)	Returns the index of last occurrence of specified character in given string
(3)	indexOf()	(c)	Returns the characters you specified
(4)	lastIndexOf()	(d)	Returns the index of the first occurrence of the character
(5)	substr()	(e)	Returns a number of character in a string

Ans. : (1) – (e), (2) – (a), (3) – (d), (4) – (b), (5) – (c)

(IV) **(March 2022)**

	A		B
(1)	substring()	(a)	Returns the characters you specified
(2)	trim()	(b)	Converts a string into lower case
(3)	toLowerCase()	(c)	Returns length of string
(4)	toUpperCase()	(d)	Converts the string into Upper case
		(e)	Removes white spaces from both sides of a string

Ans. : (1) – (a), (2) – (e), (3) – (b), (4) – (d)

(V)

	A		B
(1)	abs()	(a)	Returns next integer greater than or equal to a given number
(2)	cbrt()	(b)	Returns the highest valued number in a list of Numbers
(3)	ceil()	(c)	Returns the absolute value of a number
(4)	floor()	(d)	Returns the cube root of a number
(5)	max()	(e)	Returns next integer less than or equal to a given number

Ans. : (1) – (c), (2) – (d), (3) – (a), (4) – (e), (5) – (b)

(VI)

	A		B
(1)	min()	(a)	Returns the base to the exponent power
(2)	pow()	(b)	Returns the square root of a number
(3)	random()	(c)	Returns the cube root of a number
(4)	sqrt()	(d)	Returns a random number between 0 and 1
		(e)	Returns the lowest valued number in a list of number

Ans. : (1) – (e), (2) – (a), (3) – (d), (4) – (b)

(VII)

	A		B
(1)	getDate()	(a)	Returns the year
(2)	getDay()	(b)	Returns the Time
(3)	getFullYear()	(c)	Returns the day of the week
(4)	getHours()	(d)	Returns the hour
		(e)	Returns the day of the month

Ans. : (1) – (e), (2) – (c), (3) – (a), (4) – (d)

(VIII)

	A		B
(1)	getMinutes()	(a)	Returns the number of miliseconds
(2)	getMonth()	(b)	Returns the minutes
(3)	getSeconds()	(c)	Sets the month
(4)	getTime()	(d)	Returns the month
		(e)	Returns the seconds

Ans. : (1) – (b), (2) – (d), (3) – (e), (4) – (a)

(IX)

	A		B
(1)	setHours()	(a)	Sets the days of the month of a date object
(2)	now()	(b)	Returns the year
(3)	setDate()	(c)	Set the full year of a date object
(4)	setFullYear()	(d)	Returns the number of miliseconds
		(e)	Sets the hours of a date object

Ans. : (1) – (e), (2) – (d), (3) – (a), (4) – (c)

(X)

	A		B
(1)	setMinutes()	(a)	Sets the month of a date object
(2)	setMonth()	(b)	Set a date to a specified number of miliseconds
(3)	setSeconds()	(c)	Returns day of the week
(4)	setTime()	(d)	Set the minutes of a date object
		(e)	Sets the seconds of a date object

Ans. : (1) – (d), (2) – (a), (3) – (e), (4) – (b)

(XI)

	A		B
(1)	concat()	(a)	Returns the value of the first element in an array that satisfies a test
(2)	copyWithin()	(b)	Search the array for an element and return its position
(3)	find()	(c)	Calls a function for each array element
(4)	forEach()	(d)	Joint two or more arrays and returns a copy of joined array
(5)	indexOf()	(e)	Copies array elements within the array, to and from specified positions.

Ans. : (1) – (d), (2) – (e), (3) – (a), (4) – (c), (5) – (b)

(XII)

	A		B
(1)	isArray()	(a)	Adds a new elements to the end of an array, and returns the new length
(2)	pop()	(b)	Sorts the elements of an array
(3)	Push()	(c)	Reverse the order of the elements in an array
(4)	reverse()	(d)	Checks whether an object is an array
(5)	sort( )	(e)	Removes the last element of an array, and returns the element

Ans. : (1) – (d), (2) – (e), (3) – (a), (4) – (c), (5) – (b)

Answer in Brief

3.1 Introduction

1. Explain the features of JavaScript.

Ans. :

i) JavaScript is light weight scripting language.

ii) No special software is required to run JavaScript programs.

iii) JavaScript is object oriented scripting language.

vi) JavaScript helps the browser to perform input validation without wasting time by the server access.

v) JavaScript can hadle date and time very effectively.

vi) JavaScript has ability to create new functions within scripts.

2. Explain difference between Client Side and Server Side Scripting.

Ans. :

	Server Side Scripting		Client Side Scripting
i)	Server side scripting is used at the backend.	i)	Client side scripting is used at the frontend.
ii)	Server side script communicates to the server.	ii)	Client side script does not need any server interaction.
iii)	Server side scripting involves languages such as ASP.NET, PHP, Ruby C#.	iii)	Client side scripting involves languages such as HTML5, JavaScript etc.
iv)	Server side scripting is useful in customizing the web pages and implements the dynamic changes in the web site.	iv)	Client side scripts are used for validation purpose and minimize the load to the server.
v)	Web server software is required to execute server side script.	v)	Client side scripts requires web browser as an interface.

3.2 Switch case and Looping Structures

3. Explain Switch case.

Ans. :

JavaScript has decision control statement known as switch. The switch statement test the value of given expression against the list of case values and when the match is found a block of statement associated with that case is executed.

Syntax of switch case is :

```
switch(expression)
{
 case x:
  // code block
  break;
 case y:
  // code block
  break;
 default:
  // code block
}
```


4. Explain For loop with syntax.

Ans. :

The JavaScript for loop iterates the elements for the fixed number of times. It should be used if number of iteration is known. The syntax of for loop is given below.

```
for (initialization; condition; increment)
{
  code to be executed
}
```

5. Explain while loop with syntax.

Ans. :

The JavaScript while loop *iterates the elements for the infinite number of times*. It should be used if number of iteration is not known. The syntax of while loop is given below.

```
while (condition)
{
  code to be executed
}
```

6. Explain Do while loop with syntax.

Ans. :

The JavaScript do while loop iterates the elements for the infinite number of times like while loop. But, code is executed at least once whether condition is true or false. The syntax of do while loop is given below.

```
do{
  code to be executed;
} while (condition);
```

7. Explain Break and Continue statement in JavaScript.

Ans. :

Break Statement

i) Break statement is used to jump out of loop.

ii) It is used to make early exit from a loop.

iii) When keyword break is encountered inside the loop, control automatically passes to the next statement after the loop.

Continue Statement

i) Sometimes in looping it may be necessary to skip statement block and take the control at the beginning for next iteration.

ii) This is done by using continue statement.

3.3 Objects in JavaScript

8. Explain DOM (Document Object Model).

Ans.

i) JavaScript is an object based scripting language.

ii) A JavaScript object is an entity having properties and methods.

iii) Properties and methods of object's are accessed with "." operator. JavaScript supports two types of objects built-in objects and user defined objects

iv) DOM(Document Object Model) :

v) When a web page is loaded, the browser creates Document Object Model of the page.

vi) The DOM is a W3C (World Wide Web Consortium) standard.

vii) "The W3C Document Object Model (DOM) is a platform and language-neutral interface that allows programs and scripts to dynamically access and update the content, structure, and style of a document."

9. Explain the predefined methods and properties for DOM object.

Ans. :

Following are the predefined methods and properties for DOM object :

Property	Description
Head	Returns the <head> element of the document
Title	Sets or returns title of the document.
URL	Returns full URL of the HTML document.
body, img	Returns <body>, <img> elements respectively.
write()	Writes HTML expressions or JavaScript code to a document.
writeln()	Same as write(), but adds a newline character after each statement.
getElementById()	There are many ways of accessing form elements, of which the easiest is by getElementById() method. In which id property is used to find an element.

10. Explain Window Object.

Ans. :

i) Window object is the parent object of all other objects.

ii) It is at the top of object hierarchy.

iii) It represents an open window in a browser.

iv) An object of a window is created automatically by the browser.

11. Explain methods and properties of Window object.

Ans. : Following are methods and properties of Window object :-

Property	Description
name	Sets or returns the name of a window.
location	Returns the Location object for the window.
document	Returns the Document object for the window.
status	Sets or returns the text in the status bar of a window.
closed	Returns a Boolean value indicating whether a window has been closed or not.
Method	Description
alert()	Displays the alert box containing message with ok button.
confirm()	Displays the confirm dialog box containing message with ok and cancel button.
prompt()	Displays a dialog box to get input from the user.
open()	Opens the new window.
close()	Closes the current window.
blur()	Removes focus from the current window.
focus()	Sets focus to the current window.
print()	Prints the content of current window.
setTimeout()	Calls a function or evaluates an expression after a specified number of milliseconds.

3.4 JavaScript Events

12. Explain JavaScript Events.

Ans. : Events are the actions done by user that occurs on the web page.

13. Explain EventHandler in JavaScript.

Ans. : Following are some of the events used with form objects.

Event handler	Description
onblur	It occurs when user leaves field or losses focus of an element.
onfocus	It occurs when an element gets focus.
onchange	It occurs when user changes content of an element or selects dropdown value. E.g. for textbox, password, select box, textarea etc.
onselect	It occurs when user selects some text of an element.
onsubmit	It occurs when user clicks submit button.
onreset	It occurs when user clicks reset button.
onload	It occurs when page/image has been loaded.
onunload	It occurs when document/page has been unloaded or closes.

3.5 JavaScript built-in Objects

14. Explain JavaScript built in Object.

Ans. :

i) JavaScript has several built-in objects.

ii) These objects provide different properties and methods that are useful while creating web pages.

15. Explain String Object in JavaScript ?

Ans. : String is used to store characters of text with single or double quotes. It is used to store and manipulate text.

Property	Description
Length	Returns the number of characters in a string
Method	**Description**
charAt()	Returns the character at the specified position (in Number).
indexOf()	Returns the index of the first occurence of specified character in given string, or -1 if it never occurs, so with that index you can determine if the string contains the specified character.
lastIndexOf()	Returns the index of the last occurrence of specified character in given string.
substr()	Returns the characters you specified: (14,7) returns 7 characters, from the 14th character.
substring()	Returns the characters you specified: (7,14) returns all characters between the 7th and the 14th.
trim()	The trim() method removes whitespace from both sides of a String
toLowerCase()	Converts a string to lower case
toUpperCase()	Converts a string to upper case

16. Explain Math Object in JavaScript.

Ans. : The math object includes mathematical constants and functions. Following table consists list of Math objects.

Method	Description
abs()	Returns the absolute value of a number.
cbrt(x)	Returns the cube root of a number.
ceil(x)	Returns the next integer greater than or equal to a given number (rounding up).

Method	Description
floor(x)	Returns the next integer less than or equal to a given number (rounding down).
max(x, y, ...)	Returns the highest-valued number in a list of numbers.
min(x, y, ...)	Returns the lowest-valued number in a list of numbers.
pow(x, y)	Returns the base to the exponent power, that is, xy.
random(x)	Returns a random number between 0 and 1 (including 0, but not 1).
sqrt(x)	Returns the square root of a number.

17. Explain Date Object in JavaScript.

Ans. : Date object is used to create date and time values. It is created using **new** keyword.

Method	Description
getDate()	Returns the day of the month (from 1-31)
getDay()	Returns the day of the week (from 0-6)
getFullYear()	Returns the year (four digits).
getHours()	Returns the hour (from 0-23).
getMinutes()	Returns the minutes (from 0-59).
getMonth()	Returns the month (from 0-11).
getSeconds()	Returns the seconds (from 0-59).
getTime()	Returns the number of milliseconds since midnight Jan 1, 1970.
now()	Returns the number of milliseconds since midnight Jan 1, 1970.
setDate()	Sets the day of the month of a date object.
setFullYear()	Sets the full year of a date object.
setHours()	Sets the hours of a date object.
setMinutes()	Set the minutes of a date object.
setMonth()	Sets the month of a date object.
setSeconds()	Sets the seconds of a date object.
setTime()	Sets a date to a specified number of milliseconds after/before Jan 1, 1970.

18. Explain Number Object in JavaScript.

Ans. : It helps in working with numbers. Property and Methods of number object are :

Property	Description
MIN_VALUE	Returns the largest minimum value.
MAX_VALUE	Returns the largest maximum value.

Property	Description
NaN	It represents 'Not a Number' value.
Method	Description
isInteger()	It determines whether the given value is a Integer
parseFloat()	It converts the given string into a floating point number.
parseInt()	It converts the given string into a integer number.
isFixed()	It returns the string that represents a number with exact digits after a decimal point.

19. Explain Array Object in JavaScript.

Ans. : An array is an object that can store a collection of items. Arrays are used to store multiple values in single variable. Array is a special variable which can hold more than one values at a time. Property and Methods of array object are :

Property	Description
Index	The property represents the zero-based index of the match in the string
Length	Reflect number of elements in array.
Method	**Description**
concat()	Joins two or more arrays, and returns a copy of the joined arrays
Method	**Description**
copyWithin()	Copies array elements within the array, to and from specied positions.
find()	Returns the value of the first element in an array that satisfies a test in testing.
forEach()	Calls a function for each array element.
indexOf()	Search the array for an element and returns its position.
isArray()	Checks whether an object is an array.
pop()	Removes the last element of an array, and returns that element.
push()	Adds new elements to the end of an array, and returns the new length.
reverse()	Reverses the order of the elements in an array.
sort()	Sorts the elements of an array.

Extra JavaScript Examples

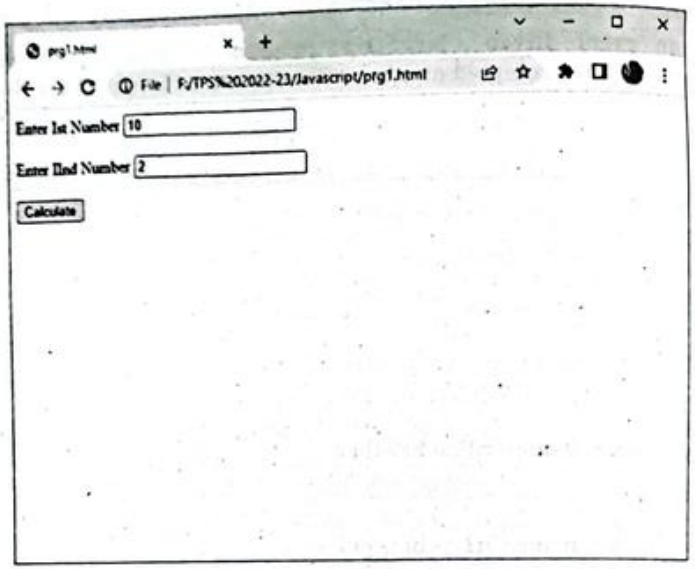
1. Write an event driven JavaScript program to display multiplication and subtraction of two accepted number when the mouse is moved over the button.

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Ist Number
<input type="number" name="n1"><br><br>
Enter IInd Number
<input type="number" name="n2"><br><br>
<input type="button" name="b1" value="Calculate" onMouseMove="chk()">
</form>
</body>
<script type="text/javascript">
function chk()
{
var a,b,mul,sub;
a=parseInt(f1.n1.value);
b=parseInt(f1.n2.value);
mul=a*b;
sub=a-b;
alert("Multiplication is "+mul);
alert("Substraction is "+sub);
}
</script>
</html>
```

Output :



2. Write an event driven JavaScript program to display whether accepted number is even or odd.

Ans. :

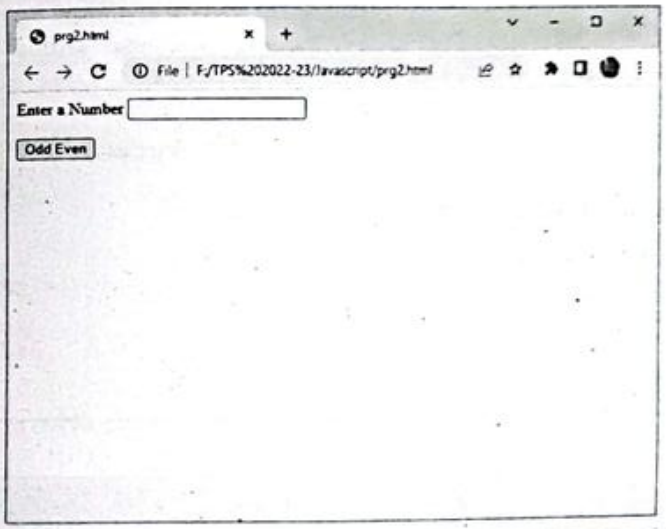
Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter a Number
<input type="number" name="n1"><br><br>
<input type="button" name="b1" value="Odd Even" onClick="even()">
</form>
</body>
<script type="text/javascript">
function even()
{
var a;
a=parseInt(f1.n1.value);
if(a%2==0)
```



```
alert("Number is Even number");
else
alert("Number is Odd number");
}
</script>
</html>
```

Output :



3. Write an event driven JavaScript program to display addition and subtraction of two accepted numbers on click of a button.

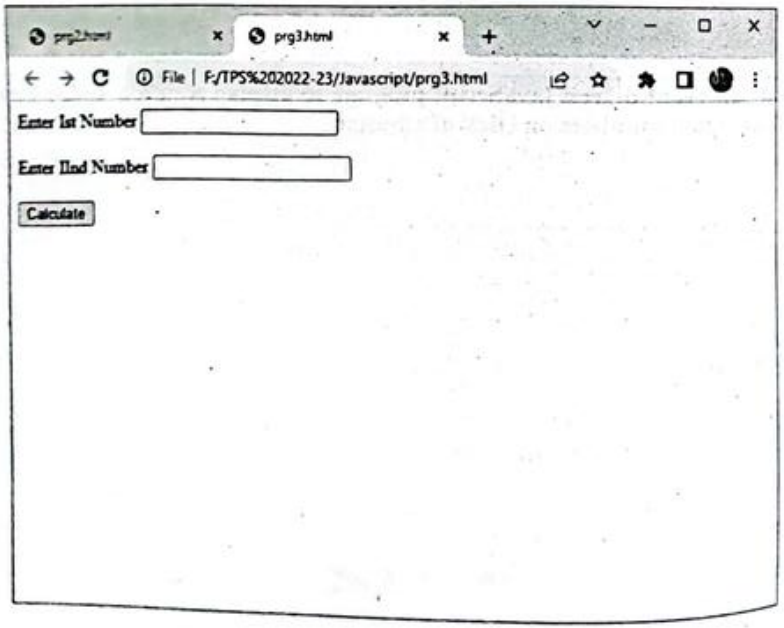
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Ist Number
<input type="number" name="n1"><br><br>
Enter IInd Number
<input type="number" name="n2"><br><br>
<input type="button" name="b1" value="Calculate" onClick="chk()">
```

```
</form>
</body>
<script type="text/javascript">
function chk()
{
var a,b,add,sub;
a=parseInt(f1.n1.value);
b=parseInt(f1.n2.value);
add=a+b;
sub=a-b;
alert("Addition is "+add);
alert("Substraction is "+sub);
}
</script>
</html>
```

Output :



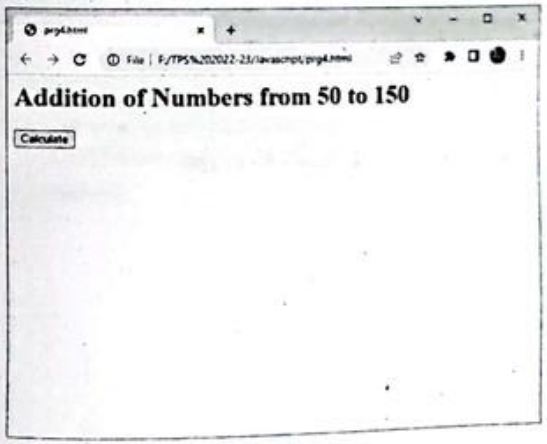
4. Write an event driven JavaScript program to display addition of numbers from 50 to 150 when the mouse is moved over a button.

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
<h1>Addition of Numbers from 50 to 150</h1>
<input type="button" name="b1" value="Calculate" onMouseOver="chk()">
</form>
</body>
<script type="text/javascript">
function chk()
{
var i,sum=0;
for(i=50;i<=150;i++)
{
sum=sum+i;
}
alert("Addition of Numbers from 50 to 150 is "+sum);
}
</script>
</html>
```

Output :



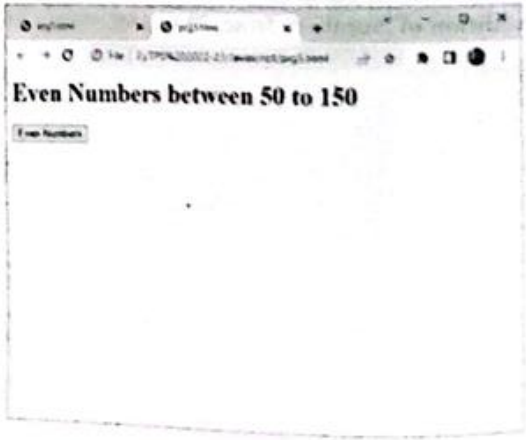
5. Write an event driven JavaScript program to display even numbers between 50 to 150.

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
<h1>Even Numbers between 50 to 150</h1>
<input type="button" name="b1" value="Even Numbers" onMouseOver="chk()">
</form>
</body>
<script type="text/javascript">
function chk()
{
var i;
for(i=50;i<=150;i=i+2)
{
document.write(i+"<br>");
}
}
</script>
</html>
```

Output :



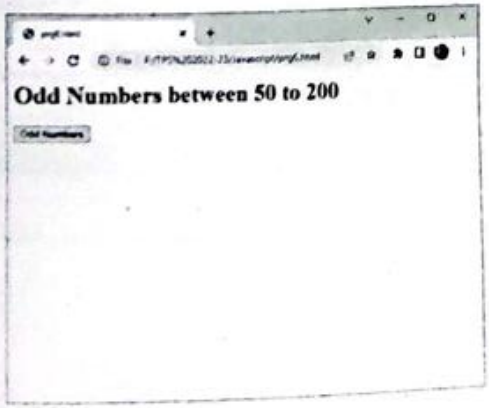
6. Write an event driven JavaScript program to display odd numbers between 50 to 200.

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
<h1>Odd Numbers between 50 to 200</h1>
<input type="button" name="b1" value="Odd Numbers" onClick="chk()">
</form>
</body>
<script type="text/javascript">
function chk()
{
var i;
for(i=51;i<=150;i=i+2)
{
document.write(i+"<br>");
}
}
</script>
</html>
```

Output :



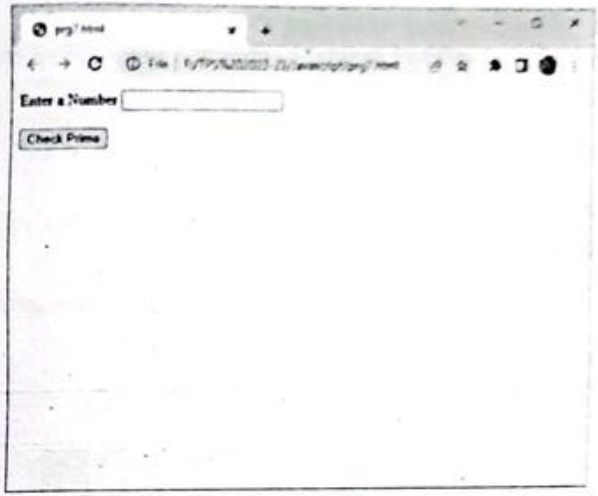
7. **Write an event driven JavaScript program to display whether the accepted number is prime or not.**

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter a Number
<input type="number" name="n1"><br><br>
<input type="button" name="b1" value="Check Prime" onClick="pr()">
</form>
</body>
<script type="text/javascript">
function pr()
{
var a,p=1,i;
a=parseInt(f1.n1.value);
for(i=2;i<a;i++)
{
if(a%i==0)
p=0;
}
if(p==1)
alert("Number is Prime Number");
else
alert("Number is not a Prime Number");
}
</script>
</html>
```


Output :



8. **Write an event driven JavaScript program to display number of vowels from the accepted string. Write an event driven JavaScript program to display number of characters from the accepted string.**

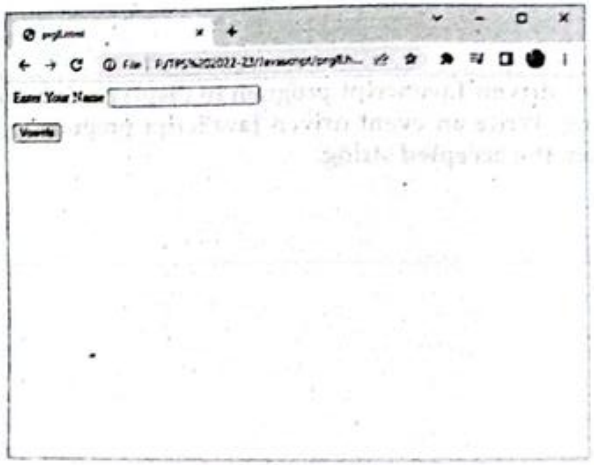
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Your Name
<input type="text" name="t1"><br><br>
<input type="button" name="b1" value="Vowels" onClick="vw()">
</form>
</body>
<script type="text/javascript">
function vw()
{
var s,i,ch,c;
c=0;
s=f1.t1.value;
for(i=0;i<=s.length;i++)
```

```
{
ch=s.charAt(i);
if(ch=="A" || ch=="a" || ch=="E" || ch=="e" || ch=="I" || ch=="i" || ch=="O" ||
ch=="o" || ch=="U" || ch=="u")
c++;
}
alert("Number of Vowels in string are "+c);
alert("Number of Characters in string are "+s.length);
}
</script>
</html>
```

Output :



9. Write an event driven JavaScript program to display cube of the accepted number when the mouse is moved over a button.

Ans. :

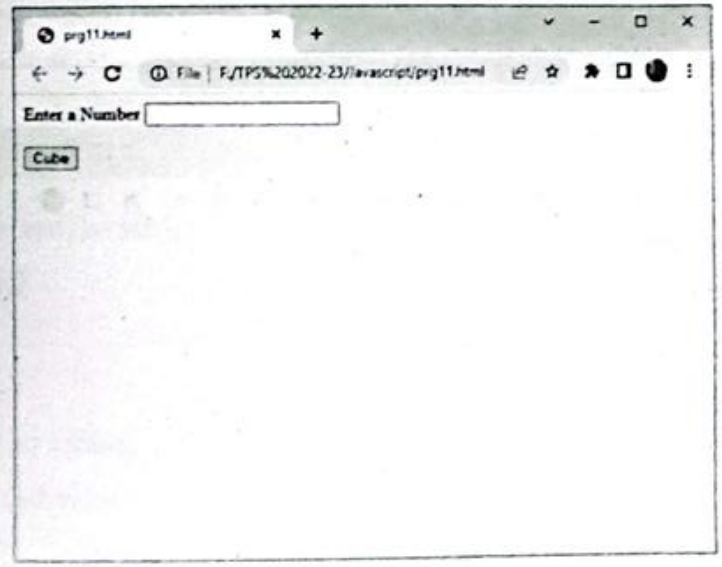
Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter a Number
<input type="number" name="n1"><br><br>
<input type="button" name="b1" value="Cube" onClick="chk()">
```



```
</form>
</body>
<script type="text/javascript">
function chk()
{
var a,c;
a=parseInt(f1.n1.value);
c=a*a*a;
alert("Cube of a Number is "+c);
}
</script>
</html>
```

Output :



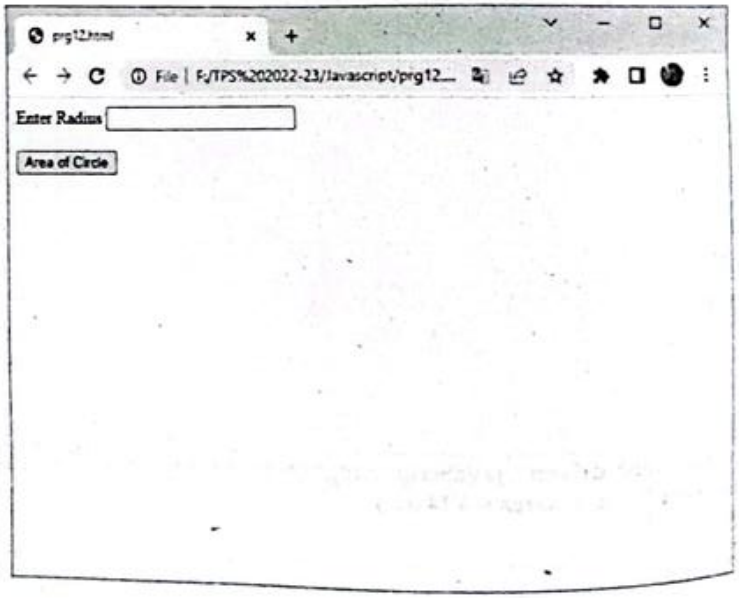
10. Write an event driven JavaScript program to display area of a circle. Accept radius from the user (area = 3.14xrxr).

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
```

```
Enter Radius
<input type="number" name="n1"><br><br>
<input type="button" name="b1" value="Area of Circle" onClick="area()">
</form>
</body>
<script type="text/javascript">
function area()
{
var a,r,p=3.14;
r=parseInt(f1.n1.value);
a=p*r*r;
alert("Area of Circle is "+a);
}
</script>
</html>
```

Output :



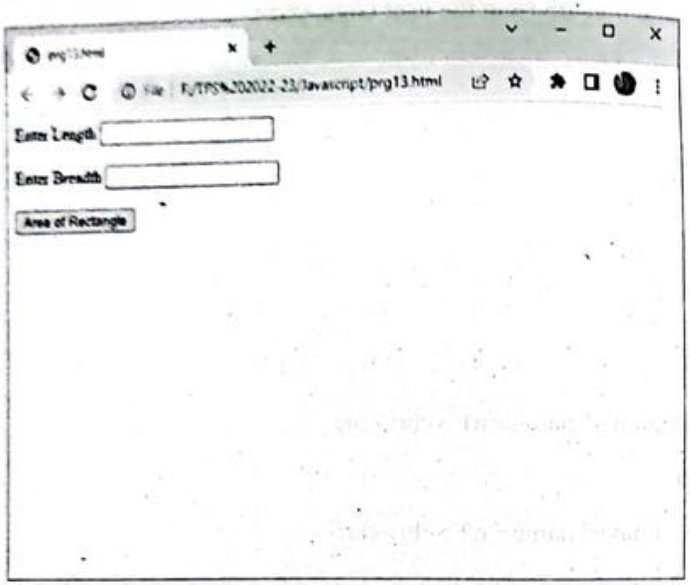
11. Write an event driven JavaScript program to display area of a rectangle. Accept length and breadth from the user (area = lxb).

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Length
<input type="number" name="n1"><br><br>
Enter Breadth
<input type="number" name="n2"><br><br>
<input type="button" name="b1" value="Area of Rectangle" onClick="area()">
</form>
</body>
<script type="text/javascript">
function area()
{
var l,b,a;
l=parseInt(f1.n1.value);
b=parseInt(f1.n2.value);
a=l*b;
alert("Area of Rectangle is "+a);
}
</script>
</html>
```

Output :



12. Write an event driven JavaScript program to display area of a triangle. Accept base and height from the user (area = 1/2xbxh).

Ans. :

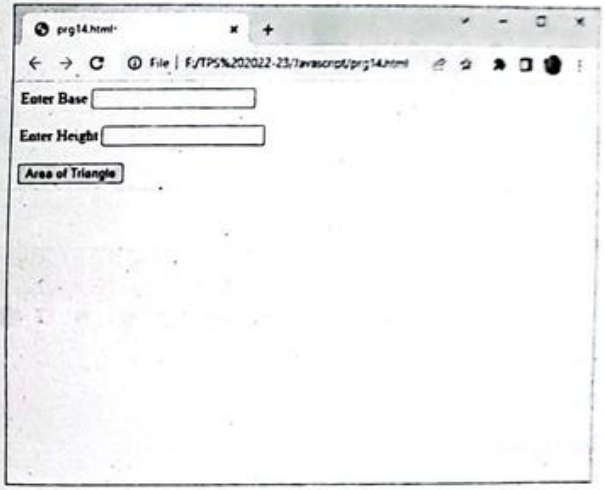
Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Base
<input type="number" name="n1"><br><br>
Enter Height
<input type="number" name="n2"><br><br>
<input type="button" name="b1" value="Area of Triangle" onClick="area()">
</form>
</body>
<script type="text/javascript">
function area()
{
var b,h,a;
```



```
b=parseInt(f1.n1.value);
h=parseInt(f1.n2.value);
a=1/2*b*h;
alert("Area of Triangle is "+a);
}
</script>
</html>
```

Output :



13. Write an event driven JavaScript program to display perimeter of a rectangle. Accept length and breadth from the user. (Perimeter = 2 X length + 2 X Breadth)

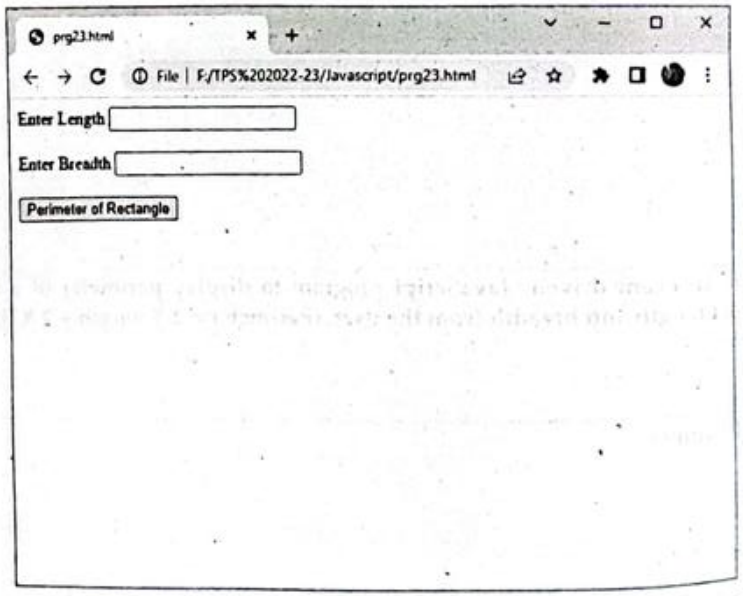
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Length
<input type="number" name="n1"><br><br>
Enter Breadth
<input type="number" name="n2"><br><br>
<input type="button" name="b1" value="Perimeter of Rectangle" onClick="per()">
```

```
</form>
</body>
<script type="text/javascript">
function per()
{
var l,b,p;
l=parseInt(f1.n1.value);
b=parseInt(f1.n2.value);
p=((2*l)+(2*b));
alert("Perimeter of Rectangle is "+p);
}
</script>
</html>
```

Output :



14. Write an event driven JavaScript program to display day of the week for variable day = 5. If the value of the day is changed it should be displayed. Use switch statement.

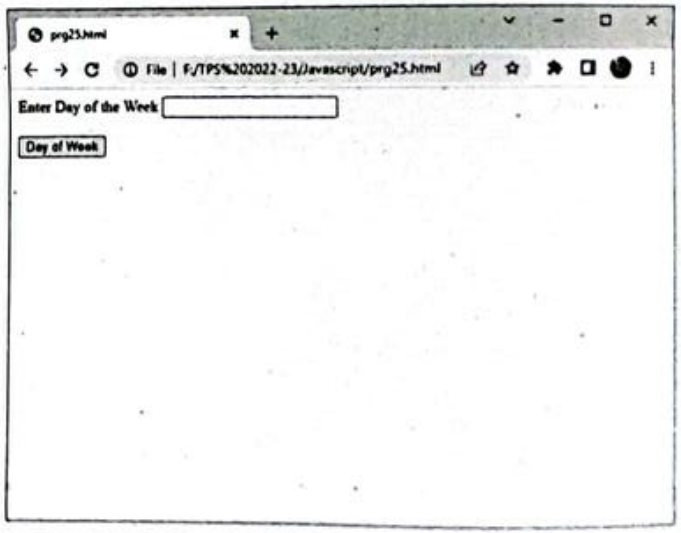
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Day of the Week
<input type="number" name="n1"><br><br>
<input type="button" name="b1" value="Day of Week" onClick="dow()">
</form>
</body>
<script type="text/javascript">
function dow()
{
var day=5;
day=parseInt(f1.n1.value);
switch(day)
{
case 1:
alert("Monday");
break;
case 2:
alert("Tuesday");
break;
case 3:
alert("Wednesday");
break;
case 4:
alert("Thursday");
break;
case 5:
alert("Friday");
break;
```

```
case 6:
alert("Saturday");
break;
case 7:
alert("Sunday");
break;
default:
alert("Invalid Day");
}
}
</script>
</html>
```

Output :



15. Write an event driven JavaScript program to display factorial of number 35 when mouse is moved on the button.

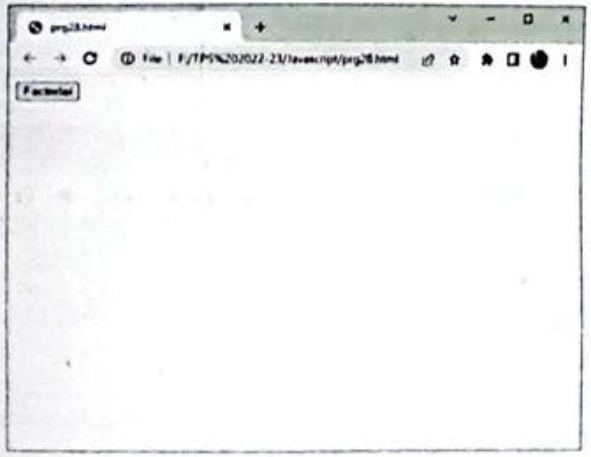
Ans. : Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
<input type="button" name="b1" value="Factorial" onMouseOver="fact()">
```



```
</form>
</body>
<script type="text/javascript">
function fact()
{
var a,f=1,i;
a=35;
for(i=a;i>=1;i--)
{
f=f*i;
}
alert("Factorial of a Number is "+f);
}
</script>
</html>
```

Output :



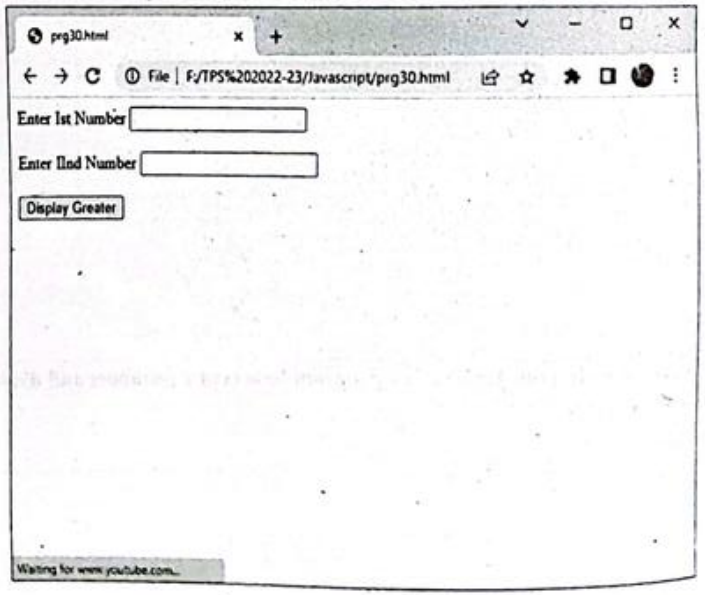
16. Write an event driven JavaScript program to accept 2 numbers and display which number is greater.

Ans. : Coding :

```
<!DOCTYPE html>
<html>
<body>
```

```
<form name="f1">
Enter Ist Number
<input type="number" name="n1"><br><br>
Enter IInd Number
<input type="number" name="n2"><br><br>
<input type="button" name="b1" value="Display Greater" onClick="grt()">
</form>
</body>
<script type="text/javascript">
function grt()
{
var a,b;
a=parseInt(f1.n1.value);
b=parseInt(f1.n2.value);
if(a>b)
alert("Greater Number is "+a);
else
alert("Greater Number is "+b);
}
</script>
</html>
```

Output :



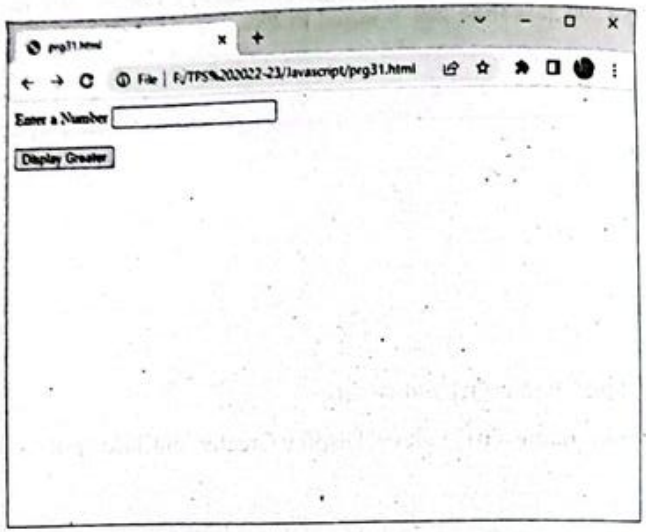
17. Write an event driven JavaScript program to accept number and determine if it is greater than 200 ,less than 200 or equal to 200.

Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter a Number
<input type="number" name="n1"><br><br>
<input type="button" name="b1" value="Display Greater" onClick="grt()">
</form>
</body>
<script type="text/javascript">
function grt()
{
var a;
a=parseInt(f1.n1.value);
if(a>200)
alert("Number is greater than 200 ");
else
{
if(a<200)
alert("Number is less than 200 ");
else
alert("Number is equal to 200 ");
}
}
</script>
</html>
```

Output :



18. Write an event driven JavaScript program to accept pin code and check whether it is an integer.

Ans. :

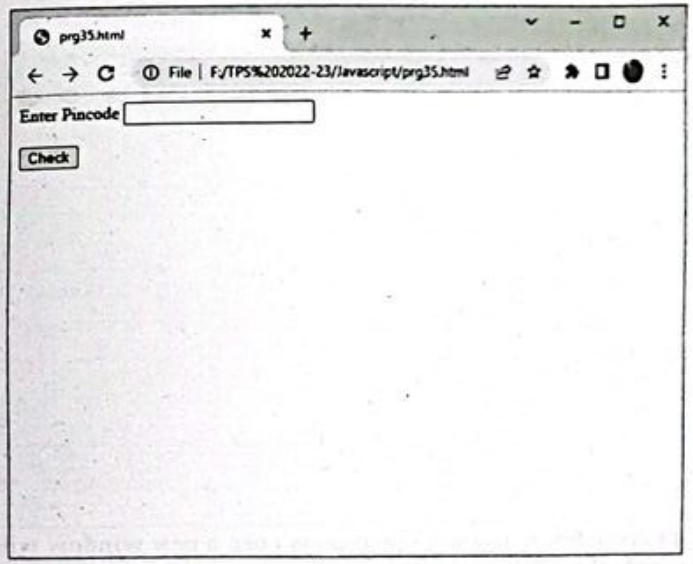
Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
Enter Pincode
<input type="text" name="t1"><br><br>
<input type="button" name="b1" value="Check" onClick="pin()">
</form>
</body>
<script type="text/javascript">
function pin()
{
var a;
a=f1.t1.value;
if(isNaN(a))
alert("Pincode contains characters");
```



```
else
alert("Pincode contains Integer");
}
</script>
</html>
```

Output :



19. Write an event driven JavaScript program to call a function after 3000 milliseconds and display text 'Hello'.

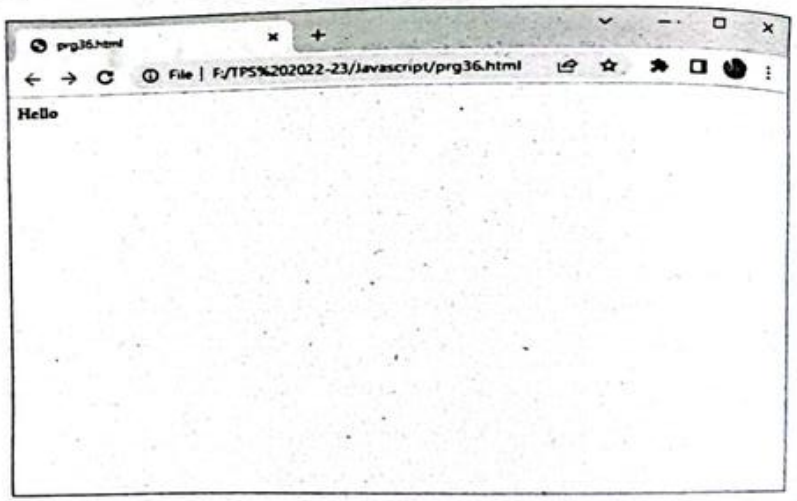
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body onLoad="f1()">
</body>
<script type="text/javascript">
function f1()
{
window.setTimeout("f2()",3000);
}
```

```
function f2()
{
document.write("Hello");
}
</script>
</html>
```

Output :



20. Write an event driven JavaScript program to open a new window with skyblue background color.

Ans. :

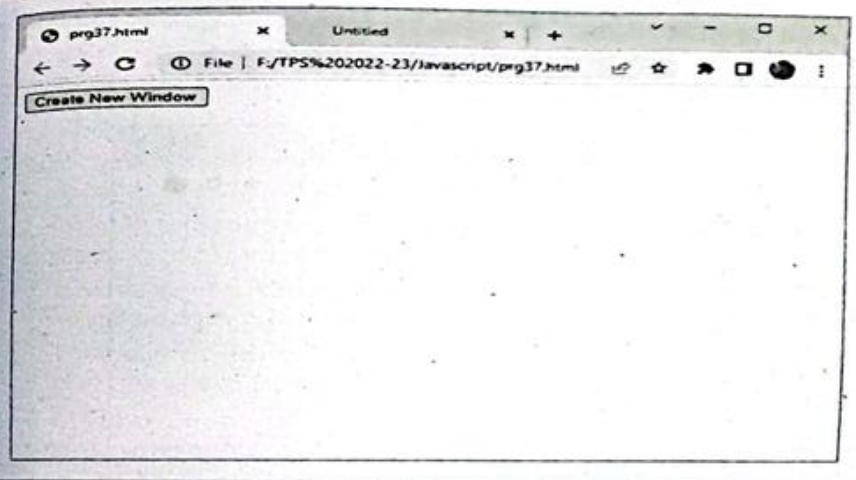
Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="f1">
<input type="button" name="b1" value="Create New Window" onClick="nw()">
</form>
</body>
<script type="text/javascript">
function nw()
{
var a=window.open();
```



```
a.document.write("<h1> This is a new window</h1>");
a.document.bgColor="skyblue";
}
</script>
</html>
```

Output :



21. Write an event driven JavaScript program to display 2 colors after every 3000 milliseconds on click of a button.

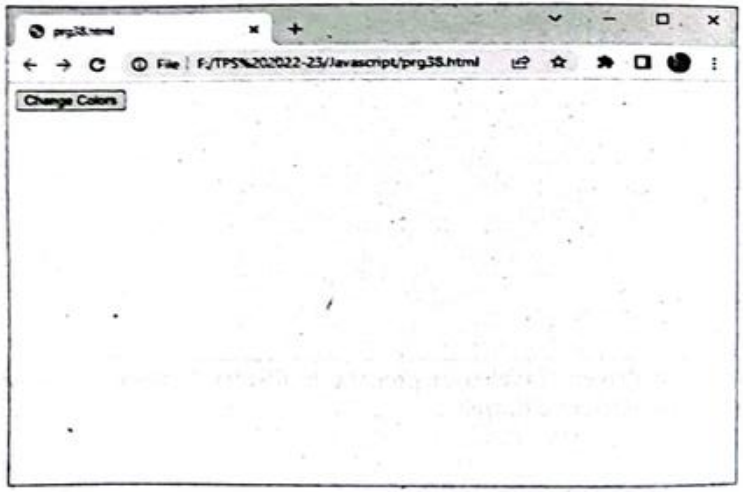
Ans. :

Coding :

```
<!DOCTYPE html>
<html>
<body>
<form name="frm1">
<input type="button" name="b1" value="Change Colors" onClick="f1()">
</form>
</body>
<script type="text/javascript">
function f1()
{
document.bgColor="pink";
window.setTimeout("f2()",3000);
```

```
}
function f2()
{
document.bgColor="cyan";
window.setTimeout("f1()",3000);
}
</script>
</html>
```

Output :



Extra Programs for Self-Practice

1. Write an event driven JavaScript program to display addition and remainder of two accepted numbers when the mouse is moved over a button.
2. Write an event driven JavaScript program to display number sequence from 100 to 500 when the mouse is moved over a button
3. Write an event driven JavaScript program to display numbers from 1 to 150 when the mouse is moved over a button.
4. Write an event driven JavaScript program to accept a number and determine if it is odd or even.
5. Write an event driven JavaScript program to display sum of numbers from 100 to 1000 when the mouse is moved over a button.
6. Write an event driven JavaScript program to display length of the text "Information Technology" on click of a button.

7. Write an event driven JavaScript program to display square of a number and cube of an accepted number on click of a button.
8. Write an event driven JavaScript program to accept three numbers and display addition and multiplication of the numbers.
9. Write an event driven JavaScript program to display circumference of a circle by accepting radius from the user. (Circumference =2 X 3.14 X R).
10. Write an event driven JavaScript program to display perimeter of a square. Accept side value from the user. Perimeter = 2 X Side X Side.
11. Write an event driven JavaScript program to display perimeter of a square. Accept side value from the user. Perimeter = 2 X Side X Side.
12. Write an event driven JavaScript program to display square of numbers from 2 to 20 on click of a button.
13. Write an event driven JavaScript program to accept 3 numbers and display average of that number. (Average=number1+number2+number3 divided by 3).
14. Write an event driven JavaScript program to accept number and determine if it is greater than 100, less than 100 or equal to 100.
15. Write an event driven JavaScript program to accept age and display message "hello " if age is greater than 18.
16. Write an event driven JavaScript program to accept a base and height and display perimeter of a parallelogram. (Hint-Perimeter of parallelogram= 2 X Base + 2 X Height.)
17. Write an event driven JavaScript program to display 2 colors after every 3000 milliseconds when the mouse is moved over a button.

Self Assessment

Time : 1 Hour **Marks: - 40**

1. **Fill in the blanks** (05)
   1. _____means increment or decrement value of a running variable.
   2. The way in which HTML document content is accessed and modified is called as_____.
   3. _____ is an object based scripting language.
   4. _____ property of window object sets or returns the fact in the status bar of a window.
   5. _____ event handler occurs when user clicks reset button.
2. **True or False** (05)
   1. onload event handler occurs when document/page has been loaded.
   2. NaN property of number of object returns Not a Number value.
   3. cbrt() method of math object returns the cube root of a number.
   4. Date object is created using new keyword.
   5. getMinutes() method of Date object returns minutes.

3. MCQ one correct alternative. (05)

1. In _____ scripting the script resides on the web server.
   (a) Client side (b) My side (c) Server Side (d) Website
2. _____ is a JavaScript based open source frontend web framework developed for single page application.
   (a) Vue JS (b) Response (c) Angular JS (d) React
3. _____ property of DOM object sets or returns title of the document.
   (a) head (b) body (c) source (d) Title
4. _____ method of window object calls a function or evaluates an expression after a specified number of milliseconds.
   (a) open() (b) setTimeout() (c) blur() (d) focus()
5. _____ method of Array object sorts the elements of an array.
   (a) concat() (b) sort() (c) pop() (d) find ()

4. MCQ two correct alternative. (06)

1. Types of loops in JavaScript are _____
   (a) for .....loop (b) while .....loop
   (c) for ....next (d) while.....wend
2. JavaScript supports following types of objects _____
   (a) Variables (b) built-in objects
   (c) User defined objects (d) loops
3. Following are the event handlers in JavaScript _____
   (a) onsubmit (b) write()
   (c) onchange (d) writeln()

5. MCQ three correct alternative. (06)

1. Popular Framework /libraries _____
   (a) HTML (b) Angular JS (c) Network Frame
   (d) Vue Js (e) React (f) Switch Case
2. Methods of Window Object are _____
   (a) Name (b) blur() (c) location
   (d) focus() (e) closed (f) setTimeOut()

6. Match the Following. (03)

A	B
1. Length	(a) Converts the string into Upper case
2. toUpperCase()	(b) Returns the character at the specified position
3. getMinutes()	(c) Converts the given string into integer number
4. chartAt()	(d) Returns the largest minimum value
5 parseInt()	(e) Returns a number of character in a string
6 MIN_VALUE	(f) Returns the minutes

7. Answer the following (10)

a) Explain Break statement **(Refer Page No 3.45)**
b) Short note on DOM **(Refer Page No 3.46)**
c) Explain properties of Window object **(Refer Page No 3.47)**
d) Explain onfocus and onblur event handler in JavaScript **(Refer Page No 3.52)**
e) Explain properties on String object in JavaScript **(Refer Page No 3.52)**

8. Write a JavaScript program (10)

a) Program to check whether the number is prime number of not. **(Refer Page No 3.56)**
b) Program to print numbers from 1 to 10 using for loop. **(Refer Page No 3.53)**

Answer of Questions

1. Fill in the blanks
1) Iteration 2) Document Object Model
3) JavaScript 4) status 5) onreset

2. True or False
1) True 2) True 3) True 4) True 5) True

3. MCQ one correct alternative
1) C 2) C 3) D 4) B 5) B

4. MCQ two correct alternative
1) (a, b) 2) (b, c) 3) (a, c)

5. MCQ three correct alternative
1) (b, d, e) 2) (b, d, f)

6. Match the Following
1. – (e), 2. – (a), 3. – (f), 4 – (b) 5 – (c) 6 – (d)

Continue from Page No. 4-23

7. Answer the following (10)

a) Define IOT **(Refer Page No 4.16)**
b) Short note on Applications of IoT **(Refer Page No 4.17)**
c) Explain models of Cloud computing. **(Refer Page No 4. 18)**
d) Explain Hybrid cloud. **(Refer Page No 4. 19)**
e) Short Note on Artificial Intelligence. **(Refer Page No 4.20)** □□□

Answer of Questions

1. Fill in the blanks
1) Cloud computing 2) IaaS 3) Web
4) Data 5) Artificial Intelligence

2. True or False
1) True 2) True 3) True 4) False 5) True

3. MCQ one correct alternative
1) A 2) A 3) B 4) B 5) B

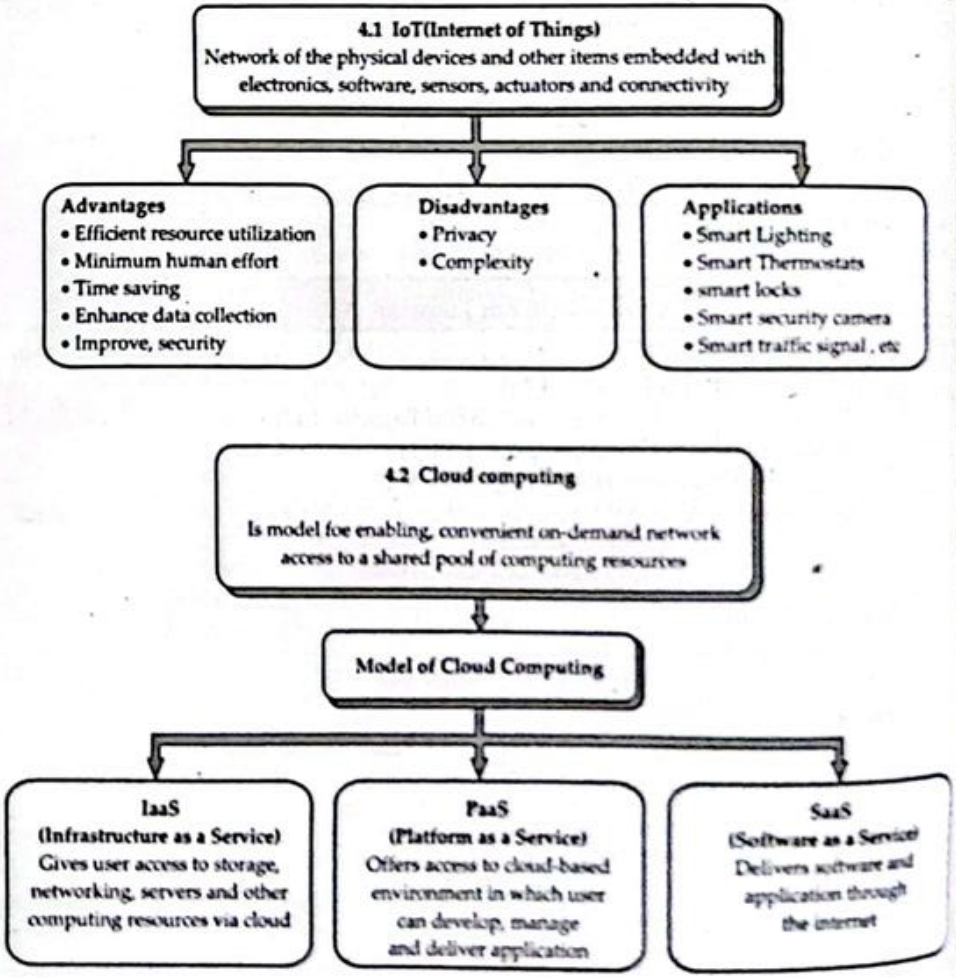
4. MCQ two correct alternative
1) (a, b) 2) (a, b) 3) (c, d)

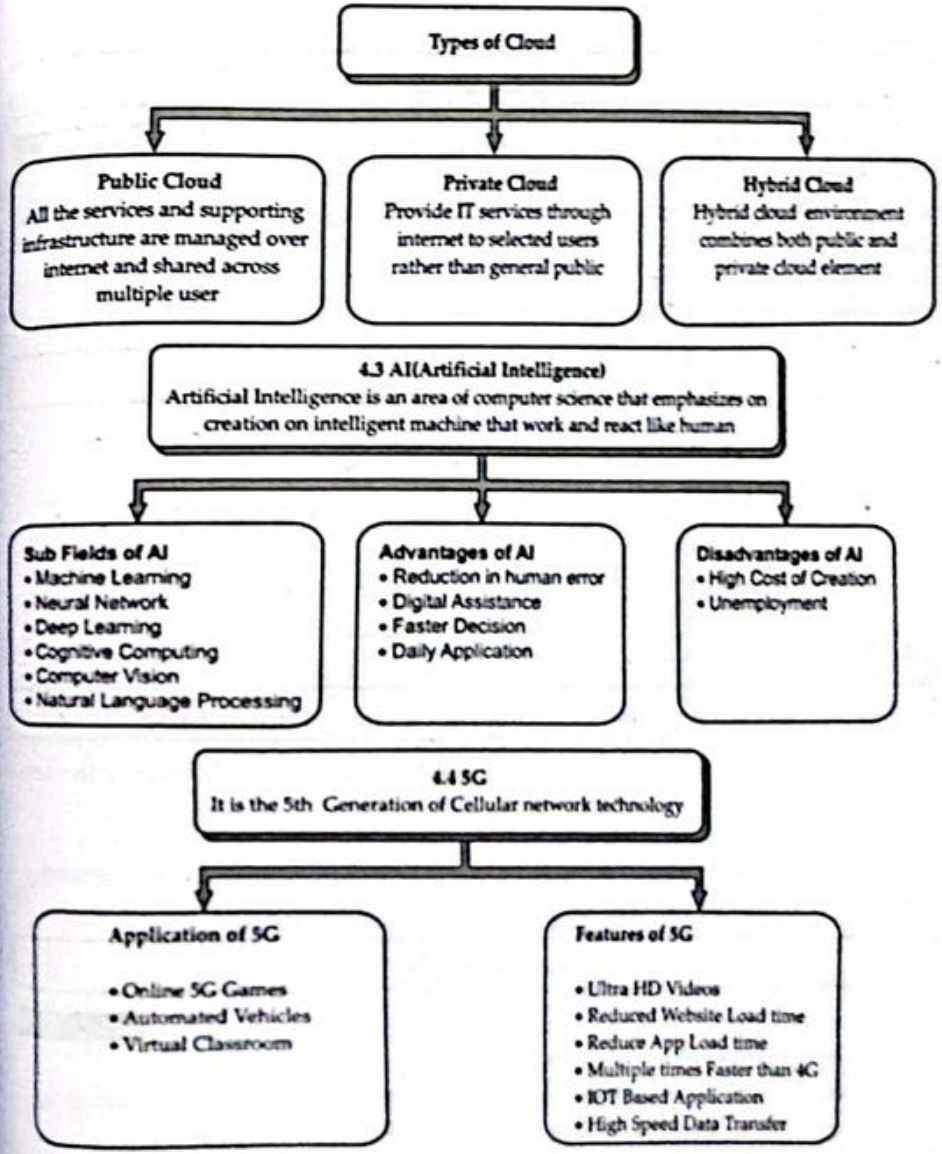
5. MCQ three correct alternative
1) (a, b, c) 2) (a, b, d)

6. Match the Following
1. - (b) 2. - (c) 3. - (a) 4. - (e) 5. - (f) 6. - (d)

4 Emerging Technologies

Chapter at a Glance





Exercise

Fill in the Blanks

1. IoT is Referred as _____. (March 2023)

Ans. : Internet of Things

2. Smart Home is the application of _____ Technology.

Ans. : IoT

3. Amazon is the _____ service provider.

Ans : Saas

Match the Following

(1)

'A'		'B'	
1.	Smart City	(a)	Software as a Service
2.	Amazon Web Server	(b)	Platform as a Service
3.	PaaS	(c)	Cloud Computing
4.	SaaS	(d)	IOT

Ans. : 1. - (d), 2. – (c), 3. – (b), 4. – (a)

State whether the following statement is True or False

4. PaaS provides a platform tools to test, develop and host application in the same environment.

Ans. : True

5. Cloud computing means to store and access data and programs over the internet.

Ans. : True

Answer Briefly

6. Give some application of IoT. (March 2023)

Ans. : Following are some applications of IoT :

(i) **Smart thermostats** - Allows users to schedule monitor and remotely control home.

(ii) **Smart lighting** - Illumination of light is controlled on the basis of day light.

(iii) Smart locks and garage door openers, Password based or facial recognition based doors and locks.

(iv) **Smart security cameras** – Security cameras that can identify known and unknown person and raise alarm, in case of security threat.

(v) **Smart traffic signals** – Signal that can adjust their timing to accommodate commutes and holiday traffic and keep cars moving.

7. Explain detail about Cloud Computing.

Ans. :

(i) Cloud computing is the on-demand availability of computer system without direct active management by the user.

(ii) There are three primary service models of cloud computing that are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

(iii) **Infrastructure as a service (IaaS) :** IaaS provides automated and scalable environments that provide a high degree of control and flexibility for the user.

(iv) **Platform as a service (PaaS) :** A service provider offers access to a cloud-based environment in which users can develop, manage and deliver applications.

(v) **Software as a service (SaaS) :** A service provider delivers software and applications through the internet or an API.

(vi) There are three basic types of deployment of cloud computing that are Public, Private and Hybrid.

(vii) In public cloud, all the services and supporting infrastructure are managed off-site over the internet and shared across multiple users.

(viii) Private cloud provides I.T services through the internet or a private network to selected users rather than to general public.

(ix) Hybrid cloud environments combines both Public and Private cloud elements.

8. Explain models of Cloud Computing. (March 2022)

Ans. :

(i) Cloud Computing is the on-demand availability of computer system without direct active management by the user.

(ii) There are three primary service models of cloud computing that are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

(iii) **Infrastructure as a service (IaaS)**

- IaaS provides automated and scalable environments that provide a high degree of control and flexibility for the user.
- Instead of purchasing hardware , users pay for IaaS on demand.
- **Examples :** Amazon web services (AWS), Microsoft Azure VM, Google Compute Engine (GCE)

(iv) **Platform as a service (PaaS)**

- A service provider offers access to a cloud-based environment in which users can develop, manage and deliver applications.
- PaaS provides a platform with tools to test, develop and host applications in the same environment
- **Examples :** Google App Engine

(v) **Software as a service (SaaS)**

- A service provider delivers software and applications through the internet or an API.
- Applications are accessible from almost any internet-connected device, from virtually anywhere in the world.
- **Examples :** Google's G suite, GitHub , Dropbox.

9. Give brief idea about AI. (March 2022)

Ans. :

(i) Artificial Intelligence is the study of making intelligent machine's that can perform operation like human being.

(ii) AI is different from robotics, but related to some extent, in which machines sense their environment, perform calculations and do physical tasks either by themselves.

Different fields of AI are as follows :

(iii) **Machine learning** is the study of computer algorithms that improve automatically through experience.

(iv) **Neural networks** imitate the brain's array of interconnected neurons, and relay information between various units to find connections and derive meaning from data.

(v) **Deep learning** utilizes really big neural networks and a lot of computing power to find complex patterns in data, for applications such as image and speech recognition.

(vi) **Natural language** processing involves analyzing and understanding human language and responding to it.

10. Explain concept of 5G.

Ans. :

(i) 5G is the fifth generation of cellular network technology.

(ii) 5G is the next generation of wireless communications. It is expected to provide Internet connections that are multiple times faster than 4G LTE (Long Term Evolution).

(iii) 5G technology may use a variety of spectrum bands, including millimeter wave (mmWave) radio spectrum, which can carry very large amounts of data at a short distance.

(iv) The drawback of the higher frequencies is that they are more easily obstructed by the walls of buildings, trees and other foliage, and even changes in the weather.

(v) The new 5G networks will be able to transmit very large amounts of data—but only a few blocks at a time.

(vi) 5G networks are digital cellular networks, in which the service area covered by providers is divided into small geographical areas called cells.

(vii) 5G can support up to a million devices per square kilometer, compared to 4G.

Fill in the Blanks

4.1 IoT(Internet of Things)

1. IoT stands for _____. (March 2022)

Ans. : Internet of Things (IoT)

2. The internet of things is the _____ of physical devices ,vehicles , home appliances etc.

Ans. : network

3. _____ helps reduce human efforts.

Ans. : Internet of Things (IoT)

4. _____ is the primary factor in automation which can save through IoT platform.

Ans. : Time

5. IoT devices can collect data from _____.

Ans. : Environment

6. The data collected using IoT devices can be used to take _____. (March 2022)

Ans. : decisions

7. _____ based security systems can make home or office environment secure.

Ans. : IoT

8. _____ Allows users to schedule, monitor and remotely control home temperatures.

Ans. : Smart thermostats

9. Illumination of light is controlled on the basis of day light is example of _____.

Ans. : Smart Lighting

10. Smart Home is the application of _____ Technology. (March 2022)

Ans. : IoT

4.2 Cloud Computing

11. _____ is a model for enabling, convenient or demand network access to a shared pool of computing resources. (March 2022)

Ans. : Cloud Computing

12. SaaS stands for_____. (March 2022)

Ans. : Software as a Service

13. IaaS Stands for_____. (March 2022)

Ans. : Infrastructure as a Service

14. PaaS Stands for_____. (March 2022)

Ans. : Platform as a Service

15. _____ gives user access to storage, networking, servers and other computing resources via the cloud.

Ans. : IaaS

16. There are _____ primary service models of cloud computing.
Ans. : three
17. Instead of purchasing hardware outright, users pay for _____ on demand.
Ans. : IaaS
18. Amazon Web Services (AWS) is an example of_____. (March 2022)
Ans. : IaaS
19. In IaaS _____ is Scalable depending on processing and storage needs.
Ans. : Infrastructure
20. IaaS provides automated and scalable _____ that provide a high degree of control and flexibility for the user.
Ans. : Environment
21. PaaS provide a _____ with tools to test , develop and host application in the same environment.
Ans. : Platform
22. _____ provide a platform with tools to test , develop and host application in the same environment. (March 2022)
Ans. : PaaS
23. _____ enables organization to focus on development without having a worry about underlying infrastructure.
Ans. : PaaS
24. SaaS service provider delivers _____ and application through internet.
Ans. : Software
25. In _____ user do not install application on their local devices.
Ans. : SaaS
26. In SaaS the application reside on a remote cloud network accessed through the web or an _____.
Ans. : API
27. In SaaS the application reside on a remote cloud network accessed through the _____ or an API. (March 2022)
Ans. : web
28. _____ vendors provide users with software and application via a subscription model.
Ans. : SaaS
29. Users do not have to manage , install or upgrade software , _____ manage this.
Ans. : SaaS providers
30. _____ is secure on cloud.
Ans. : Data
31. GitHub is an example of _____.
Ans. : SaaS

32. In _____ all the services and supporting infrastructure are shared across multiple users.
Ans. : public cloud
33. _____ is more efficient and inexpensive than other cloud solution. (March 2022)
Ans. : public cloud
34. In _____ cloud all data is protected behind the firewall.
Ans. : Private
35. _____ solutions are preferred for enhanced security and privacy by the users.
Ans. : Private cloud
36. _____ cloud environment combines both Public and Private element.
Ans. : Hybrid
37. _____ cloud allows greater flexibility.
Ans. : Hybrid
38. Hybrid cloud allow greater _____.
Ans. : Flexibility
39. _____ computing solutions are inexpensive than the actual overall infrastructure set up for the I.T services. (March 2022)
Ans. : Cloud
40. Cloud computing solutions are_____ than the actual overall infrastructure set up for the I.T services.
Ans. : Inexpensive
41. _____ computing solution are more reliable than in-house I.T infrastructure.
Ans. : Cloud
42. _____ computing solutions are more portable
Ans. : Cloud

4.3 AI (Artificial Intelligence)

43. AI stands for_____. (March 2022)
Ans. : Artificial Intelligence
44. _____ is an area of computer science that emphasizes on creation of intelligent machines that work and react like human.
Ans. : Artificial intelligence
45. Machine learning is a sub field of_____. (March 2022)
Ans. : Artificial Intelligence
46. Deep Learning is a sub field of_____.
Ans. : Artificial Intelligence
47. _____ imitate the brain's array of interconnected neurons. (March 2022)
Ans. : Neural networks

48. _____ is used for application like image and speech recognition.
Ans. : Deep Learning
49. _____ computing is about creating a "natural ,human-like interaction", including using the abilities to interpret speech and respond to it.
Ans. : Cognitive
50. _____. employs pattern recognition and deep learning to understand content of pictures and videos.
Ans. : Computer Vision
51. _____ involves analyzing and understanding human language and responding it.
Ans. : Natural Language Processing
52. Artificial Intelligence helps to _____ human errors.
Ans. : reduce
53. Chatbot are example of _____.
Ans. : Digital Assistance

4.4 5G

54. 5G Stands for_____. (March 2022)
Ans. : fifth generation
55. 5G is the next generation of_____ communication.
Ans. : wireless
56. _____ is the next generation of wireless communication.
Ans. : 5G
57. LTE stand for_____.
Ans. : Long Term Evolution
58. 5G technologies may use a variety of _____.
Ans. : Spectrum Bands
59. _____ is the faster method of data transfer. (March 2022)
Ans. : 5G
60. _____ networks are digital cellular networks.
Ans. : 5G
61. _____ is the fifth generation of cellular network technology.
Ans. : 5G

True or False

4.1 IoT(Internet of Things)

1. The internet of thing is the network of physical devices only. (March 2022)
Ans. : False
2. With the help of IoT devices can communicate and interact over the internet and they can be remotely monitored and controlled.
Ans. : True

3. IoT helps to minimize the human efforts. (March 2022)
Ans. : True
4. Time is the primary factors in automation which can be saved through IoT platforms.
Ans. : True
5. The data collected using IoT devices is used for taking decisions.
Ans. : True
6. The designing ,developing, maintaining and enabling the large technology to IoT system is easy.
Ans. : False

4.2 Cloud Computing

7. IaaS provide automated and static environment. (March 2022)
Ans. : False
8. In IaaS enterprises saves the cost of buying and maintaining their own hardware.
Ans. : True
9. PaaS provides a platform tools to test, develop and host application in the same environment.
Ans. : True
10. PaaS Do not provide collaborative work if teams works remotely. (March 2022)
Ans. : False
11. In SaaS a service provider delivers software and application through the internet.
Ans. : True
12. PaaS vendors provides users with software and application via a subscription model.
Ans. : False
13. SaaS provider manage, install and ,update the software. (March 2022)
Ans. : True
14. Public cloud are more efficient and inexpensive than private and hybrid cloud.
Ans. : True
15. In Private cloud all the services and supporting infrastructure is shared among across the multiple users.
Ans. : False
16. Private cloud is accessible to selected public rather than general public.
Ans. : True
17. Hybrid cloud allows greater flexibility. (March 2022)
Ans. : True
18. Cloud computing solution are inexpensive than the actual overall infrastructure.
Ans. : True

19. Cloud computing solution are not reliable than In-house I.T infrastructure.

Ans. : False

20. Cloud computing solutions are more portable. (March 2022)

Ans. : True

4.3 AI(Artificial Intelligence)

21. Artificial intelligence is an area of computer science that emphasizes on creation of intelligent machines that work and react like human.

Ans. : True

22. Computer is programmed properly with artificial intelligence gives 100% accuracy.

Ans. : True

23. Chatbot are example of Digital Assistance

Ans. : True

24. AI helps to take faster decision as compare to humans.

Ans. : True

25. AI creates Unemployment.

Ans. : True

26. Creation cost is high in AI.

Ans. : True

4.4 5G

27. 4G is the next generation of wireless communication

Ans. : False

28. 5G technologies may use variety of spectrum brands. (March 2022)

Ans. : True

29. 5G is the fastest method of a data transfer.

Ans. : True

MCQ (One Correct Answers)

4.1 IoT(Internet of Things)

1. The internet of things is the _____ of physical devices ,vehicles , home appliances etc. (March 2022)
   (a) collection (b) network
   (c) relation (d) group

Ans. : (a) network

2. IoT helps to _____ human efforts.
   (a) remove (b) reduce (c) increase (d) stop

Ans. : (b) reduce

3. IoT device can collect data from _____.
   (a) environment (b) internet
   (c) books (d) documents

Ans. : (a) environment

4.2 Cloud Computing

4. SaaS Stand for _____. (March 2022)
   (a) Software as a Service (b) Software as a self
   (c) Software as Secure (d) Software as a Section

Ans. : (a) Software as a service

5. PaaS Stand for _____. (March 2022)
   (a) Platform as a Service (b) Platform as a self
   (c) Platform as Secure (d) Platform as a Section

Ans. : (a) Platform as a service

6. IaaS Stand for _____. (March 2022)
   (a) Infrastructure as a Service (b) Infrastructure as a self
   (c) Infrastructure as Secure (d) Infrastructure as a Section

Ans. : (a) Infrastructure as a service

7. In _____ cloud services and supporting infrastructure are shared across multiple users.
   (a) public (b) private (c) secure (d) hybrid

Ans. : (a) public

8. In _____ cloud services all the data is protected behind the firewall.
   (a) public (b) private (c) secure (d) hybrid (March 2022)

Ans. : (a) private

4.3 AI(Artificial Intelligence)

9. _____ imitate the brain's array of interconnected neurons.
   (a) Machine learning (b) Neural network
   (c) Deep learning (d) Computer vision

Ans. : (b) Neural network

10. _____ is about creating a "natural, human-like interaction".
(a) Machine learning (b) Neural network
(c) Cognitive computing (d) Computer vision

Ans. : (b) Cognitive computing

4.4 5G

11. _____ is the next generation of wireless communication. (March 2019)
(a) 5G (b) 4G (c) 3G (d) 6G

Ans. : (b) 5G

12. LTE stands for_____
(a) Long Term Evolution (b) Long Term Evolve
(c) Large Term Evolution (d) Long Test Evolution

Ans. : (b) Long Term Evolution

MCQ (Two Correct Answers)

4.2 Cloud Computing

1. Primary service models of cloud computing are _____.
(a) SaaS (b) PaaS (c) FaaS (d) DaaS

Ans. : (a), (b)

2. Example of IAAS are _____.
(a) Amazon Web Services (b) Microsoft Azure VM
(c) GitHub (d) Google's G Suit

Ans. : (a), (b)

3. Example of SaaS are _____.
(a) GitHub (b) SAP (c) Dropbox (d) AWS

Ans. : (a), (b)

4. Following are the types of cloud.
(a) Public Cloud (b) Private Cloud
(c) Personal Cloud (d) secure Cloud

Ans. : (a), (b)

5. In SaaS the application reside on a remote cloud network accessed through the _____ or an _____
(a) web (b) file (c) API (d) wifi

Ans. : (a), (c)

4.3 AI(Artificial Intelligence)

6. Disadvantages of AI are _____.
(a) Digital Assistance (b) Faster Decisions
(c) Cost of creation (d) Unemployment

Ans. : (c), (d)

MCQ (Three Correct Answers)

4.1 IoT(Internet of Things)

1. Internet of Things (IoT) is the network of _____.
(a) Physical devices (b) sensors (c) actuators
(d) wires (e) Time saving (f) security

Ans. : (a), (b), (c)

2. Advantages of IoT are _____.
(a) Time saving (b) Enhance data collection
(c) Improve security (d) Privacy
(e) Complexity (f) Privacy

Ans. : (a), (b), (c)

4.2 Cloud Computing

3. Primary service models of cloud computing are _____.
(a) SaaS (b) PaaS (c) FaaS
(d) IaaS (e) Xaas (f) Zaas

Ans. : (a), (b), (d)

4. Example of IAAS are _____.
(a) Amazon Web Services (b) Microsoft Azure VM
(c) Google Compute Engine (d) Google's G Suit
(e) Smart locks (f) No locks

Ans. : (a), (b), (c)

5. Example of SaaS are _____.
   (a) GitHub (b) SAP (c) Dropbox
   (d) AWS (e) VM (f) Lock

Ans. : (a), (b), (c)

6. Types of Cloud are _____.
   (a) Public Cloud (b) Private Cloud (c) Personal Cloud
   (d) Hybrid Cloud (e) My Cloud (f) House Cloud

Ans. : (a), (b), (d)

Match the Following

4.1 IoT(Internet of Things)

A		B	
1.	Smart City	(a)	Network of physical devices, sensors, connectivity ,etc
2.	Internet Of Things	(b)	IoT

Ans. : 1. = (b) 2. = (a)

4.2 Cloud Computing

A		B	
1.	Public Cloud	(a)	Software as a Service
2.	IaaS	(b)	Shared across multiple users
3.	SaaS	(c)	Infrastructure as a Service
4.	PaaS	(d)	All data is protected behind the firewall
5.	Hybrid Cloud	(e)	Platform as a service
6.	Private Cloud	(f)	Allows greater flexibility

Ans. :

1. = (b) 2. = (c) 3. = (a) 4. = (e) 5. = (f) 6. = (d)

4.3 AI (Artificial Intelligence)

A		B	
1.	Artificial Intelligence	(a)	imitate the brain's array of interconnected neurons
2.	Machine Learning	(b)	image and speech recognition
3.	Neural Network	(c)	creating a "natural, human-like interaction"
4.	Deep Learning	(d)	Area of Computer Science

A		B	
5.	Cognitive computing	(e)	Analyze and understand human language
6.	Computer Vision	(f)	Digital assistance
7.	Natural Language Processing	(g)	Automates analytical model building
8.	Chatbot	(h)	OK Google
9.	Google's	(i)	understanding the content of pictures and videos

Ans. :

1. = (d) 2. = (g) 3. = (a) 4. = (b) 5. = (c)

6. = (i) 7. = (e) 8.= (f) 9. = (h)

4.4 5G

A		B	
1.	5G	(a)	Long Term Evolution
2.	LTE	(b)	fifth generation

Ans. : 1. = (b) 2. = (a)

Answer in Briefly

4.1 IoT (Internet of Things)

1. Define IoT with example. (March 2022)

Ans. :

Definition :

(i) IoT stands for Internet of Things, which is a way of connecting all the devices and objects around us into a common cloud network.

(ii) It helps in creating opportunities for more direct integration of the physical world into computer based systems.

(iii) It improves efficiency, reduces cost and human efforts.

(iv) IOT extends Internet connectivity beyond standard devices, like desktops, laptops, smart phones and tablets, to any range of traditionally dumb or non-internet enabled devices and everyday object.

(v) **Following are some examples of IoT :**

(vi) **Smart lighting** - Illumination of light is controlled on the basis of day light.

(vii) **Smart thermostats** - Allows users to schedule monitor and remotely control home temperatures.

(viii) Smart locks and garage door openers, Password based or facial recognition based doors and locks.

(ix) **Smart security cameras** – Security cameras that can identify known and unknown person and raise alarm, in case of security threat.

(x) **Smart traffic signals** – Signal that can adjust their timing to accommodate commutes and holiday traffic and keep cars moving.

2. Explain advantages and disadvantages of IoT. (March 2024)

Ans. :

Advantages of IoT are as follows :

(i) **Efficient resource utilization :** If we know the functionality and the way that how each device works, we definitely increase the efficient resource utilization.

(ii) **Minimize human effort :** As the devices of IoT interact and communicate with each other and do lot of task for us, then they minimize the human effort.

(iii) **Time saving :** Time is the primary factors in automation which can be saved through IOT platform.

(iv) **Enhance Data Collection :** IoT devices can collect data from environment like weather, sound, pollution etc. This data then can be used to take decisions.

(v) **Improve security :** IoT based security systems helps in increasing security.

Disadvantages of IoT are as follows :

(i) **Privacy :** IoT uses internet due to which personal data may get hack.

(ii) **Complexity :** The designing, developing, maintaining of IoT system is quite complicated.

3. Give some application of IoT. (March 2024)

Ans. :

Following are some applications of IoT :

(i) **Smart thermostats** - Allows users to schedule monitor and remotely control home

(ii) **Smart lighting** - Illumination of light is controlled on the basis of day light.

(iii) Smart locks and garage door openers, Password based or facial recognition based doors and locks.

(iv) **Smart security cameras** – Security cameras that can identify known and unknown person and raise alarm, in case of security threat.

(v) **Smart traffic signals** – Signal that can adjust their timing to accommodate commutes and holiday traffic and keep cars moving.

4.2 Cloud Computing

4. Explain detail about Cloud Computing.

Ans. :

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(ii) There are three primary service models of cloud computing that are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

(iii) **Infrastructure as a service (IaaS) :** IaaS provides automated and scalable environments that provide a high degree of control and flexibility for the user.

(iv) **Platform as a service (PaaS) :** A service provider offers access to a cloud-based environment in which users can develop, manage and deliver applications.

(v) **Software as a service (SaaS) :** A service provider delivers software and applications through the internet or an APL

(vi) There are three basic types of deployment of cloud computing that are Public, Private and Hybrid.

(vii) In public cloud, all the services and supporting infrastructure are managed off-site over the internet and shared across multiple users.

(viii) Private cloud provides I.T services through the internet or a private network to selected users rather than to general public.

(ix) Hybrid cloud environments combines both Public and Private cloud elements.

5. Explain models of Cloud Computing. (March 2022)

Ans. :

(i) Cloud Computing is the on-demand availability of computer system without direct active management by the user.

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- Instead of purchasing hardware , users pay for IaaS on demand.
- **Examples :** Amazon web services (AWS), Microsoft Azure VM, Google Compute Engine (GCE)

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- A service provider offers access to a cloud-based environment in which users can develop, manage and deliver applications.

- PaaS provides a platform with tools to test, develop and host applications in the same environment
- **Examples :** Google App Engine

(v) **Software as a service (SaaS)**

- A service provider delivers software and applications through the internet as an API.
- Applications are accessible from almost any internet-connected device, from virtually anywhere in the world.
- **Examples :** Google's G suite, GitHub , Dropbox.

6. Explain types of Cloud Computing. March 2022

Ans. :

(i) There are three basic types of deployment of cloud computing that are Public, Private and Hybrid.

(ii) **Public Cloud :** In public cloud, all the services and supporting infrastructure are managed off-site over the internet and shared across multiple users.

(iii) **Public Cloud :** These are more efficient and inexpensive than private and hybrid cloud solutions.

(iv) **Private Cloud :** As the name suggests private cloud provides I.T services through the internet or a private network to selected users rather than to general public.

(v) In private cloud data is protected behind the firewall.

(vi) Private cloud solutions are preferred for enhanced security and privacy by the users.

(vii) **Hybrid Cloud :** Hybrid cloud environments combines both Public and Private cloud elements.

(viii) Communication in hybrid cloud is done over an encrypted connection and allows the portability of data and applications.

(ix) Hybrid cloud allows greater flexibility as compared to that of public and private cloud solutions

7. Explain some benefits of Cloud Computing. March 2022

Ans. :

Benefits of Cloud Computing are as follows :

(i) **Cost saving :** Cloud computing solutions are inexpensive than physical infrastructure.

(ii) **Reliable :** Cloud computing solutions are more reliable than In-house IT infrastructure.

(iii) **Mobility :** Cloud computing solutions are more portable because user can access data anytime, anywhere as required.

(iv) **Scalability :** Cloud computing is more scalable than physical storage.

(v) **Automatic update :** Data on cloud gets updated automatically

4.3 AI(Artificial Intelligence)

8. Give brief idea about AI. March 2022

Ans. :

(i) Artificial Intelligence is the study of making intelligent machine's that can perform operation like human being.

(ii) AI is different from robotics, but related to some extent, in which machines sense their environment, perform calculations and do physical tasks either by themselves.

Different fields of AI are as follows :

(iii) **Machine learning** is the study of computer algorithms that improve automatically through experience.

(iv) **Neural networks** imitate the brain's array of interconnected neurons, and relay information between various units to find connections and derive meaning from data.

(v) **Deep learning** utilizes really big neural networks and a lot of computing power to find complex patterns in data, for applications such as image and speech recognition.

(vi) **Natural language** processing involves analyzing and understanding human language and responding to it.

9. Define AI with sub fields of AI.

Ans. :

(i) Artificial intelligence (AI) is an area of computer science that emphasizes on creation of intelligent machines that work and reacts like humans.

(ii) AI is different from robotics, but related to some extent, in which machines sense their environment, perform calculations and do physical tasks either by themselves or under the direction of people.

(iii) Different fields of AI are as follows:-

(iv) Machine learning is the study of computer algorithms that improve automatically through experience.

(v) Neural networks imitate the brain's array of interconnected neurons, and relay information between various units to find connections and derive meaning from data.

(vi) Deep learning utilizes really big neural networks and a lot of computing power to find complex patterns in data, for applications such as image and speech recognition.

(vii) Natural language processing involves analyzing and understanding human language and responding to it.

(viii) Cognitive computing is about creating a "natural, human-like interaction", including using the ability to interpret speech and respond to it.

(ix) Computer vision employs pattern recognition and deep learning to understand the content of pictures and videos, and to enable machines to use real-time images to make sense of what's around them.

10. What are the advantages and disadvantages of AI?

Ans. :

Advantages of Artificial Intelligence are as follows :

(i) **Reduction in human error :** Computers if programmed properly with artificial Intelligence gives 100% accuracy as compared to task performed by human as there is always a chance for human mistakes.

(ii) **Digital Assistance :** Some of the highly advanced organizations use digital assistants to interact with users which save the need for human resources. Example Chabot

(iii) **Faster Decisions :** Using AI alongside other technologies, we can make machines take decisions faster than a human and carry out actions quicker.

(iv) **Daily Applications :** Daily applications such as Apple's Siri, Window's Cortara and Google's OK Google are frequently used in our daily routine whether it is for searching a location, taking a selfie, making a phone call, replying to a mail and many more.

Disadvantages of AI are as follows :

(i) **High Costs of Creation :** As the machines used in AI based environments are very complex and high in price, it increases the cost for overall set up.

(ii) **Unemployment :** As AI is replacing the majority of the repetitive tasks and other works with robots. This will reduce human interference but cause major problems in the employment standards.

4.4 5G

11. Explain concept of 5G. (March 2022)

Ans. :

(i) 5G is the fifth generation of cellular network technology.

(ii) 5G is the next generation of wireless communications. It is expected to provide Internet connections that are multiple times faster than 4G LTE (Long Term Evolution).

(iii) 5G technology may use a variety of spectrum bands, including millimeter wave (mmWave) radio spectrum, which can carry very large amounts of data at a short distance.

(iv) The drawback of the higher frequencies is that they are more easily obstructed by the walls of buildings, trees and other foliage, and even changes in the weather.

(v) The new 5G networks will be able to transmit very large amounts of data—but only a few blocks at a time.

(vi) 5G networks are digital cellular networks, in which the service area covered by providers is divided into small geographical areas called cells.

(vii) 5G can support up to a million devices per square kilometer, compared to 4G.

12. Explain features of 5G. (March 2022)

Ans. :

(i) It is much faster than 4G.

(ii) It reduces website load time.

(iii) It enables high speed data transfer.

(iv) It can stream high resolution videos easily.

(v) It is more reliable than 4G.

(vi) IoT based application are easy to build using 5G.

Self Assessment

Time : 1 Hour **Marks: - 40**

1. Fill in the blanks (05)

1. _____ is a model for enabling, convenient or demand network access to a shared pool of computing resources.
2. _____ gives user access to storage, networking, servers and other computing resources via the cloud.
3. In SaaS the application reside on a remote cloud network accessed through the _____ or an API.
4. _____ is secure on cloud.
5. Machine learning is a sub field of_____.

2. True or False (05)

1. The data collected using IoT devices is used for taking decisions.
2. Public cloud are more efficient and inexpensive than private and hybrid cloud.
3. Computer is programmed properly with artificial intelligence gives 100% accuracy.
4. 4G is the next generation of wireless communication.
5. AI helps to take faster decision as compare to humans.

3. MCQ one correct alternative. 05

1. IoT device can collect data from _____.
   (a) environment (b) internet
   (c) books (d) documents
2. SaaS Stand for _____.
   (a) Software as a Service (b) Software as a self
   (c) Software as Secure (d) Software as a Section
3. _____ imitate the brain's array of interconnected neurons.
   (a) Machine learning (b) Neural network
   (c) Deep learning (d) Computer vision
4. _____ is the next generation of wireless communication.
   (a) 5G (b) 4G (c) 3G (d) 6G
5. _____ is about creating a "natural, human-like interaction".
   (a) Machine learning (b) Neural network
   (c) Cognitive computing (d) Computer vision

4. MCQ two correct alternative. 06

1. Primary service models of cloud computing are _____.
   (a) SaaS (b) PaaS (c) FaaS (d) DaaS
2. Following are the types of cloud.
   (a) Public Cloud (b) Private Cloud
   (c) Personal Cloud (d) secure Cloud
3. Disadvantages of AI are _____.
   (a) Digital Assistance (b) Faster Decisions
   (c) Cost of creation (d) Unemployment

5. MCQ three correct alternative. 06

1. Internet of Things (IoT) is the network of _____.
   (a) Physical devices (b) sensors (c) actuators
   (d) wires (e) Time saving (f) security
2. Types of Cloud are _____.
   (a) Public Cloud (b) Private Cloud (c) Personal Cloud
   (d) Hybrid Cloud (e) My Cloud (f) House Cloud

6. Match the Following. 05

A	B
1. Public Cloud	(a) Software as a Service
2. IaaS	(b) Shared across multiple users
3. SaaS	(c) Infrastructure as a Service
4. PaaS	(d) All data is protected behind the firewall
5. Hybrid Cloud	(e) Platform as a service
6. Private Cloud	(f) Allows greater flexibility

(Note : Please see Q. 7 and answer of Questions 1 to 6 on Page No. 3-80) 20

5 Server-Side Scripting (PHP)

Chapter at a Glance

- 5.1 Introduction to PHP
- 5.2 Server Side Scripting
- 5.3 Features of PHP
  - PHP Features : - Simple, Faster, Open Source, Platform Independent, Case Sensitive, Error reporting, Interpreted etc.
- 5.4 First Sample code of
  - PHP file extension, php, how to execute PHP program, how to save code in specific folder
  - PHP Case Sensitivity, PHP variables, rules for declaring variables, different variable scopes in PHP with examples
  - PHP Data Types : - String, Integer, Float, Boolean, Array, NULL Comments in PHP, Single line and multi-line comments.
  - Control structures in PHP : - If statement, if else statement Loop structure in PHP : - For, foreach with syntax and examples
- 5.5 PHP String Functions
  - strlen(), strrev(), substr(), strtolower(), trim(), ucwords(), str_replace(), str_word_count(), strops(), substr_count()
- 5.6 PHP Arrays
  - Create an array, types of array, indexed array, multi-dimensional arrays, PHP associative arrays with syntax and examples.
- 5.7 PHP User Defined Functions
  - Function meaning, syntax, function arguments, returning a value to a function with examples
- 5.8 PHP form Handling
  - Simple HTML form with php code, GET vs POST with examples, create database connection object, SQL statement and its execution, cookies and sessions in PHP

3. MCQ one correct alternative. (03)

1. IoT device can collect data from _____.
   (a) environment (b) internet
   (c) books (d) documents
2. SaaS Stand for _____.
   (a) Software as a Service (b) Software as a self
   (c) Software as Secure (d) Software as a Section
3. _____ imitate the brain's array of interconnected neurons.
   (a) Machine learning (b) Neural network
   (c) Deep learning (d) Computer vision
4. _____ is the next generation of wireless communication.
   (a) 5G (b) 4G (c) 3G (d) 6G
5. _____ is about creating a "natural, human-like interaction".
   (a) Machine learning (b) Neural network
   (c) Cognitive computing (d) Computer vision

4. MCQ two correct alternative. (06)

1. Primary service models of cloud computing are _____.
   (a) SaaS (b) PaaS (c) FaaS (d) DaaS
2. Following are the types of cloud.
   (a) Public Cloud (b) Private Cloud
   (c) Personal Cloud (d) secure Cloud
3. Disadvantages of AI are _____.
   (a) Digital Assistance (b) Faster Decisions
   (c) Cost of creation (d) Unemployment

5. MCQ three correct alternative. (06)

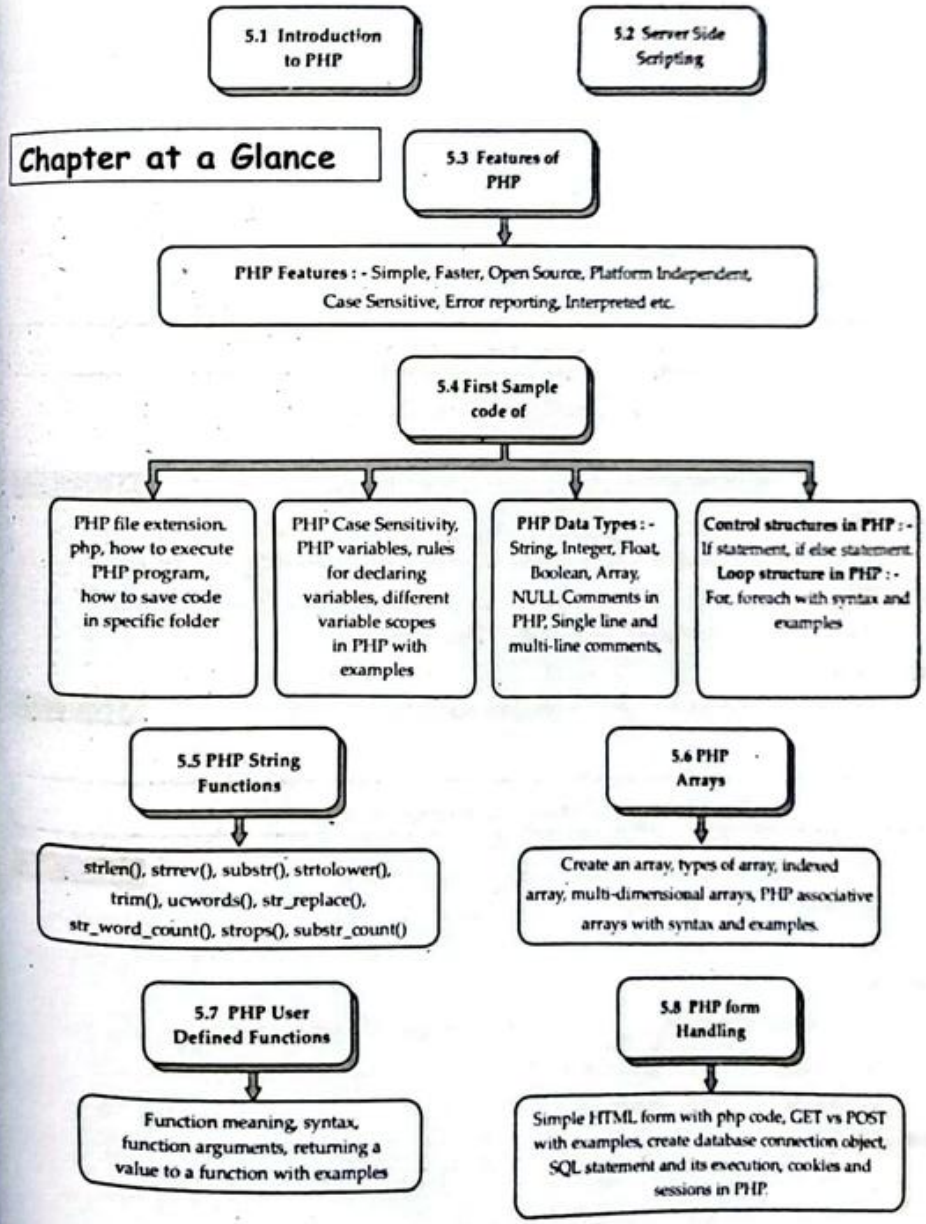
1. Internet of Things (IoT) is the network of _____.
   (a) Physical devices (b) sensors (c) actuators
   (d) wires (e) Time saving (f) security
2. Types of Cloud are _____.
   (a) Public Cloud (b) Private Cloud (c) Personal Cloud
   (d) Hybrid Cloud (e) My Cloud (f) House Cloud

6. Match the Following. (03)

A	B
1. Public Cloud	(a) Software as a Service
2. IaaS	(b) Shared across multiple users
3. SaaS	(c) Infrastructure as a Service
4. PaaS	(d) All data is protected behind the firewall
5. Hybrid Cloud	(e) Platform as a service
6. Private Cloud	(f) Allows greater flexibility

(Note : Please see Q. 7 and answer of Questions 1 to 6 on Page No. 3-80)

5 Server-Side Scripting (PHP)



Exercise

Fill in the blanks

1. PHP is _____ scripting language. (March 2022)

Ans. : Server side

2. PHP is _____ language i.e. there is no need for compilation.

Ans. : interpreted

3. A variable starts with _____ sign followed by variable name.

Ans. : $

4. An _____ is a variable, which can hold more than one value at a time.

Ans. : array

5. Information can be passed to functions through _____.

Ans. : arguments

State True/False

1. PHP is platform dependent scripting language.

Ans. : False

2. $_POST is an array of variables passed via the URL parameters. (March 2022)

Ans. : False

3. A Function is a block of statements that can be used repeatedly in a program.

Ans. : True

4. PHP cannot be embedded along with HTML tags.

Ans. : False

5. GET should NEVER be used for sending sensitive information. (March 2022)

Ans. : True

Multiple Choice Question (1 correct)

1. The program file of PHP have _____ extension. (March 2022)

(a) .asp (b) .php

(c) .js (d) .txt

Ans. : (b)

2. A variable declared _____ a function has global scope.

(a) outside (b) anywhere

(c) inside (d) none

Ans. : (a)

3. The _____ function returns a part of a string. (March 2022)

(a) trim() (b) ucwords()

(c) substr() (d) strpos()

Ans. : (c)

Multiple Choice Question. (2 correct)

1. The _____ & _____ are valid datatype in PHP.

(a) Double (b) Varchar

(c) Integer (d) Array (e) BigInt

Ans. : (c,d)

2. Single line comment in PHP is possible using _____.

(a) // (b) /* */

(c) # (d) <!> (e) $

Ans. : (a,b)

Multiple Choice Question. (3 correct)

1. In PHP, three types of arrays are _____. (March 2022)

(a) Indexed (b) Simple

(c) Associative (d) Multidimensional

(e) Complex (f) General

Ans. : (a,c,d)

2. The scope of variable can be _____. (March 2022)

(a) local (b) global (c) universal

(d) static (e) final (f) outside

Ans. : (a,b,d)

Brief Questions

1. Explain any two features of PHP?

Ans. : Refer answer the following Q. 3, 5.22

2. What are the rules to declare variable in PHP?

Ans. : Refer answer the following Q.6, 5.23

3. What is server sidescripting?

Ans. : Refer answer the following Q. 2, 5.22

4. List the supported datatypes in PHP

Ans. : Refer answer the following Q. 8, 5.23

5. Explain any two string manipulation function.

Ans. : Refer answer the following Q. 15, 5.26

Write Programs for the following

1. Write a PHP code which calculates square of any number using form.

Ans. : Refer PHP Extra programs for practice Program No 3

2. Write a PHP code to count no. of words in the given string.

Ans. : Refer PHP Extra programs for practice Program No 13

3. Create a website with two PHP webpage in which each webpage is connected. The first page of the website contains two form fields for taking 'name' and 'password' from users. On onclick event, details of forms should be displayed on second webpage.

Ans. : Refer PHP Extra programs for practice Program No 14

Fill in the Blanks

5.2 Server Side Scripting

1. A .....is a Computer system that serves as a central repository of data and programs.

Ans. : Server

2. The server side environment that runs a scripting language is called .....

Ans. : Web Server

3. PHP is .....side scripting language. (March 2022)

Ans. : Server

5.3 Features of PHP

4. PHP is .....language, there is no need for compilation.

Ans. : Interpreted

5.4 First Sample code of PHP

5. PHP files have extension ..... (March 2022)

Ans. : .php

6. .....keyword is used to display text on web page. (March 2022)

Ans. : echo

7. .....is a symbol or name that stands for a value.

Ans. : Variable

8. A variable starts with the .....sign followed by the name of the variable. (March 2022)

Ans. : $

9. A variable name cannot start with a .....

Ans. : Number

10. A variable declared outside a function has a .....scope and can only be accessed outside a function. (March 2022)

Ans. : Global

11. A variable declared within a function has a .....scope and can only be accessed within that function. (March 2022)

Ans. : Local

12. .....are the statements in PHP code, which are not visible in the output of the program.

Ans. : Comments

13. The .....statement allows programmer to make decision based on one more conditions.

Ans. : if

14. The .....statement allows programmer to make decision based on either this or that condition.

Ans. : if else

15. .....are used to execute the same block of code repeatedly as long as certain condition is satisfied.

Ans. : loop

16. .....loop works only on arrays.

Ans. : foreach

17. .....functions returns the length of a string. (March 2022)

Ans. : strlen()

18. .....functions counts the number of words in a string.

Ans. : str_word_count()

19. .....function reverses a string. (March 2022)

Ans. : strrev()

20. .....function searches for a specific text within a string.

Ans. : strpos()

21. .....function replaces some characters with some other characters in a string.

Ans. : str_replace()

22. .....function returns a part of a string. (March 2022)

Ans. : substr()

23. .....function converts a string to lowercase. (March 2022)

Ans. : strtolower()

24. .....function counts the number of times a substring occurs in a string.

Ans. : substr_count()

25. .....converts the first character of each word in a string to uppercase. (March 2022)

Ans. : ucwords()

26. .....functions removes whitespace and other predefined characters from both sides of a string. (March 2022)

Ans. : trim()

5.5 PHP String Functions

27. A .....is series of characters. (March 2022)

Ans. : string

28. A .....is a block of statements that can be used repeatedly in a program.

Ans. : function

5.6 PHP Arrays

29. An ..... is a special variable which can hold more than one value at a time.

Ans. : array

30. An ..... stores multiple values in single variable. (March 2022)

Ans. : array

31. The ..... PHP functions is used to create an array variable.

Ans. : array()

32. There are ..... types of arrays in PHP.

Ans. : Three

33. The ..... arrays are the arrays with numeric index. (March 2022)

Ans. : Indexed

34. The ..... arrays are the arrays with named keys. (March 2022)

Ans. : Associative

35. The ..... arrays are the arrays containing one or more arrays. (March 2022)

Ans. : Multi-dimensional

36. The ..... function is used to return the length of an array.

Ans. : count()

5.7 PHP User Defined Functions

37. A ..... is a block of statements that can be used repeatedly in a program. (March 2022)

Ans. : function

38. Information can be passed to functions through .....

Ans. : Arguments

39. ..... are specified after the function name, inside the parentheses.

Ans. : Arguments

40. The ..... statement is used to let a function return a value.

Ans. : return

5.8 PHP Form Handling

41. The form data is sent with the ..... method.

Ans. : HTTP POST

42. ..... is an array of variables passed via the URL parameters. (March 2022)

Ans. : $_GET

43. ..... is an array of variables passed via the URL parameters. (March 2022)

Ans. : $_POST

44. Information sent from a form with ..... method is visible to everyone.

Ans. : GET

45. ..... method may be used for sending non sensitive data. (March 2022)

Ans. : GET

46. Information sent from a form with ..... method is invisible to everyone.

Ans. : POST

47. ..... method has no limits on the amount of information to send.

Ans. : POST

48. ..... object helps to connect PHP code with any database server.

Ans. : PDO()

49. ..... method in PHP is used to check whether variable has value or not.

Ans. : isset()

50. A ..... is a small text file that the server sends on the user's computer.

Ans. : Cookie

51. ..... is used to identify user or its machine and track activities created on the user computer.

Ans. : Cookie

52. When browser requests server page ..... are sent along with the request. (March 2022)

Ans. : cookie

53. ..... is used to store user information on server to track user activities.

Ans. : session

54. ..... helps web application to maintain user information on all the pages. (March 2022)

Ans. : Session

True or False

5.1 Introduction to PHP

1. PHP is widely used open source general purpose scripting language.

Ans. : True

2. PHP does not supports databases.

Ans. : False

3. PHP runs on client side. (March 2022)

Ans. : False

5.2 Server Side Scripting

4. The server side environment that runs a scripting language is termed web server.

Ans. : True

5. PHP hides the code from the user.

Ans. : True

5.3 Features of PHP

6. PHP is very difficult to use. (March 2022)

Ans. : False

7. There is no need of compilation for PHP code.

Ans. : True

8. PHP is faster than other scripting language.

Ans. : True

9. You need to pay for use of PHP. (March 2022)

Ans. : False

10. PHP is platform independent.

Ans. : True

11. PHP is case sensitive only at the time of variable declaration.

Ans. : True

5.4 First Sample code of PHP

12. The PHP code can also be embedded with HTML tags using <? PHP and ? >

Ans. : True

13. PHP files are saved with .html extension. (March 2022)

Ans. : False

14. echo keyword is used to display text on web page.

Ans. : True

15. Variables are used to store values. (March 2022)

Ans. : True

16. A variable starts with $ sign, followed by name of variable.

Ans. : True

17. A variable name can starts with a number.

Ans. : False

18. A variable name can contain only alpha-numeric characters and underscores. (March 2022)

Ans. : True

19. A variable declared outside a function has a GLOBAL SCOPE. (March 2022)

Ans. : True

20. A variable declared within a function has a LOCAL SCOPE. (March 2022)

Ans. : True

21. Variables can store data of different types.

Ans. : True

22. To check data type of variable var _ put ( ) method is used. (March 2022)

Ans. : False

23. Comments are the non executable statement in PHP code. (March 2022)

Ans. : True

24. If statement allows to make decision based on one or more conditions.

Ans. : True

25. For each loop works only on arrays. (March 2022)

Ans. : True

26. StrLength() function returns the length of a string.

Ans. : False

27. Str_word_count() function counts the number of words in a string.

Ans. : True

28. Strrev() function reverses a string. (March 2022)

Ans. : True

29. Strpos() function returns a string. (March 2022)

Ans. : True

30. Str_replace() function replaces some characters with some other characters in a string.

Ans. : True

31. Substr() function returns a part of a string. (March 2022)

Ans. : True

32. Strlower() function converts a string to lower case. (March 2022)

Ans. : False

33. Substr_count() function counts the number of times a substring occurs in a string.

Ans. : True

34. Ucwords() function converts the first characters of each word in a string to uppercase. (March 2022)

Ans. : True

35. trim() function adds white space and other pre defined characters from both sides of a string.

Ans. : False

5.5 PHP String Functions

36. A string is a series of characters. (March 2022)

Ans. : True

37. A function is a block of statements that can be used repeatedly in a program.

Ans. : True

5.6 PHP Arrays

38. Array can store multiple values at a time.

Ans. : True

39. In PHP, array() function is used to create a array. (March 2022)

Ans. : True

40. Syntax of creating array variable is a = arr (values).

Ans. : False

41. In PHP there are six types of arrays.

Ans. : False

42. An array with a numeric index is called indexed arrays.

Ans. : True

43. Multi-dimensional arrays are the arrays with named keys. (March 2022)

Ans. : False

44. Associative array contains one or more arrays.

Ans. : False

45. Array index always starts with zero.

Ans. : True

46. The len() function is used to return the length of an array.

Ans. : false

47. PHP can handle multiple levels of multi dimensional arrays.

Ans. : True

48. We can create our own functions in PHP.

Ans. : True

5.7 PHP User Defined Functions

49. A function is a block of statements that can be used repeatedly in a program.

Ans. : True

50. A function name cannot start with a number. (March 2022)

Ans. : True

51. Information can be passed to functions through methods.

Ans. : False

52. Arguments are specified after the function name, inside the parentheses. (March 2022)

Ans. : True

53. Return statement is used to return a value of a function. (March 2022)

Ans. : True

5.8 PHP Form Handling

54. The form data is sent with the HTTP Post method.

Ans. : True

55. The PHP superglobals $_GET and $_POST are used to collect form data. (March 2022)

Ans. : True

56. $_GET and $_POST syperglobals which means that they are never accessible.

Ans. : False

57. $_GET is an array of variables passed via the URL parameters.

Ans. : True

58. $_POST is an array of variables passed via the HTTP POST method. (March 2022)

Ans. : True

59. Information sent from a form with GET method is not visible to everyone. (March 2022)

Ans. : False

60. GET has limits on the amount of information to send. (March 2022)

Ans. : True

61. GET should NEVER be used for sending passwords or other sensitive information.

Ans. : True

62. Information sent from a form with POST method is invisible to everyone. (March 2022)

Ans. : True

63. POST has limits on the amount of information to send. (March 2022)

Ans. : False

64. isset() method in PHP is used to check whether variable has value or not. (March 2022)

Ans. : True

65. A cookie is a small text file the client sends on the user's computer.

Ans. : False

66. When browser requests server page, cookies are sent along with the request.

Ans. : True

67. Cookies store visited page on browser to optimize search. (March 2022)

Ans. : True

68. Session is used to store user information on server to track user activities.

Ans. : True

69. Session helps web application to maintain user information on all the pages.

Ans. : True

70. PHP Data Object helps us to connect PHP code in Uniform Method of access to multiple databases. (March 2022)

Ans. : True

MCQ (One Correct Answers)

5.1 Introduction to PHP

1. ..... is a widely used open source server side programming language which runs on various platforms.
   (a) Word (b) PHP (c) Excel (d) PowerPoint

Ans. : (b)

5.2 Server Side Scripting

2. A .....is a Computer system that serves as a central repository of data and programs.
   (a) Server (b) Network (c) Client (d) Object

Ans. : (a)

3. The server side environment that runs a scripting language is termed .....
   (a) Browser (b) Computer (c) Web server (d) Client (March 2022)

Ans. : (c)

5.3 Features of PHP

4. .....is a script executed on server which generate dynamic HTML pages.
   (a) PHP (b) MS Word (c) MSExcel (d) Client

Ans. : (a)

5.4 First Sample code of PHP

5. The PHP code can also be embedded with HTML tags using .....
   (a) <!Php!> (b) <?php and ?>
   (c) <=php> (d) <#php#>

Ans. : (b)

6. PHP files are saved with ..... extension. (March-2022)
   (a) .html (b) .xml (c) .js (d) .php

Ans. : (d)

7. .....keyword is used to display text on web page.
   (a) echo (b) print (c) write (d) cin

Ans. : (a)

8. .....is a symbol or name that stands for a value.
   (a) Method (b) condition (c) Variable (d) Value

Ans. : (c)

9. A variable in PHP starts with the .....sign followed by name of the variable.
   (a) # (b) @ (c) = (d) $ (March 2022)

Ans. : (d)

10. PHP variable name cannot starts with a ..... (March 2022)
    (a) Number (b) Character
    (c) Alphanumeric (d) Dollar

Ans. : (a)

11. Correct method for declaring PHP variable is ..... (March 2022)
    (a) # age (b) $ age (c) @ age (d) age

Ans. : (b)

12. A variable declaration .....a function has a GLOBAL SCOPE. (March 2022)
    (a) inside (b) middle (c) outside (d) topside

Ans. : (c)

13. A variable declared .....a function has a LOCAL SCOPE (March 2022)
    (a) Inside (b) outside (c) between (d) within

Ans. : (d)

14. PHP stores all .....variables in an array called $ GLOBAL[Index] (March 2022)
    (a) local (b) global (c) Static (d) within

Ans. : (b)

15. To check data type of variable in PHP .....method is used.
    (a) var_dump() (b) var _del()
    (c) var_remove() (d) var_delete()

Ans. : (a)

16. .....are the statements in PHP code which are not visible in the output of the program.
    (a) Methods (b) comments (c) Objects (d) values

Ans. : (b)

17. How to add a single line comment in PHP ?
(a) // (b) # (c) $ (d) @
Ans. : (a)

18. How to a multiline comment in PHP ?
(a) // (b) # (c) $ (d) /* and */
Ans. : (d)

19. ..... statement allows programmer to make decision, based on one or more conditions. (March 2022)
(a) if (b) loop (c) variable (d) method
Ans. : (a)

20. .....statement allows programmer to make decisions based on either this or that decision. (March 2022)
(a) loop (b) if else
(c) variable (d) method
Ans. : (b)

21. .....are used to execute the same block of code repeatedly as long as a certain condition is satisfied. (March 2022)
(a) Variable (b) Method (c) Object (d) loops
Ans. : (d)

22. .....loop works only on arrays. (March 2022)
(a) for next (b) for (c) foreach (d) forone
Ans. : (c)

23. .....function returns the length of a string. (March 2022)
(a) strrev() (b) strlen() (c) strlength() (d) substr()
Ans. : (b)

24. .....function counts the number of words in a string.
(a) strlen() (b) str_word_count()
(c) substr() (d) trim()
Ans. : (b)

25. .....function reverses a string. (March 2022)
(a) strrev() (b) trim() (c) UCwords() (d) strops()
Ans. : (a)

26. .....function searches for a specific text within a string. (March 2022)
(a) substr() (b) strrev() (c) strpos() (d) str_replace()
Ans. : (c)

27. .....function replaces some characters with same other character in a string.
(a) strrev() (b) str _replace()
(c) trim() (d) strpos()
Ans. : (b)

28. .....function returns a part of a string. (March 2022)
(a) substr() (b) trim() (c) strrev() (d) str_len()
Ans. : (a)

29. .....function converts a string to lowercase. (March 2022)
(a) UCwords() (b) strtolower()
(c) strlower() (d) trim()
Ans. : (b)

30. .....function counts the number of times substring occurs in a string.
(a) substr() (b) substr_count()
(c) trim() (d) strlen() (March 2022)
Ans. (b)

31. .....function converts the first character of each word in a string to upper case.
(a) substr() (b) strtolower()
(c) trim() (d) ucwords() (March 2022)
Ans.: (d)

32. .....function removes whitespace and other predefined characters from both sides of a string. (March 2022)
(a) trim() (b) strlen() (c) straps() (d) strrev()
Ans. : (a)

5.5 PHP String Functions

33. A ..... is series of characters. (March 2022)
(a) String (b) Number (c) Object (d) Value
Ans. : (a)

34. A ..... is a block of statements that can be used repeatedly in a program.
(a) function (b) class (c) method (d) variable (March 2022)
Ans. : (a)

5.6 PHP Arrays

35. .....is a variable which can hold more than one value at a time. (March 2022)
(a) Array (b) Method (c) Object (d) Constant
Ans. : (a)

36. In PHP ..... function is used to create array.
(a) arr() (b) array() (c) value() (d) variable()
Ans. : (b)

37. Syntax of creating array variable is ..... (March 2022)
(a) @ a = array() (b) #a = arr()
(c) = array() (d) $a = array (values)
Ans. : (d)

38. There are .....types of arrays in PHP. (March 2022)
(a) Two (b) One (c) Three (d) Four
Ans. : (c)

39. .....arrays are the arrays with a numeric index. (March 2022)
(a) Associative (b) Indexed
(c) Multi-dimensional (d) Normal
Ans. : (b)

40. .....arrays are the arrays with named keys.
(a) Indexed (b) Associative
(c) Dimensional (d) Normal
Ans. : (b)

41. .....arrays are arrays containing one or more arrays.
(a) Multi-dimensional (b) Indexed
(c) Associative (d) Numbered
Ans. : (a)

42. Array index always starts with .....
(a) one (b) two (c) zero (d) four
Ans. : (c)

5.7 PHP User Defined Functions

43. A .....is a block of statements that can be repeatedly in a program. (March 2022)
(a) function (b) condition (c) object (d) value
Ans. : (a)

44. Information can be passed to functions through ..... (March 2022)
(a) Condition (b) arguments (c) array (d) variable
Ans. : (b)

5.8 PHP Form Handling

45. The form data is sent with the .....method.
(a) HTTP POST (b) FTP POST
(c) STP POST (d) SMPT POST
Ans. : (a)

46. .....is an array of variables passed via the URL parameters. (March 2022)
(a) $_GET (b) #GET (c) ._GET (d) @_GET
Ans. : (a)

47. .....is an array of variables passed via the HTTP POST method. (March 2022)
(a) $POST (b) $_POST (c) @_POST (d) #_POST
Ans. : (b)

48. Information sent from a form with .....method is visible to everyone.
(a) POST (b) GET (c) HTTP (D) FTP (March 2022)
Ans. : (b)

49. Information sent from a form with .....method is invisible to everyone.
(a) POST (b) GET (c) HTTP (d) FTP
Ans. : (a)

50. .....NEVER be used for sending sensitive information. (March 2022)
(a) POST (b) HTTP (c) GET (d) FTP
Ans. : (c)

51. .....method in PHP is used to check whether variable has value or not.
(a) outset() (b) inset() (c) noset() (d) isset() (March 2022)
Ans. : (d)

52. A .....is a small text file that the server sends on the user's computer.
(a) Cookie (b) session
(c) Presentation (d) Method
Ans.: (a)

53. .....are sent along when browser requests server pages.
(a) Cookie (b) Session (c) Method (d) Server
Ans. : (a)

54. .....is used to store information on server to track user activities.
(a) Cookie (b) session (c) Presentation (d) Object
Ans. : (b)

55. .....helps web application to maintain user information on all the pages.
(a) Session (b) presentation
(c) Cookie (d) object
Ans. : (a)

MCQ (Two Correct Answers)

5.1 Introduction to PHP

1. PHP runs on following platform.
(a) Linux (b) Excel (c) Word (d) Windows
Ans. : (a), (d)

2. PHP is compatible with following servers.
(a) Unix (b) XAMMP (c) Linux (d) Apache
Ans. : (b), (d)

5.2 Server Side Scripting

3. Programming languages for server side programming are .....
(a) PHP (b) Tally (c) Excel (d) Python
Ans. : (a), (d)

5.3 Features of PHP

4. Features of PHP are ..... (March 2021)
(a) Interpreted (b) Open source
(c) Difficult (d) Platform dependent
Ans.: (a), (b)

5.4 First Sample code of PHP

5. Popular PHP Frameworks are .....
(a) Echo (b) Laravel (c) ServerSide (d) Symfony
Ans. : (b), (d)

6. Variables are used for storing values such as .....and .....
(a) Methods (b) Numeric values
(c) Character Strings (d) Functions
Ans. : (b), (c)

7. A variable starts with the .....sign followed by .....of the variable.
(a) $ (b) # (c) name (d) value
Ans. : (a), (c)

8. A variable declared .....a function has GLOBAL SCOPE and a variable declared .....a function has LOCAL SCOPE.
(a) inside (b) outside (c) within (d) between
Ans. : (b), (c)

9. Following are the data types used in PHP .....
(a) Boolean (b) Text (c) Number (d) Array
Ans. : (a), (d)

10. Control structures in PHP are ..... (March 2022)
(a) for (b) if statement
(c) for each (d) if else statement
Ans. : (b), (d)

11. Loop structure in PHP are ..... (March 2022)
(a) for (b) for each (c) if (d) if else
Ans. : (a), (b)

12. Comments in PHP is possible in following ways. (March 2022)
(a) // (b) \\ (c) /* and */ (d) /# and #/
Ans. : (a), (c)

13. Comments are of .....and .....type.
(a) Single line (b) No line (c) Two line (d) Multi-line
Ans. : (a), (d)

14. Following are the pre defined functions in PHP to manipulate string. (March 2022)
(a) trim() (b) substr() (c) length() (D) strreverse()
Ans. : (a), (b)

5.7 PHP User Defined Functions

15. A function name can start with a .....or ..... (March 2022)
(a) symbol (b) Number (c) Letter (d) Underscore
Ans. : (c), (d)

5.8 PHP Form Handling

16. Form data can be submitted by .....or .....method. (March 2022)
(a) Return (b) forward (c) GET (d) POST
Ans. : (c), (d)

17. The PHP super globals .....and .....are used to collect form data.
(a) $_GET (b) $POST (c) $GET (d) $_POST

Ans. : (a), (d)

18. .....is an array of variables passed via URL parameters and .....is an array of variables passed via HTTP POST method. (March 2022)
(a) $GET (b) $_GET (c) $_POST (d) $POST

Ans. : (b), (c)

19. .....has limits on the amount of information to sends whereas .....has no limits on the amount of information to send.
(a) Value (b) Method (c) GET (d) POST

Ans. : (c), (d)

MCQ (Three Correct Answers)

5.1 Introduction to PHP

1. PHP runs on following platforms ..... (March 2022)
(a) Systems (b) Words (c) Linux (d) Excel
(e) Unix (f) Windows

Ans. : (c), (e), (f)

2. PHP is compatible with following servers.
(a) XAMMP (b) Linux (c) Apache
(d) Unix (e) Lightpad (f) Word

Ans. : (a), (c), (e)

5.2 Server Side Scripting

3. Programming languages for server side programming are (March 2022)
(a) PHP (b) Windows (c) Linux (d) Python
(e) Unix (f) Java and JSP

Ans. : (a), (d), (f)

5.3 Features of PHP

4. Features of PHP are ..... (March 2022)
(a) Difficult (b) Simple (c) Error Creating
(d) Platform Independent (e) Error Reporting
(f) Platform Dependent

Ans. : (b), (d), (e)

5.4 First Sample code of PHP

5. Variables are used for storing values such as ....., ..., and .....
(a) Character (b) Numeric (c) Objects (d) Methods
(e) Functions (f) Memory addresses (March 2022)

Ans. : (a), (b), (f)

6. The scope of variable in PHP are ..... (March 2022)
(a) Local (b) Regional (c) National (d) Global
(e) Static (f) International

Ans. : (a), (d), (e)

7. PHP data types are ..... (March 2022)
(a) Text (b) String (c) Number (d) Integer
(e) Empty (f) Boolean

Ans. : (b), (d), (f)

8. Following are the predefined functions in PHP to manipulate string (March 2022)
(a) Lower() (b) strops() (c) substr() (d) strlen()
(e) upper() (f) strlength()

Ans. : (b), (c), (d)

9. Rules for declaring php variables are _____, _____, _____. (March 2022)
(a) A variable name start with $ sign
(b) A variable name can start with a number
(c) Variable names are case sensitive
(d) A variable same start with * sign
(e) A variable name cannot start with number
(f) Variable names are no case sensitivity

Ans. : (a), (c), (e)

10. Comments in PHP can be given using _____, _____ and _____.
(a) # (b) * (c) // (d) $ (e) /* (f) ^

Ans. : (a), (c), (e)

5.6 PHP Arrays

11. In PHP three types of arrays are ....., ..., ... (March 2022)
(a) inside (b) multiple
(c) indexed (d) associate
(e) outside (f) multi-dimensional

Ans. : (c), (d), (f)

Answer in Briefly

5.1 Introduction to PHP

1. Explain PHP.

Ans. :

i) PHP (Hypertext Pre-processor) is a widely-used open source general-purpose scripting language.

ii) PHP runs on various platforms (Linux, Unix, Mac OS X, Windows etc.).

iii) PHP supports a wide range of databases. PHP is easy to learn and runs efficiently on the server side.

5.2 Server Side Scripting

2. Explain Server Side Scripting?

Ans. :

(i) A server is a computer system that serves as a central control of data and programs shared by clients.

(ii) The server side environment that runs a scripting language is termed as web server.

(iii)A user's request is fulfilled by running a script directly on the web server.

(iv) It is used to provide interactive web sites.

(v) Programming language for server side programming are PHP, Python JSP.

5.3 Features of PHP

3. Explain features of PHP (any 5).

Ans. : PHP is most popular and frequently used worldwide server side scripting language. Following are features of PHP :

(i) Simple : It is very simple and easy to use, as compared to other scripting languages.

(ii) Interpreted : It is an interpreted language, i.e. there is no need for compilation.

(iii) Faster : It is faster than other scripting language e.g. JSP and ASP.

(iv) Open Source : Open source means you need not pay for use of PHP. You can freely download and use.

(v) Platform Independent : PHP code will be run on every platform, Linux, Unix, Mac OS X, Windows.

5.4 First sample code of PHP

4. Explain how to save and execute PHP program.

Ans. :

(i) Type the PHP code using any text editor (Notepad).

(ii) Save with .php extension in appropriate folder.

(iii) Go to browser and type http://localhost/php/first.php

5. What is a variable?

Ans. :

(i) Variable is a symbol or name that contains value.

(ii) Variables are used for storing values such as numbers or characters.

(iii) The stored values can be used in any part of the program.

6. What are the rules for declaring PHP variables?

Ans. :

Following are the rules for declaring variables in PHP:-

(i) A variable starts with the $ sign, followed by the name of the variable

(ii) A variable name must start with a letter or the underscore character

(iii) A variable name cannot start with a number

(iv) A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)

(v) Variable names are case-sensitive ($age and $AGE are two different variables)

7. Explain variable scopes in PHP.

Ans. :

There are three different variables scopes in PHP :

- local
- global
- static

(i) A variable declared outside a function has a **GLOBAL SCOPE** and can only be accessed outside a function.

(ii) A variable declared within a function has a **LOCAL SCOPE** and can only be accessed within that function.

(iii) If we want a local variable not to be deleted then we have to use of static keyboard.

8. Explain PHP data types.

Ans. :

(i) String: - A string is a sequence of characters. A string can be any text inside quotes. You can use single or double quotes:

(ii) Integer:- An integer data type is a non-decimal number between -2,147,483,648 and 2,147,483,647.

(iii) Float (floating point numbers) :- A float (floating point number) is a number with a decimal point or a number in exponential form.

(iv) Boolean : A Boolean represents two possible states: TRUE or FALSE.

(v) Array:- An array stores multiple values in one single variable.

(vi) NULL:- Null is a special data type which can have only one value: NULL. A variable of data type NULL is a variable that has no value assigned to it.

9. Explain Single line and multi line comments in PHP.

Ans. :

(i) A comment in PHP code is a line that is not executed as a part of the program.

(ii) Its only purpose is to be read by someone who is looking at the code.

(iii) PHP supports several ways of commenting: Single line comment and multi line comment.

(iv) Single line comment can be written in two ways, they are :-

// This is a single-line comment

This is also a single-line comment

(v) Multi line comment can be written as:-

/* This is a multiple-lines comment block that spans over multiple lines */

10. Explain the operators used in PHP.

Ans. : Operators used in PHP are :

1. Arithmetic Operators

Operator	Definition	Example
+	Addition	a+b
-	Subtraction	a-b
*	Multiplication	a*b
/	Division	a/b
%	Modulus	a%b

2. Relational Operators

Operator	Definition	Example
>	Greater than	a>b
<	Less than	a<b
>=	Greater than or equal to	a>=b
<=	Less than or equal to	a<=b
==	Equal to	a==b

11. Explain PHP If statement with syntax.

Ans. :

if statement - executes some code if one condition is true

Syntax of if statement is

```
if (condition) {
  code to be executed if condition is true;
}
```

12. Explain PHP If else statement with syntax.

Ans. :

The if...else statement executes some code if a condition is true and another code if that condition is false.

Syntax of if else is

```
if (condition) {
  code to be executed if condition is true;
} else {
  code to be executed if condition is false;
}
```

13. Explain PHP for loop with syntax.

Ans. :

(i) Loops are used to execute the same block of code again and again, as long as a certain condition is true.

(ii) The for loop is used when you know in advance how many times the script should run.

Syntax

```
for (init counter; test counter; increment counter) {
  code to be executed for each iteration;
}
```

14. Explain PHP foreach with syntax.

Ans. :

The foreach loop works only on arrays, and is used to loop through each key/value pair in an array.

Syntax

```
foreach ($array as $value) {
  code to be executed;
}
```

5.5 PHP String Functions

15. Explain String functions used in PHP (any 5).

Function	Description
strlen()	Returns the length of a string (i.e. total no. of characters)
str_word_count()	Counts the number of words in a string
strrev()	Reverses a string
strpos()	Searches for a specific text within a string and returns the character position of the first match and if no match is found, then it will return false
str_replace()	Replaces some characters with some other characters in a string
substr()	Returns a part of a string
strtolower()	Converts a string to lowercase
substr_count()	Counts the number of times a substring occurs in a string
ucwords()	Converts the first character of each word in a string to uppercase
trim()	Removes whitespace and other predefined characters from both sides of a string

5.6 PHP Arrays

16. Explain Array and how to create arrays in PHP with syntax.

Ans. :

(i) An array is a special variable, which can hold more than one value at a time.

(ii) An array stores multiple values in one single variable

(iii) In PHP the array() function is used to create an array.

(iv) Syntax to create array is :- $a=array(value)

17. Explain types of Arrays in PHP.

Ans. :

In PHP, there are three types of arrays:

(i) **Indexed arrays** - Arrays with a numeric index. For example:- $cars = array("Volvo", "BMW", "Toyota");

(ii) **Associative arrays** - Arrays with named keys. For example :- $age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");

(iii) **Multidimensional arrays** - Arrays containing one or more arrays. For example :- $cars = array

```
(
array("Volvo",22,18),
array("BMW",15,13),
array("Saab",5,2),
array("Land Rover",17,15)
);
```

5.7 PHP User Defined Functions

18. Explain how to declare user defined function in PHP.

Ans. :

(i) A function is a block of statements that can be used repeatedly in a program.

(ii) A function will not execute automatically when a page loads.

(iii) A function will be executed by a call to the function.

(iv) A user-defined function declaration starts with the word function :

Syntax

```
function functionName() {
  code to be executed;
}
```

19. Explain PHP function arguments.

Ans. :

(i) Information can be passed to functions through arguments. An argument is just like a variable.

(ii) Arguments are specified after the function name, inside the parentheses. You can add as many arguments as you want, just separate them with a comma.

(iii) Example is :-

```
<?php
function familyName($fname) {
  echo "$fname Refsnes.<br>";
}
familyName("Jani");
familyName("Hege");
familyName("Stale");
familyName("Kai Jim");
familyName("Borge");
?>
```

5.8 PHP Form Handling

20. Explain difference between GET and POST.

Ans. :

(i) Both GET and POST are treated as superglobals which means that they are always accessible regardless of scope.

(ii) It can be accessed from any function, class or file without having to do anything special.

(iii) $_GET is an array of variables passed via the URL parameters.

(iv) $_POST is an array of variables passed via the HTTP POST method.

21. Explain isset() method in PHP.

Ans. :

(i) isset() method is used in PHP to check whether variable has value or not.

(ii) This helps us to know if the button is clicked or not.

22. Short note on Cookies and d Session in PHP.

Ans. :

(a) Cookies:-

(i) A cookie is often used to identify a user.

(ii) A cookie is a small file that the server embeds on the user's computer.

(iii) Each time the same computer requests a page with a browser, it will send the cookie too.

(iv) With PHP, you can both create and retrieve cookie values.

(b) Session

(i) Session is used to store user information on server to track user activities.

(ii) It helps in web application to maintain user information on all the pages.

(iii) For example, if you login to gmail account, the session helps to access youtube account also.

23. Explain steps to create Database & Tables.

Ans. :

Steps to create database and Table :

1) For database creation → open postgres sql terminal -→ create database college; and press enter.

2) For Table creation -→ create table student(name text, gender Boolean); press enter table will be created.

Programs for Practice

5.4 First sample code of PHP

1. Program to display "Hello world" message on the web page.

Coding :

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Web Page</h1>
<?php
echo "Hello World";
?>
</body>
</html>
```

Steps to save and execute PHP programs:-

1. Type the above program and save it as "demo.php" using any text editor. (for eg Notepad, gedit).
2. Create a folder with your name (for example php).

   Note : Create folder in server's root directory. For Ubuntu the path of root directory is var/www/html. For Windows the path of root directory is c:/XAMPP/htdocs.
3. Save the "demo.php" file in your folder.
4. Open browser and type in address bar http://localhost/php/. Click on demo.php.

Output :

2. PHP Program to print sum of two numbers.

Coding :

```
<?php
$a=10;
$b=20;
$c=$a+$b;
echo "<br><br> Sum of two numbers is $c";
?>
```

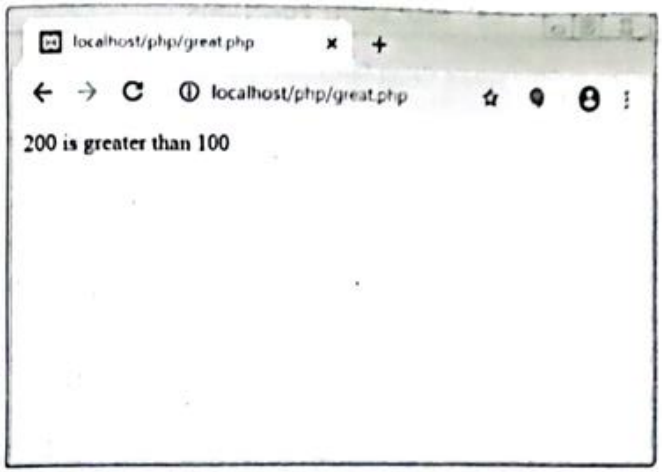
Output :

3. PHP program to check and print greatest among two numbers.

Coding :

```
<?php
$a=100;
$b=200;
if($a>$b)
echo "$a is greater than $b";
else
echo "$b is greater than $a";
?>
```


Output:-

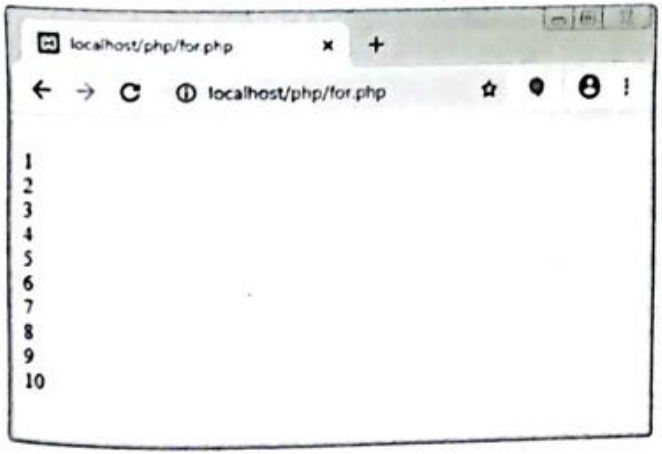


4. PHP program to print numbers from 1 to 10 using for loop.

Coding :

```
<?php
for($i=1;$i<=10;$i++)
{
echo "<br> $i";
}
?>
```

Output :-



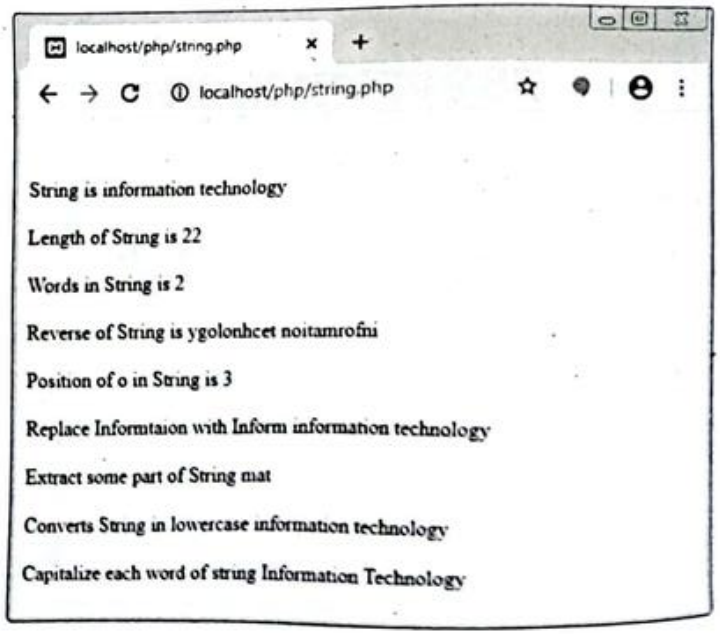
5.5 PHP String Functions

5. PHP program to demonstrate string functions

Coding :

```
<?php
$a="information technology";
echo "<br><br> String is ".$a;
echo "<br><br> Length of String is ".strlen($a);
echo "<br><br> Words in String is ".str_word_count($a);
echo "<br><br> Reverse of String is ".strrev($a);
echo "<br><br> Position of o in String is ".strpos($a,"o");
echo "<br><br> Replace Informtaion with Inform ".str_replace
("Information","Inform",$a);
echo "<br><br> Extract some part of String ".substr($a,5,3);
echo "<br><br> Converts String in lowercase ".strtolower($a);
echo "<br><br> Capitalize each word of string ".ucwords($a);
?>
```

Output :



5.6 PHP Arrays

6. PHP program to print the name of colors store in array variable.

Coding

```
<?php
$x=array("Red","Blue","Brown","Pink","Purple");
echo "Color names are <br>";
foreach($x as $value)
{
echo "$value <br>";
}
?>
```

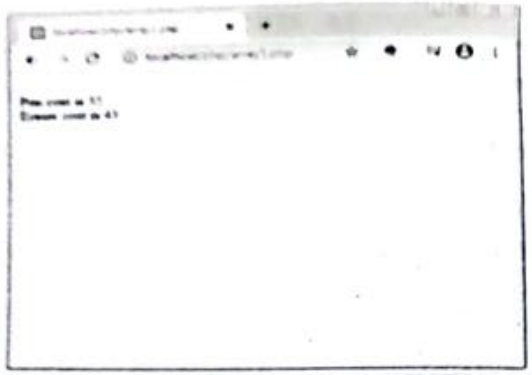
Output :

7. PHP program to print the cost of product store in array variable.

Coding :

```
<?php
$p = array("Pen"=>"35", "Ball"=>"37", "eraser"=>"43");
echo "<br>Pen cost is ".$p['Pen'];
echo "<br>Eraser cost is ".$p['eraser'];
?>
```

Output :



8. PHP program to print the information about cars sold and cars in stock.

Coding :

```
<?php
$cars = array
 (
 array("Volvo",22,18),
 array("BMW",15,13),
 array("Saab",5,2),
 array("Land Rover",17,15)
 );
echo $cars[0][0].": In stock: ".$cars[0][1].", sold: ".$cars[0][2].".<br>";
echo $cars[1][0].": In stock: ".$cars[1][1].", sold: ".$cars[1][2].".<br>";
echo $cars[2][0].": In stock: ".$cars[2][1].", sold: ".$cars[2][2].".<br>";
echo $cars[3][0].": In stock: ".$cars[3][1].", sold: ".$cars[3][2].".<br>";
?>
```

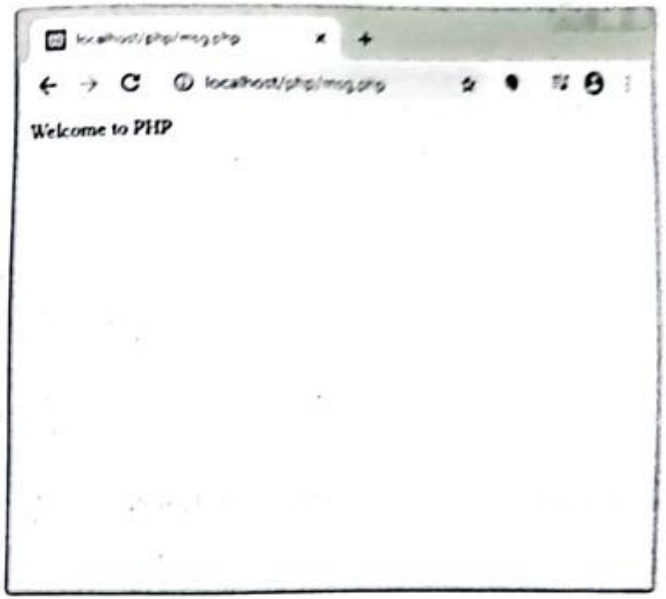
5.7 PHP User Defined Functions

9. PHP program to print message "Welcome to PHP" using function.

Coding :

```
<?php
function msg()  // declaring a function
{
echo "Welcome to PHP";  //defining function
}
msg(); // calling function
?>
```


Output :



10. PHP program to print Name and Price of the product by passing arguments to function.

Coding :

```
<?php
function pr($n,$price)
{
echo "Product name is $n and Price is $price <br><br>";
}
pr("Notebook",45);
pr("Calculator",400);
pr("Pen",30);
?>
```

Output :

11. PHP program to print sum of two numbers using function.

Coding :

```
<?php
function sum(int $a,int $b)
{
$c=$a+$b;
return $c;
}
echo "10+20= " .sum(10,20). "<br><br>";
echo "11+2= " .sum(11,2). "<br><br>";
echo "31+9= " .sum(31,9). "<br><br>";
?>
```


Output :

5.8 PHP Form Handling

12. Create a HTML form with two input fields to accept number from the user and a button to print addition of two numbers.

Coding :

Form.html

```
<html>
<body>
<form action="add.php" method =post">
Enter First Number
<input type="text" name="t1"><br><br>
Enter Second Number
<input type="text" name="t2"><br><br>
<input type="submit" value="Add">
</form>
</body>
</html>
```

add.php

```
<?php
$t1=$_GET["t1"];
```

```
$t2=$_GET["t2"];
$c=$t1+$t2;
echo "<br><br> Sum of two numbers is $c";
?>
```

Output :

13. PHP program to create admission form for student. Accept name and gender from the student. Create a college database and create table student having the fields name and gender.

Steps to create database and Table :

1) For database creation → open postgres sql terminal -→ create database college; and press enter.
2) For Table creation -→ create table student(name text, gender Boolean); press enter, table will be created.

Now type code in admission.php

```
<!DOCTYPE html>
<html>
<body>
<h1 align="center">Addmission Form</h1>
<form method="post">
<label>Enter Name</label>
<input type="text" name="name" id="id_name"><br><br>
<label>Gender</label>
<input type="radio" name="gender" id="id_gender" value="male">Male<br><br>
<input type="radio" name="gender" id="id_gender" value="female">FeMale<br><br>
```



```
<input type="radio" name="gender" id="id_gender" value="other">Other<br><br>
<input type="submit" name="submit" value="submit" id="submit">
</form>
</body>
</html>
<?php
$servername="pgsql:host=localhost;dbname=college";
$username="postgres";
$password="laxmi";
$conn=new PDO($servername,$username,$password);
if(isset($_POST['submit']))
{
$name=$_POST["name"];
$gender=$_POST["gender"];
$sql="INSERT INTO student(name,gender) VALUES ('".$name."','".$gender."')";
$conn->exec($sql);
echo "New record added successfully";
}
?>
```

PHP Extra Programs for Practice

1. Write a PHP code which calculates and prints addition, subtraction, multiplication, division of two numbers using form.

Ans. :

Coding :-

```
<html>
<body>
<form method="post">
Enter First Number
<input type="text" name="first"><br><br>
Enter Second Number
<input type="text" name="second"><br><br>
<input type="submit" name="submit" value="Submit">
</form>
```

```
</body>
</html>
<?php
 if(isset($_POST['submit']))
{
$first = $_POST['first'];
$second = $_POST['second'];
$sum =  $first+$second;
$sub =  $first-$second;
$mul =  $first*$second;
$div =  $first/$second;
echo "<br><br>The Addition  is: ".$sum;
echo "<br><br>The Subtraction  is: ".$sub;
echo "<br><br>The Multiplication  is: ".$mul;
echo "<br><br>The Division  is: ".$div;
}
?>
```

2. Write a PHP code which calculates and prints area and circumference of a circle. Hint(area=3.14x r^2 and c=2x3.14xr) using form.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter Radius
<input type="text" name="radius"><br><br>
<input type="submit" name="submit" value="Submit">
</form>
</body>
</html>
<?php
 if(isset($_POST['submit']))
{
```



```
$p=3.14;
$radius = $_POST['radius'];
$a =  $p*$radius*$radius;
$c =  $p*2*$radius;
echo "<br><br>The Area of circle  is: ".$a;
echo "<br><br>The Circumference of circle  is: ".$c;
}
?>
```

3. Write a PHP code which calculates square of a number using form.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter a Number
<input type="text" name="sq"><br><br>
<input type="submit" name="submit" value="Sqaure">
</form>
</body>
</html>
<?php
 if(isset($_POST['submit']))
{
$sq = $_POST['sq'];
$s=$sq*$sq;
echo "<br><br>The Square of number is: ".$s;
}
?>
```

4. Write a PHP code which calculates cube of a number using form.

Ans. :

```
<html>
<body>
<form method="post">
Enter a Number
```

```
<input type="text" name="sq"><br><br>
<input type="submit" name="submit" value="Cube">
</form>
</body>
</html>
<?php
 if(isset($_POST['submit']))
{
$sq = $_POST['sq'];
$s=$sq*$sq*$sq;
echo "<br><br>The Cube of number is: ".$s;
}
?>
```

5. Write a PHP code to check and print greatest among two numbers using form an function.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter First Number
<input type="text" name="f"><br><br>
Enter Second Number
<input type="text" name="s"><br><br>
<input type="submit" name="submit" value="Greatest">
</form>
</body>
</html>
<?php
function grt()
{
 if(isset($_POST['submit']))
{
$f = $_POST['f'];
```



```
$s = $_POST['s'];
if($f>$s)
echo "<br><br>The greater number is " .$f;
else
echo "<br><br>The greater number is " .$s;
}
}
grt();
?>
```

6. Write a PHP code to check and print greatest among three numbers using form.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter First Number
<input type="text" name="f"><br><br>
Enter Second Number
<input type="text" name="s"><br><br>
Enter Third Number
<input type="text" name="t"><br><br>
<input type="submit" name="submit" value="Greatest 3">
</form>
</body>
</html>
<?php
function grt()
{
 if(isset($_POST['submit']))
{
$f = $_POST['f'];
$s = $_POST['s'];
$t = $_POST['t'];
if($f>$s && $f>$t)
```

```
echo "<br><br>The greater number is " .$f;
else
{
if($s>$f && $s>$t)
echo "<br><br>The greater number is " .$s;
else
{
if($t>$f && $t>$s)
echo "<br><br>The greater number is " .$t;
}
}
}
}
grt();
?>
```

7. Write a PHP code to check and print whether the number is Odd or Even using form.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter a Number
<input type="text" name="f"><br><br>
<input type="submit" name="submit" value="Even Odd">
</form>
</body>
</html>
<?php
 if(isset($_POST['submit']))
{
$f = $_POST['f'];
if($f%2==0)
echo "<br><br>The number is Even number";
```



```
else
echo "<br><br>The number is Odd number";
}
?>
```

8. Write a PHP code to check and print whether the number is positive, negative or Zero using form.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter a Number
<input type="text" name="f"><br><br>
<input type="submit" name="submit" value="Positive Negative Zero">
</form>
</body>
</html>
<?php
 if(isset($_POST['submit']))
{
$f = $_POST['f'];
if($f>0)
echo "<br><br>The number is Positive number";
else
{
if($f<0)
echo "<br><br>The number is Negative number";
else
echo "<br><br>The number is Zero";
}
}
?>
```

9. Write a PHP code to print even numbers between 1-20.

Ans. :

Coding :

```
<?php
for($i=2;$i<=20;$i=$i+2)
{
echo "<br>".$i;
}
?>
```

10. Write a PHP code to input a number and print the factorial of a number.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter a Number
<input type="text" name="f'><br><br>
<input type="submit" name="submit" value="Factorial">
</form>
</body>
</html>
<?php
$fc=1;
 if(isset($_POST['submit']))
{
$f = $_POST['f'];
for($i=$f;$i>=1;$i--)
{
$fc=$fc*$i;
}
echo "<br><br>The Factorial is ".$fc;
}
?>
```


11. Write a PHP code to print multiplication table of a number.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter a Number
<input type="text" name="f"><br><br>
<input type="submit" name="submit" value="Table">
</form>
</body>
</html>
<?php
 if(isset($_POST['submit']))
{
$f = $_POST['f'];
for($i=1;$i<=10;$i++)
{
$m=$f*$i;
echo "<br><br> ".$f. "x".$i."=".$m;
}
}
?>
```

12. Write a PHP code to check whether the number is prime number or not.

Ans. :

Coding :

```
<html>
<body>
<form method="post">
Enter a Number
<input type="text" name="f"><br><br>
<input type="submit" name="submit" value="Prime number">
</form>
```

```
</body>
</html>
<?php
$p=1;
if(isset($_POST['submit']))
{
$f = $_POST['f'];
for($i=2;$i<$f;$i++)
{
if($f%$i==0)
$p=0;
}
if($p==1)
echo "<br><br>Number is Prime Number";
else
echo "<br><br>Number is not a Prime Number";
}
?>
```

13. Write a PHP code to count the number of words in the given string.

Ans. :

Coding :

```
<?php
$a="Hypertext Preprocessor";
echo "<br><br> String is ".$a;
echo "<br><br> Words in String is ".str_word_count($a);
?>
```

14. Create a website with two PHP webpage in which each webpage is connected. The first page of the website contains two form fields for taking 'name' and 'password' from users. On onclick event, details of forms should be displayed on second web page.

Ans. :

Coding :

Form.html

```
<html>
<body>
<form action="wc.php" method="post">
Enter Name
<input type="text" name="name"><br><br>
Enter Password
<input type="password" name="pass"><br><br>
<input type="submit" name="submit" value="Submit">
</form>
</body>
</html>
```

wc.php

```
<html>
<body>
Welcome
<?php
echo $_POST["name"]; ?> <br><br>
Your Password is
<?php
echo $_POST["pass"];
?>
```

Self Assessment

Time : 1 Hour Marks: - 40

1. **Fill in the blanks** (05)
   1. File extension of php is _____.
   2. _____are used to execute the same block of code repeatedly as long as certain condition is satisfied.
   3. A _____ is series of characters.
   4. _____ is an array of variables passed via the URL parameters.
   5. _____is used to store user information on server to track user activities.
2. **True or False** (05)

1. PHP does not supports databases.
2. PHP is case sensitive only at the time of variable declaration.
3. Variables can store data of different types.
4. Ucwords() function converts the first characters of each word in a string to uppercase.
5. We can create our own functions in PHP.

3. MCQ one correct alternative. (05)

1. _____ are the statements in PHP code which are not visible in the output of the program.
   (a) Methods (b) comments
   (c) Objects (d) values
2. How to a multiline comment in PHP ?
   (a) // (b) # (c) $ (d) /* and */
3. A _____ is series of characters.
   (a) String (b) Number
   (c) Object (d) Value
4. A _____ is a block of statements that can be used repeatedly in a program.
   (a) function (b) condition
   (c) object (d) value
5. Information sent from a form with _____ method is visible to everyone.
   (a) POST (b) GET
   (c) HTTP (D) FTP

4. MCQ two correct alternative. (06)

1. Loop structure in PHP are _____
   (a) for (b) for each (c) if (d) if else
2. Form data can be submitted by _____ or _____ method.
   (a) Return (b) forward (c) GET (d) POST
3. A function name can start with a _____ or _____
   (a) symbol (b) Number (c) Letter (d) Underscore

5. MCQ three correct alternative. (09)

1. Variables are used for storing values such as _____, _____, and _____
   (a) Character (b) Numeric (c) Objects
   (d) Methods (e) Functions (f) Memory addresses
2. Following are the predefined functions in PHP to manipulate string _____
   (a) Lower() (b) strops() (c) substr()
   (d) strlen() (e) upper() (f) strlength()

6. Match the Following. (03)

A	B
1. Variable	(a) Series of characters
2. for each loop	(b) Block of statements that can be used repeatedly in a program
3. String	(c) Allows programmer to make decisions
4. Function	(d) Method visible to everyone
5. If Statement	(e) works only with arrays
6. GET	(f) can store data of different types

7. Answer the following (10)

a) Explain features of PHP. **(Refer No 5.22 Q.3)**
b) Explain steps to save and execute PHP programs. **(Refer No 5..23 Q.4)**
c) Explain comments in PHP. **(Refer No 5.24 Q.9)**
d) Short note on Cookies in PHP. **(Refer No 5.28 Q.22)**
e) Explain For loop in PHP. **(Refer No 5.25 Q.13)**

8. Write a PHP Code (10)

a) Write a PHP code which calculates square of a number using form. **(Refer Page No 5.43)**
b) Write a PHP code which calculates and prints area and circumference of a circle. Hint(area = $3.14 \times r^2$ and $c = 2 \times 3.14 \times r$) using form. **(Refer Page No 5.43)**

Answer of Questions

1. Fill in the blanks
1) .php 2) loop 3) String 4) $_GET 5) Session

2. True or False
1) False 2) True 3) True 4) True 5) True

3. MCQ one correct alternative
1) B 2) D 3) A 4) A 5) B

4. MCQ two correct alternative
1) (a, b) 2) (c, d) 3) (c, d)

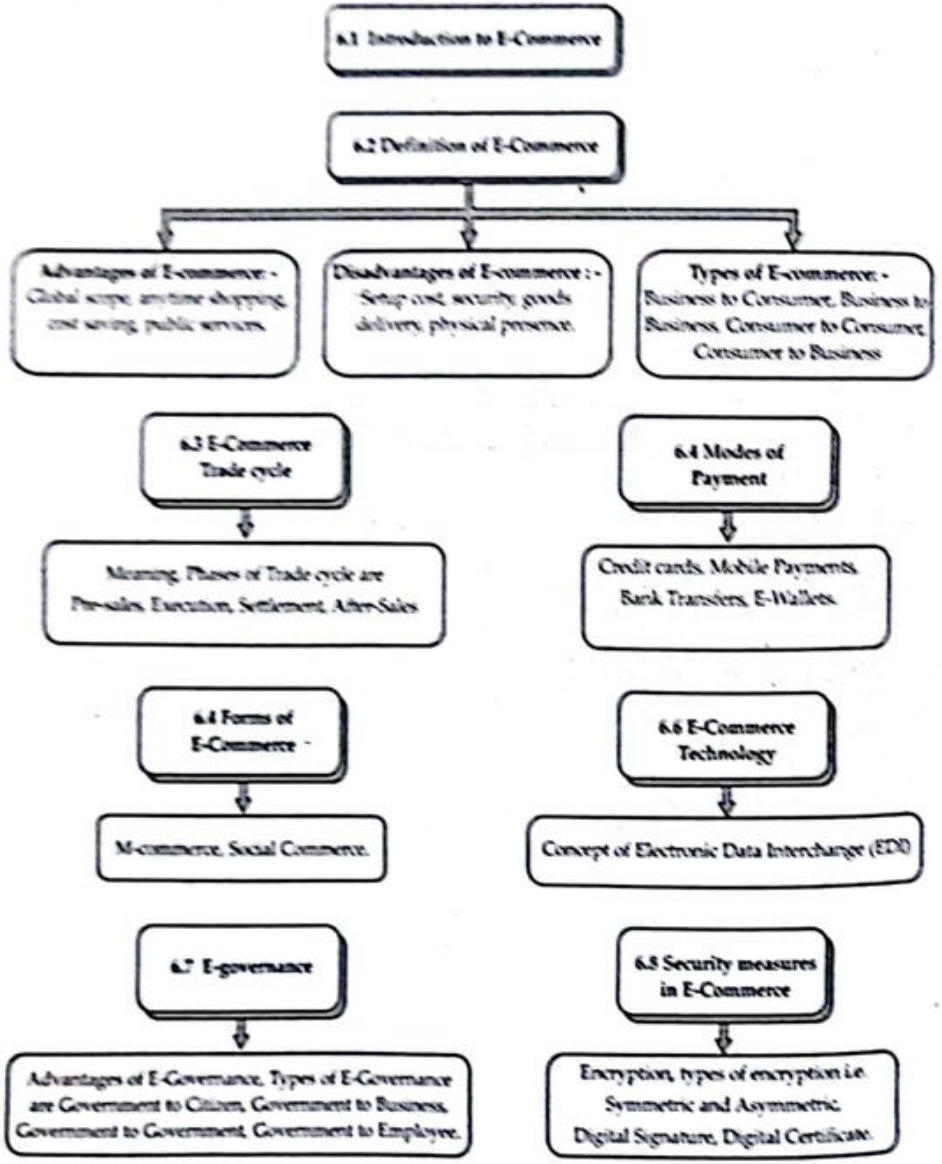
5. MCQ three correct alternative
1) (a, b, f) 2) (b, c ,d)

6. Match the Following
1. - f 2. - e 3. - a 4. - b 5. - c 6. - d

□□□

6 E-Commerce & E-Governance

Chapter at a Glance



Exercise

Fill in the blanks.

1. E-Commerce's scope is ..... March 2024
   Ans. global
2. A customer can do shopping ..... online using type of E-Commerce. March 2024
   Ans. : B2C
3. The .....phase consist of Order and Delivery
   Ans. : execution
4. E-wallet is a type of ..... account in which a user can store his/her money for any future online transaction.
   Ans. : prepaid
5. EDI is .....exchange of information. March 2024
   Ans. : electronic
6. The ..... type of e-governance refers to the government services which enable citizens to get access to wide variety of public services.
   Ans. : G2C
7. The e-governance which provides safe and secure inter-relationship between domestic or foreign government is..... March 2024
   Ans. : G2G

State True/False.

1. C2C type of E-commerce deals with Business and Customer. March 2024
   Ans. : False
2. The lack of a personal touch can be a disadvantage for many types of services and products in E-commerce.
   Ans. : True
3. Checking the balance of holiday is an example of G2C.
   Ans. : False
4. E-commerce provides more options to compare and select the cheaper and better options.
   Ans. : True
5. M-commerce can be used through desktop computer.
   Ans. : False

Multiple Choice Question. (1 correct answer)

1. Invoice and payment are included in .....phase of trade cycle. March 2024
   (a) Presale (b) execution (c) settlement (d) After sale
   Ans. : (c)

2. License renewal is an example of ..... e-governance.
(a) G2C (b) G2B (c) G2G (d) G2E

Ans. : (a)

Multiple Choice Question. (2 correct answer)

1. Encryption consist of two processes and.....
(a) encryption (b) signature (c) decryption
(d) digitization (e) security

Ans. : (a, c)

2. Social commerce is a subset of electronic commerce that involves social media like ..... and.....
(a) Facebook (b) Instagram (c) gmail
(d) whatsapp (e) software

Ans. : (a, b)

Match the following.

A	B
1. M-commerce (March 2022)	(a) Coded form of data
2. Cipher Text	(b) B2B
3. EDI (March 2022)	(c) E-bill payment
4. Wholesaler-to	(d) Paperless exchange of information
5. License renewal services	(e) G2G
6. Online facility to employees like leave	(f) G2C
7. Government agencies share same database	(g) G2E

Ans. : 1–c , 2-a, 3-d, 4-b , 5-f, 6-g, 7-e

Answer in Brief

1. Explain phases of trade cycle. (March 2022)

Ans. : Refer 6.29, Q. 5

2. Explain M-Commerce. (March 2022)

Ans. : Refer 6.30, Q. 7

3. Describe process of encryption.

Ans. : Refer 6.33, Q. 12

4. What is E-governance.

Ans. : Refer 6.31, Q. 9

5. List out advantages of E-commerce. (March 2022)

Ans. : Refer 6.27, 28, Q. 2

6. Which are the different types of e-governance?

Ans. : Refer 6.32, Q. 11

7. State two examples of G2E services. (March 2022)

Ans. : Refer 6.33, Q. 11

8. Write any four advantages of e-governance.

Ans. : Refer 6.31, 32, Q. 8

Fill in the Blanks

6.1 Introduction

1. E-commerce stands for..... (March 2022)

Ans. : Electronic Commerce

2. .....is nothing but buying and selling of goods.

Ans. : Commerce

6.2 Definition of E-Commerce

3. .....is the process of buying and selling of goods and services using electronic medium. (March 2022)

Ans. : Electronic Commerce

4. .....is also referred as paperless exchange of business information.

Ans. : Electronic Commerce

5. ..... Commerce focuses on the exchange of products and services through personal interactions. (March 2022)

Ans. : Traditional

6. .....Commerce has limited business hours.

Ans. : Traditional

7. .....Commerce is limited to a particular geographical location. (March 2022)

Ans. : Traditional

8. Modes of payment in.....commerce include cash, cheques and credits cards.

Ans. : Traditional

9. In.....Commerce scope is local. (March 2022)

Ans. : Traditional

10. .....Commerce trading activities are online via the internet and can be considered automatic.

Ans. : Electronic

11. .....Commerce is 24 × 7, it can be done anytime day and night. (March 2022)

Ans. : Electronic

12. In.....Commerce modes of payment are bank transfer, credit card, e-wallet etc.

Ans. : Electronic

13. In.....Commerce scope is global.

Ans. : Electronic

14. B2C stands for..... (March 2022)

Ans. : Business to Consumer

15. In.....model business sells it's product directly to a customer.

Ans. : B2C (Business to Consumer)

16. B2B stands for.....

Ans. : Business to Business

17. In.....model business sells its products to an intermediate buyer.

Ans. : B2B (Business to Business)

18. C2C stands for..... (March 2022)

Ans. : Consumer to Consumer

19. In ..... model, consumer helps consumer to sell their assets by publishing their information on the website.

Ans. : C2C (Consumer to Consumer)

20. C2B Stands for..... (March 2022)

Ans. : Consumer to Business

21. In.....model, consumers have products or services of value that can be consumed by businesses.

Ans. : C2B (Consumer to Business)

6.3 E-Commerce Trade cycle

22. A.....is the series of exchanges between a customer and supplier that takes place when a commercial exchange is executed. (March 2022)

Ans. : Trade Cycle

23. The.....phase of Trade cycle consist of two steps like search and negotiate. (March 2022)

Ans. : Pre-sales

24. The.....phase of Trade cycle consists of Order and Delivery. (March 2022)

Ans. : Execution

25. The.....phase of Trade Cycle consist of Invoice and Payment. (March 2022)

Ans. : Settlement

26. The.....phase of Trade Cycle consist of warranty and After Sale Services. (March 2022)

Ans. : After sales

6.4 Modes of Payment

27. .....transfer is used when money is sent from one bank account to another.

Ans. : Bank

28. .....is a type of electronic card which is used for transactions made online through a computer or a smartphone. (March 2022)

Ans. : E-Wallet

6.5 Forms of E-commerce

29. .....is the buying and selling of goods and services through wireless handheld devices such as smartphones and tablets. (March 2022)

Ans. : M-Commerce (Mobile Commerce)

30. .....Commerce is a form of electronic, commerce that involves social media that supports social interaction.

Ans. : Social

31. .....Commerce is the use of networking websites such as facebook, Instagram and Twitter as vehicles to promote and sell products and services.

Ans. : Social

6.6 E-Commerce Technology

32. EDI stands for..... (March 2022)

Ans. : Electronic Data Interchange

33. ..... is the electronic interchange of business information using a standardized format.

Ans. : EDI (Electronic Data Interchange)

34. The..... is a process which allows one company to send information to another company electronically rather than a paper. (March 2022)

Ans. : EDI (Electronic Data Interchange)

35. Business entities conducting business electronically are called .....

Ans. : Trading Partners

36. E-wallet is a type of.....account in which a user-can store money for any future online transaction. (March 2022)

Ans. : Prepaid

37. ..... is computer-to-computer interchange of strictly formatted documents via telecommunication or physically transported on electronic storage media. (March 2022)

Ans. : EDI (Electronic Data Interchange)

6.7 E-governance

38. ..... signifies the implementation of information technology in the government processes and function.

Ans. : E-Governance

39. E-Governance is of.....types. (March 2022)

Ans. : 4 (Four)

40. G2C stands for.....

Ans. : Government-to-Citizen

41. The .....refers to the government services which enable citizens to get access to wide variety of public services. (March 2022)

Ans. : G2C (Government to Citizen)

42. G2B stands for.....

Ans. : Government to Business

43. The ..... is the exchange of services between Government and Business Organization. (March 2022)

Ans. : G2B (Government to Business)

44. The ..... consists of many services exchanged between business sectors and government.

Ans. : G2B (Government to Business)

45. G2G stands for..... (March 2022)

Ans. : Government to Government

46. The ..... refers to the interaction between different government departments, organizations and agencies. (March 2022)

Ans. : G2G (Government to Government)

47. The E-Governance which provides safe and secure inter-relationship between domestic and foreign government is .....

Ans. : G2G (Government to Government)

48. G2E stands for..... (March 2022)

Ans. : Government to Employee

49. The.....is the internal part to G2G sector.

Ans. : G2E (Government to Employee)

50. The.....governance aims to bring employees together and improvise knowledge sharing.

Ans. : G2E (Government to Employee)

6.8 Security measures in E-Commerce

51. .....is widely used on the internet to protect user information being sent between a browser and a server. (March 2022)

Ans. : Encryption

52. .....converts plain text into Cipher text. (March 2022)

Ans. : Encryption

53. .....converts Cipher text into plain text.

Ans. : Decryption

54. A.....is also known as an electronic signature. (March 2022)

Ans. : Digital Signature

55. A.....guarantees the authenticity of an electronic document.

Ans. : Digital Signature

56. A ..... is an electronic password that allows a person to exchange data securely over the Internet using public key infrastructure. (March 2022)

Ans. : Digital Certificate

57. .....is also known as a public key certificate or identity certificate.

Ans. : Digital Certificate

58. PKI stands for..... (March 2022)

Ans. : Public Key Infrastructure

True or False

6.1 Introduction

1. E-Commerce stands for Electronic commerce. (March 2022)

Ans. : True

2. Commerce is an important part of business.

Ans. : True

3. Commerce means buying and selling of goods and services. (March 2022)

Ans. : True

6.2 Definition of E-Commerce

4. E-Commerce is a process of buying and selling of goods or services using Internet. (March 2022)

Ans. : True

5. E-Commerce focuses on the exchange of products and services through personal interactions.

Ans. : False

6. Traditional Commerce is manual. (March 2022)

Ans. : True

7. E-Commerce trading activities are online via the internet. (March 2022)

Ans. : True

8. Traditional Commerce is limited to business hours.

Ans. : True

9. E-Commerce is 24 × 7.

Ans. : True

10. E-Commerce provides face to face interaction. (March 2022)

Ans. : False

11. Traditional Commerce is limited to a particular geographical location.

Ans. : True

12. E-Commerce is local. (March 2022)

Ans. : false

13. Modes of payments in E-commerce are bank transfer, e-wallet etc.

Ans. : True

14. Goods and delivery of services is instant with Traditional commerce.

Ans. : True

15. Traditional Commerce's Scope is global. (March 2022)

Ans. : false

16. E-Commerce reduces paper work.

Ans. : True

17. E-commerce increases the cost of searching a product. (March 2022)

Ans. : False

18. E-commerce does not allow intermediaries.

Ans. : True

19. Set-up cost of E-Commerce is expensive. (March 2022)

Ans. : True

20. In B2C Model business sells its products to an intermediate buyer who then sells product to the final customer. (March 2022)

Ans. : False

21. In B2C model business sells its product directly to customer.

Ans. : True

22. In C2C model, consumer helps consumer to sell their assets by publishing their information on website. (March 2022)

Ans. : True

23. OLX, Quikr, online auction are the examples of B2B model. (March 2022)

Ans. : False

24. In C2B model consumers have products or services of value that can be consumed by businesses.

Ans. : True

25. Types of E-commerce are B2B, C2B, C2C etc. (March 2022)

Ans. : True

6.3 E-Commerce Trade cycle

26. Trade Cycle is the series of exchanges between a customer and supplier that take place when a commercial exchange is executed. (March 2022)

Ans. : True

27. Execution phase consists of two steps like search and negotiate.

Ans. : False

28. The execution phase consists of Order and Delivery. (March 2022)

Ans. : True

29. The After sales phase consists of Invoice and Payment.

Ans. : False

30. Invoice means customer will receive a bill for purchased product. (March 2022)

Ans. : True

31. After sales phase consists of warranty and after sales services.

Ans. : True

32. Settlement means customer will do complaints about the performance of product and get maintenance service from supplier.

Ans. : False.

6.4 Modes of Payment

33. Credit cards are most common way for customers to pay online.

Ans. : True

34. Mobile payment apps are UPI, Paytm, Paypal etc. (March 2022)

Ans. : True

35. Bank transfer cannot be used when money is sent from one bank account to another.

Ans. : False

36. NEFT, IMPS etc. are the examples of Bank transfer. (March 2022)

Ans. : True

37. E-Wallet is electronic card which is used for transactions made online through a computer or smartphone.

Ans. : True

38. E-wallet is a type of post paid account. (March 2022)

Ans. : False

39. State bank Buddy, Paytm wallets are examples of E-wallets.

Ans. : True

6.5 Forms of E-commerce

40. Some common forms of E-commerce are C2B, B2B, C2C etc. (March 2022)

Ans. : False

41. M-Commerce is buying and selling of goods and services through smartphones and tablets.

Ans. : True

42. M-Commerce enables users to access online shopping by using a desktop computer. (March 2022)

Ans. : False

43. The applications of M-Commerce are mobile banking, E-bill payment, online auctions etc.

Ans. : True

44. Social Commerce is a form of electronic commerce that involves online media that supports social interaction. (March 2022)

Ans. : True

6.6 E-Commerce Technology

45. EDI stands for Electronic Data Information. (March 2022)

Ans. : False

46. EDI is the non-electronic interchange of business information using a standardised format.

Ans. : False

47. EDI is a process which allows one company to send information to another electronically rather than paper. (March 2022)

Ans. : True

48. EDI is paperless exchange of electronic information. (March 2022)

Ans. : True

49. Business entities conducting business electronically are called trading partners.

Ans. : True

50. Two most common documents exchanged using EDI are purchase orders and invoices. (March 2022)

Ans. : True

6.7 E-governance

51. E-Governance is the implementation of information technology in the government processes and function. (March 2022)

Ans. : True

52. E-Governance delivers SMART governance.

Ans. : True

53. Types of E-Governance are B2B, B2C, C2C. (March 2022)

Ans. : False

54. The G2C refers to the government services which enable citizens to get access to wide variety of public services.

Ans. : True

55. License renewals and paying tax are the examples of G2C. (March 2022)

Ans. : True

56. The G2B is the internal part of G2G sector.

Ans. : False

57. G2B provides access to relevant forms needed to comply.

Ans. : True

58. The G2G refers to the interaction between different government departments, organizations and agencies.

Ans. : True

59. In G2G type of e-governance, government agencies can share the same database using online communication. (March 2022)

Ans. : True

60. The G2E is the internal part of G2G sector.

Ans. : True

61. Checking balance of holiday is example of G2E. (March 2022)

Ans. : True

62. Reviewing salary payment records, applying for leave is the example of G2C.

Ans. : False

63. Digital India is a campaign launched by the Government of India is order to make Government's Services available to citizens electronically.

Ans. : True

64. e-mitra project, e-Seva project, CET are the examples of successful implementation of E-Governance.

Ans. : True

6.8 Security measures in E-Commerce

65. Encryption is used to protect user information being sent between browser and a server. (March 2022)

Ans. : True

66. Encryption converts plain text into coded form of data.

Ans. : True

67. Conversion of plain text into Cipher text is called decryption. (March 2022)

Ans. : False

68. Decryption converts Cipher text into plain text.

Ans. : True

69. Encryption is of two types Symmetric and Asymmetric. (March 2022)

Ans. : True.

70. Digital Signature is also known as electronic signature.

Ans. : True

71. Digital Signature is also known as public key certificate. (March 2022)

Ans. : False

72. Digital Signature is on electronic passwords that allows organization to exchange data securely over the Internet using PKI.

Ans. : True

MCQ (One Correct Answers)

6.1 Introduction

1. E-Commerce stands for ..... (March 2022)

(a) Engage Commerce (b) Evolve Commerce

(c) Electronic Commerce (d) Easy Commerce

Ans. : (c)

6.2 Definition of E-Commerce

2. .....is the process of buying and selling of goods and services using electronic medium.

(a) E-Commerce (b) Digital Certificate

(c) E-Governance (d) Digital Signature

Ans. : (a)

3. ..... Commerce focuses on the exchange of products and services through personal interactions. (March 2022)

(a) E-Governance (b) Electronic

(c) Digital Signature (d) Traditional

Ans. : (d)

4. .....Commerce has limited to business hours.

(a) Traditional (b) Electronic

(c) Digital (d) E-governance

Ans. : (a)

5. .....Commerce provides face to face interaction. (March 2022)

(a) Electronic (b) Traditional

(c) Digital (d) Web hasting

Ans. : (b)

6. In .....Commerce scope is local. (March 2022)

(a) Electronic (b) Divisional

(c) Digital (d) Traditional

Ans. : (d)

7. .....Commerce has no time limits, it is available 24 × 7.

(a) Electronic (b) Divisional

(c) Traditional (d) Digital

Ans. : (a)

8. In .....Commerce trading activities are online via the internet.

(a) Digital (b) Traditional

(c) Electronic (d) Divisional

Ans. : (c)

9. In .....Commerce scope is a global. (March 2022)

(a) Digital (b) Electronic

(c) Traditional (d) Divisional

Ans. : (b)

10. .....Commerce reduces the paper work and lower the transaction cost.

(a) Electronic (b) Divisional

(c) Traditional (d) local (March 2022)

Ans. : (a)

11. .....application provides users with more options to compare and select the cheaper and better option. (March 2022)

(a) Z-Commerce (b) G – Commerce

(c) E-Commerce (d) V-Commerce

Ans. : (c)

12. .....Commerce allows the customers and the business to be in touch directly without any intermediaries.

(a) Traditional (b) Electronic (c) Divisional (d) Digital

Ans. : (b)

13. In ..... model business sells its product directly to a customer. (March 2022)

(a) C2B (b) B2C (c) B2B (d) C2C

Ans. : (b)

14. In .....model, business sells its products to an intermediate buyer who then sells the product to the final customer. (March 2022)

(a) C2B (b) B2C (c) B2B (d) C2C

Ans. : (c)

15. In .....model, consumer helps consumer to sells their assets by publishing their information on the website. (March 2022)

(a) C2B (b) B2C (c) B2B (d) C2C

Ans. : (d)

16. On-line auction is example of.....model. (March 2022)

(a) C2B (b) B2C (c) B2B (d) C2C

Ans. : (d)

17. In.....model consumers have products or services of value that can be consumed by businesses.

(a) C2B (b) B2B (c) C2C (d) B2C

Ans. : (a)

6.3 E-Commerce Trade cycle

18. A.....is the series of exchanges between a customer and supplier that takes place when a commercial exchange is executed. (March 2022)

(a) EDI (b) Trade Cycle

(c) Encryption (d) Decryption

Ans. : (b)

19. .....phase of Trade cycle consist of two steps like search and negotiate.

(a) Presale (b) Execution (c) Settlement (d) After sales (March 2022)

Ans. : (a)

20. .....phase of Trade cycle consists of Order and Delivery.

(a) Presale (b) Execution (c) Settlement (d) After sales

Ans. : (b)

21. In.....phase of Trade Cycle the customer sends an order for the selected product and receives delivery of the product. (March 2022)

(a) Presale (b) After sales (c) Execution (d) Settlement

Ans. : (c)

22. The.....phase of Trade Cycle consist of warranty and After Sale Services.

(a) Presale (b) Execution (c) Settlement (d) After sales (March 2022)

Ans. : (d)

23. ..... services means customer will do complaints about the performance of product and get maintenance service from the suppliers.

(a) Presale (b) After sales (c) Execution (d) Settlement

Ans. : (b)

6.4 Modes of Payment

24. .....offer a quick solution for customers to purchase on e-commerce website.

(a) Mobile Payments (b) Digital Signature

(c) Cost Payments (d) Cheque Payments

Ans. : (a)

25. Apps the BHIM, UPI, paytm are the examples of..... (March 2022)

(a) cheque payment (b) cash payment

(c) EDI (d) Mobile payments

Ans. : (d)

26. .....is used when money is sent from one bank account to another.

(a) EDI (b) Bank Transfer

(c) Digital Signature (d) Trade Cycle

Ans. : (b)

27. NEFT, IMPS are the examples of..... (March 2022)

(a) Credit Card (b) Trade Cycle

(c) Debit Card (d) Bank Transfer

Ans. : (d)

28. .....is a type of electronic card which is used for transactions made online through a Computer or a smart-phone. (March 2022)

(a) E-Wallets (b) E-Book (c) E-Pocket (d) E-Draw

Ans. : (a)

29. State Bank Buddy, paytm wallets are the examples of.....

(a) E-Book (b) E-Pocket (c) E-Wallet (d) E-Draw

Ans. : (c)

30. E-wallet is a type of.....account in which a user can store money for any future online transaction. (March 2022)

(a) Pre-paid (b) Cash-paid (c) Bank paid (d) Cheque paid

Ans. : (a)

6.5 Forms of E-commerce

31. .....is the buying and selling of goods and services through wireless handheld devices such as smartphones and tablets.

(a) Trade Cycle (b) M-Commerce

(c) Social Commerce (d) EDI

Ans. : (b)

32. .....enables users to access online shopping platforms without using desktop computer. (March 2022)
(a) M-Commerce (b) EDI
(c) T-Commerce (d) Trade Cycle

Ans. : (a)

33. Mobile banking, E-bill payment, ticket booking are the examples of .....
(a) EDI (b) Trade Cycle
(c) M-Commerce (d) Digital Certificate

Ans. : (c)

34. .....Commerce is a form of electronic commerce that involves social media that supports social interaction.
(a) Social (b) National (c) Local (d) Divisional

Ans. : (a)

35. .....Commerce is a subset of electronic commerce that involves social media like facebook, Instagram and Twitter. (March 2022)
(a) Local (b) Divisional (c) National (d) Social

Ans. : (d)

6.6 E-Commerce Technology

36. EDI stands for ..... (March 2022)
(a) Easy Data Interchange (b) Economic Data Interchange
(c) Electronic Data Interchange (d) Enable Data Interchange

Ans. : (c)

37. .....is the electronic interchange of business information using a standardized format.
(a) EDI (b) Trade Cycle
(c) Digital Signature (d) Digital Certificate

Ans. : (a)

38. .....is a process which allows one company to send information to another company electronically rather than a paper. (March 2022)
(a) Trade Cycle (b) EDI
(c) Social Commerce (d) Mobile Commerce

Ans. : (b)

39. Business entities conducting business electronically are called .....
(a) Global partners (b) Local partners
(c) Visiting partners (d) Trading partners

Ans. : (d)

6.7 E-governance

40. .....is the implementation of information technology in the government processes and functions.
(a) E-Governance (b) E-Commerce
(c) Social Commerce (d) M-Commerce

Ans. : (a)

41. The .....type of e-governance refers to the government services which enable citizens to get access to wide variety of public services. (March 2022)
(a) G2E (b) G2C (c) G2B (d) G2G

Ans. : (b)

42. License renewals and paying tax can be done through ..... type of E-Governance.
(a) G2E (b) G2G (c) G2C (d) G2B

Ans. : (c)

43. The .....type of E-governance is the exchange of services between government and business organizations. (March 2022)
(a) G2C (b) G2B (c) G2G (d) G2E

Ans. : (b)

44. The ..... type of E-governance refers to the interaction between different government departments, organizations and agencies. (March 2022)
(a) G2C (b) G2B (c) G2G (d) G2E

Ans. : (c)

45. In ..... type of E-governance, government agencies can share the same database using on line communication.
(a) G2C (b) G2B (c) G2G (d) G2E

Ans. : (c)

46. The .....type of E-governance is the internet part of G2G sector. (March 2022)
(a) G2C (b) G2B (c) G2G (d) G2E

Ans. : (d)

47. ..... type of E-Governance provides facilities to employers like applying for leave, reviewing salary payment record and checking balance of holiday. (March 2022)
(a) G2C (b) G2B (c) G2G (d) G2E

Ans. : (d)

6.8 Security measures in E-Commerce

48. ..... is used on the internet to protect user information being sent between a browser and a server. (March 2022)
(a) Encryption (b) Decryption
(c) Digital Signature (d) Digital Certificate
Ans. : (a)

49. .....converts plain text into Cipher text. (March 2022)
(a) Decryption (b) Encryption
(c) Digital Signature (d) Digital Certificate
Ans. : (b)

50. .....converts Cipher text into plain text.
(a) Decryption (b) Encryption
(c) Digital Signature (d) Digital Certificate
Ans. : (a)

51. .....is of two types symmetric and Asymmetric. (March 2022)
(a) Encryption (b) Private key
(c) Public key (d) Decryption
Ans. : (a)

52. A.....is also known as an electronic signature.
(a) Digital Certificate (b) Digital Signature
(c) Cryptography (d) EDI
Ans. : (b)

53. PKI stands for..... (March 2022)
(a) Public Key Infrastructure (b) Personal Key Certificate
(c) Publish Key Certificate (d) Person Key Certificate
Ans. : (a)

54. ..... is also known as Public key certificate or identity certificate.
(a) EDI (b) Digital Certificate
(c) Trade Cycle (d) Digital Signature
Ans. : (b)

55. A ..... is an electronic password that allows organization to exchange data securely over the Internet using PKI. (March 2022)
(a) Digital Signature (b) EDI
(c) Trade Cycle (d) Digital Certificate
Ans. : (d)

MCQ (Two Correct Answers)

6.2 Definition of E-Commerce

1. E-Commerce is also referred as paperless exchange of business information using .....
(a) EDI, (b) Electronic Fund Transfer
(c) Private Key (d) Public Key
Ans. : (a), (b)

2. Advantages of E-Commerce are..... (March 2022)
(a) Security (b) Global scope
(c) Cost saving (d) High setup cost
Ans. : (b), (c)

3. Disadvantages of E-Commerce are.....
(a) High set up cost (b) Late delivery
(c) Public serves (d) Any time shopping
Ans. : (a), (b)

4. Types of E-Commerce are..... (March 2022)
(a) B2C (b) C2B (c) G2G (d) G2E
Ans. : (a), (b)

5. Examples of B2C model are.....
(a) Linux (b) Flipkart (c) Amazon (d) Window
Ans. : (b), (c)

6. Examples of C2C model are..... (March 2022)
(a) Network provider (b) OLX
(c) EDI (d) Quikr
Ans. : (b), (d)

6.3 E-Commerce Trade cycle

7. Following are the phases of Trade cycle. (March 2022)
(a) Pre-sales (b) EDI (c) Trade Cycle (d) Execution
Ans. : (a), (d)

8. Pre-sales consists of two steps.....and..... (March 2022)
(a) Order (b) Delivery (c) Search (d) Negotiate
Ans. : (c), (d)

9. The execution phase consists of .....and .....
(a) Order (b) Delivery (c) Search (d) Negotiate
Ans. : (a), (b)

10. The settlement phase consist of ..... and ..... (March 2022)
(a) Order (b) Invoice (c) Search (d) Payment
Ans. : (b), (d)

11. The After sales phase consists of ..... and .....
(a) Order (b) Delivery (c) warranty (d) After sale service
Ans. : (c), (d)

6.4 Modes of Payment

12. Modes of payments are ..... (March 2022)
(a) M-Commerce (b) Mobile Payments
(c) E-Commerce (d) E-Wallets
Ans. : (b), (d)

13. Examples of Mobile payments apps are .....
(a) BHIM (b) EDI (c) Paytm (d) Trade Cycle
Ans. : (a), (c)

14. Examples of Bank Transfers are ..... (March 2022)
(a) EDI (b) NEFT (c) IMPS (d) Cash Transfer
Ans. : (b), (c)

15. Examples of E-Wallets are .....
(a) Paytm Wallets (b) State Bank Buddy
(c) EDI (d) Trade Cycle
Ans. : (a), (b)

6.5 Forms of E-commerce

16. Common forms of E-commerce are ....., ...
(a) Local Commerce (b) M-Commerce
(c) National Commerce (d) Social Commerce
Ans. : (b), (d)

17. Application of M-Commerce are ....., ... (March 2022)
(a) E-bill payment (b) Local Commerce
(c) Mobile Banking (d) Social Commerce
Ans. : (a), (c)

18. Social Commerce is the use of networking websites such as ....., ... as vehicles to promote and sell products and services.
(a) G-mail (b) Google Docs
(c) Facebook (d) Twitter
Ans. : (c), (d)

6.6 E-Commerce Technology

19. Two common documents can be exchanged using EDI are ....., ...
(a) Purchase orders (b) DOCs
(c) Trading (d) Invoices (March 2022)
Ans. : (a), (d)

6.7 E-governance

20. Advantages of E-governance are ..... (March 2022)
(a) Increase of overall cost (b) Decrease convenience
(c) High transparency (d) Reduced corruption
Ans. : (c), (d)

21. Type of E-Governance are ..... (March 2022)
(a) B2C (b) C2C (c) G2E (d) G2G
Ans. : (c), (d)

22. G2C type of E-governance provides services like ..... And .....
(a) License renewals (b) Paying tax
(c) Trade Cycle (d) EDI
Ans. : (a), (b)

23. G2E type of E-governance provides ....., ...Online facilities to employees.
(a) License renewals (b) Salary payment record
(c) Booking Tickets (d) applying for leave
Ans. : (b), (d)

24. Examples of successful implementation of E-Governance projects are ..... and ..... (March 2022)
(a) e-seva (b) e-Mitra (c) e-tax (d) e-friend
Ans. : (a), (b)

6.8 Security measures in E-Commerce

25. Encryption is of two types.....and ..... (March 2022)
(a) Plaintext (b) symmetric (c) Coded text (d) asymmetric
Ans. : (b), (d)

26. Encryption consists of two processes .....and.....
(a) EDI (b) Trade cycle
(c) Encryption (d) Decryption
Ans. : (c), (d)

27. Encryption converts.....text into.....text. (March 2022)
(a) Plain (b) Cipher (c) Random (d) General
Ans. : (a), (b)

28. Decryption converts.....text into.....text.
(a) Cipher (b) Plain (c) Random (d) General
Ans. : (a), (b)

29. Following are the security measure in E-commerce.
(a) EDI (b) Trade cycle
(c) Digital signature (d) Digital certificate
Ans. : (c), (d)

MCQ (Three Correct Answers)

6.2 Definition of E-Commerce

1. Advantages of E-Commerce are..... (March 2022)
(a) Global scope (b) Local scope
(c) Cost saving (d) Time restriction
(e) Anytime shopping (f) Involve intermediaries
Ans. : (a), (c), (e)

2. Disadvantages of E-Commerce are..... (March 2022)
(a) Global scope (b) Expensive
(c) Cost saving (d) Lack of personal touch
(e) Anytime shopping (f) Late deliveries
Ans. : (b), (d), (f)

3. Types of E-Commerce are..... (March 2022)
(a) B2C (b) G2E (c) B2B
(d) G2G (e) G2C (f) C2C
Ans. : (a), (c), (f)

6.3 E-Commerce Trade cycle

4. Following are the phases of Trade Cycle. (March 2022)
(a) C2B (b) Presale (c) Execution (d) Settlement
(e) B2C (f) C2C
Ans. : (b), (c), (d)

6.4 Modes of Payment

5. Following are the Modes of Payment ..... (March 2022)
(a) Presale (b) Execution
(c) Settlement (d) Mobile payments
(e) Bank transfer (f) E-wallets
Ans. : (d), (e), (f)

6. Examples of Mobile payment apps are.....
(a) BHIM (b) Trade Cycle (c) EDI
(d) UPI (e) Google pay (f) Google Docs
Ans. : (a), (d), (e)

6.5 Forms of E-commerce

7. Applications of M-Commerce are..... (March 2022)
(a) Ticket booking (b) RTGS (c) NEFT
(d) E-Bill payment (e) IMPS (f) Online auctions
Ans. : (a), (d), (f)

8. Social Commerce is the use of networking websites such as ....., ... and .....as vehicles to promote and sell products and services.
(a) Facebook (b) Instagram (c) Twitter (d) Gmail
(e) Google Docs (f) Google pay
Ans. : (a), (b), (c)

6.8 Security measures in E-Commerce

9. Types of E-Governance are..... (March 2022)
(a) G2C (b) G2G (c) C2B
(d) B2B (e) C2C (f) G2E
Ans. : (a), (b), (f)

10. Advantages of E-Governance are .....

(a) Increased convenience (b) No transparency
(c) Reduction in overall cost (d) Increase in cost
(e) Expanded reach of government (f) More corruption

Ans. : (a), (c), (e)

11. Examples of successful implementation of E-Governance projects are .....

(a) e-Mitra (b) B2B (c) C2B (d) e-Seva
(e) C2G (f) CET **(March 2022)**

Ans. : (a), (d), (f)

6.8 Security measures in E-Commerce

12. Security measure in E-Commerce are ..... **(March 2022)**

(a) EDI (b) Trade Cycle
(c) Encryption (d) Digital Signature
(e) E-commerce (f) Digital Certificate

Ans. : (c), (d), (f)

Match the Following

6.2 Definition of E-Commerce

(I)

	A		B
(1)	Traditional Commerce **(March 2022)**	(a)	Scope is Global
(2)	E-Commerce	(b)	Sells products to an intermediate buyer
(3)	B2C **(March 2022)**	(c)	Helps consumer to sell their assets by publishing information on website
(4)	B2B	(d)	Scope is local
(5)	C2C **(March 2022)**	(e)	Consumers have products or services of value that can be consumed by business
(6)	C2B	(f)	Sells products directly to a customer

Ans. : (1) – (d), (2) – (a), (3) – (f), (4) – (b), (5) – (c), (6) – (e)

6.3 E-Commerce Trade cycle

(I)

	A		B
(1)	Presale	(a)	Consists of Order and Delivery
(2)	Execution	(b)	Consists of Warranty and After sales
(3)	Settlement	(c)	Consists of Invoice and Payment
(4)	After sales	(d)	Consists of Search and negotiate

Ans. : (1) – (d), (2) – (a), (3) – (c), (4) – (b)

6.4 Modes of Payment

(I)

	A		B
(1)	Credit cards **(March 2022)**	(a)	Money sent from one bank to another
(2)	Mobile payments	(b)	Type of prepaid account in which user can store money
(3)	Bank transfer	(c)	Cards used by customers to pay online
(4)	E-Wallets **(March 2022)**	(d)	Forms of E-commerce
		(e)	Offers quick solution for customers to purchase on e-commerce websites

Ans. : (1) – (c), (2) – (e), (3) – (a), (4) – (b)

6.5 Forms of E-commerce, 6.6 E-Commerce Technology

(I)

	A		B
(1)	M-Commerce **(March 2022)**	(a)	Form of E-commerce that involves social media that supports social interaction
(2)	Social Commerce	(b)	Electronic interchange of business information
(3)	EDI	(c)	Buying and selling of goods and services through smart phones and tablets

Ans. : (1) – (c), (2) – (a), (3) – (b)

6.7 E-governance

(I)

	A		B
(1)	G2C (March 2022)	(a)	Exchange of services between Government and Business organizations
(2)	G2B (March 2022)	(b)	Is the internal part of G2G sector
(3)	G2G (March 2022)	(c)	Refers to the interaction between different government departments, organization and agencies.
(4)	G2E (March 2022)	(d)	Refers to government services which enable citizens to access wide variety of public services.

Ans. : (1) – (d), (2) – (a), (3) – (c), (4) – (b)

6.8 Security measures in E-Commerce

(I)

	A		B
(1)	Encryption	(a)	Also known as electronic signature
(2)	Decryption	(b)	Converts plain text into cipher text
(3)	Digital Signature (March 2022)	(c)	Is a electronic password
(4)	Digital Certificate (March 2022)	(d)	Converts cipher text into plain text

Ans. : (1) – (b), (2) – (d), (3) – (a), (4) – (c)

Answer in Brief

6.2 Definition of E-Commerce

1. Define E-Commerce. (March 2022)

Ans. :

(i) E-Commerce is defined as the process of buying and selling of goods or services using an electronic medium such as Internet.

(ii) E-commerce is also referred as a paperless exchange of business information using EDI, E-mail, Electronic fund transfer etc.

2. Explain Advantages and Disadvantages of E-Commerce. (March 2022)

Ans. :

Advantages of E-Commerce :

(i) **Global scope :** E-commerce provides the sellers with a global reach. Now sellers and buyers can meet in the virtual world, without barrier of place.

(ii) **Electronic transaction :** E-commerce reduces the paper work and significantly lower the transaction cost.

(iii) **Anytime shopping :** The great advantage of E-Commerce is the convenience. A customer can shop 24 × 7.

(iv) **No intermediaries :** Electronic commerce also allows the customer and the business to be in touch directly, without any intermediaries.

Disadvantages of E-Commerce :

(i) **Setup Cost :** The setup of the hardware and the software, the training cost of employees, the constant maintenance and upkeep are all quite expensive.

(ii) **Security :** Security is another area of concern. Credit card theft, identity theft etc. remain big concerns with the customers.

(iii) **Goods Delivery :** There may arrive some problem with fulfillment of order. Even after the order is placed there can be problems with shipping, delivery, mix-ups etc. This leaves the customers unhappy and dissatisfied.

3. Difference between Traditional Commerce and E-Commerce.

Ans. :

Traditional Commerce	E-commerce (March 2022)
Traditional commerce focuses on the exchange of products and services through personal interactions so it is manual.	E-commerce trading activities are online via the internet.
Traditional commerce is limited to business hours.	E-commerce is 24 × 7, it can be done anytime day and night.
Traditional commerce provides face to face interaction.	E-commerce can be termed as screen to face interaction.
Traditional commerce is limited to a particular geographical location.	E-commerce is global and has no physical limitation.
Modes of payment in traditional commerce include cash, cheques and credit cards.	In E-commerce modes of payments are bank transfer, credit card, e-wallet, mobile payment and many more.
Goods and delivery of services is instant with traditional commerce.	In E-commerce delivery of goods or services takes some time.
Traditional Commerce's scope is local.	E-commerce's scope is global.

4. Explain type of E-Commerce. (March 2022)

Ans. :

Types of E-Commerce are as follows :

(i) Business to Consumer (B2C) :

(a) In B2C model, business sells it's products directly to customer.

(b) Customer can view and choose to order the products shown on the website.

(c) The website will send notification and organization will dispatch the product to customer.

(d) Examples Amazon, Flipkart etc.

(ii) Business to Business (B2B) :

(a) In B2B model, business sells products to an intermediate buyer.

(b) Buyer then sells the product to final customer.

(c) Example Tata communications.

(iii) Consumer to Consumer (C2C) :

(a) In C2C model, consumer helps consumer to sell their assets like cars, bikes, rent a room etc by publishing their information on websites.

(b) Example OLX, Quikr online auction.

(iv) Consumer to Business (C2B) :

(a) In C2B model, consumers have products or services of value that can be consumed by businesses.

(b) For example A blog can be written by an author for a business to improve sale of product ebay.

6.3 E-Commerce Trade cycle

5. Explain phases of Trade Cycle. (March 2022)

Ans. :

A trade cycle is the series of exchanges, between a customer and supplier that take place when a commercial exchange is executed. A general trade cycle consists of following phases:

(i) Pre-Sales : It consist of two steps like Search and Negotiate. Customer search for required website for product to be purchased. In Negotiate step customer find a supplier who offers good quality product at cheaper price and then customer agrees the terms forwarded by supplier.

(ii) Execution : This phase consist of Order and Delivery. Customer sends an order for the selected product and after processing the order, customer receives delivery of the product.

Settlement : This phase consist of Invoice (if any) and Payment. Invoice means customer will receive a bill for purchased product and after confirmation of received product, customer will pay for the same.

After-Sales : This phase consists of warranty and After Sale Services. In warranty period, customer will get all maintenance services for free or at minimum cost. After sale services means customer will do complaints (if any) about the performance of product and get maintenance service from the supplier.

6.4 Modes of Payment

Explain various Modes of Payment. (March 2022)

Ans. :

Credit Cards : Credit cards are the most common ways for customers to pay online. Merchants can reach out to an international market with credit cards by integrating a payment gateway into their business.

Mobile Payments : Mobile payments offer a quick solution for customers to purchase on e-commerce websites. Examples are apps like Paytm, Google Pay, BHIM etc.

Bank Transfers : Bank transfer is used when money is send from one bank account to another. Transferring money from bank account is fast and safe then cash withdrawal. Example NEFT, IMPS etc.

E-wallets : E-wallet is a type of electronic card which is sued for transactions made online through a computer or smart phone. It is a type of prepaid account in which user can store money for future transaction. Examples are State Bank Buddy, Paytm Wallets.

6.5 Forms of E-commerce

Write Short notes on :

(i) M-commerce (Mobile Commerce) (March 2022)

(ii) Social Commerce

Ans. :

M-commerce (Mobile Commerce) :

(a) M-commerce is buying and selling of goods and services through wireless devices such as smart phones and tablets.

(b) M-Commerce enables the user to access online shopping platforms without using desktop computers.

(c) Applications of M-Commerce are Mobile banking, E-bill payment, ticket booking etc.

(ii) Social Commerce :

(a) Social Commerce is a form of electronic commerce that involves social media that supports social interaction.

(b) It enables shoppers to get advice from trusted individuals, find good and services and then purchase them.

(c) Social commerce is the use of networking websites such as Facebook, Instagram and Twitter to promote and sell products and services.

(d) The success is measured by the degree to which consumers interact with company's marketing through retweets, likes and shares.

6.6 E-Commerce Technology

8. Short note on Electronic Data Interchange (EDI). (March 2022)

Ans. :

Electronic Data Interchange (EDI)

(i) EDI is the electronic interchange of business information using a standardized format.

(ii) It is a process which allows one company to send information to another company electronically rather than on paper.

(iii) Business entities conducting business electronically are called trading partners.

(iv) In EDI, two most common documents which are exchanged are purchase order and invoices.

6.7 E-governance

9. What is E-Governance? (March 2022)

Ans. :

(i) E-Governance signifies the implementation of Information Technology in the Government processes.

(ii) The basic purpose of E-Governance is to simply processes for all, i.e. government, citizens, businesses etc. at all levels.

(iii) E-Governance delivers SMART (S-Simple, M-Moral, A-Accessible, R-Responsive, T-Transparent Government).

10. List the advantages of E-Governance. (March 2022)

Ans. :

Advantages of E-Governance are :

1. Improves delivery and efficiency of government services
2. Improved government interactions with business and industry

3. Citizen empowerment through access to information
4. More efficient government management
5. Less corruption in the administration
6. Increased transparency in administration
7. Greater convenience to citizens and businesses
8. Cost reductions and revenue growth
9. Increased legitimacy of government
10. Improved relations between the public authorities and civil society

11. Explain types of E-Governance. (March 2022)

Ans. :

E-Governance is of 4 types depending on the specific types of services :

1. **Government to Citizen (G2C) :**

(i) The Government to citizen refers to the government services which enable citizens to get access to wide variety of public services.

(ii) Most of the government services fall under G2C.

(iii) A citizen can have access to the services anytime from anywhere.

(iv) Services like license renewals and paying tax are essential in G2C.

(v) It also focuses on geographic land barriers.

2. **Government to Business (G2B) :**

(i) G2B is the exchange of services between Government and Business organizations.

(ii) G2B provides access to relevant forms needed to comply.

(iii) The G2B consists of many services exchanged between business sectors and government.

(iv) It aims at eliminating of paper work, cost and establish transparency in the business environment while interacting with government.

3. **Government to Government (G2G) :**

(i) The Government to Government refers to the interaction between different government departments, organizations and agencies.

(ii) In G2G government agencies can share the same database using online communication.

(iii) The government departments can work together.

(iv) G2G services can be at the local level or international level.

(v) It provides safe and secure inter relationship between domestic or foreign government.

4. Government to Employee (G2E) :

(i) The Government to Employee is the internal part of G2G sector.

(ii) G2E aims to bring employees together and improvise knowledge sharing.

(iii) G2E provides online facilities to the employees like applying for leave, reviewing salary payment record and checking the balance of holiday.

(iv) This sector provides human resource training and development.

(v) Examples of successful implementation of E-governance are e-Mitra, e-Seva project, CET (Common Entrance Test).

6.8 Security measures in E-Commerce

12. Explain process of Encryption.

Ans. :

(i) Encryption is widely used on the internet to protect user information being sent between a browser and a server.

(ii) This includes passwords, payment information and other personal information that should be considered private.

(iii) Encryption converts Plain text into Cipher text means non readable form of data.

(iv) Decryption is opposite of encryption i.e. it converts Cipher text into Plain text.

(v) **Encryption is of two types :** Symmetric and Asymmetric.

13. Write short notes on :

(a) Digital Signature (March 2022) **(b) Digital Certificate**

Ans. :

(a) Digital Signature :

(i) Digital Signature is also known as electronic signature.

(ii) Digital Signature guarantees the authenticity of an electronic document or message in digital communication and uses encryption technique to provide proof of original and unmodified documentation.

(iii) Digital Signatures are used in E-commerce, financial transactions.

(iv) This is the direct transfer of information between two partners.

(b) Digital Certificate :

(i) Digital Certificate is an electronic record "password" that allows a person, organization to exchange data securely over the internet using the public key infrastructure (PKI).

(ii) It is also known as public key certificate or identity certificate.

(iii) Using Digital Certificate information is transferred between two authorized partners who have digital certificates issued by supreme authority.

Self Assessment

Time : 1 Hour Marks: - 40

1. Fill in the blanks (05)

1. E-commerce is ____ and has no physical limitation.
2. _____ reduces the paper work and lower the transaction cost.
3. In _____model business sells its products to an intermediate buyer.
4. The _____ phase of Trade cycle consists of Order and Delivery.
5. Decryption converts _____ text into plain text.

2. True or False (05)

1. E-commerce application provides users with more options to compare and select cheaper and better option.
2. A blog can be written by an author for a business to improve sale of products.
3. M-commerce is buying and selling of goods and services through wireless device.
4. E-governance is of 6 types.
5. Digital Signature is also called as electronic signature.

3. MCQ one correct alternative. (05)

1. _____ Commerce has limited to business hours.
   (a) Traditional (b) Electronic
   (c) Digital (d) E-governance
2. On-line auction is example of _____model.
   (a) C2B (b) B2C (c) B2B (d) C2C
3. _____phase of Trade cycle consists of Order and Delivery
   (a) Presale (b) Execution (c) Settlement (d) After sales
4. Apps the BHIM, UPI, paytm are the examples of _____.
   (a) cheque payment (b) cash payment
   (c) EDI (d) Mobile payments
5. _____enables users to access online shopping platforms without using desktop computer.
   (a) M-Commerce (b) EDI
   (c) T-Commerce (d) Trade Cycle

4. MCQ two correct alternative. (06)

1. Examples of C2C model are _____
   (a) Network provider (b) OLX
   (c) EDI (d) Quikr
2. The After sales phase consists of _____ and _____
   (a) Order (b) Delivery (c) warranty (d) After sale service
3. Two common documents can be exchanged using EDI are ____ and _____.
   (a) Purchase orders (b) DOCs (c) Trading (d) Invoices

5. **MCQ three correct alternative.** (06)

1. Disadvantages of E-Commerce are _____
   (a) Global scope (b) Expensive
   (c) Cost saving (d) Lack of personal touch
   (e) Anytime shopping (f) Late deliveries
2. Applications of M-Commerce are _____.
   (a) Ticket booking (b) RTGS (c) NEFT
   (d) E-Bill payment (e) IMPS (f) Online auctions

6. **Match the Following.** (03)

A	B
1. M-Commerce	a) Form of E-commerce that involves social media that supports social interaction
2. Social Commerce	b) Electronic interchange of business information
3. EDI	c) Buying and selling of goods and services through smart phones and tablets
4. Credit cards	d) Consists of Search and negotiate
5. Presale	e) Cards used by customers to pay online

7. **Answer the following** (10)

a) Explain disadvantages of E-commerce **(Refer Page No 6.28 Q.2)**
b) Explain Traditional Commerce **(Refer Page No 6.28 Q.3)**
c) Short note on Consumer to Business **(Refer Page No 6.29 Q.4)**
d) Explain any two Modes of Payment. **(Refer Page No 6.30 Q.6)**
e) Short note on Electronic Data Interchange **(Refer Page No 6.31 Q.8)**

Answer of Questions

1. **Fill in the blanks**
   1) global 2) Electronic Commerce 3) Business to Business
   4) Execution 5) Cipher
2. **True or False**
   1) True 2) True 3) True
   4) False 5) True
3. **MCQ one correct alternative**
   1) A 2) D 3) B 4) D 5) B
4. **MCQ two correct alternative**
   1) (b, d) 2) (c, d) 3) (a, d)
5. **MCQ three correct alternative**
   1) (b, d, f) 2) (a, d, f)
6. **Match the Following**
   1. - (c), 2. - (a), 3. - (b), 4 - (f) 5 - (d) 6 - (e)

□□□

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INFORMATION TECHNOLOGY

PRACTICAL HANDBOOK SOP

SKILL ORIENTED PRACTICALS

Note :-
Students should file minimum 12 skill sets from the SOPs as follows :

Advanced Web Designing	- Any 05
JavaScript	- Any 04
Server side scripting	- Any 03

Practical Examination

Marks : 20 Duration : 3 Hrs

The marking scheme for practical examination is as follows for the annual exam of XII Science IT Std.

Q. No.	Description		Marks
1.	Write a Programme and Execute it		15
	For writing correct code	10 Marks	
	Execution of the code	05 Marks	
2.	Complete and Certified Journal		05
	Total		20

- Students should perform any one Practical based on the skill sets taught during the academic year.
- Practical slips should contain two questions on two different skill sets out of which students will opt any one.